

ISC Class XII Mathematics

Chapter 6 Applications of Derivatives

168 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 6.1 – Rate of Change of Quantities
- II. 6.2 – Tangents and Normals
- III. 6.3 – Solving Inequalities via Sign-Scheme Method
- IV. 6.4 – Increasing and Decreasing Functions
- V. 6.5 – Maxima and Minima Without Derivative Tests
- VI. 6.6 – Maxima and Minima Using Derivative Tests
- VII. 6.7 – Applications of Maxima and Minima

Compiled Answer Key – Step-by-step ISC board-style solutions

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Q. No.	Topic / Statement Summary	Type No.	Kind	Section
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Exercise 6.1 – Rate of Change of Quantities

1	Find the rate of change of the area of a circle with respect to its radius when the radius ...	1a	Numerical	6.1
2	If the radius of a circle is increasing at the rate of 0.5 cm/sec, at what rate ...	1b	Numerical	6.1
3	If the radius of a circle is increasing at the rate of 3 cm/sec, at what rate is ...	1c	Numerical	6.1
4	If the sides of a square are decreasing at the rate of 1.5 cm/sec, at what rate ...	1d	Numerical	6.1
5	If an edge of a variable cube is increasing at the rate of 10 cm/sec, at what ...	1e	Numerical	6.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
6	A balloon which always remains spherical has a variable radius. Find the rate at which its ...	1f	Numerical	6.1
7	If the radius of a balloon which always remains spherical is increasing at the rate of ...	1g	Numerical	6.1
8	If the radius of a soap bubble is increasing at the rate of $\frac{1}{2}$ cm/sec ...	1h	Numerical	6.1
9	A cylindrical tank of radius 10 m is being filled with wheat at the rate of ...	1i	Numerical	6.1
10	(i) The total revenue in rupees received from the sale of x units of a product is given ...	10	Numerical	6.1
11	Find the rate of change of area of a circle of radius r with regard to its radius. How ...	1a	Numerical	6.1
12	Find the rate of change of the whole surface of a closed circular cylinder of radius r ...	1j	Numerical	6.1
13	A stone is dropped into a quiet lake and waves move in circles at the speed of ...	1c	Numerical	6.1
14	A stone is dropped into a quiet lake and waves move in circles at a speed of ...	1c	Numerical	6.1
15	If the circumference of circle is increasing at the constant rate, prove that rate of ...	12a	Proof	6.1
16	Find the point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the ...	13	Numerical	6.1
17	The median of an equilateral triangle is increasing at the rate of ...	1k	Numerical	6.1
18	The length x of a rectangle is decreasing at the rate of 5 cm/minute and the ...	1l	Numerical	6.1
19	The volume of a cube is increasing at the rate of 9 cubic centimetres per second ...	1m	Numerical	6.1
20	A particle moves along the curve $y = \frac{2}{3}x^3 + 1$. Find the points on the curve ...	12c	Numerical	6.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
21	A man of height 2 metres walks at a uniform speed of 5 km/h away from a lamp post which is ...	18a	Numerical	6.1
22	A girl of height 1.6 m walks at the rate of 50 metres per minute away from a lamp which is ...	18a	Numerical	6.1
23	A kite is 120 m high and 130 m of string is out. If the kite is moving horizontally at the ...	18b	Numerical	6.1
24	A circular cone, with semi-vertical angle 45° , is fixed with its axis vertical and ...	18c	Numerical	6.1
25	Sand is pouring from a pipe at the rate of $12 \text{ cm}^3/\text{sec}$. The falling sand ...	18d	Numerical	6.1
26	A conical vessel whose height is 4 metres and of base radius 2 metres is being filled with ...	18e	Numerical	6.1
27	Water is dripping out at a steady rate of 1 cu cm/sec through a tiny hole at the ...	18f	Numerical	6.1
28	The population of rabbits in a forest is modelled by the function below: ...	24	Proof	6.1

Exercise 6.2 – Tangents and Normals

1	What is the slope of the tangent to the following curves:	25a	Numerical	6.2
(i)	$y = x^3 - x$ at $x = 2$?	25a	Numerical	6.2
(ii)	$y = 3x^4 - 4x$ at $x = 4$?	25a	Numerical	6.2
(iii)	$x^2 + 3y + y^2 = 5$ at the point $(1, 1)$?	25a	Numerical	6.2
(iv)	$y = x^3 - 3x + 2$ at the point whose x -coordinate is 3?	25a	Numerical	6.2
(v)	$y = 3x^2 + 4x$ at the point whose x -coordinate is -2 ?	25a	Numerical	6.2
(vi)	$y = 2 \sin^2(3x)$ at $x = \frac{\pi}{6}$?	25a	Numerical	6.2
2	What is the slope of the normal to the following curves:	25b	Numerical	6.2
(i)	$y = x^3 - 5x^2 - x + 1$ at the point $(1, -4)$?	25b	Numerical	6.2
(ii)	$y = 2x^2 + 3 \sin x$ at $x = 0$?	25b	Numerical	6.2
3	Find the point on the curve $y = 3x^2 - 2x + 1$ at which the slope of the tangent is 4.	27a	Numerical	6.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
4	Determine the point on the curve $y = 3x^2 - 1$ at which the slope of the tangent is 3.	27a	Numerical	6.2
5	At what point on the curve $y = x^2$ does the tangent make an angle of 45° with the ...	27b	Numerical	6.2
6	See parts below:	27c	Numerical	6.2
(i)	Find the point on the curve $y = x^2 - 2x + 3$ at which the tangent is parallel ...	27c	Numerical	6.2
(ii)	Find the point on the curve $y = x^2 + 2x - 1$, where the tangent is parallel to ...	27c	Numerical	6.2
7	Find the slope of the normal to the curve $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ at ...	25c	Numerical	6.2
8	Find the slope of the normal to the curve $x = 1 - a \sin \theta$, $y = b \cos^2 \theta$...	25c	Numerical	6.2
9	Write the equation of the normal to the curve $y^2 = 8x$ at the origin.	37a	Numerical	6.2
10	Find the slopes of the tangents at the three points where the curve $y = x(x^2 - 3x - 4)$...	25d	Numerical	6.2
11	Show that the tangents to the curve $y = 7x^3 + 11$ at the points where $x = 2$ and ...	27d	Proof	6.2
12	If the tangent to the curve $y = x^3 + ax + b$ at $(1, -6)$ is parallel to the line ...	27e	Numerical	6.2
13	Determine the point of the curve $y = 3x^2 - 5$ at which the tangent is perpendicular to a ...	27f	Numerical	6.2
14	Determine the points on the curve $y = x^3 - 3x^2 - 9x + 7$ at which the tangents are ...	27c	Numerical	6.2
15	Find the points on the curve $x^2 + y^2 - 2x - 3 = 0$ at which the tangents are parallel ...	27c	Numerical	6.2
16	Find the points on the curve $\frac{x^2}{4} + \frac{y^2}{25} = 1$ at which the ...	27c	Numerical	6.2
(i)	parallel to x -axis	27c	Numerical	6.2
(ii)	parallel to y -axis.	27c	Numerical	6.2
17	Find the points on the curve $x^2 + y^2 = 13$, the tangent at each one of which is ...	27g	Numerical	6.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
18	Find the points on the curve $y = x^3 - 3x^2 - 4x$ at which the tangent lines are parallel ...	27g	Numerical	6.2
19	Find the equations of the tangent and the normal to each of the following curves at ...	37b	Numerical	6.2
(i)	$y = x^3$ at $(1, 1)$	37b	Numerical	6.2
(ii)	$y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(1, 3)$	37b	Numerical	6.2
(iii)	$y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(0, 5)$	37b	Numerical	6.2
(iv)	$y = x^2$ at $(0, 0)$	37b	Numerical	6.2
(v)	$y = \sin^2 x$ at $x = \frac{\pi}{2}$	37b	Numerical	6.2
20	Find the equation of the normal to the curve $ay^2 = x^3$ at the point (am^2, am^3) .	37c	Numerical	6.2
21	See parts below:	37d	Numerical	6.2
(i)	Find the equation of the tangent to the curve $\sqrt{x} + \sqrt{y} = a$ at the ...	37d	Numerical	6.2
(ii)	Find the equations of the tangent and the normal to the curve ...	37d	Numerical	6.2
22	See parts below:	37e	Numerical	6.2
(i)	Find the equation of tangent to the curve given by $x = a \sin^3 t$...	37e	Numerical	6.2
(ii)	Find the equation of the tangent to the curve $x = \sin 3t, y = \cos 2t$ at ...	37e	Numerical	6.2
23	See parts below:	37f	Numerical	6.2
(i)	Find the equations of all lines having slope 2 and that are tangents to the ...	37f	Numerical	6.2
(ii)	Find the equation of the tangent to the curve $y = x + \frac{4}{x^2}$ which is ...	37f	Numerical	6.2
24	Find the equation of the tangent to the curve $y = x^2 - 2x + 7$ which is parallel to the line $2x - y + 9 = 0$	42a	Numerical	6.2
(i)	perpendicular to the line	42a	Numerical	6.2
(ii)	$5y - 15x = 13$.	42a	Numerical	6.2
25	See parts below:	42a	Numerical	6.2
(i)	Find the equations of the tangents to the curve $y = 4x^3 - 3x + 5$ which are ...	42a	Numerical	6.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Find the equations of the tangents to the curve $3x^2 - y^2 = 2$ which are ...	42a	Numerical	6.2
26	Find the equation of the tangent line to the curve $y = \sqrt{5x - 3} - 5$, which is ...	42a	Numerical	6.2
27	Find the equations of the normals to the curve $3x^2 - y^2 = 8$ which are parallel to the ...	42b	Numerical	6.2
28	Find the equation of the normal to the curve $y = x + \frac{1}{x}$, $x > 0$...	42b	Numerical	6.2
29	Find the equation of the normal to the curve $x^2 = 4y$ which passes through the point ...	37g	Numerical	6.2
30	(see parts below)	45a	Numerical	6.2
(i)	Find the angle of intersection between the curves $xy = 6$ and $x^2y = 12$.	45a	Numerical	6.2
(ii)	Find the angle of intersection between the curves $y = 4 - x^2$ and $y = x^2$.	45a	Numerical	6.2
31	Find the angle of intersection of the curves $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$ at ...	45a	Numerical	6.2
32	Prove that the curves $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ intersect at right angles at ...	45b	Proof	6.2
33	Find the condition that the curves $2x = y^2$ and $2xy = k$ intersect orthogonally.	45b	Numerical	6.2
34	Prove that the curves $xy = 4$ and $x^2 + y^2 = 8$ touch each other.	45c	Proof	6.2
35	Find the angle of intersection between the curves $y^2 = 4x$ and $x^2 = 4y$.	45a	Numerical	6.2

Exercise 6.3 – Solving Inequalities via Sign-Scheme Method

1	Solve the following inequalities for $x \in \mathbb{R}$:	48a	Numerical	6.3
(i)	$x(x - 2)(x - 5)(x + 3) > 0$	48a	Numerical	6.3
(ii)	$x^4 - 5x^2 + 4 \geq 0$	48a	Numerical	6.3
2	Find all real values of x which satisfy:	48b	Numerical	6.3
(i)	$x^3(x - 1)(x - 2) > 0$	48b	Numerical	6.3
(ii)	$x^2(x - 1)(x - 2) \leq 0$	48b	Numerical	6.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
3	Solve the following rational inequalities for $x \in \mathbb{R}$:	48c	Numerical	6.3
(i)	$\frac{1}{x-2} \leq 1$	48c	Numerical	6.3
(ii)	$\frac{(x+1)(x-3)}{x+2} \geq 0$	48c	Numerical	6.3

Exercise 6.4 – Increasing and Decreasing Functions

1	Show that the following functions are strictly increasing on $\mathbb{R}$:	51a	Proof	6.4
(i)	$f(x) = 3x + 17$	51a	Proof	6.4
(ii)	$f(x) = 7x - 3$	51a	Proof	6.4
2	See parts below:	51b	Proof	6.4
(i)	Show that the function f given by $f(x) = 3 - 7x$ is strictly decreasing.	51b	Proof	6.4
(ii)	Show that the function $f(x) = 1 - \frac{1}{x}$ is strictly increasing.	51b	Proof	6.4
3	See parts below:	51c	Proof	6.4
(i)	Prove that the function $f(x) = 3 + \frac{1}{x}$ is strictly decreasing.	51c	Proof	6.4
(ii)	Show that the function $f(x) = x - \frac{1}{x}$ is strictly increasing.	51c	Proof	6.4
4	See parts below:	51d	Proof	6.4
(i)	Show that the function $f(x) = e^{-2x}$ is strictly decreasing on $\mathbb{R}$.	51d	Proof	6.4
(ii)	Show that the function $f(x) = 5^x$ is strictly increasing.	51d	Proof	6.4
(iii)	Show that the function $f(x) = \log x$ is strictly increasing.	51d	Proof	6.4
(iv)	Show that the function $f(x) = -x + \cos x$ is decreasing.	51d	Proof	6.4
5	Find the values of 'a' for which the function $f(x) = b - ax + \sin x$ is increasing on ...	55a	Numerical	6.4
6	See parts below:	56a	Numerical	6.4
(i)	Find the interval in which the function f given by $f(x) = 7 - 4x - x^2$ is ...	56a	Numerical	6.4
(ii)	If $f(x) = x^3 - 48x$, find the interval where $f : \mathbb{R} \rightarrow \mathbb{R}$ is ...	56a	Numerical	6.4
(iii)	Find the interval in which the function $f(x) = 2x^3 - 3x$ is strictly ...	56a	Numerical	6.4
7	Show that the function f defined by $f(x) = (x - 1)e^x + 1$ is an increasing function ...	51e	Proof	6.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
8	For what value of k , the function $f(x) = kx^3 + 5$ is decreasing?	55b	Numerical	6.4
9	See parts below:	51f	Proof	6.4
(i)	Prove that the function $f(x) = ax + b$ is strictly decreasing iff $a < 0$.	51f	Proof	6.4
(ii)	Prove that the function $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.	51f	Proof	6.4
10	Prove that the function $f(x) = x^3 - 6x^2 + 12x + 5$ is increasing on $\mathbb{R}$.	51g	Proof	6.4
11	Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ is strictly increasing on $\dots$	51g	Proof	6.4
12	Prove that the function $f(x) = 5 - 3x + 3x^2 - x^3$ is decreasing on $\mathbb{R}$.	51g	Proof	6.4
13	See parts below:	51h	Proof	6.4
(i)	Show that $f(x) = 2x - \sin 2x$ is an increasing function.	51h	Proof	6.4
(ii)	Show that the function $f(x) = \frac{16 \sin x}{4 + \cos x} - x$ is strictly $\dots$	51h	Proof	6.4
14	See parts below:	62a	Numerical	6.4
(i)	Prove that the function f given by $f(x) = \log \cos x$ is strictly decreasing $\dots$	62a	Numerical	6.4
(ii)	If $f(x) = a(\tan x - \cot x)$, where $a > 0$, then find whether $f(x)$ is $\dots$	62a	Numerical	6.4
15	Determine whether the following functions are increasing or decreasing for the stated $\dots$	62b	Numerical	6.4
(i)	$f(x) = \frac{1}{x}, \quad x < 0$	62b	Numerical	6.4
(ii)	$f(x) = \frac{1}{1+x^2}, \quad x \geq 0$	62b	Numerical	6.4
(iii)	$f(x) = \cos^2 x$ in $[0, \frac{\pi}{2}]$	62b	Numerical	6.4
(iv)	$f(x) = -\frac{x}{2} + \sin x, \quad -\frac{\pi}{3} < x < \frac{\pi}{3}$	62b	Numerical	6.4
16	Which of the following functions are strictly decreasing on $(0, \frac{\pi}{2})$?	62c	Numerical	6.4
(i)	$\cos x$	62c	Numerical	6.4
(ii)	$\cos 2x$	62c	Numerical	6.4
(iii)	$\sin^2 x$	62c	Numerical	6.4
(iv)	$\tan x$	62c	Numerical	6.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
17	Consider the statement "There exists at least one value of $b \in \mathbb{R}$ for which ...	55c	Numerical	6.4
18	Prove that the function $f(x) = \sin(2x + \frac{\pi}{4})$ is decreasing for ...	51i	Proof	6.4
19	Prove that the function $f(x) = \sin x$ is:	51j	Proof	6.4
(i)	strictly increasing in $(0, \frac{\pi}{2})$	51j	Proof	6.4
(ii)	strictly decreasing in $(\frac{\pi}{2}, \pi)$	51j	Proof	6.4
(iii)	neither increasing nor decreasing in $(0, \pi)$	51j	Proof	6.4
20	Prove that the function f defined by $f(x) = \cos x$ is:	51k	Proof	6.4
(i)	strictly increasing in $(\pi, 2\pi)$	51k	Proof	6.4
(ii)	strictly decreasing in $(0, \pi)$	51k	Proof	6.4
(iii)	neither increasing nor decreasing in $(0, 2\pi)$	51k	Proof	6.4
21	Prove that the following functions are neither increasing nor decreasing:	69	Proof	6.4
(i)	$f(x) = x^2 - x + 1$ on $(-1, 1)$	69	Proof	6.4
(ii)	$f(x) = \frac{1}{x^2+1}$ on $\mathbb{R}$	69	Proof	6.4
22	Prove that the function $f(x) = x $ is:	51l	Proof	6.4
(i)	strictly increasing in $(0, \infty)$	51l	Proof	6.4
(ii)	strictly decreasing in $(-\infty, 0)$	51l	Proof	6.4
23	Prove that the function f given by $f(x) = \frac{4\sin x}{2+\cos x} - x$ is ...	51m	Proof	6.4
24	(see parts below)	56b	Numerical	6.4
(i)	Find the intervals in which the function $f(x) = 2x^3 - 3x^2 - 36x + 7$ is strictly ...	56b	Numerical	6.4
(ii)	Find the intervals in which the function $f(x) = 4x^3 - 6x^2 - 72x + 30$ is strictly ...	56b	Numerical	6.4
(iii)	Find the intervals in which the function $f(x) = 2x^3 - 9x^2 + 12x + 15$ is strictly ...	56b	Numerical	6.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(iv)	Find the intervals in which the function $f(x) = -2x^3 - 9x^2 - 12x + 1$ is strictly ...	56b	Numerical	6.4
(v)	Find the intervals in which the function $f(x) = x + \frac{1}{x}, x \neq 0$ is strictly ...	56c	Numerical	6.4
(vi)	Find the intervals in which the function $f(x) = \frac{4x^2+1}{x}, x \neq 0$ is ...	56c	Numerical	6.4
(vii)	Find the intervals in which the function $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$ is ...	56d	Numerical	6.4
(viii)	Find the intervals in which the function $f(x) = \frac{1}{4}x^4 - x^3 - 5x^2 + 24x + 12$...	56d	Numerical	6.4
(ix)	Find the intervals in which the function $f(x) = x^4 - 4x$ is strictly increasing or ...	56d	Numerical	6.4
(x)	Find the intervals in which the function $f(x) = (x-1)^3(x-2)^2$ is strictly ...	56e	Numerical	6.4
25	(see parts below)	56f	Numerical	6.4
(i)	Find the interval in which the function $f(x) = x^2e^{-x}$ is increasing.	56f	Numerical	6.4
(ii)	Find the intervals in which the function given by $f(x) = x^2e^x$ is strictly increasing ...	56f	Numerical	6.4
26	Find the intervals in which the function $f(x) = 2\log(x-2) - x^2 + 4x - 5$ is strictly ...	56g	Numerical	6.4
27	Find the intervals in which the function f given by ...	56h	Numerical	6.4
(i)	strictly increasing	56h	Numerical	6.4
(ii)	strictly decreasing	56h	Numerical	6.4

Exercise 6.5 – Maxima and Minima Without Derivative Tests

1	Find the maximum and the minimum values (if any) of the following functions:	79a	Numerical	6.5
(i)	$f(x) = x^2$	79a	Numerical	6.5
(ii)	$f(x) = (2x-1)^2 + 3$	79a	Numerical	6.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
2	Find the maximum and the minimum values (if any) of the following functions:	79a	Numerical	6.5
(i)	$f(x) = -(x-1)^2 + 10$	79a	Numerical	6.5
(ii)	$f(x) = 9x^2 + 12x + 2$	79a	Numerical	6.5
3	Find the maximum and the minimum values (if any) of the following functions:	79b	Numerical	6.5
(i)	$f(x) = x $	79b	Numerical	6.5
(ii)	$f(x) = x^3 + 1$	79b	Numerical	6.5
4	Find the maximum and the minimum values (if any) of the following functions:	79b	Numerical	6.5
(i)	$f(x) = x+2 - 1$	79b	Numerical	6.5
(ii)	$f(x) = 3 + x $	79b	Numerical	6.5
5	Find the maximum and the minimum values (if any) of the following functions:	79c	Numerical	6.5
(i)	$f(x) = - x+1 + 3$	79c	Numerical	6.5
(ii)	$f(x) = \sin 2x$	79c	Numerical	6.5
6	Find the maximum and the minimum values (if any) of the following functions:	79c	Numerical	6.5
(i)	$f(x) = \sin 2x + 5$	79c	Numerical	6.5
(ii)	$f(x) = \sin 4x + 3 $	79c	Numerical	6.5
7	Find the maximum and the minimum values (if any) of the following functions:	79c	Numerical	6.5
(i)	$f(x) = \sin 3x - 3$	79c	Numerical	6.5
(ii)	$f(x) = 4 - \cos 3x $	79c	Numerical	6.5
8	See parts below:	79c	Numerical	6.5
(i)	Is the minimum value of $\cos x$ zero?	79c	Numerical	6.5
(ii)	What are the minimum and maximum values of $2 - 3 \cos x$?	79c	Numerical	6.5
9	What are the maximum and minimum values of $3 \sin x + 4 \cos x$?	79d	Numerical	6.5
10	Find the maximum and minimum values, if any, of the following functions:	79e	Numerical	6.5
(i)	$f(x) = x$ in $(0, 1)$	79e	Numerical	6.5
(ii)	$f(x) = x$ in $[0, 1]$	79e	Numerical	6.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
11	Prove that the function $f(x) = x^3 + x^2 + x + 1$ does not have maxima or minima.	84	Proof	6.5
12	Find the (absolute) maximum and the (absolute) minimum values of the following ...	85a	Numerical	6.5
(i)	$f(x) = (x - 1)^2 + 3$ in $[-3, 1]$	85a	Numerical	6.5
(ii)	$f(x) = 2x^3 - 15x^2 + 36x + 1$ on the interval $[1, 5]$	85a	Numerical	6.5
(iii)	$f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$ in $[0, 3]$	85a	Numerical	6.5
(iv)	$f(x) = x + \sin x$ in $[0, 2\pi]$	85a	Numerical	6.5
(v)	$f(x) = \sin x + \sqrt{3} \cos x$ in $[0, \pi]$	85a	Numerical	6.5
(vi)	$f(x) = 3 + x + 1 $ in $[-2, 3]$	85a	Numerical	6.5
13	Find the maximum and minimum values of $x + \sin 2x$ on $[0, 2\pi]$. Also find points of ...	85b	Numerical	6.5
14	Find the absolute maximum and absolute minimum values of the function ...	85c	Numerical	6.5

Exercise 6.6 – Maxima and Minima Using Derivative Tests

1	Find the turning points of the following functions and distinguish between them. Also ...	88a	Numerical	6.6
(i)	$f(x) = 2x^3 - 21x^2 + 36x - 20$	88a	Numerical	6.6
(ii)	$f(x) = x^3 - 3x^2 + 3x$	88a	Numerical	6.6
2	Find the coordinates of stationary points on the curve $y = x^3 - 3x^2 - 9x + 7$, and ...	88a	Numerical	6.6
3	Find the points of local maxima and minima (if any) of each of the following ...	88b	Numerical	6.6
(i)	$f(x) = x^2$	88b	Numerical	6.6
(ii)	$f(x) = x^3 - 3x$	88b	Numerical	6.6
(iii)	$f(x) = x^3 - 12x$	88b	Numerical	6.6
(iv)	$f(x) = x^3 - 6x^2 + 9x + 15$	88b	Numerical	6.6
(v)	$f(x) = x^3 - 3x + 3$	88b	Numerical	6.6
(vi)	$f(x) = 2x^3 - 6x^2 + 6x + 5$	88b	Numerical	6.6
(vii)	$f(x) = \frac{x}{2} + \frac{2}{x}, \quad x > 0$	88b	Numerical	6.6
(viii)	$f(x) = \frac{1}{x^2 + 2}$	88b	Numerical	6.6
(ix)	$f(x) = 2 \sin x - x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$	88b	Numerical	6.6
(x)	$f(x) = \sin x + \cos x, \quad 0 < x < \frac{\pi}{2}$	88b	Numerical	6.6
(xi)	$f(x) = 3x^4 + 4x^3 - 12x^2 + 12$	88b	Numerical	6.6

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
4	For what values of a and b , the function $f(x) = x^3 + ax^2 + bx - 3$ has local ...	90a	Numerical	6.6
5	Given $2f(x) = \log x ^a - bx^2 + x$, if $f(x)$ has extreme values at $x = -1$ and ...	90b	Numerical	6.6
6	Aditya hires a speed boat in an island to do water sports, which is moving along a curve ...	92	Numerical	6.6
7	Anu enjoys a roller coaster ride by first going downwards and then upwards to the ...	93	Numerical	6.6
(i)	What is the rate of change of displacement with respect to the time?	93	Numerical	6.6
(ii)	How many seconds will it take her to go to its maximum height?	93	Numerical	6.6

Exercise 6.7 – Applications of Maxima and Minima

1	Find two numbers whose sum is 24 and whose product is as large as possible.	94	Numerical	6.7
2	The sum of two positive numbers is 20. Find the numbers	95	Numerical	6.7
(i)	if the sum of their squares is minimum.	95	Numerical	6.7
(ii)	if the product of the square of one and the cube of the other is maximum.	95	Numerical	6.7
3	The product of two positive numbers is 16. Find the numbers	96	Numerical	6.7
(i)	if their sum is least.	96	Numerical	6.7
(ii)	if the sum of one and the square of the other is least.	96	Numerical	6.7
4	Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.	97	Numerical	6.7
5	Find two positive numbers x and y such that their sum is 35 and the product x^2y^5 ...	97	Numerical	6.7
6	Sum of two numbers is 5. If the sum of the cubes of these numbers is least, then find the ...	98	Numerical	6.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
7	Three numbers are given whose sum is 180 and the ratio of first two of them is 1 : 2. If ...	99	Numerical	6.7
8	Find the maximum profit that a company can make, if the profit function is given by ...	100	Numerical	6.7
9	The cost (in Rs.) per article C , of manufacturing a certain article is given by the ...	101	Numerical	6.7
10	Prove that of all rectangles with given area, the square has the smallest perimeter.	102	Proof	6.7
11	Of all rectangles each of which has perimeter 24 cm, find the one having maximum area ...	103	Numerical	6.7
12	Prove that of all rectangles with given perimeter, the square has the largest area.	103	Proof	6.7
13	Prove that the perimeter of a right-angled triangle of given hypotenuse is maximum when ...	104	Proof	6.7
14	Two sides of a triangle have lengths a and b , and the angle between them is θ ...	105	Numerical	6.7
15	A right-angled triangle with constant area is given. Prove that the hypotenuse of the ...	106	Proof	6.7
16	The lengths of the sides of a triangle are $9 + x^2$, $9 + x^2$ and $18 - 2x^2$...	107	Numerical	6.7
(i)	the area of the triangle in terms of x .	107	Numerical	6.7
(ii)	the value of x for which this area is maximum.	107	Numerical	6.7
17	The perimeter of a triangle is 8 cm. If one of the sides is 3 cm, what are the lengths of ...	108	Numerical	6.7
18	A sheet of paper is to contain 18 cm^2 of printed matter. The margins at the top ...	109	Numerical	6.7
19	Show that of all rectangles inscribed in a given fixed circle the square has the maximum ...	110	Proof	6.7
20	See parts below:	111	Proof	6.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	The sum of perimeters of a circle and a square is k , where k is some ...	111	Proof	6.7
(ii)	Given the sum of perimeters of a circle and a square, show that the sum of areas ...	111	Proof	6.7
21	A wire 10 metres long is cut into two parts. One part is bent into the shape of a circle ...	112	Numerical	6.7
22	A wire of length 36 cm is cut into two pieces. One of the piece is turned into the form of ...	113	Numerical	6.7
23	A window is in the form of a rectangle above which there is a semicircle. If the perimeter ...	114	Proof	6.7
24	A rectangular area of 9000 m^2 is to be surrounded by a fence with two opposite ...	116	Numerical	6.7
25	See parts below:	117	Numerical	6.7
(i)	Find the point on the parabola $y^2 = 4x$ which is nearest to the point ...	117	Numerical	6.7
(ii)	Find the point on the curve $x^2 = 8y$ which is nearest to the point $(2, 4)$.	117	Numerical	6.7
26	A running track of 440 m is to be laid out enclosing a football field. The ...	118	Numerical	6.7
27	Find the maximum area of an isosceles triangle inscribed in the ellipse ...	119	Numerical	6.7
28	Show that the surface area of a closed cuboid with square base and given volume is minimum ...	120	Proof	6.7
29	An open box with a square base is to be made of given iron sheet of area ...	121	Proof	6.7
30	The volume of a closed metal box with a square base is 4096 cm^3 . The cost of ...	122	Numerical	6.7
31	An open tank with a square base and vertical sides is to be constructed from a metal sheet ...	123	Proof	6.7
32	A tank with rectangular base and rectangular sides, open at the top, is to be constructed ...	124	Numerical	6.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
33	A rectangular sheet of tin 45 cm by 24 cm is to be made into a box ...	125	Numerical	6.7
34	Show that the semi-vertical angle of a cone of the maximum volume and of given slant ...	126	Proof	6.7
35	Show that a right circular cylinder of given volume, which is open at the top, has minimum ...	127	Proof	6.7
36	See parts below:	128	Numerical	6.7
(i)	Show that a closed cylindrical vessel of given volume has the least (total) ...	128	Numerical	6.7
(ii)	A closed right circular cylinder has a volume of 2156 cu. cm. What will ...	128	Numerical	6.7
37	Show that the radius of a closed right circular cylinder of given surface area and maximum ...	129	Proof	6.7
38	Find the maximum volume of the cylinder which can be inscribed in a sphere of radius ...	130	Numerical	6.7
39	A cone is inscribed in a sphere of radius 12 cm. If the volume of the cone is ...	131	Numerical	6.7
40	Show that the altitude of a right circular cone of maximum curved surface area which can ...	132	Proof	6.7
41	Show that the height of a right circular cylinder of greatest volume which can be ...	133	Proof	6.7
42	The sum of the surface areas of a rectangular parallelopiped with sides x , $2x$ and ...	115	Numerical	6.7
43	A manufacturer plans to construct a cylindrical can to hold one cubic metre of oil. If the ...	134	Numerical	6.7

Formula & Standard Results Sheet

I. Rate of Change of Quantities (Ex 6.1)

Rate of Change

If $y = f(x)$, the rate of change of y w.r.t. x is $\frac{dy}{dx}$

For a quantity changing with time, differentiate w.r.t. t instead

Related Rates via Chain Rule

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

The core tool linking the rate of one quantity to the rate of another through a shared relation

Standard Mensuration Formulas Needed for Rates

$A = \pi r^2$, $C = 2\pi r$, $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$ (sphere); $V = \pi r^2 h$, $S = 2\pi r h + 2\pi r^2$ (cylinder); $V = \frac{1}{3}\pi r^2 h$ (cone)

Write the known geometric formula first, then differentiate both sides w.r.t. t

Related Rates with a Geometric Setup

Use similar triangles or a trigonometric ratio to relate the two changing quantities

Common in shadow/kite/leaking-vessel problems, where the direct formula is not given outright and must be derived first

Marginal Cost and Marginal Revenue

$$MC = \frac{dC}{dx}, \quad MR = \frac{dR}{dx}$$

The derivative of total cost/revenue gives the approximate cost/revenue of producing one additional unit

II. Tangents and Normals (Ex 6.2)

Slope of Tangent

$$m_T = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$$

Evaluate the derivative at the specific point of contact

Slope of Normal

$$m_N = -\frac{1}{m_T}$$

The normal is perpendicular to the tangent at the point of contact

Equation of Tangent

$$y - y_1 = m_T(x - x_1)$$

Point-slope form using the tangent's slope

Equation of Normal

$$y - y_1 = m_N(x - x_1)$$

Point-slope form using the normal's slope

Horizontal and Vertical Tangents

$$\begin{aligned} \text{Tangent parallel to } x\text{-axis} &\iff \frac{dy}{dx} = 0; \\ \text{tangent parallel to } y\text{-axis} &\iff \frac{dx}{dy} = 0 \end{aligned}$$

Solve the appropriate derivative equation to locate these points

Parallel and Perpendicular Lines

$$\text{Parallel: } m_1 = m_2; \quad \text{Perpendicular: } m_1 m_2 = -1$$

Match the tangent/normal slope against the given line's slope

Tangent/Normal to a Parametric Curve

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Compute the slope in terms of the parameter, then substitute the specific parameter value

Angle of Intersection of Two Curves

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

m_1, m_2 are the tangent slopes of the two curves at their common point

Orthogonal Curves

$$m_1 m_2 = -1$$

Condition for two curves to intersect at right angles

III. Solving Inequalities via Sign-Scheme Method (Ex 6.3)

Sign-Scheme (Wavy Curve) Method

$$f(x) = (x - a_1)(x - a_2) \cdots (x - a_n)$$

Sign alternates across each root on the number line; determine the sign in the rightmost interval first, then alternate

Repeated Factors

At a root of even multiplicity, the sign does NOT change; at odd multiplicity, the sign changes

Always check the multiplicity of each factor before alternating signs

Rational Inequalities

Bring all terms to one side, combine into a single fraction, then apply the sign-scheme method

Denominator roots are always excluded from the solution set, even if the inequality is non-strict

IV. Increasing and Decreasing Functions (Ex 6.4)

Monotonicity Test

$$f'(x) > 0 \implies f \text{ increasing}; \quad f'(x) < 0 \implies f \text{ decreasing}$$

Applied throughout an interval; equality at isolated points does not break strict monotonicity

Finding Intervals of Increase/Decrease

Find $f'(x)$, locate its roots, then use the sign-scheme method on $f'(x)$ across the domain

The roots of $f'(x)$ partition the domain into intervals of constant sign

Parameter Range for Monotonicity

$$\text{Require } f'(x) \geq 0 \text{ (or } \leq 0) \forall x \text{ in the given interval, then solve the resulting inequality for the parameter}$$

Often reduces to a condition on the discriminant or sign of a quadratic in the parameter

V. Maxima and Minima Without Derivative Tests (Ex 6.5)

Extrema by Inspection

For functions like $|x|$, $a \sin x + b \cos x$, or piecewise-defined functions, use the known range/behaviour of the function. Faster than calculus when the function's shape is already familiar, e.g. $-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$

No Extrema on an Open Interval

If f is strictly monotonic throughout an interval, it has no local maximum or minimum. A strictly increasing or decreasing function only attains extreme values at the endpoints, if at all

Absolute Extrema on a Closed Interval

Evaluate f at all critical points inside $[a, b]$ and at both endpoints a, b ; the maximum/minimum is the largest/smallest value. Critical points alone are not enough on a closed interval – the endpoints must always be checked too

VI. Maxima and Minima Using Derivative Tests (Ex 6.6)

First Derivative Test

$f'(x)$ changes sign from $+$ to $-$ at $c \implies$ local max at c ; $-$ to $+$ $\implies$ local min at c
Check the sign of $f'(x)$ just before and just after the critical point

Second Derivative Test

$f'(c) = 0$ and $f''(c) < 0 \implies$ local max; $f'(c) = 0$ and $f''(c) > 0 \implies$ local min
Faster than the first derivative test when $f''(c) \neq 0$; if $f''(c) = 0$, fall back to the first derivative test

Stationary Points

$f'(x) = 0$
Every local extremum of a differentiable function occurs at a stationary point, but not every stationary point is an extremum

VII. Applications of Maxima and Minima (Ex 6.7)**Constrained Optimization Method**

- (1) Express the quantity to be optimized as a function of one variable using the constraint.
0. (3) Confirm max/min using the first or second derivative test.

The recurring three-step structure behind every applied optimization problem, regardless of the specific geometric or physical context

Common Constraint Substitutions

Use the given fixed quantity (perimeter, surface area, volume, or a geometric relationship) to express one variable in terms of the others.

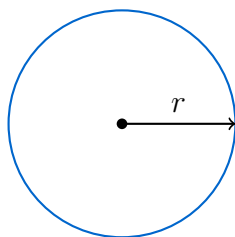
Reduces a two-variable objective function down to one variable so ordinary differentiation applies

6.1 – Rate of Change of Quantities

Question 1

Find the rate of change of the area of a circle with respect to its radius when the radius is 6 cm.

TYPE 1a Rate of Change of Standard Geometric Quantities – Rate of Change of Circle Area with respect to Radius



Circle of radius r

Given: $r = 6$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let A be the area and r be the radius of the circle. The formula for the area of a circle is:

$$A = \pi r^2$$

Differentiating A with respect to radius r :

$$\begin{aligned} \frac{dA}{dr} &= \frac{d}{dr}(\pi r^2) \\ &= \pi \frac{d}{dr}(r^2) \\ &= 2\pi r \end{aligned}$$

Using the standard derivative formula $\frac{d}{dr}(r^n) = nr^{n-1}$ with $n = 2$.

Now, substitute the given value $r = 6$ cm:

$$\begin{aligned} \left. \frac{dA}{dr} \right|_{r=6} &= 2\pi(6) \\ &= 12\pi \text{ cm}^2/\text{cm} \end{aligned}$$

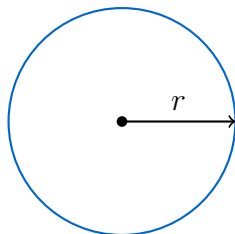
Final Answer

$12\pi \text{ cm}^2/\text{cm}$

Question 2

If the radius of a circle is increasing at the rate of 0.5 cm/sec, at what rate is its circumference increasing?

TYPE 1b Rate of Change of Standard Geometric Quantities – Time Rate of Change of Circle Circumference from Radius



Circle with radius r changing with time

Given: $\frac{dr}{dt} = 0.5$ cm/sec

Concept / Formula Used: $C = 2\pi r$

Solution:

Let C be the circumference and r be the radius of the circle at any time t . The circumference of a circle is given by:

$$C = 2\pi r$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dC}{dt} &= \frac{d}{dt}(2\pi r) \\ &= 2\pi \frac{dr}{dt}\end{aligned}$$

Given that $\frac{dr}{dt} = 0.5$ cm/sec, substituting this value:

$$\begin{aligned}\frac{dC}{dt} &= 2\pi \times 0.5 \\ &= \pi \text{ cm/sec}\end{aligned}$$

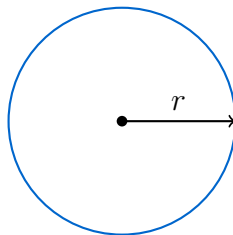
Final Answer

π cm/sec

Question 3

If the radius of a circle is increasing at the rate of 3 cm/sec, at what rate is its area increasing when its radius is 2 cm?

TYPE 1c Rate of Change of Standard Geometric Quantities – Time Rate of Change of Circle Area from Radius Rate



Expanding circle with radius r

Given: $\frac{dr}{dt} = 3$ cm/sec, $r = 2$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let A be the area and r be the radius of the circle at time t . The area of a circle is given by:

$$A = \pi r^2$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dA}{dt} &= \frac{d}{dt}(\pi r^2) \\ &= \pi \cdot 2r \frac{dr}{dt} \\ &= 2\pi r \frac{dr}{dt}\end{aligned}$$

Substitute $r = 2$ cm and $\frac{dr}{dt} = 3$ cm/sec:

$$\begin{aligned}\frac{dA}{dt} &= 2\pi(2)(3) \\ &= 12\pi \text{ cm}^2/\text{sec}\end{aligned}$$

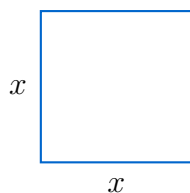
Final Answer

$$12\pi \text{ cm}^2/\text{sec}$$

Question 4

If the sides of a square are decreasing at the rate of 1.5 cm/sec, at what rate is its perimeter decreasing?

TYPE 1d Rate of Change of Standard Geometric Quantities – Time Rate of Change of Square Perimeter from Side Rate



Square of side x

Given: $\frac{dx}{dt} = -1.5$ cm/sec

Concept / Formula Used: $P = 4x$

Solution:

Let x be the length of a side of the square and P be its perimeter at time t . The perimeter of a square is given by:

$$P = 4x$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dP}{dt} &= \frac{d}{dt}(4x) \\ &= 4\frac{dx}{dt}\end{aligned}$$

Given that the side is decreasing at 1.5 cm/sec, we have $\frac{dx}{dt} = -1.5$ cm/sec. Substituting this value:

$$\begin{aligned}\frac{dP}{dt} &= 4(-1.5) \\ &= -6 \text{ cm/sec}\end{aligned}$$

The negative sign indicates a decrease in perimeter. Thus, the rate at which the perimeter is decreasing is 6 cm/sec.

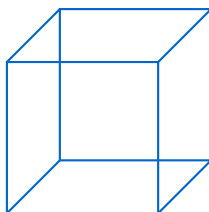
Final Answer

6 cm/sec

Question 5

If an edge of a variable cube is increasing at the rate of 10 cm/sec, at what rate is its volume increasing when its edge is 5 cm?

TYPE 1e Rate of Change of Standard Geometric Quantities – Time Rate of Change of Cube Volume from Edge Rate



x

Cube with edge x

Given: $\frac{dx}{dt} = 10$ cm/sec, $x = 5$ cm

Concept / Formula Used: $V = x^3$

Solution:

Let x be the side edge length and V be the volume of the cube at time t . The volume of a cube is given by:

$$V = x^3$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt}(x^3) \\ &= 3x^2 \frac{dx}{dt}\end{aligned}$$

Substitute $x = 5$ cm and $\frac{dx}{dt} = 10$ cm/sec:

$$\begin{aligned}\frac{dV}{dt} &= 3(5)^2(10) \\ &= 3(25)(10) \\ &= 750 \text{ cm}^3/\text{sec}\end{aligned}$$

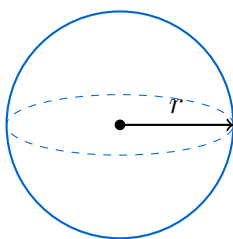
Final Answer

750 cm³/sec

Question 6

A balloon which always remains spherical has a variable radius. Find the rate at which its volume is increasing with respect to radius when the radius is 10 cm.

TYPE 1f Rate of Change of Standard Geometric Quantities – Rate of Change of Sphere Volume with respect to Radius



Spherical balloon of radius r

Given: $r = 10$ cm

Concept / Formula Used: $V = \frac{4}{3}\pi r^3$

Solution:

Let V be the volume and r be the radius of the spherical balloon. The volume of a sphere is given by:

$$V = \frac{4}{3}\pi r^3$$

Differentiating V with respect to radius r :

$$\begin{aligned}\frac{dV}{dr} &= \frac{d}{dr} \left(\frac{4}{3} \pi r^3 \right) \\ &= \frac{4}{3} \pi \cdot \frac{d}{dr} (r^3) \\ &= \frac{4}{3} \pi \cdot (3r^2) \\ &= 4\pi r^2\end{aligned}$$

Substitute $r = 10$ cm:

$$\begin{aligned}\left. \frac{dV}{dr} \right|_{r=10} &= 4\pi(10)^2 \\ &= 4\pi(100) \\ &= 400\pi \text{ cm}^3/\text{cm}\end{aligned}$$

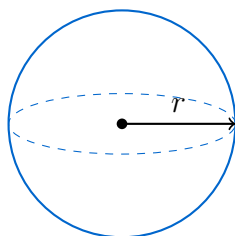
Final Answer

$$400\pi \text{ cm}^3/\text{cm}$$

Question 7

If the radius of a balloon which always remains spherical is increasing at the rate of 1.5 cm/sec, at what rate is its surface area increasing when its radius is 12 cm?

TYPE 1g Rate of Change of Standard Geometric Quantities – Time Rate of Change of Sphere Surface Area from Radius



Sphere of radius r

Given: $\frac{dr}{dt} = 1.5$ cm/sec, $r = 12$ cm

Concept / Formula Used: $S = 4\pi r^2$

Solution:

Let S be the surface area and r be the radius of the spherical balloon at time t . The surface area of a sphere is given by:

$$S = 4\pi r^2$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dS}{dt} &= \frac{d}{dt}(4\pi r^2) \\ &= 4\pi \cdot 2r \frac{dr}{dt} \\ &= 8\pi r \frac{dr}{dt}\end{aligned}$$

Substitute $r = 12$ cm and $\frac{dr}{dt} = 1.5$ cm/sec:

$$\begin{aligned}\frac{dS}{dt} &= 8\pi(12)(1.5) \\ &= 96\pi \times 1.5 \\ &= 144\pi \text{ cm}^2/\text{sec}\end{aligned}$$

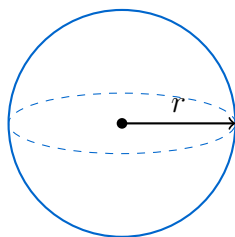
Final Answer

$144\pi \text{ cm}^2/\text{sec}$

Question 8

If the radius of a soap bubble is increasing at the rate of $\frac{1}{2}$ cm/sec, at what rate is its volume increasing when the radius is 1 cm?

TYPE 1h Rate of Change of Standard Geometric Quantities – Time Rate of Change of Sphere Volume from Radius Rate



Spherical soap bubble of radius r

Given: $\frac{dr}{dt} = \frac{1}{2}$ cm/sec, $r = 1$ cm

Concept / Formula Used: $V = \frac{4}{3}\pi r^3$

Solution:

Let V be the volume and r be the radius of the soap bubble at time t . The volume of a sphere is given by:

$$V = \frac{4}{3}\pi r^3$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt} \left(\frac{4}{3} \pi r^3 \right) \\ &= \frac{4}{3} \pi \cdot 3r^2 \frac{dr}{dt} \\ &= 4\pi r^2 \frac{dr}{dt}\end{aligned}$$

Substitute $r = 1$ cm and $\frac{dr}{dt} = \frac{1}{2}$ cm/sec:

$$\begin{aligned}\frac{dV}{dt} &= 4\pi(1)^2 \left(\frac{1}{2} \right) \\ &= 4\pi \cdot \frac{1}{2} \\ &= 2\pi \text{ cm}^3/\text{sec}\end{aligned}$$

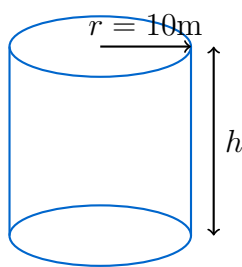
Final Answer

$2\pi \text{ cm}^3/\text{sec}$

Question 9

A cylindrical tank of radius 10 m is being filled with wheat at the rate of 314 cubic metre per hour. Find the rate at which the depth of the wheat is increasing, take $\pi = 3.14$.

TYPE 1i Rate of Change of Standard Geometric Quantities – Time Rate of Change of Cylinder Height with Constant R



Cylindrical tank filled with wheat

Given: $r = 10$ m, $\frac{dV}{dt} = 314 \text{ m}^3/\text{hr}$, $\pi = 3.14$

Concept / Formula Used: $V = \pi r^2 h$

Solution:

Let r be the radius, h be the depth of wheat, and V be the volume of wheat in the cylindrical tank at time t . The volume of the cylindrical tank is given by:

$$V = \pi r^2 h$$

Since the radius of the tank remains constant at $r = 10$ m:

$$\begin{aligned} V &= \pi(10)^2 h \\ &= 100\pi h \end{aligned}$$

Differentiating both sides with respect to time t :

$$\begin{aligned} \frac{dV}{dt} &= \frac{d}{dt}(100\pi h) \\ &= 100\pi \frac{dh}{dt} \end{aligned}$$

Given that $\frac{dV}{dt} = 314 \text{ m}^3/\text{hr}$ and $\pi = 3.14$:

$$\begin{aligned} 314 &= 100(3.14) \frac{dh}{dt} \\ 314 &= 314 \frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{314}{314} \\ \frac{dh}{dt} &= 1 \text{ m/hr} \end{aligned}$$

Final Answer

1 m/hr

Question 10

- (i) The total revenue in rupees received from the sale of x units of a product is given by $R(x) = 3x^2 + 36x + 5$. Find the marginal revenue when $x = 5$.
(ii) The total cost $C(x)$ associated with the production of x units of an item is given by $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$. Find the marginal cost when 3 units are produced.

TYPE 10 Marginal Cost and Marginal Revenue Calculations

- Given:** (i) $R(x) = 3x^2 + 36x + 5$, $x = 5$
(ii) $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$, $x = 3$

Concept / Formula Used: $MR = \frac{dR}{dx}$ $MC = \frac{dC}{dx}$

Solution:

Part (i): Calculation of Marginal Revenue Marginal Revenue (MR) is defined as the rate of change of total revenue with respect to the number of units sold:

$$MR = \frac{d}{dx}R(x)$$

Given $R(x) = 3x^2 + 36x + 5$, differentiate with respect to x :

$$\begin{aligned}\frac{d}{dx}R(x) &= \frac{d}{dx}(3x^2 + 36x + 5) \\ &= 3(2x) + 36(1) + 0 \\ &= 6x + 36\end{aligned}$$

Substitute $x = 5$:

$$\begin{aligned}MR &= 6(5) + 36 \\ &= 30 + 36 \\ &= 66\end{aligned}$$

Part (ii): Calculation of Marginal Cost Marginal Cost (MC) is defined as the rate of change of total cost with respect to the output produced:

$$MC = \frac{d}{dx}C(x)$$

Given $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$, differentiate with respect to x :

$$\begin{aligned}\frac{d}{dx}C(x) &= \frac{d}{dx}(0.005x^3 - 0.02x^2 + 30x + 5000) \\ &= 0.005(3x^2) - 0.02(2x) + 30(1) + 0 \\ &= 0.015x^2 - 0.04x + 30\end{aligned}$$

Substitute $x = 3$:

$$\begin{aligned}MC &= 0.015(3)^2 - 0.04(3) + 30 \\ &= 0.015(9) - 0.12 + 30 \\ &= 0.135 - 0.12 + 30 \\ &= 30.015\end{aligned}$$

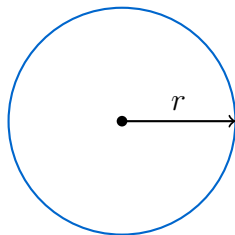
Final Answer

- (i) Marginal Revenue = Rs. 66
(ii) Marginal Cost = Rs. 30.015

Question 11

Find the rate of change of area of a circle of radius r with regard to its radius. How fast is the area changing with respect to the radius when the radius is 5 cm?

TYPE 1a Rate of Change of Standard Geometric Quantities – Rate of Change of Circle Area with respect to Radius

Circle with radius r **Given:** $r = 5$ cm**Concept / Formula Used:** $A = \pi r^2$ **Solution:**

Let A be the area of a circle of radius r . The area formula is:

$$A = \pi r^2$$

Differentiating A with respect to radius r :

$$\begin{aligned}\frac{dA}{dr} &= \frac{d}{dr}(\pi r^2) \\ &= \pi \cdot 2r \\ &= 2\pi r\end{aligned}$$

This gives the general rate of change of area with respect to the radius as $\frac{dA}{dr} = 2\pi r$.

When the radius $r = 5$ cm:

$$\begin{aligned}\left. \frac{dA}{dr} \right|_{r=5} &= 2\pi(5) \\ &= 10\pi \text{ cm}^2/\text{cm}\end{aligned}$$

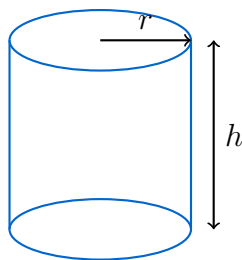
Final Answer

$$\frac{dA}{dr} = 2\pi r; \text{ when } r = 5 \text{ cm, the rate of change is } 10\pi \text{ cm}^2/\text{cm}$$

Question 12

Find the rate of change of the whole surface of a closed circular cylinder of radius r and height h with respect to change in radius.

TYPE 1j Rate of Change of Standard Geometric Quantities – Rate of Change of Cylinder Surface Area with respect to r

*Closed circular cylinder*

Given: Closed circular cylinder with radius r and height h

Concept / Formula Used: $S = 2\pi rh + 2\pi r^2$

Solution:

Let S be the total (whole) surface area of a closed circular cylinder. The formula for total surface area is:

$$S = 2\pi rh + 2\pi r^2$$

Differentiating S with respect to radius r , treating height h as constant:

$$\begin{aligned} \frac{dS}{dr} &= \frac{d}{dr}(2\pi rh + 2\pi r^2) \\ &= 2\pi h \frac{d}{dr}(r) + 2\pi \frac{d}{dr}(r^2) \\ &= 2\pi h(1) + 2\pi(2r) \\ &= 2\pi h + 4\pi r \\ &= 2\pi(h + 2r) \end{aligned}$$

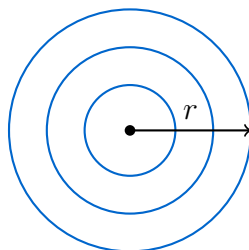
Final Answer

$$2\pi(h + 2r)$$

Question 13

A stone is dropped into a quiet lake and waves move in circles at the speed of 5 cm/sec. At the instant when the radius of the circular wave is 8 cm, how fast is the enclosed area increasing?

TYPE 1c Rate of Change of Standard Geometric Quantities – Time Rate of Change of Circle Area from Radius Rate

*Concentric circular ripples in water*

Given: $\frac{dr}{dt} = 5$ cm/sec, $r = 8$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let r be the radius of the circular wave and A be its enclosed area at time t . The area of a circle is given by:

$$A = \pi r^2$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dA}{dt} &= \frac{d}{dt}(\pi r^2) \\ &= 2\pi r \frac{dr}{dt}\end{aligned}$$

Substitute $r = 8$ cm and $\frac{dr}{dt} = 5$ cm/sec:

$$\begin{aligned}\frac{dA}{dt} &= 2\pi(8)(5) \\ &= 80\pi \text{ cm}^2/\text{sec}\end{aligned}$$

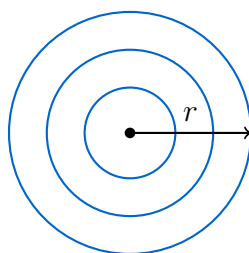
Final Answer

$80\pi \text{ cm}^2/\text{sec}$

Question 14

A stone is dropped into a quiet lake and waves move in circles at a speed of 4 cm/sec. At the instant when the radius of the circular wave is 10 cm, how fast is the enclosed area increasing?

TYPE 1c Rate of Change of Standard Geometric Quantities – Time Rate of Change of Circle Area from Radius Rate



Concentric circular ripples in water

Given: $\frac{dr}{dt} = 4$ cm/sec, $r = 10$ cm

Concept / Formula Used: $A = \pi r^2$

Solution:

Let r be the radius of the circular wave and A be its enclosed area at time t . The area of a circle is given by:

$$A = \pi r^2$$

Differentiating both sides with respect to time t using the chain rule:

$$\begin{aligned}\frac{dA}{dt} &= \frac{d}{dt}(\pi r^2) \\ &= 2\pi r \frac{dr}{dt}\end{aligned}$$

Substitute $r = 10$ cm and $\frac{dr}{dt} = 4$ cm/sec:

$$\begin{aligned}\frac{dA}{dt} &= 2\pi(10)(4) \\ &= 80\pi \text{ cm}^2/\text{sec}\end{aligned}$$

Final Answer

$$80\pi \text{ cm}^2/\text{sec}$$

Question 15

If the circumference of circle is increasing at the constant rate, prove that rate of change of area of circle is directly proportional to its radius.

TYPE 12a Relations Between Coordinate Rates of Change – Proving Proportionality of Rate of Change of Circle Area

To Prove: $\frac{dA}{dt} \propto r$

Concept / Formula Used: $C = 2\pi r$ $A = \pi r^2$

Solution:

Let r be the radius, C be the circumference, and A be the area of the circle at time t . Since the circumference is increasing at a constant rate, let:

$$\frac{dC}{dt} = k \quad (\text{where } k \text{ is a constant})$$

We know that the formula for circumference is:

$$C = 2\pi r$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dC}{dt} &= 2\pi \frac{dr}{dt} \\ k &= 2\pi \frac{dr}{dt} \\ \frac{dr}{dt} &= \frac{k}{2\pi}\end{aligned}$$

Now, the formula for the area of the circle is:

$$A = \pi r^2$$

Differentiating both sides with respect to time t :

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Substituting $\frac{dr}{dt} = \frac{k}{2\pi}$ into the expression for $\frac{dA}{dt}$:

$$\begin{aligned} \frac{dA}{dt} &= 2\pi r \left(\frac{k}{2\pi} \right) \\ \frac{dA}{dt} &= kr \end{aligned}$$

Since k is a constant, we conclude:

$$\frac{dA}{dt} \propto r$$

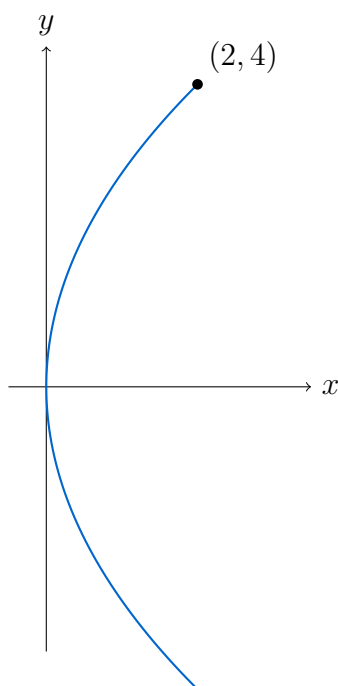
Hence Proved

LHS = RHS, hence proved that the rate of change of area is directly proportional to its radius.

Question 16

Find the point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the same rate.

TYPE 13 Finding Curve Points with Equal Coordinate Rates of Change



Parabola $y^2 = 8x$ with required point

Given: Curve: $y^2 = 8x$, rate condition: $\frac{dx}{dt} = \frac{dy}{dt} \neq 0$

Concept / Formula Used: t

Solution:

The equation of the curve is given as:

$$y^2 = 8x$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{d}{dt}(y^2) &= \frac{d}{dt}(8x) \\ 2y \frac{dy}{dt} &= 8 \frac{dx}{dt}\end{aligned}$$

Given that the abscissa (x) and ordinate (y) change at the same rate:

$$\frac{dx}{dt} = \frac{dy}{dt}$$

Substitute $\frac{dy}{dt} = \frac{dx}{dt}$ into the differentiated equation:

$$2y \frac{dx}{dt} = 8 \frac{dx}{dt}$$

Dividing both sides by $\frac{dx}{dt}$ (since $\frac{dx}{dt} \neq 0$):

$$\begin{aligned}2y &= 8 \\ y &= \frac{8}{2} \\ y &= 4\end{aligned}$$

Now, substitute $y = 4$ back into the original curve equation $y^2 = 8x$:

$$\begin{aligned}(4)^2 &= 8x \\ 16 &= 8x \\ x &= \frac{16}{8} \\ x &= 2\end{aligned}$$

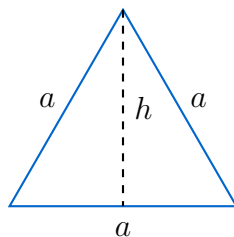
Therefore, the required point on the curve is $(2, 4)$.

Final Answer

$(2, 4)$

Question 17

The median of an equilateral triangle is increasing at the rate of $2\sqrt{3}$ cm/sec. Find the rate at which the side is increasing.



Equilateral triangle showing median h and side a

Given: $\frac{dh}{dt} = 2\sqrt{3}$ cm/sec

Concept / Formula Used: $h = \frac{\sqrt{3}}{2}a$

Solution:

Let a be the side length and h be the median (altitude) of the equilateral triangle at time t . The height/median of an equilateral triangle in terms of its side a is:

$$h = \frac{\sqrt{3}}{2}a$$

Differentiating both sides with respect to time t :

$$\frac{dh}{dt} = \frac{\sqrt{3}}{2} \frac{da}{dt}$$

Substitute the given value $\frac{dh}{dt} = 2\sqrt{3}$ cm/sec:

$$2\sqrt{3} = \frac{\sqrt{3}}{2} \frac{da}{dt}$$

Dividing both sides by $\sqrt{3}$:

$$\begin{aligned} 2 &= \frac{1}{2} \frac{da}{dt} \\ \frac{da}{dt} &= 2 \times 2 \\ \frac{da}{dt} &= 4 \text{ cm/sec} \end{aligned}$$

Final Answer

4 cm/sec

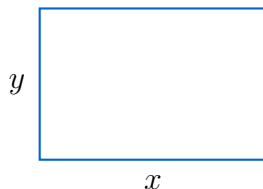
Question 18

The length x of a rectangle is decreasing at the rate of 5 cm/minute and the width y is increasing at the rate of 4 cm/minute. When $x = 8$ cm and $y = 6$ cm, find the rate of change of

(i) the perimeter

(ii) the area of the rectangle.

TYPE 11 Rate of Change of Standard Geometric Quantities – Time Rate of Change of Rectangle Perimeter and Area



Rectangle of length x and width y

Given: $\frac{dx}{dt} = -5$ cm/min, $\frac{dy}{dt} = 4$ cm/min, $x = 8$ cm, $y = 6$ cm

Concept / Formula Used: $P = 2(x + y)$ $A = xy$

Solution:

Part (i): Rate of change of perimeter Let P be the perimeter of the rectangle.

$$P = 2(x + y)$$

Differentiating with respect to time t :

$$\frac{dP}{dt} = 2 \left(\frac{dx}{dt} + \frac{dy}{dt} \right)$$

Substitute $\frac{dx}{dt} = -5$ cm/min and $\frac{dy}{dt} = 4$ cm/min:

$$\begin{aligned} \frac{dP}{dt} &= 2(-5 + 4) \\ &= 2(-1) \\ &= -2 \text{ cm/min} \end{aligned}$$

Thus, the perimeter is decreasing at the rate of 2 cm/min.

Part (ii): Rate of change of area Let A be the area of the rectangle.

$$A = xy$$

Differentiating with respect to time t using the product rule:

$$\frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt}$$

Substitute $x = 8$ cm, $y = 6$ cm, $\frac{dx}{dt} = -5$ cm/min, and $\frac{dy}{dt} = 4$ cm/min:

$$\begin{aligned} \frac{dA}{dt} &= (8)(4) + (6)(-5) \\ &= 32 - 30 \\ &= 2 \text{ cm}^2/\text{min} \end{aligned}$$

Thus, the area is increasing at the rate of $2 \text{ cm}^2/\text{min}$.

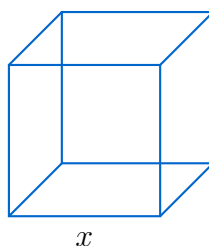
Final Answer

- (i) Rate of change of perimeter = -2 cm/min (decreasing at 2 cm/min)
 (ii) Rate of change of area = $2 \text{ cm}^2/\text{min}$

Question 19

The volume of a cube is increasing at the rate of 9 cubic centimetres per second. How fast is the surface area increasing when the length of the edge is 10 centimetres?

TYPE 1m Rate of Change of Standard Geometric Quantities – Time Rate of Change of Cube Surface Area from Volume



Cube of edge x

Given: $\frac{dV}{dt} = 9 \text{ cm}^3/\text{sec}$, $x = 10 \text{ cm}$

Concept / Formula Used: $V = x^3$ $S = 6x^2$

Solution:

Let x be the edge length, V be the volume, and S be the total surface area of the cube at time t . The volume of a cube is:

$$V = x^3$$

Differentiating with respect to time t :

$$\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$$

Given that $\frac{dV}{dt} = 9 \text{ cm}^3/\text{sec}$:

$$\begin{aligned} 9 &= 3x^2 \frac{dx}{dt} \\ \frac{dx}{dt} &= \frac{9}{3x^2} \\ \frac{dx}{dt} &= \frac{3}{x^2} \end{aligned}$$

Now, the surface area of the cube is given by:

$$S = 6x^2$$

Differentiating with respect to time t :

$$\frac{dS}{dt} = 12x \frac{dx}{dt}$$

Substitute $\frac{dx}{dt} = \frac{3}{x^2}$ into the derivative of surface area:

$$\begin{aligned}\frac{dS}{dt} &= 12x \left(\frac{3}{x^2} \right) \\ &= \frac{36}{x}\end{aligned}$$

Substitute $x = 10$ cm:

$$\begin{aligned}\frac{dS}{dt} &= \frac{36}{10} \\ &= 3.6 \text{ cm}^2/\text{sec}\end{aligned}$$

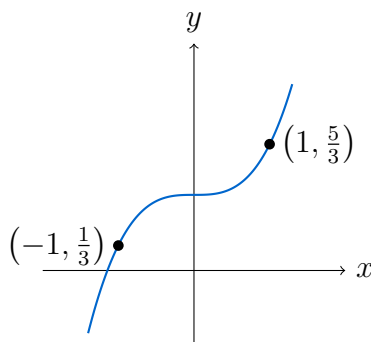
Final Answer

3.6 cm²/sec

Question 20

A particle moves along the curve $y = \frac{2}{3}x^3 + 1$. Find the points on the curve at which the y -coordinate is changing twice as fast as the x -coordinate.

TYPE 12c Relations Between Coordinate Rates of Change – Finding Curve Points with Proportional Coordinate Rates



Graph of curve $y = \frac{2}{3}x^3 + 1$ showing the two points

Given: Curve $y = \frac{2}{3}x^3 + 1$, rate condition $\frac{dy}{dt} = 2\frac{dx}{dt}$

Concept / Formula Used: $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$

Solution:

The given curve equation is:

$$y = \frac{2}{3}x^3 + 1$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dy}{dt} &= \frac{d}{dt} \left(\frac{2}{3}x^3 + 1 \right) \\ &= \frac{2}{3} \cdot 3x^2 \frac{dx}{dt} + 0 \\ &= 2x^2 \frac{dx}{dt}\end{aligned}$$

We are given that the y -coordinate changes twice as fast as the x -coordinate:

$$\frac{dy}{dt} = 2 \frac{dx}{dt}$$

Substituting $\frac{dy}{dt} = 2 \frac{dx}{dt}$ into the equation:

$$2 \frac{dx}{dt} = 2x^2 \frac{dx}{dt}$$

Assuming $\frac{dx}{dt} \neq 0$, dividing both sides by $2 \frac{dx}{dt}$:

$$\begin{aligned}1 &= x^2 \\ x^2 &= 1 \\ x &= \pm 1\end{aligned}$$

Case 1: When $x = 1$ Substitute $x = 1$ into the curve equation $y = \frac{2}{3}x^3 + 1$:

$$\begin{aligned}y &= \frac{2}{3}(1)^3 + 1 \\ &= \frac{2}{3} + 1 \\ &= \frac{5}{3}\end{aligned}$$

This gives the first point $(1, \frac{5}{3})$.

Case 2: When $x = -1$ Substitute $x = -1$ into the curve equation $y = \frac{2}{3}x^3 + 1$:

$$\begin{aligned}y &= \frac{2}{3}(-1)^3 + 1 \\ &= -\frac{2}{3} + 1 \\ &= \frac{1}{3}\end{aligned}$$

This gives the second point $(-1, \frac{1}{3})$.

Therefore, the required points on the curve are $(1, \frac{5}{3})$ and $(-1, \frac{1}{3})$.

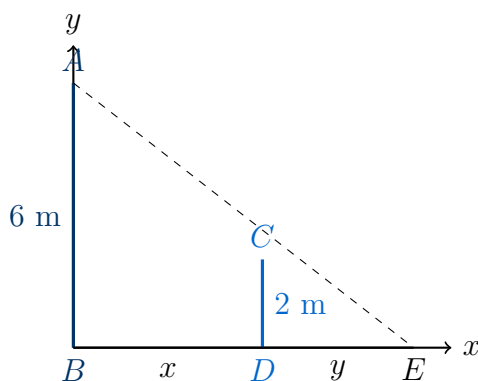
Final Answer

$$\left(1, \frac{5}{3}\right) \text{ and } \left(-1, \frac{1}{3}\right)$$

Question 21

A man of height 2 metres walks at a uniform speed of 5 km/h away from a lamp post which is 6 metres high. Find the rate at which the length of shadow increases.

TYPE 18a Related Rates via Geometric/Trigonometric Setup – Rate of Shadow Length Increase for a Walking Person



Man walking away from a lamp post

Given: Height of lamp post $AB = 6$ m, height of man $CD = 2$ m, rate of walking $\frac{dx}{dt} = 5$ km/h.

To Find: The rate at which the length of the shadow increases $\left(\frac{dy}{dt}\right)$.

Concept / Formula Used: $\frac{CD}{AB} = \frac{DE}{BE}$

Solution:

Let AB be the lamp post of height 6 m, and CD be the man of height 2 m.

Let x be the distance of the man from the lamp post ($BD = x$), and let y be the length of his shadow ($DE = y$).

The total distance from the base of the lamp post to the tip of the shadow is:

$$BE = BD + DE = x + y$$

From the geometric arrangement, $\triangle ECD$ and $\triangle EAB$ are similar triangles. Setting up the ratio of corresponding sides:

$$\begin{aligned}\frac{CD}{AB} &= \frac{DE}{BE} \\ \frac{2}{6} &= \frac{y}{x+y} \\ \frac{1}{3} &= \frac{y}{x+y}\end{aligned}$$

Cross-multiplying gives:

$$\begin{aligned}x + y &= 3y \\x &= 3y - y \\x &= 2y \\y &= \frac{1}{2}x\end{aligned}$$

Differentiating both sides with respect to time t :

$$\frac{dy}{dt} = \frac{1}{2} \frac{dx}{dt}$$

Substitute the given walking speed $\frac{dx}{dt} = 5$ km/h:

$$\begin{aligned}\frac{dy}{dt} &= \frac{1}{2} \times 5 \\ \frac{dy}{dt} &= \frac{5}{2} \text{ km/h} = 2.5 \text{ km/h}\end{aligned}$$

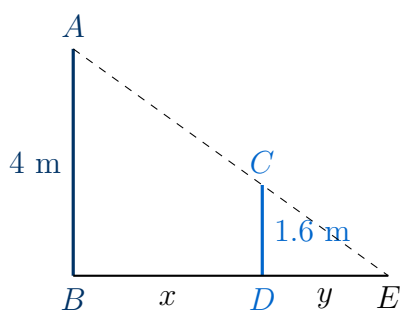
Final Answer

The length of the shadow increases at a rate of $\frac{5}{2}$ km/h (or 2.5 km/h).

Question 22

A girl of height 1.6 m walks at the rate of 50 metres per minute away from a lamp which is 4 m above the ground. How fast is the girl's shadow lengthening?

TYPE 18a Related Rates via Geometric/Trigonometric Setup – Rate of Shadow Length Increase for a Walking Person



Girl walking away from a lamp

Given: Height of lamp $AB = 4$ m, height of girl $CD = 1.6$ m, speed of girl $\frac{dx}{dt} = 50$ m/min.

To Find: The rate at which the girl's shadow is lengthening $\left(\frac{dy}{dt}\right)$.

Concept / Formula Used: $\triangle ECD \sim \triangle EAB$ $\frac{CD}{AB} = \frac{DE}{BE}$

Solution:

Let AB be the height of the lamp (4 m) and CD be the height of the girl (1.6 m).

Let x be the distance of the girl from the lamp post ($BD = x$), and let y be the length of her shadow ($DE = y$).

The total length BE is $x + y$.

Using the similar triangles $\triangle ECD$ and $\triangle EAB$:

$$\begin{aligned}\frac{CD}{AB} &= \frac{DE}{BE} \\ \frac{1.6}{4} &= \frac{y}{x+y} \\ \frac{2}{5} &= \frac{y}{x+y}\end{aligned}$$

Cross-multiplying:

$$\begin{aligned}2(x+y) &= 5y \\ 2x + 2y &= 5y \\ 2x &= 5y - 2y \\ 2x &= 3y \\ 3y &= 2x \\ y &= \frac{2}{3}x\end{aligned}$$

Differentiating both sides with respect to time t :

$$\frac{dy}{dt} = \frac{2}{3} \frac{dx}{dt}$$

Substitute the given rate $\frac{dx}{dt} = 50$ m/min:

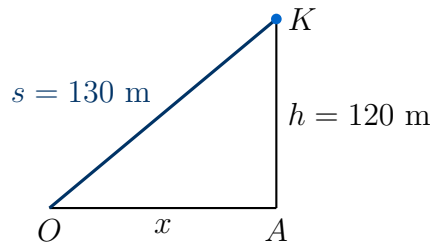
$$\begin{aligned}\frac{dy}{dt} &= \frac{2}{3} \times 50 \\ \frac{dy}{dt} &= \frac{100}{3} \text{ m/minute} = 33\frac{1}{3} \text{ m/minute}\end{aligned}$$

Final Answer

The girl's shadow is lengthening at the rate of $33\frac{1}{3}$ m/minute (or $\frac{100}{3}$ m/min).

Question 23

A kite is 120 m high and 130 m of string is out. If the kite is moving horizontally at the rate of 5.2 m/sec, find the rate at which the string is being paid out at that instant.



Kite flying horizontally

Given: Height of kite $h = 120$ m (constant), length of string $s = 130$ m, rate of horizontal movement $\frac{dx}{dt} = 5.2$ m/sec.

To Find: The rate at which the string is being paid out $\left(\frac{ds}{dt}\right)$.

Concept / Formula Used: $x^2 + h^2 = s^2$ h

Solution:

Let x be the horizontal distance of the kite from the observer, $h = 120$ m be the constant height, and s be the length of the string.

Applying Pythagoras Theorem to the right-angled triangle formed:

$$\begin{aligned}x^2 + h^2 &= s^2 \\x^2 + 120^2 &= s^2\end{aligned}$$

When the length of the string $s = 130$ m, we determine the horizontal distance x :

$$\begin{aligned}x^2 + 120^2 &= 130^2 \\x^2 + 14400 &= 16900 \\x^2 &= 16900 - 14400 \\x^2 &= 2500 \\x &= 50 \text{ m}\end{aligned}$$

Now, differentiating the equation $x^2 + 120^2 = s^2$ with respect to time t :

$$\begin{aligned}\frac{d}{dt}(x^2 + 120^2) &= \frac{d}{dt}(s^2) \\2x \frac{dx}{dt} + 0 &= 2s \frac{ds}{dt} \\x \frac{dx}{dt} &= s \frac{ds}{dt}\end{aligned}$$

Substitute $x = 50$ m, $\frac{dx}{dt} = 5.2$ m/sec, and $s = 130$ m:

$$50 \times 5.2 = 130 \times \frac{ds}{dt}$$

$$260 = 130 \frac{ds}{dt}$$

$$\frac{ds}{dt} = \frac{260}{130}$$

$$\frac{ds}{dt} = 2 \text{ m/sec}$$

Final Answer

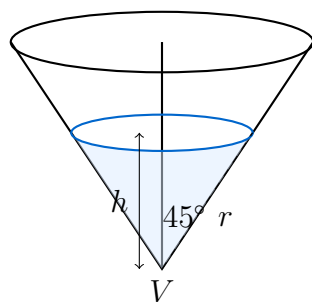
The rate at which the string is being paid out is 2 m/sec.

Question 24

A circular cone, with semi-vertical angle 45° , is fixed with its axis vertical and its vertex downwards. Water is poured into the cone at the rate of 2 cm^3 per second. Find the rate at which the depth of the water is increasing when the depth is 4 cm.

TYPE 18c Related Rates via Geometric/Trigonometric Setup – Rate of Water Level Rise in a Cone with Given Semi-Ve

★ Likely Exam Question



Inverted conical vessel containing water

Given: Semi-vertical angle $\alpha = 45^\circ$, rate of volume increase $\frac{dV}{dt} = 2 \text{ cm}^3/\text{sec}$, depth $h = 4$ cm.

To Find: The rate of increase of depth of water $\left(\frac{dh}{dt}\right)$ when $h = 4$ cm.

Concept / Formula Used: $V = \frac{1}{3}\pi r^2 h$ $\tan \alpha = \frac{r}{h}$

Solution:

Let r be the radius and h be the depth of water in the conical vessel at any time t .
Given semi-vertical angle $\alpha = 45^\circ$, we relate r and h :

$$\begin{aligned}\tan 45^\circ &= \frac{r}{h} \\ 1 &= \frac{r}{h} \\ r &= h\end{aligned}$$

Substituting $r = h$ into the volume formula of the cone:

$$\begin{aligned}V &= \frac{1}{3}\pi r^2 h \\ V &= \frac{1}{3}\pi (h)^2 h \\ V &= \frac{1}{3}\pi h^3\end{aligned}$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt} \left(\frac{1}{3}\pi h^3 \right) \\ \frac{dV}{dt} &= \frac{1}{3}\pi \cdot 3h^2 \frac{dh}{dt} \\ \frac{dV}{dt} &= \pi h^2 \frac{dh}{dt}\end{aligned}$$

Substitute $\frac{dV}{dt} = 2 \text{ cm}^3/\text{sec}$ and $h = 4 \text{ cm}$:

$$\begin{aligned}2 &= \pi(4)^2 \frac{dh}{dt} \\ 2 &= 16\pi \frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{2}{16\pi} \\ \frac{dh}{dt} &= \frac{1}{8\pi} \text{ cm/sec}\end{aligned}$$

Final Answer

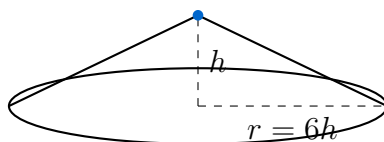
The depth of the water is increasing at the rate of $\frac{1}{8\pi} \text{ cm/sec}$.

Question 25

Sand is pouring from a pipe at the rate of $12 \text{ cm}^3/\text{sec}$. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-sixth of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm ?

TYPE 18d Related Rates via Geometric/Trigonometric Setup – Rate of Height Increase of a Sand Cone with Proportion

★ Likely Exam Question



Cone of falling sand

Given: Rate of volume increase $\frac{dV}{dt} = 12 \text{ cm}^3/\text{sec}$, relation $h = \frac{1}{6}r$, height $h = 4 \text{ cm}$.

To Find: The rate at which the height of the sand cone is increasing $\left(\frac{dh}{dt}\right)$.

Concept / Formula Used: $V = \frac{1}{3}\pi r^2 h$

Solution:

Let r be the base radius and h be the height of the sand cone at any time t .

Given relation between height and radius:

$$h = \frac{1}{6}r$$

$$r = 6h$$

Substitute $r = 6h$ into the volume formula of the cone:

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi(6h)^2 h$$

$$V = \frac{1}{3}\pi(36h^2)h$$

$$V = 12\pi h^3$$

Differentiating with respect to time t :

$$\frac{dV}{dt} = \frac{d}{dt}(12\pi h^3)$$

$$\frac{dV}{dt} = 12\pi \cdot 3h^2 \frac{dh}{dt}$$

$$\frac{dV}{dt} = 36\pi h^2 \frac{dh}{dt}$$

Substitute the given values $\frac{dV}{dt} = 12 \text{ cm}^3/\text{sec}$ and $h = 4 \text{ cm}$:

$$12 = 36\pi(4)^2 \frac{dh}{dt}$$

$$12 = 36\pi(16) \frac{dh}{dt}$$

$$12 = 576\pi \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{12}{576\pi}$$

$$\frac{dh}{dt} = \frac{1}{48\pi} \text{ cm/sec}$$

Final Answer

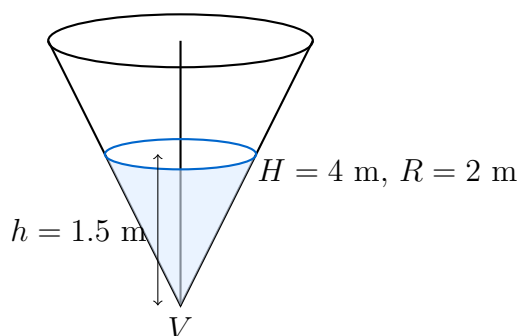
The height of the sand cone is increasing at the rate of $\frac{1}{48\pi} \text{ cm/sec}$.

Common Mistake: Failing to substitute $r = 6h$ before differentiating introduces $\frac{dr}{dt}$ needlessly, complicating the derivative.

Question 26

A conical vessel whose height is 4 metres and of base radius 2 metres is being filled with water at the rate of 0.75 cubic metres per minute. Find the rate at which the level of the water is rising when the depth of water is 1.5 metres.

TYPE 18e Related Rates via Geometric/Trigonometric Setup – Rate of Water Level Rise in a Conical Vessel of Given D



Water level in a conical vessel

Given: Height of conical vessel $H = 4 \text{ m}$, base radius $R = 2 \text{ m}$, rate of filling $\frac{dV}{dt} = 0.75 \text{ m}^3/\text{min}$, water depth $h = 1.5 \text{ m}$.

To Find: The rate of rise of water level $\left(\frac{dh}{dt}\right)$.

Concept / Formula Used: $\frac{r}{h} = \frac{R}{H}$ $V = \frac{1}{3}\pi r^2 h$

Solution:

Let r and h be the radius of the top surface of water and the depth of water at time t , respectively.

By similar vertical cross-sectional triangles:

$$\begin{aligned}\frac{r}{h} &= \frac{R}{H} \\ \frac{r}{h} &= \frac{2}{4} = \frac{1}{2} \\ r &= \frac{1}{2}h\end{aligned}$$

Substitute $r = \frac{1}{2}h$ into the volume equation:

$$\begin{aligned}V &= \frac{1}{3}\pi r^2 h \\ V &= \frac{1}{3}\pi \left(\frac{1}{2}h\right)^2 h \\ V &= \frac{1}{3}\pi \left(\frac{1}{4}h^2\right) h \\ V &= \frac{1}{12}\pi h^3\end{aligned}$$

Differentiating both sides with respect to time t :

$$\begin{aligned}\frac{dV}{dt} &= \frac{d}{dt} \left(\frac{1}{12}\pi h^3 \right) \\ \frac{dV}{dt} &= \frac{1}{12}\pi \cdot 3h^2 \frac{dh}{dt} \\ \frac{dV}{dt} &= \frac{1}{4}\pi h^2 \frac{dh}{dt}\end{aligned}$$

Convert the given values to fractions: $\frac{dV}{dt} = 0.75 = \frac{3}{4}$ m³/min, and $h = 1.5 = \frac{3}{2}$ m.

Substitute these into the differentiated equation:

$$\begin{aligned}\frac{3}{4} &= \frac{1}{4}\pi \left(\frac{3}{2}\right)^2 \frac{dh}{dt} \\ \frac{3}{4} &= \frac{1}{4}\pi \cdot \frac{9}{4} \frac{dh}{dt} \\ \frac{3}{4} &= \frac{9\pi}{16} \frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{3}{4} \times \frac{16}{9\pi} \\ \frac{dh}{dt} &= \frac{4}{3\pi} \text{ m/min}\end{aligned}$$

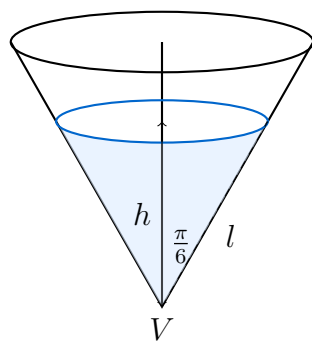
Final Answer

The level of the water is rising at a rate of $\frac{4}{3\pi}$ m/min.

Question 27

Water is dripping out at a steady rate of 1 cu cm/sec through a tiny hole at the vertex of the conical vessel, whose axis is vertical. When the slant height of water in the vessel is 4 cm, find the rate of decrease of slant height, where the semi-vertical angle of the cone is $\frac{\pi}{6}$.

TYPE 18f Related Rates via Geometric/Trigonometric Setup – Rate of Decrease of Slant Height in a Leaking Conical Vessel

★ Likely Exam Question

Water leaking from vertex of conical vessel

Given: Rate of volume change $\frac{dV}{dt} = -1$ cm³/sec, semi-vertical angle $\alpha = \frac{\pi}{6}$, slant height $l = 4$ cm.

To Find: The rate of decrease of slant height $\left(-\frac{dl}{dt}\right)$.

Concept / Formula Used: α l $r = l \sin \alpha$ $h = l \cos \alpha$ $V = \frac{1}{3}\pi r^2 h$

Solution:

Let l be the slant height, r be the radius, and h be the height of the cone of water at time t .

Expressing radius r and height h in terms of slant height l and semi-vertical angle $\alpha = \frac{\pi}{6}$:

$$r = l \sin \left(\frac{\pi}{6}\right) = l \cdot \frac{1}{2} = \frac{l}{2}$$

$$h = l \cos \left(\frac{\pi}{6}\right) = l \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}l}{2}$$

Substitute r and h into the volume formula $V = \frac{1}{3}\pi r^2 h$:

$$V = \frac{1}{3}\pi \left(\frac{l}{2}\right)^2 \left(\frac{\sqrt{3}l}{2}\right)$$

$$V = \frac{1}{3}\pi \cdot \frac{l^2}{4} \cdot \frac{\sqrt{3}l}{2}$$

$$V = \frac{\sqrt{3}\pi}{24} l^3$$

Differentiating with respect to time t :

$$\frac{dV}{dt} = \frac{d}{dt} \left(\frac{\sqrt{3}\pi}{24} l^3 \right)$$

$$\frac{dV}{dt} = \frac{\sqrt{3}\pi}{24} \cdot 3l^2 \frac{dl}{dt}$$

$$\frac{dV}{dt} = \frac{\sqrt{3}\pi}{8} l^2 \frac{dl}{dt}$$

Since water is dripping out at $1 \text{ cm}^3/\text{sec}$, $\frac{dV}{dt} = -1 \text{ cm}^3/\text{sec}$. Substitute $\frac{dV}{dt} = -1$ and $l = 4 \text{ cm}$:

$$-1 = \frac{\sqrt{3}\pi}{8} (4)^2 \frac{dl}{dt}$$

$$-1 = \frac{\sqrt{3}\pi}{8} (16) \frac{dl}{dt}$$

$$-1 = 2\sqrt{3}\pi \frac{dl}{dt}$$

$$\frac{dl}{dt} = -\frac{1}{2\sqrt{3}\pi} \text{ cm/sec}$$

The negative sign signifies a decrease in slant height.

Final Answer

The rate of decrease of slant height is $\frac{1}{2\sqrt{3}\pi} \text{ cm/sec}$.

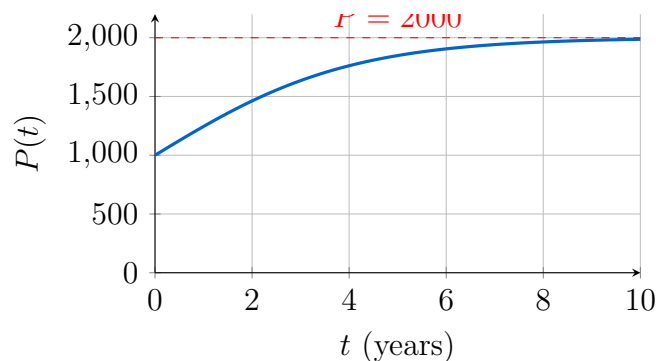
Question 28

The population of rabbits in a forest is modelled by the function below:

$$P(t) = \frac{2000}{1 + e^{-0.5t}}$$

where P represents the population of rabbits in t years. Determine whether the rabbit population is increasing or not, and justify your answer.

TYPE 24 Determining Growth Behavior of a Logistic Population Model



$$\text{Rabbit population } P(t) = \frac{2000}{1+e^{-0.5t}} \text{ over time}$$

Given: Population model $P(t) = \frac{2000}{1+e^{-0.5t}}$ for time $t \geq 0$.

To Find: Determine if $P(t)$ is increasing by analyzing the sign of the first derivative $\frac{dP}{dt}$.

Concept / Formula Used: $f(t) \quad f'(t) > 0 \quad t$

Solution:

To determine whether the rabbit population is increasing, we evaluate the derivative of $P(t)$ with respect to t .

Rewrite $P(t)$ in index form:

$$P(t) = 2000 (1 + e^{-0.5t})^{-1}$$

Differentiating with respect to t using the chain rule:

$$\frac{dP}{dt} = 2000 \cdot (-1) (1 + e^{-0.5t})^{-2} \cdot \frac{d}{dt} (1 + e^{-0.5t})$$

Evaluating the inner derivative $\frac{d}{dt} (1 + e^{-0.5t})$:

$$\begin{aligned} \frac{d}{dt} (1 + e^{-0.5t}) &= 0 + e^{-0.5t} \cdot (-0.5) \\ &= -0.5e^{-0.5t} \end{aligned}$$

Substituting this back into the expression for $\frac{dP}{dt}$:

$$\begin{aligned} \frac{dP}{dt} &= -2000 (1 + e^{-0.5t})^{-2} \cdot (-0.5e^{-0.5t}) \\ &= \frac{2000 \times 0.5 e^{-0.5t}}{(1 + e^{-0.5t})^2} \\ &= \frac{1000e^{-0.5t}}{(1 + e^{-0.5t})^2} \end{aligned}$$

Now, analyze the components of $\frac{dP}{dt}$ for all $t \geq 0$:

$$e^{-0.5t} > 0 \quad \text{for all real } t$$

$$(1 + e^{-0.5t})^2 > 0 \quad \text{for all real } t$$

$$1000 > 0$$

Therefore:

$$\frac{dP}{dt} > 0 \quad \text{for all } t \geq 0$$

Since the derivative $\frac{dP}{dt}$ is strictly positive for all $t \geq 0$, the function $P(t)$ is strictly increasing.

Hence Proved

Since $\frac{dP}{dt} > 0$ for all $t \geq 0$, the rabbit population is increasing.

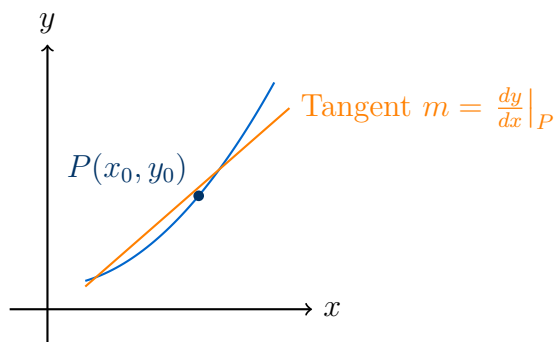
6.2 – Tangents and Normals

Question 1

What is the slope of the tangent to the following curves:

- (i) $y = x^3 - x$ at $x = 2$?
- (ii) $y = 3x^4 - 4x$ at $x = 4$?
- (iii) $x^2 + 3y + y^2 = 5$ at the point $(1, 1)$?
- (iv) $y = x^3 - 3x + 2$ at the point whose x -coordinate is 3?
- (v) $y = 3x^2 + 4x$ at the point whose x -coordinate is -2 ?
- (vi) $y = 2 \sin^2(3x)$ at $x = \frac{\pi}{6}$?

TYPE 25a Finding Slope of Tangent or Normal at a Point – Finding Slope of Tangent to Curve at Given Point



Slope of tangent to a curve $y = f(x)$ at a given point $P(x_0, y_0)$

Given: Curves $y = f(x)$ and specific points or x -coordinates as given in parts (i) through (vi).

Concept / Formula Used: m $y = f(x)$ (x_0, y_0) $m = \left. \frac{dy}{dx} \right|_{(x_0, y_0)}$

Solution:

(i) Given curve: $y = x^3 - x$

Differentiating with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^3 - x) \\ &= 3x^2 - 1\end{aligned}$$

Evaluating at $x = 2$:

$$\begin{aligned}m &= 3(2)^2 - 1 \\ &= 3(4) - 1 \\ &= 12 - 1 = 11\end{aligned}$$

(ii) Given curve: $y = 3x^4 - 4x$

Differentiating with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(3x^4 - 4x) \\ &= 12x^3 - 4\end{aligned}$$

Evaluating at $x = 4$:

$$\begin{aligned}m &= 12(4)^3 - 4 \\ &= 12(64) - 4 \\ &= 768 - 4 = 764\end{aligned}$$

(iii) Given curve: $x^2 + 3y + y^2 = 5$

Differentiating implicitly with respect to x :

$$\begin{aligned}\frac{d}{dx}(x^2) + \frac{d}{dx}(3y) + \frac{d}{dx}(y^2) &= \frac{d}{dx}(5) \\ 2x + 3\frac{dy}{dx} + 2y\frac{dy}{dx} &= 0 \\ (3 + 2y)\frac{dy}{dx} &= -2x \\ \frac{dy}{dx} &= -\frac{2x}{3 + 2y}\end{aligned}$$

Evaluating at $(x, y) = (1, 1)$:

$$\begin{aligned} m &= -\frac{2(1)}{3+2(1)} \\ &= -\frac{2}{3+2} = -\frac{2}{5} \end{aligned}$$

(iv) Given curve: $y = x^3 - 3x + 2$

Differentiating with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^3 - 3x + 2) \\ &= 3x^2 - 3 \end{aligned}$$

Evaluating at $x = 3$:

$$\begin{aligned} m &= 3(3)^2 - 3 \\ &= 3(9) - 3 \\ &= 27 - 3 = 24 \end{aligned}$$

(v) Given curve: $y = 3x^2 + 4x$

Differentiating with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(3x^2 + 4x) \\ &= 6x + 4 \end{aligned}$$

Evaluating at $x = -2$:

$$\begin{aligned} m &= 6(-2) + 4 \\ &= -12 + 4 = -8 \end{aligned}$$

(vi) Given curve: $y = 2 \sin^2(3x)$

Using chain rule to differentiate with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= 2 \cdot 2 \sin(3x) \cdot \frac{d}{dx}(\sin 3x) \\ &= 4 \sin(3x) \cdot \cos(3x) \cdot 3 \\ &= 6(2 \sin 3x \cos 3x) \\ &= 6 \sin(6x) \end{aligned}$$

Evaluating at $x = \frac{\pi}{6}$:

$$\begin{aligned} m &= 6 \sin \left(6 \cdot \frac{\pi}{6} \right) \\ &= 6 \sin(\pi) \\ &= 6(0) = 0 \end{aligned}$$

Final Answer

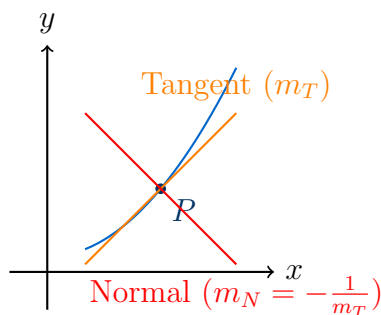
- (i) Slope $m = 11$
- (ii) Slope $m = 764$
- (iii) Slope $m = -\frac{2}{5}$
- (iv) Slope $m = 24$
- (v) Slope $m = -8$
- (vi) Slope $m = 0$

Question 2

What is the slope of the normal to the following curves:

- (i) $y = x^3 - 5x^2 - x + 1$ at the point $(1, -4)$?
- (ii) $y = 2x^2 + 3 \sin x$ at $x = 0$?

TYPE 25b Finding Slope of Tangent or Normal at a Point – Finding Slope of Normal to Curve at Given Point



Relationship between tangent and normal at point P

Given: Curves and points for parts (i) and (ii).

Concept / Formula Used: $m_N \quad m_T \quad m_N = -\frac{1}{m_T} = -\frac{1}{\left.\frac{dy}{dx}\right|_{(x_0, y_0)}}$

Solution:

- (i) Given curve: $y = x^3 - 5x^2 - x + 1$

Differentiating with respect to x :

$$\frac{dy}{dx} = 3x^2 - 10x - 1$$

At the point $(1, -4)$, the slope of the tangent is:

$$\begin{aligned} m_T &= 3(1)^2 - 10(1) - 1 \\ &= 3 - 10 - 1 = -8 \end{aligned}$$

Therefore, the slope of the normal is:

$$\begin{aligned} m_N &= -\frac{1}{m_T} \\ &= -\frac{1}{-8} = \frac{1}{8} \end{aligned}$$

(ii) Given curve: $y = 2x^2 + 3 \sin x$

Differentiating with respect to x :

$$\frac{dy}{dx} = 4x + 3 \cos x$$

At $x = 0$, the slope of the tangent is:

$$\begin{aligned} m_T &= 4(0) + 3 \cos(0) \\ &= 0 + 3(1) = 3 \end{aligned}$$

Therefore, the slope of the normal is:

$$\begin{aligned} m_N &= -\frac{1}{m_T} \\ &= -\frac{1}{3} \end{aligned}$$

Final Answer

- (i) Slope of normal $m_N = \frac{1}{8}$
 (ii) Slope of normal $m_N = -\frac{1}{3}$

Question 3

Find the point on the curve $y = 3x^2 - 2x + 1$ at which the slope of the tangent is 4.

TYPE 27a Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Given Tangent Slope

Given: Curve equation $y = 3x^2 - 2x + 1$ and slope of tangent $m = 4$.

Concept / Formula Used: $m = \frac{dy}{dx}$ $\frac{dy}{dx} = m$ x

Solution:

The given curve equation is:

$$y = 3x^2 - 2x + 1$$

Differentiating with respect to x :

$$\frac{dy}{dx} = 6x - 2$$

Given that the slope of the tangent is 4:

$$\begin{aligned}\frac{dy}{dx} = 4 &\implies 6x - 2 = 4 \\ 6x &= 6 \\ x &= 1\end{aligned}$$

Substituting $x = 1$ back into the curve equation to find y :

$$\begin{aligned}y &= 3(1)^2 - 2(1) + 1 \\ &= 3 - 2 + 1 \\ &= 2\end{aligned}$$

Final Answer

The required point on the curve is $(1, 2)$.

Question 4

Determine the point on the curve $y = 3x^2 - 1$ at which the slope of the tangent is 3.

TYPE 27a Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Given Tangent Slope

Given: Curve $y = 3x^2 - 1$ and target tangent slope $m = 3$.

Concept / Formula Used: $\frac{dy}{dx} = m$

Solution:

Given curve:

$$y = 3x^2 - 1$$

Differentiating with respect to x :

$$\frac{dy}{dx} = 6x$$

We are given that the slope of the tangent is 3:

$$\begin{aligned}6x &= 3 \\ x &= \frac{3}{6} = \frac{1}{2}\end{aligned}$$

Substituting $x = \frac{1}{2}$ into the original equation:

$$\begin{aligned}y &= 3\left(\frac{1}{2}\right)^2 - 1 \\ &= 3\left(\frac{1}{4}\right) - 1 \\ &= \frac{3}{4} - 1 = -\frac{1}{4}\end{aligned}$$

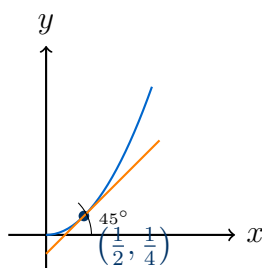
Final Answer

The required point on the curve is $\left(\frac{1}{2}, -\frac{1}{4}\right)$.

Question 5

At what point on the curve $y = x^2$ does the tangent make an angle of 45° with the x -axis?

TYPE 27b Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Given Tangent In



Tangent line making angle $\theta = 45^\circ$ with positive x -axis

Given: Curve equation $y = x^2$ and angle $\theta = 45^\circ$ made by tangent with the positive x -axis.

Concept / Formula Used: $m = \tan \theta = \frac{dy}{dx}$

Solution:

Given angle $\theta = 45^\circ$, the slope of the tangent is:

$$m = \tan(45^\circ) = 1$$

Differentiating the curve $y = x^2$ with respect to x :

$$\frac{dy}{dx} = 2x$$

Equating the derivative to the slope:

$$\begin{aligned} 2x &= 1 \\ x &= \frac{1}{2} \end{aligned}$$

Substituting $x = \frac{1}{2}$ into $y = x^2$:

$$y = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Final Answer

The point on the curve is $\left(\frac{1}{2}, \frac{1}{4}\right)$.

Question 6

- (i) Find the point on the curve $y = x^2 - 2x + 3$ at which the tangent is parallel to x -axis.
 (ii) Find the point on the curve $y = x^2 + 2x - 1$, where the tangent is parallel to x -axis.

TYPE 27c Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to x -axis

Given: Curves given in (i) and (ii), with tangent condition being parallel to x -axis.

Concept / Formula Used: x $m = 0$ $\frac{dy}{dx} = 0$

Solution:

- (i) Given curve: $y = x^2 - 2x + 3$

Differentiating with respect to x :

$$\frac{dy}{dx} = 2x - 2$$

Since the tangent is parallel to the x -axis:

$$\begin{aligned}\frac{dy}{dx} &= 0 \\ 2x - 2 &= 0 \\ 2x &= 2 \implies x = 1\end{aligned}$$

Substituting $x = 1$ into $y = x^2 - 2x + 3$:

$$\begin{aligned}y &= (1)^2 - 2(1) + 3 \\ &= 1 - 2 + 3 = 2\end{aligned}$$

Thus, the required point is $(1, 2)$.

- (ii) Given curve: $y = x^2 + 2x - 1$

Differentiating with respect to x :

$$\frac{dy}{dx} = 2x + 2$$

Since the tangent is parallel to the x -axis:

$$\begin{aligned}\frac{dy}{dx} &= 0 \\ 2x + 2 &= 0 \\ 2x &= -2 \implies x = -1\end{aligned}$$

Substituting $x = -1$ into $y = x^2 + 2x - 1$:

$$\begin{aligned} y &= (-1)^2 + 2(-1) - 1 \\ &= 1 - 2 - 1 = -2 \end{aligned}$$

Thus, the required point is $(-1, -2)$.

Final Answer

- (i) Point is $(1, 2)$
- (ii) Point is $(-1, -2)$

Question 7

Find the slope of the normal to the curve $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ at $\theta = \frac{\pi}{4}$.

TYPE 25c Finding Slope of Tangent or Normal at a Point – Finding Slope of Normal to Parametric Curve

Given: Parametric equations $x = a \cos^3 \theta$, $y = a \sin^3 \theta$, and parameter value $\theta = \frac{\pi}{4}$.

Concept / Formula Used: $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \quad m_N = -\frac{1}{m_T}$

Solution:

Differentiating x with respect to θ :

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{d}{d\theta}(a \cos^3 \theta) \\ &= a \cdot 3 \cos^2 \theta \cdot (-\sin \theta) \\ &= -3a \cos^2 \theta \sin \theta \end{aligned}$$

Differentiating y with respect to θ :

$$\begin{aligned} \frac{dy}{d\theta} &= \frac{d}{d\theta}(a \sin^3 \theta) \\ &= a \cdot 3 \sin^2 \theta \cdot (\cos \theta) \\ &= 3a \sin^2 \theta \cos \theta \end{aligned}$$

Now calculating $\frac{dy}{dx}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy/d\theta}{dx/d\theta} \\ &= \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} \\ &= -\frac{\sin \theta}{\cos \theta} = -\tan \theta \end{aligned}$$

Slope of tangent m_T at $\theta = \frac{\pi}{4}$:

$$m_T = -\tan\left(\frac{\pi}{4}\right) = -1$$

Therefore, the slope of the normal m_N is:

$$\begin{aligned} m_N &= -\frac{1}{m_T} \\ &= -\frac{1}{-1} = 1 \end{aligned}$$

Final Answer

The slope of the normal at $\theta = \frac{\pi}{4}$ is 1.

Question 8

Find the slope of the normal to the curve $x = 1 - a \sin \theta$, $y = b \cos^2 \theta$ at $\theta = \frac{\pi}{2}$.

TYPE 25c Finding Slope of Tangent or Normal at a Point – Finding Slope of Normal to Parametric Curve

Given: Parametric equations $x = 1 - a \sin \theta$, $y = b \cos^2 \theta$ at $\theta = \frac{\pi}{2}$.

Concept / Formula Used: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$ $m_N = -\frac{1}{m_T}$

Solution:

Differentiating x with respect to θ :

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{d}{d\theta}(1 - a \sin \theta) \\ &= -a \cos \theta \end{aligned}$$

Differentiating y with respect to θ :

$$\begin{aligned} \frac{dy}{d\theta} &= \frac{d}{d\theta}(b \cos^2 \theta) \\ &= b \cdot 2 \cos \theta \cdot (-\sin \theta) \\ &= -2b \sin \theta \cos \theta \end{aligned}$$

Now finding $\frac{dy}{dx}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{-2b \sin \theta \cos \theta}{-a \cos \theta} \\ &= \frac{2b \sin \theta}{a} \end{aligned}$$

Evaluating the slope of tangent m_T at $\theta = \frac{\pi}{2}$:

$$\begin{aligned} m_T &= \frac{2b \sin(\pi/2)}{a} \\ &= \frac{2b(1)}{a} = \frac{2b}{a} \end{aligned}$$

Therefore, the slope of the normal m_N is:

$$\begin{aligned} m_N &= -\frac{1}{m_T} \\ &= -\frac{1}{\frac{2b}{a}} = -\frac{a}{2b} \end{aligned}$$

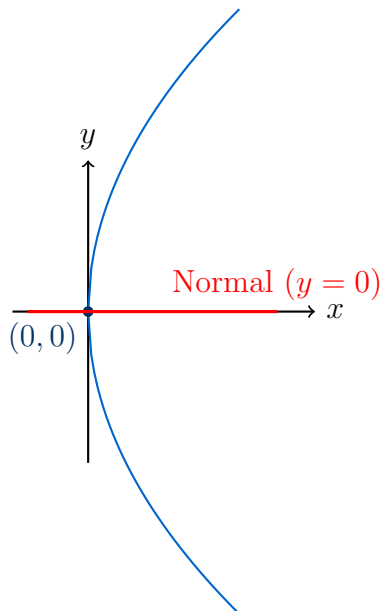
Final Answer

The slope of the normal at $\theta = \frac{\pi}{2}$ is $-\frac{a}{2b}$.

Question 9

Write the equation of the normal to the curve $y^2 = 8x$ at the origin.

TYPE 37a Finding Equation of Tangent or Normal – Finding Equation of Normal at Origin with Vertical Tangent



Parabola $y^2 = 8x$ with tangent (vertical line $x = 0$) and normal (horizontal line $y = 0$) at origin

Given: Curve equation $y^2 = 8x$ and point of contact $(0, 0)$.

Concept / Formula Used: (x_1, y_1) m $y - y_1 = m(x - x_1)$

Solution:

Differentiating $y^2 = 8x$ implicitly with respect to x :

$$2y \frac{dy}{dx} = 8$$

$$\frac{dy}{dx} = \frac{4}{y}$$

At the origin $(0, 0)$, as $y \rightarrow 0$, $\frac{dy}{dx} \rightarrow \infty$. This means the tangent line at $(0, 0)$ is vertical ($x = 0$, the y -axis).

Since the normal is perpendicular to the tangent line at $(0, 0)$, the normal line must be horizontal. A horizontal line passing through the origin $(0, 0)$ has slope $m_N = 0$.

Equation of normal line passing through $(0, 0)$ with slope $m_N = 0$:

$$y - 0 = 0(x - 0)$$

$$y = 0$$

Final Answer

The equation of the normal at the origin is $y = 0$ (the x -axis).

Question 10

Find the slopes of the tangents at the three points where the curve $y = x(x^2 - 3x - 4)$ cuts the x -axis.

TYPE 25d Finding Slope of Tangent or Normal at a Point – Finding Tangent Slopes at Curve x -Intercepts

Given: Curve equation $y = x(x^2 - 3x - 4)$.

Concept / Formula Used: x $y = 0$ $m = \frac{dy}{dx}$

Solution:

First, find the points where the curve cuts the x -axis by setting $y = 0$:

$$x(x^2 - 3x - 4) = 0$$

$$x(x - 4)(x + 1) = 0$$

This gives three roots:

$$x = 0, \quad x = 4, \quad x = -1$$

So the curve cuts the x -axis at $(0, 0)$, $(4, 0)$, and $(-1, 0)$.

Next, expand the curve equation for differentiation:

$$y = x^3 - 3x^2 - 4x$$

Differentiating with respect to x :

$$\frac{dy}{dx} = 3x^2 - 6x - 4$$

Now, evaluate the slope of tangent at each of the three points:

(i) At $(0, 0)$:

$$m_1 = 3(0)^2 - 6(0) - 4 = -4$$

(ii) At $(4, 0)$:

$$\begin{aligned} m_2 &= 3(4)^2 - 6(4) - 4 \\ &= 3(16) - 24 - 4 \\ &= 48 - 28 = 20 \end{aligned}$$

(iii) At $(-1, 0)$:

$$\begin{aligned} m_3 &= 3(-1)^2 - 6(-1) - 4 \\ &= 3(1) + 6 - 4 \\ &= 9 - 4 = 5 \end{aligned}$$

Final Answer

The slopes of the tangents at the three points are -4 , 20 , and 5 .

Question 11

Show that the tangents to the curve $y = 7x^3 + 11$ at the points where $x = 2$ and $x = -2$ are parallel.

TYPE 27d Finding Curve Points Satisfying a Tangent/Normal Condition – Proving Tangents at Two Points are Parallel

To Prove: The slopes of the tangents to $y = 7x^3 + 11$ at $x = 2$ and $x = -2$ are equal.

Concept / Formula Used: $m_1 = m_2$

Solution:

The equation of the given curve is:

$$y = 7x^3 + 11$$

Differentiating with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(7x^3 + 11) \\ &= 21x^2 \end{aligned}$$

Let m_1 be the slope of the tangent at $x = 2$:

$$\begin{aligned} m_1 &= \left. \frac{dy}{dx} \right|_{x=2} \\ &= 21(2)^2 \\ &= 21(4) = 84 \end{aligned}$$

Let m_2 be the slope of the tangent at $x = -2$:

$$\begin{aligned} m_2 &= \left. \frac{dy}{dx} \right|_{x=-2} \\ &= 21(-2)^2 \\ &= 21(4) = 84 \end{aligned}$$

Since $m_1 = m_2 = 84$, the slopes of the tangents at $x = 2$ and $x = -2$ are equal.

Hence Proved

Since $m_1 = m_2 = 84$, the tangents to the curve at $x = 2$ and $x = -2$ are parallel.

Question 12

If the tangent to the curve $y = x^3 + ax + b$ at $(1, -6)$ is parallel to the line $2x - 2y + 7 = 0$, find a and b .

TYPE 27e Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Parameters from Tangent Par

★ Likely Exam Question

Given: Curve $y = x^3 + ax + b$, point of contact $(1, -6)$, and line $2x - 2y + 7 = 0$.

Solution:

First, since $(1, -6)$ lies on the curve $y = x^3 + ax + b$:

$$\begin{aligned} -6 &= (1)^3 + a(1) + b \\ -6 &= 1 + a + b \\ a + b &= -7 \quad \text{--- (Equation 1)} \end{aligned}$$

Next, find the slope of the given line $2x - 2y + 7 = 0$:

$$\begin{aligned} 2y &= 2x + 7 \\ y &= x + \frac{7}{2} \end{aligned}$$

Comparing with $y = mx + c$, slope of the line $m_L = 1$.

Now differentiate the curve $y = x^3 + ax + b$ with respect to x :

$$\frac{dy}{dx} = 3x^2 + a$$

The slope of the tangent at $(1, -6)$ is:

$$m_T = 3(1)^2 + a = 3 + a$$

Since the tangent is parallel to the line $2x - 2y + 7 = 0$:

$$m_T = m_L$$

$$3 + a = 1$$

$$a = 1 - 3 = -2$$

Substitute $a = -2$ into Equation 1:

$$-2 + b = -7$$

$$b = -7 + 2 = -5$$

Final Answer

$$a = -2, \quad b = -5$$

Question 13

Determine the point of the curve $y = 3x^2 - 5$ at which the tangent is perpendicular to a line whose slope is $-\frac{1}{3}$.

TYPE 27f Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Perpendicular

Given: Curve $y = 3x^2 - 5$ and slope of line $m_L = -\frac{1}{3}$.

Concept / Formula Used: $-1 \quad m_T \cdot m_L = -1 \implies m_T = -\frac{1}{m_L}$

Solution:

Given slope of the line $m_L = -\frac{1}{3}$. Since the tangent is perpendicular to this line, the slope of the tangent m_T is:

$$m_T = -\frac{1}{-\frac{1}{3}} = 3$$

Differentiating the curve equation $y = 3x^2 - 5$ with respect to x :

$$\frac{dy}{dx} = 6x$$

Set $\frac{dy}{dx}$ equal to m_T :

$$6x = 3$$

$$x = \frac{3}{6} = \frac{1}{2}$$

Substitute $x = \frac{1}{2}$ into $y = 3x^2 - 5$:

$$\begin{aligned} y &= 3 \left(\frac{1}{2} \right)^2 - 5 \\ &= 3 \left(\frac{1}{4} \right) - 5 \\ &= \frac{3}{4} - 5 = -\frac{17}{4} \end{aligned}$$

Final Answer

The required point on the curve is $\left(\frac{1}{2}, -\frac{17}{4} \right)$.

Question 14

Determine the points on the curve $y = x^3 - 3x^2 - 9x + 7$ at which the tangents are parallel to x -axis.

TYPE 27c Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to x -axis

Given: Curve $y = x^3 - 3x^2 - 9x + 7$ and tangent condition $\frac{dy}{dx} = 0$.

Concept / Formula Used: $\Rightarrow \frac{dy}{dx} = 0$

Solution:

Differentiating $y = x^3 - 3x^2 - 9x + 7$ with respect to x :

$$\frac{dy}{dx} = 3x^2 - 6x - 9$$

Setting $\frac{dy}{dx} = 0$:

$$\begin{aligned} 3x^2 - 6x - 9 &= 0 \\ 3(x^2 - 2x - 3) &= 0 \\ x^2 - 2x - 3 &= 0 \\ (x - 3)(x + 1) &= 0 \end{aligned}$$

This gives two values for x :

$$x = 3 \quad \text{or} \quad x = -1$$

Case 1: When $x = 3$,

$$\begin{aligned} y &= (3)^3 - 3(3)^2 - 9(3) + 7 \\ &= 27 - 27 - 27 + 7 \\ &= -20 \end{aligned}$$

Case 2: When $x = -1$,

$$\begin{aligned}y &= (-1)^3 - 3(-1)^2 - 9(-1) + 7 \\&= -1 - 3(1) + 9 + 7 \\&= -1 - 3 + 9 + 7 = 12\end{aligned}$$

Final Answer

The required points are $(3, -20)$ and $(-1, 12)$.

Question 15

Find the points on the curve $x^2 + y^2 - 2x - 3 = 0$ at which the tangents are parallel to x -axis.

TYPE 27c Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to x -axis

Given: Circle $x^2 + y^2 - 2x - 3 = 0$ and horizontal tangent condition $\frac{dy}{dx} = 0$.

Concept / Formula Used: $\frac{d}{dx}(x^2 + y^2 - 2x - 3) = 0$

Solution:

Differentiating $x^2 + y^2 - 2x - 3 = 0$ implicitly with respect to x :

$$\begin{aligned}2x + 2y \frac{dy}{dx} - 2 &= 0 \\2y \frac{dy}{dx} &= 2 - 2x \\\frac{dy}{dx} &= \frac{1 - x}{y} \quad (\text{for } y \neq 0)\end{aligned}$$

For tangents parallel to the x -axis, $\frac{dy}{dx} = 0$:

$$\frac{1 - x}{y} = 0 \implies 1 - x = 0 \implies x = 1$$

Substitute $x = 1$ into the original circle equation:

$$\begin{aligned}(1)^2 + y^2 - 2(1) - 3 &= 0 \\1 + y^2 - 2 - 3 &= 0 \\y^2 - 4 &= 0 \\y^2 = 4 &\implies y = \pm 2\end{aligned}$$

Both points $(1, 2)$ and $(1, -2)$ have non-zero y -coordinates, satisfying $y \neq 0$.

Final Answer

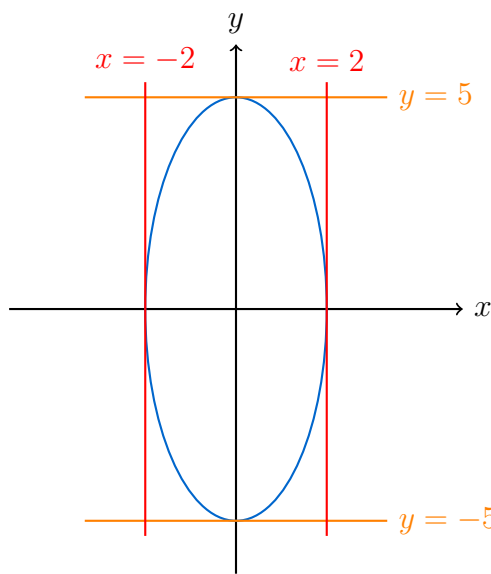
The points on the curve are $(1, 2)$ and $(1, -2)$.

Question 16

Find the points on the curve $\frac{x^2}{4} + \frac{y^2}{25} = 1$ at which the tangents are

- (i) parallel to x -axis
- (ii) parallel to y -axis.

TYPE 27c Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to



Ellipse $\frac{x^2}{4} + \frac{y^2}{25} = 1$ showing vertical and horizontal tangents

Given: Ellipse equation $\frac{x^2}{4} + \frac{y^2}{25} = 1$.

Concept / Formula Used: $x \implies \frac{dy}{dx} = 0$ $y \implies \frac{dx}{dy} = 0$

Solution:

Differentiating $\frac{x^2}{4} + \frac{y^2}{25} = 1$ implicitly with respect to x :

$$\begin{aligned}\frac{2x}{4} + \frac{2y}{25} \frac{dy}{dx} &= 0 \\ \frac{x}{2} + \frac{2y}{25} \frac{dy}{dx} &= 0 \\ \frac{2y}{25} \frac{dy}{dx} &= -\frac{x}{2} \\ \frac{dy}{dx} &= -\frac{25x}{4y}\end{aligned}$$

(i) Tangents parallel to x -axis ($\frac{dy}{dx} = 0$):

$$-\frac{25x}{4y} = 0 \implies x = 0$$

Substituting $x = 0$ into the ellipse equation:

$$\begin{aligned}\frac{0^2}{4} + \frac{y^2}{25} &= 1 \\ \frac{y^2}{25} &= 1 \implies y^2 = 25 \implies y = \pm 5\end{aligned}$$

The points are $(0, 5)$ and $(0, -5)$.

(ii) Tangents parallel to y -axis ($\frac{dx}{dy} = 0$, i.e., denominator of $\frac{dy}{dx}$ is zero):

$$4y = 0 \implies y = 0$$

Substituting $y = 0$ into the ellipse equation:

$$\begin{aligned}\frac{x^2}{4} + \frac{0^2}{25} &= 1 \\ \frac{x^2}{4} &= 1 \implies x^2 = 4 \implies x = \pm 2\end{aligned}$$

The points are $(2, 0)$ and $(-2, 0)$.

Final Answer

- (i) Points parallel to x -axis: $(0, 5)$ and $(0, -5)$
 (ii) Points parallel to y -axis: $(2, 0)$ and $(-2, 0)$

Question 17

Find the points on the curve $x^2 + y^2 = 13$, the tangent at each one of which is parallel to the line $2x + 3y = 7$.

TYPE 27g Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to a Line

Given: Circle $x^2 + y^2 = 13$ and line $2x + 3y = 7$.

Concept / Formula Used: $m_T = \frac{dy}{dx} \Big|_P \quad m_L$

Solution:

First, find slope of the line $2x + 3y = 7$:

$$\begin{aligned}3y &= -2x + 7 \\ y &= -\frac{2}{3}x + \frac{7}{3} \implies m_L = -\frac{2}{3}\end{aligned}$$

Now differentiate $x^2 + y^2 = 13$ implicitly with respect to x :

$$\begin{aligned} 2x + 2y \frac{dy}{dx} &= 0 \\ 2y \frac{dy}{dx} &= -2x \\ \frac{dy}{dx} &= -\frac{x}{y} \end{aligned}$$

Equating the slope of the tangent to the slope of the given line:

$$\begin{aligned} -\frac{x}{y} &= -\frac{2}{3} \\ \frac{x}{y} &= \frac{2}{3} \implies y = \frac{3}{2}x \end{aligned}$$

Substitute $y = \frac{3}{2}x$ into the curve $x^2 + y^2 = 13$:

$$\begin{aligned} x^2 + \left(\frac{3}{2}x\right)^2 &= 13 \\ x^2 + \frac{9}{4}x^2 &= 13 \\ \frac{13}{4}x^2 &= 13 \\ x^2 &= 4 \implies x = \pm 2 \end{aligned}$$

Finding corresponding y -values:

$$\begin{aligned} \text{If } x = 2, \quad y &= \frac{3}{2}(2) = 3 \\ \text{If } x = -2, \quad y &= \frac{3}{2}(-2) = -3 \end{aligned}$$

Final Answer

The required points are $(2, 3)$ and $(-2, -3)$.

Question 18

Find the points on the curve $y = x^3 - 3x^2 - 4x$ at which the tangent lines are parallel to the line $4x + y - 3 = 0$.

TYPE 27g Finding Curve Points Satisfying a Tangent/Normal Condition – Finding Curve Points with Tangent Parallel to a Line

Given: Curve $y = x^3 - 3x^2 - 4x$ and parallel line $4x + y - 3 = 0$.

Concept / Formula Used: $\frac{dy}{dx} = m_L$

Solution:

First, find the slope of the line $4x + y - 3 = 0$:

$$y = -4x + 3 \implies m_L = -4$$

Differentiating $y = x^3 - 3x^2 - 4x$ with respect to x :

$$\frac{dy}{dx} = 3x^2 - 6x - 4$$

Equating $\frac{dy}{dx}$ to -4 :

$$3x^2 - 6x - 4 = -4$$

$$3x^2 - 6x = 0$$

$$3x(x - 2) = 0$$

This gives two roots:

$$x = 0 \quad \text{or} \quad x = 2$$

Evaluating corresponding y -values:

$$\text{When } x = 0 : \quad y = 0^3 - 3(0)^2 - 4(0) = 0$$

$$\begin{aligned} \text{When } x = 2 : \quad y &= (2)^3 - 3(2)^2 - 4(2) \\ &= 8 - 12 - 8 = -12 \end{aligned}$$

Final Answer

The required points are $(0, 0)$ and $(2, -12)$.

Question 19

Find the equations of the tangent and the normal to each of the following curves at the given point:

- (i) $y = x^3$ at $(1, 1)$
- (ii) $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(1, 3)$
- (iii) $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(0, 5)$
- (iv) $y = x^2$ at $(0, 0)$
- (v) $y = \sin^2 x$ at $x = \frac{\pi}{2}$

TYPE 37b Finding Equation of Tangent or Normal – Finding Equations of Tangent and Normal at Given Point

Given: Curves and points specified in parts (i) through (v).

Concept / Formula Used: $y - y_1 = m_T(x - x_1)$ $m_T = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$ $y - y_1 =$
 $m_N(x - x_1)$ $m_N = -\frac{1}{m_T}$

Solution:(i) $y = x^3$ at $(1, 1)$:

$$\frac{dy}{dx} = 3x^2 \implies m_T = 3(1)^2 = 3.$$

$$\text{Slope of normal } m_N = -\frac{1}{3}.$$

$$\text{Tangent: } y - 1 = 3(x - 1)$$

$$y - 1 = 3x - 3 \implies 3x - y - 2 = 0$$

$$\text{Normal: } y - 1 = -\frac{1}{3}(x - 1)$$

$$3y - 3 = -x + 1 \implies x + 3y - 4 = 0$$

(ii) $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(1, 3)$:

$$\frac{dy}{dx} = 4x^3 - 18x^2 + 26x - 10.$$

$$\text{At } x = 1: m_T = 4(1)^3 - 18(1)^2 + 26(1) - 10 = 4 - 18 + 26 - 10 = 2.$$

$$\text{Slope of normal } m_N = -\frac{1}{2}.$$

$$\text{Tangent: } y - 3 = 2(x - 1)$$

$$y - 3 = 2x - 2 \implies 2x - y + 1 = 0$$

$$\text{Normal: } y - 3 = -\frac{1}{2}(x - 1)$$

$$2y - 6 = -x + 1 \implies x + 2y - 7 = 0$$

(iii) $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(0, 5)$:

$$\text{At } x = 0: m_T = 4(0)^3 - 18(0)^2 + 26(0) - 10 = -10.$$

$$\text{Slope of normal } m_N = -\frac{1}{-10} = \frac{1}{10}.$$

$$\text{Tangent: } y - 5 = -10(x - 0)$$

$$y - 5 = -10x \implies 10x + y - 5 = 0$$

$$\text{Normal: } y - 5 = \frac{1}{10}(x - 0)$$

$$10y - 50 = x \implies x - 10y + 50 = 0$$

(iv) $y = x^2$ at $(0, 0)$:

$$\frac{dy}{dx} = 2x \implies m_T = 2(0) = 0.$$

Since tangent has slope 0, it is horizontal; normal is vertical.

$$\text{Tangent: } y - 0 = 0(x - 0) \implies y = 0$$

$$\text{Normal: } x = 0$$

(v) $y = \sin^2 x$ at $x = \frac{\pi}{2}$:

First find y : $y = \sin^2\left(\frac{\pi}{2}\right) = (1)^2 = 1$. Point is $\left(\frac{\pi}{2}, 1\right)$.

$$\frac{dy}{dx} = 2 \sin x \cos x = \sin 2x.$$

At $x = \frac{\pi}{2}$: $m_T = \sin\left(2 \cdot \frac{\pi}{2}\right) = \sin \pi = 0$.

Since $m_T = 0$, tangent is horizontal and normal is vertical.

$$\text{Tangent: } y = 1 \implies y - 1 = 0$$

$$\text{Normal: } x = \frac{\pi}{2} \implies 2x - \pi = 0$$

Final Answer

- (i) Tangent: $3x - y - 2 = 0$; Normal: $x + 3y - 4 = 0$
- (ii) Tangent: $2x - y + 1 = 0$; Normal: $x + 2y - 7 = 0$
- (iii) Tangent: $10x + y - 5 = 0$; Normal: $x - 10y + 50 = 0$
- (iv) Tangent: $y = 0$; Normal: $x = 0$
- (v) Tangent: $y = 1$; Normal: $x = \frac{\pi}{2}$

Question 20

Find the equation of the normal to the curve $ay^2 = x^3$ at the point (am^2, am^3) .

TYPE 37c Finding Equation of Tangent or Normal – Finding Equation of Normal to Curve at Parametric Point

Given: Curve equation $ay^2 = x^3$ and point (am^2, am^3) .

Concept / Formula Used: $m_N = -\frac{1}{\frac{dy}{dx}\big|_P}$ $y - y_1 = m_N(x - x_1)$

Solution:

Differentiating $ay^2 = x^3$ implicitly with respect to x :

$$\frac{d}{dx}(ay^2) = \frac{d}{dx}(x^3)$$

$$2ay \frac{dy}{dx} = 3x^2$$

$$\frac{dy}{dx} = \frac{3x^2}{2ay}$$

Evaluating the slope of tangent m_T at (am^2, am^3) :

$$\begin{aligned} m_T &= \frac{3(am^2)^2}{2a(am^3)} \\ &= \frac{3a^2m^4}{2a^2m^3} = \frac{3m}{2} \end{aligned}$$

Therefore, the slope of the normal m_N is:

$$m_N = -\frac{1}{m_T} = -\frac{2}{3m}$$

Equation of the normal line at (am^2, am^3) :

$$\begin{aligned}y - am^3 &= -\frac{2}{3m}(x - am^2) \\3m(y - am^3) &= -2(x - am^2) \\3my - 3am^4 &= -2x + 2am^2 \\2x + 3my - 3am^4 - 2am^2 &= 0 \\2x + 3my - am^2(2 + 3m^2) &= 0\end{aligned}$$

Final Answer

The equation of the normal is $2x + 3my - am^2(2 + 3m^2) = 0$.

Question 21

- (i) Find the equation of the tangent to the curve $\sqrt{x} + \sqrt{y} = a$ at the point $\left(\frac{a^2}{4}, \frac{a^2}{4}\right)$.
- (ii) Find the equations of the tangent and the normal to the curve $x^{2/3} + y^{2/3} = 2$ at $(1, 1)$.

TYPE 37d Finding Equation of Tangent or Normal – Finding Equations of Tangent and Normal for Implicit Curves

Given: Curves and points for parts (i) and (ii).

Concept / Formula Used: $\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$

Solution:

- (i) Given curve: $\sqrt{x} + \sqrt{y} = a$

Differentiating implicitly with respect to x :

$$\begin{aligned}\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} &= 0 \\ \frac{1}{2\sqrt{y}} \frac{dy}{dx} &= -\frac{1}{2\sqrt{x}} \\ \frac{dy}{dx} &= -\frac{\sqrt{y}}{\sqrt{x}}\end{aligned}$$

Evaluating m_T at $\left(\frac{a^2}{4}, \frac{a^2}{4}\right)$:

$$m_T = -\frac{\sqrt{a^2/4}}{\sqrt{a^2/4}} = -1$$

Equation of tangent:

$$\begin{aligned}y - \frac{a^2}{4} &= -1 \left(x - \frac{a^2}{4} \right) \\y - \frac{a^2}{4} &= -x + \frac{a^2}{4} \\x + y &= \frac{a^2}{4} + \frac{a^2}{4} = \frac{a^2}{2}\end{aligned}$$

(ii) Given curve: $x^{2/3} + y^{2/3} = 2$

Differentiating implicitly with respect to x :

$$\begin{aligned}\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} &= 0 \\ \frac{2}{3}y^{-1/3}\frac{dy}{dx} &= -\frac{2}{3}x^{-1/3} \\ \frac{dy}{dx} &= -\frac{x^{-1/3}}{y^{-1/3}} = -\left(\frac{y}{x}\right)^{1/3}\end{aligned}$$

Evaluating at $(1, 1)$:

$$m_T = -\left(\frac{1}{1}\right)^{1/3} = -1$$

Slope of normal:

$$m_N = -\frac{1}{-1} = 1$$

Equation of tangent:

$$\begin{aligned}y - 1 &= -1(x - 1) \\ y - 1 &= -x + 1 \implies x + y - 2 = 0\end{aligned}$$

Equation of normal:

$$\begin{aligned}y - 1 &= 1(x - 1) \\ y - 1 &= x - 1 \implies x - y = 0\end{aligned}$$

Final Answer

- (i) Tangent: $x + y = \frac{a^2}{2}$
 (ii) Tangent: $x + y - 2 = 0$; Normal: $x - y = 0$

Question 22

- (i) Find the equation of tangent to the curve given by $x = a \sin^3 t$, $y = b \cos^3 t$ at a point where $t = \frac{\pi}{2}$.

- (ii) Find the equation of the tangent to the curve $x = \sin 3t$, $y = \cos 2t$ at $t = \frac{\pi}{4}$.

TYPE 37e Finding Equation of Tangent or Normal – Finding Equation of Tangent to Parametric Curve

Given: Parametric curves and parameter values for parts (i) and (ii).

Concept / Formula Used: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

Solution:

- (i) Given $x = a \sin^3 t$ and $y = b \cos^3 t$ at $t = \frac{\pi}{2}$:

First find coordinates of the point at $t = \frac{\pi}{2}$:

$$x = a \sin^3 \left(\frac{\pi}{2} \right) = a(1)^3 = a$$

$$y = b \cos^3 \left(\frac{\pi}{2} \right) = b(0)^3 = 0$$

The point is $(a, 0)$. Now calculate derivatives with respect to t :

$$\frac{dx}{dt} = 3a \sin^2 t \cos t$$

$$\frac{dy}{dt} = 3b \cos^2 t (-\sin t) = -3b \cos^2 t \sin t$$

$$\frac{dy}{dx} = \frac{-3b \cos^2 t \sin t}{3a \sin^2 t \cos t} = -\frac{b \cos t}{a \sin t} = -\frac{b}{a} \cot t$$

At $t = \frac{\pi}{2}$:

$$m_T = -\frac{b}{a} \cot \left(\frac{\pi}{2} \right) = -\frac{b}{a}(0) = 0$$

Equation of tangent at $(a, 0)$ with slope 0:

$$y - 0 = 0(x - a)$$

$$y = 0$$

- (ii) Given $x = \sin 3t$ and $y = \cos 2t$ at $t = \frac{\pi}{4}$:

First find coordinates of the point at $t = \frac{\pi}{4}$:

$$x = \sin \left(\frac{3\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$y = \cos \left(2 \cdot \frac{\pi}{4} \right) = \cos \left(\frac{\pi}{2} \right) = 0$$

The point is $\left(\frac{1}{\sqrt{2}}, 0\right)$. Now calculate derivatives:

$$\begin{aligned}\frac{dx}{dt} &= 3 \cos 3t \\ \frac{dy}{dt} &= -2 \sin 2t \\ \frac{dy}{dx} &= \frac{-2 \sin 2t}{3 \cos 3t}\end{aligned}$$

At $t = \frac{\pi}{4}$:

$$\begin{aligned}\sin 2t &= \sin\left(\frac{\pi}{2}\right) = 1 \\ \cos 3t &= \cos\left(\frac{3\pi}{4}\right) = -\frac{1}{\sqrt{2}} \\ m_T &= \frac{-2(1)}{3(-1/\sqrt{2})} = \frac{2\sqrt{2}}{3}\end{aligned}$$

Equation of tangent at $\left(\frac{1}{\sqrt{2}}, 0\right)$:

$$\begin{aligned}y - 0 &= \frac{2\sqrt{2}}{3} \left(x - \frac{1}{\sqrt{2}}\right) \\ y &= \frac{2\sqrt{2}}{3}x - \frac{2}{3} \\ 3y &= 2\sqrt{2}x - 2 \\ 2\sqrt{2}x - 3y - 2 &= 0\end{aligned}$$

Final Answer

- (i) Tangent equation: $y = 0$
- (ii) Tangent equation: $2\sqrt{2}x - 3y - 2 = 0$

Question 23

- (i) Find the equations of all lines having slope 2 and that are tangents to the curve $y = \frac{1}{x-3}$.
- (ii) Find the equation of the tangent to the curve $y = x + \frac{4}{x^2}$ which is parallel to x -axis.

TYPE 37f Finding Equation of Tangent or Normal – Finding Equations of Tangents with Given Slope or Parallelism

Given: Curves and slope conditions in parts (i) and (ii).

Concept / Formula Used: $\frac{dy}{dx} = m$

Solution:

(i) Curve $y = \frac{1}{x-3} = (x-3)^{-1}$:

Differentiating with respect to x :

$$\frac{dy}{dx} = -(x-3)^{-2} = -\frac{1}{(x-3)^2}$$

Set slope equal to 2:

$$\begin{aligned} -\frac{1}{(x-3)^2} &= 2 \\ (x-3)^2 &= -\frac{1}{2} \end{aligned}$$

Since $(x-3)^2 \geq 0$ for all real numbers x , there is no real solution for x . Thus, no such tangent lines exist.

(ii) Curve $y = x + \frac{4}{x^2} = x + 4x^{-2}$:

Differentiating with respect to x :

$$\frac{dy}{dx} = 1 - 8x^{-3} = 1 - \frac{8}{x^3}$$

For tangent parallel to x -axis, $\frac{dy}{dx} = 0$:

$$\begin{aligned} 1 - \frac{8}{x^3} &= 0 \\ \frac{8}{x^3} &= 1 \implies x^3 = 8 \implies x = 2 \end{aligned}$$

Substituting $x = 2$ into curve equation:

$$\begin{aligned} y &= 2 + \frac{4}{2^2} \\ &= 2 + \frac{4}{4} = 2 + 1 = 3 \end{aligned}$$

Since the tangent line has slope 0 and passes through $(2, 3)$, its equation is:

$$y - 3 = 0(x - 2) \implies y = 3$$

Final Answer

- (i) No such line exists having slope 2.
- (ii) Equation of tangent: $y = 3$

Question 24

Find the equation of the tangent to the curve $y = x^2 - 2x + 7$ which is

- (i) parallel to the line $2x - y + 9 = 0$
- (ii) perpendicular to the line $5y - 15x = 13$.

TYPE 42a Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Tangents Parallel

Given: Curve $y = x^2 - 2x + 7$ and lines in parts (i) and (ii).

Concept / Formula Used: $m_T = \frac{dy}{dx}$ $m_T = m_L$ $m_T = -\frac{1}{m_L}$

Solution:

Differentiating $y = x^2 - 2x + 7$ with respect to x :

$$\frac{dy}{dx} = 2x - 2$$

- (i) Line equation: $2x - y + 9 = 0 \implies y = 2x + 9$.

Slope of line $m_L = 2$.

Since tangent is parallel, $m_T = 2$:

$$2x - 2 = 2$$

$$2x = 4 \implies x = 2$$

When $x = 2$:

$$y = (2)^2 - 2(2) + 7 = 4 - 4 + 7 = 7$$

Equation of tangent at $(2, 7)$ with slope 2:

$$y - 7 = 2(x - 2)$$

$$y - 7 = 2x - 4 \implies 2x - y + 3 = 0$$

- (ii) Line equation: $5y - 15x = 13 \implies 5y = 15x + 13 \implies y = 3x + \frac{13}{5}$.

Slope of line $m_L = 3$.

Since tangent is perpendicular, $m_T = -\frac{1}{3}$:

$$2x - 2 = -\frac{1}{3}$$

$$2x = 2 - \frac{1}{3} = \frac{5}{3}$$

$$x = \frac{5}{6}$$

When $x = \frac{5}{6}$:

$$\begin{aligned} y &= \left(\frac{5}{6}\right)^2 - 2\left(\frac{5}{6}\right) + 7 \\ &= \frac{25}{36} - \frac{10}{6} + 7 \\ &= \frac{25 - 60 + 252}{36} = \frac{217}{36} \end{aligned}$$

Equation of tangent at $\left(\frac{5}{6}, \frac{217}{36}\right)$ with slope $-\frac{1}{3}$:

$$\begin{aligned} y - \frac{217}{36} &= -\frac{1}{3}\left(x - \frac{5}{6}\right) \\ 36y - 217 &= -12\left(x - \frac{5}{6}\right) \\ 36y - 217 &= -12x + 10 \\ 12x + 36y - 227 &= 0 \end{aligned}$$

Final Answer

- (i) $2x - y + 3 = 0$
- (ii) $12x + 36y - 227 = 0$

Question 25

- (i) Find the equations of the tangents to the curve $y = 4x^3 - 3x + 5$ which are perpendicular to the line $9y + x + 3 = 0$.
- (ii) Find the equations of the tangents to the curve $3x^2 - y^2 = 2$ which are perpendicular to the line $x + 3y = 2$.

TYPE 42a Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Tangents Parallel or Perpendicular to a Line

Given: Curves and perpendicular line equations given in parts (i) and (ii).

Concept / Formula Used: $m_T = -\frac{1}{m_L}$

Solution:

- (i) Line equation: $9y + x + 3 = 0 \implies 9y = -x - 3 \implies y = -\frac{1}{9}x - \frac{1}{3}$.

Slope of line $m_L = -\frac{1}{9}$.

Slope of perpendicular tangent $m_T = -\frac{1}{-1/9} = 9$.

Differentiating $y = 4x^3 - 3x + 5$:

$$\frac{dy}{dx} = 12x^2 - 3$$

Setting $\frac{dy}{dx} = 9$:

$$12x^2 - 3 = 9$$

$$12x^2 = 12 \implies x^2 = 1 \implies x = \pm 1$$

Case 1: $x = 1 \implies y = 4(1)^3 - 3(1) + 5 = 6$. Point is $(1, 6)$.

$$y - 6 = 9(x - 1)$$

$$y - 6 = 9x - 9 \implies 9x - y - 3 = 0$$

Case 2: $x = -1 \implies y = 4(-1)^3 - 3(-1) + 5 = -4 + 3 + 5 = 4$. Point is $(-1, 4)$.

$$y - 4 = 9(x - (-1))$$

$$y - 4 = 9x + 9 \implies 9x - y + 13 = 0$$

(ii) Line equation: $x + 3y = 2 \implies 3y = -x + 2 \implies y = -\frac{1}{3}x + \frac{2}{3}$.

Slope of line $m_L = -\frac{1}{3}$.

Slope of perpendicular tangent $m_T = -\frac{1}{-1/3} = 3$.

Differentiating $3x^2 - y^2 = 2$ implicitly with respect to x :

$$6x - 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = 6x \implies \frac{dy}{dx} = \frac{3x}{y}$$

Setting $\frac{dy}{dx} = 3$:

$$\frac{3x}{y} = 3 \implies y = x$$

Substituting $y = x$ into hyperbola $3x^2 - y^2 = 2$:

$$3x^2 - x^2 = 2$$

$$2x^2 = 2 \implies x^2 = 1 \implies x = \pm 1$$

Case 1: $x = 1 \implies y = 1$. Point is $(1, 1)$.

$$y - 1 = 3(x - 1)$$

$$y - 1 = 3x - 3 \implies 3x - y - 2 = 0$$

Case 2: $x = -1 \implies y = -1$. Point is $(-1, -1)$.

$$y - (-1) = 3(x - (-1))$$

$$y + 1 = 3x + 3 \implies 3x - y + 2 = 0$$

Final Answer

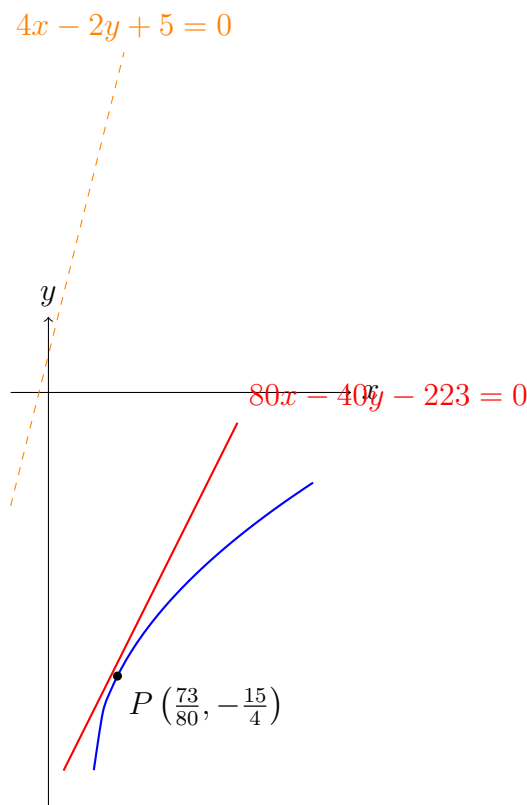
(i) $9x - y - 3 = 0$ and $9x - y + 13 = 0$

(ii) $3x - y - 2 = 0$ and $3x - y + 2 = 0$

Question 26

Find the equation of the tangent line to the curve $y = \sqrt{5x - 3} - 5$, which is parallel to the line $4x - 2y + 5 = 0$.

TYPE 42a Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Tangents Parallel



Tangent to $y = \sqrt{5x - 3} - 5$ parallel to the line $4x - 2y + 5 = 0$

Given: Curve $y = \sqrt{5x - 3} - 5$ and parallel line $4x - 2y + 5 = 0$.

Concept / Formula Used: $m = \frac{dy}{dx}$ $m_1 = m_2$ (x_1, y_1) $y - y_1 = m(x - x_1)$

Solution:

Given the equation of the line:

$$\begin{aligned} 4x - 2y + 5 &= 0 \\ 2y &= 4x + 5 \\ y &= 2x + \frac{5}{2} \end{aligned}$$

Comparing with $y = mx + c$, the slope of the given line is:

$$m_L = 2$$

Now, given the equation of the curve:

$$y = \sqrt{5x - 3} - 5$$

Differentiating both sides with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} [(5x - 3)^{1/2} - 5] \\ &= \frac{1}{2}(5x - 3)^{-1/2} \cdot \frac{d}{dx}(5x - 3) - 0 \\ &= \frac{1}{2\sqrt{5x - 3}} \cdot 5 \\ &= \frac{5}{2\sqrt{5x - 3}}\end{aligned}$$

Since the tangent line is parallel to the given line, their slopes must be equal:

$$\begin{aligned}\frac{dy}{dx} &= 2 \\ \frac{5}{2\sqrt{5x - 3}} &= 2 \\ 5 &= 4\sqrt{5x - 3} \\ \sqrt{5x - 3} &= \frac{5}{4}\end{aligned}$$

Squaring both sides:

$$\begin{aligned}5x - 3 &= \left(\frac{5}{4}\right)^2 \\ 5x - 3 &= \frac{25}{16} \\ 5x &= \frac{25}{16} + 3 \\ 5x &= \frac{25 + 48}{16} \\ 5x &= \frac{73}{16} \\ x &= \frac{73}{80}\end{aligned}$$

Substitute $x = \frac{73}{80}$ into the curve equation to find y :

$$\begin{aligned}y &= \sqrt{5\left(\frac{73}{80}\right) - 3} - 5 \\ &= \sqrt{\frac{73}{16}} - 3 - 5 \\ &= \sqrt{\frac{25}{16}} - 5 \\ &= \frac{5}{4} - 5 \\ &= -\frac{15}{4}\end{aligned}$$

Thus, the point of contact is $P\left(\frac{73}{80}, -\frac{15}{4}\right)$ and the slope of the tangent is $m = 2$.

Using the point-slope form $y - y_1 = m(x - x_1)$:

$$\begin{aligned} y - \left(-\frac{15}{4}\right) &= 2\left(x - \frac{73}{80}\right) \\ y + \frac{15}{4} &= 2x - \frac{73}{40} \end{aligned}$$

Multiplying the entire equation by 40:

$$\begin{aligned} 40y + 150 &= 80x - 73 \\ 80x - 40y - 73 - 150 &= 0 \\ 80x - 40y - 223 &= 0 \end{aligned}$$

Final Answer

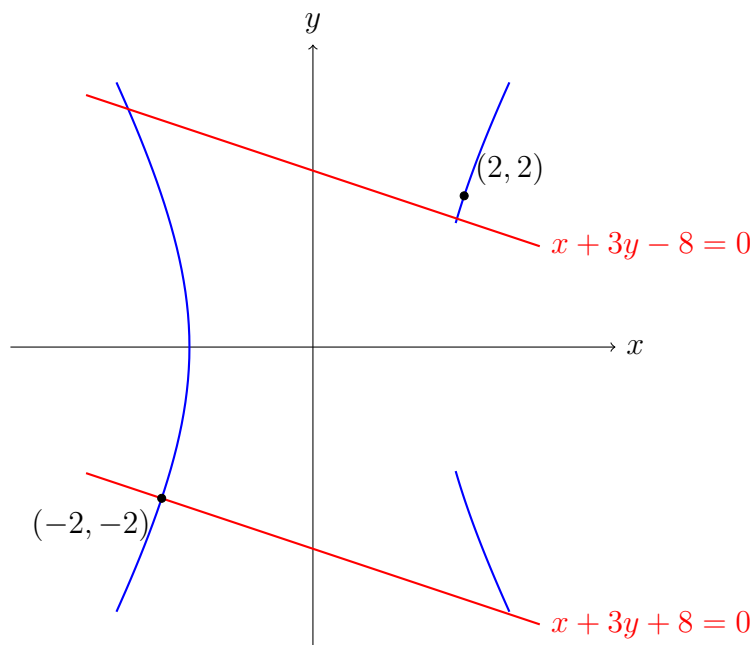
$$80x - 40y - 223 = 0$$

Question 27

Find the equations of the normals to the curve $3x^2 - y^2 = 8$ which are parallel to the line $x + 3y = 4$.

TYPE 42b Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Normals Parallel to a Line

★ Likely Exam Question



Normals to hyperbola $3x^2 - y^2 = 8$ parallel to $x + 3y = 4$

Given: Curve $3x^2 - y^2 = 8$ and parallel line $x + 3y = 4$.

Concept / Formula Used: $m_N = -\frac{1}{\frac{dy}{dx}}$

Solution:

Given the line equation:

$$\begin{aligned}x + 3y &= 4 \\3y &= -x + 4 \\y &= -\frac{1}{3}x + \frac{4}{3}\end{aligned}$$

The slope of the given line is:

$$m_L = -\frac{1}{3}$$

Given the curve equation:

$$3x^2 - y^2 = 8$$

Differentiating both sides implicitly with respect to x :

$$\begin{aligned}\frac{d}{dx}(3x^2) - \frac{d}{dx}(y^2) &= \frac{d}{dx}(8) \\6x - 2y\frac{dy}{dx} &= 0 \\2y\frac{dy}{dx} &= 6x \\\frac{dy}{dx} &= \frac{3x}{y}\end{aligned}$$

The slope of the normal at any point (x, y) is:

$$m_N = -\frac{1}{\frac{dy}{dx}} = -\frac{y}{3x}$$

Since the normal is parallel to the line $x + 3y = 4$:

$$\begin{aligned}m_N &= m_L \\-\frac{y}{3x} &= -\frac{1}{3} \\\frac{y}{x} &= 1 \\y &= x\end{aligned}$$

Substitute $y = x$ into the curve equation $3x^2 - y^2 = 8$:

$$\begin{aligned}3x^2 - x^2 &= 8 \\2x^2 &= 8 \\x^2 &= 4 \\x &= \pm 2\end{aligned}$$

When $x = 2$, $y = 2$, giving the point $(2, 2)$. When $x = -2$, $y = -2$, giving the point $(-2, -2)$.

Case 1: Normal at $(2, 2)$ with slope $m = -\frac{1}{3}$:

$$\begin{aligned}y - 2 &= -\frac{1}{3}(x - 2) \\3(y - 2) &= -(x - 2) \\3y - 6 &= -x + 2 \\x + 3y - 8 &= 0\end{aligned}$$

Case 2: Normal at $(-2, -2)$ with slope $m = -\frac{1}{3}$:

$$\begin{aligned}y - (-2) &= -\frac{1}{3}(x - (-2)) \\y + 2 &= -\frac{1}{3}(x + 2) \\3(y + 2) &= -(x + 2) \\3y + 6 &= -x - 2 \\x + 3y + 8 &= 0\end{aligned}$$

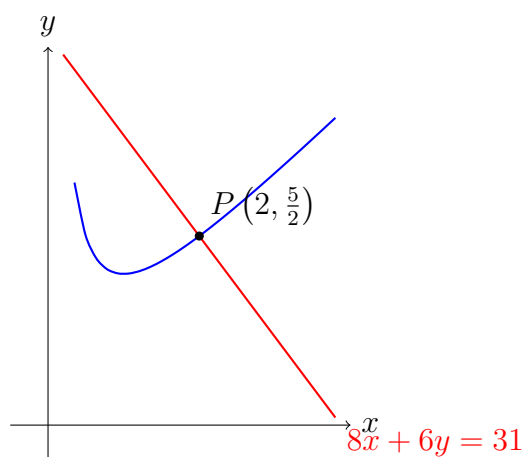
Final Answer

$$x + 3y - 8 = 0 \quad \text{and} \quad x + 3y + 8 = 0$$

Question 28

Find the equation of the normal to the curve $y = x + \frac{1}{x}$, $x > 0$, perpendicular to the line $3x - 4y = 7$.

TYPE 42b Finding Tangent/Normal Equation Parallel or Perpendicular to a Line – Finding Equations of Normals Parallel or Perpendicular to a Line



Normal to $y = x + \frac{1}{x}$ perpendicular to line $3x - 4y = 7$

Given: Curve $y = x + \frac{1}{x}$ for $x > 0$, and perpendicular line $3x - 4y = 7$.

Concept / Formula Used: $m_1 \cdot m_2 = -1$ $m_N = -\frac{1}{\frac{dy}{dx}}$

Solution:

Given the equation of the line:

$$\begin{aligned} 3x - 4y &= 7 \\ 4y &= 3x - 7 \\ y &= \frac{3}{4}x - \frac{7}{4} \end{aligned}$$

The slope of the given line is:

$$m_L = \frac{3}{4}$$

Given the curve equation for $x > 0$:

$$y = x + \frac{1}{x}$$

Differentiating with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x) + \frac{d}{dx}(x^{-1}) \\ &= 1 - \frac{1}{x^2} \\ &= \frac{x^2 - 1}{x^2} \end{aligned}$$

The slope of the normal m_N to the curve is:

$$m_N = -\frac{1}{\frac{dy}{dx}} = -\frac{x^2}{x^2 - 1}$$

Since the normal is perpendicular to the given line:

$$\begin{aligned} m_N \cdot m_L &= -1 \\ \left(-\frac{x^2}{x^2 - 1}\right) \cdot \left(\frac{3}{4}\right) &= -1 \\ \frac{3x^2}{4(x^2 - 1)} &= 1 \\ 3x^2 &= 4x^2 - 4 \\ 4x^2 - 3x^2 &= 4 \\ x^2 &= 4 \end{aligned}$$

Since $x > 0$, we choose:

$$x = 2$$

Substitute $x = 2$ into the curve equation:

$$\begin{aligned} y &= 2 + \frac{1}{2} \\ &= \frac{5}{2} \end{aligned}$$

The point of contact is $\left(2, \frac{5}{2}\right)$. The slope of the normal is:

$$m_N = -\frac{2^2}{2^2 - 1} = -\frac{4}{3}$$

Now, equation of the normal line at $\left(2, \frac{5}{2}\right)$:

$$y - \frac{5}{2} = -\frac{4}{3}(x - 2)$$

$$3\left(y - \frac{5}{2}\right) = -4(x - 2)$$

$$3y - \frac{15}{2} = -4x + 8$$

$$4x + 3y - \frac{15}{2} - 8 = 0$$

$$4x + 3y - \frac{31}{2} = 0$$

$$8x + 6y - 31 = 0$$

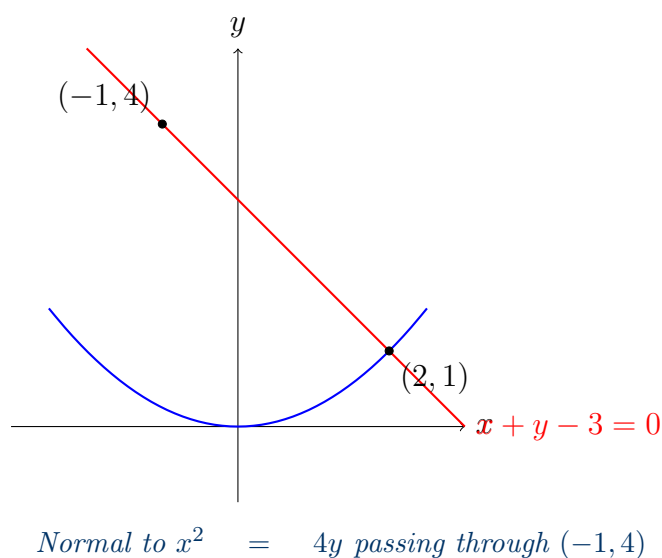
Final Answer

$$8x + 6y = 31$$

Question 29

Find the equation of the normal to the curve $x^2 = 4y$ which passes through the point $(-1, 4)$.

TYPE 37g Finding Equation of Tangent or Normal – Finding Equation of Normal Passing Through External Point



Given: Parabola $x^2 = 4y$ and external point $(-1, 4)$ through which the normal passes.

Concept / Formula Used: $m_T = \frac{dy}{dx}$ $m_N = -\frac{1}{m_T}$ (x_1, y_1) $y - y_1 = m_N(x - x_1)$

Solution:

Let (x_1, y_1) be the point of contact on the curve $x^2 = 4y$. Thus, (x_1, y_1) satisfies the curve:

$$x_1^2 = 4y_1 \implies y_1 = \frac{x_1^2}{4}$$

Differentiating the curve equation $x^2 = 4y$ with respect to x :

$$\begin{aligned} 2x &= 4 \frac{dy}{dx} \\ \frac{dy}{dx} &= \frac{x}{2} \end{aligned}$$

The slope of the tangent at (x_1, y_1) is $\left. \frac{dy}{dx} \right|_{(x_1, y_1)} = \frac{x_1}{2}$. Therefore, the slope of the normal at (x_1, y_1) is:

$$m_N = -\frac{2}{x_1}$$

The equation of the normal line at (x_1, y_1) is:

$$\begin{aligned} y - y_1 &= m_N(x - x_1) \\ y - \frac{x_1^2}{4} &= -\frac{2}{x_1}(x - x_1) \end{aligned}$$

Since this normal passes through the point $(-1, 4)$, we substitute $x = -1$ and $y = 4$:

$$\begin{aligned} 4 - \frac{x_1^2}{4} &= -\frac{2}{x_1}(-1 - x_1) \\ 4 - \frac{x_1^2}{4} &= \frac{2(1 + x_1)}{x_1} \end{aligned}$$

Multiplying both sides by $4x_1$:

$$\begin{aligned} 4x_1 \left(4 - \frac{x_1^2}{4} \right) &= 4x_1 \cdot \frac{2(1 + x_1)}{x_1} \\ 16x_1 - x_1^3 &= 8(1 + x_1) \\ 16x_1 - x_1^3 &= 8 + 8x_1 \\ x_1^3 - 8x_1 + 8 &= 0 \end{aligned}$$

To solve $x_1^3 - 8x_1 + 8 = 0$, we test integer roots: For $x_1 = 2$:

$$2^3 - 8(2) + 8 = 8 - 16 + 8 = 0$$

Thus, $x_1 = 2$ is a root. Factoring out $(x_1 - 2)$:

$$(x_1 - 2)(x_1^2 + 2x_1 - 4) = 0$$

Using $x_1 = 2$:

$$y_1 = \frac{2^2}{4} = 1$$

The point of contact is $(2, 1)$, and the slope of the normal is:

$$m_N = -\frac{2}{2} = -1$$

Substitute $(x_1, y_1) = (2, 1)$ and $m_N = -1$ into the normal line equation:

$$y - 1 = -1(x - 2)$$

$$y - 1 = -x + 2$$

$$x + y - 3 = 0$$

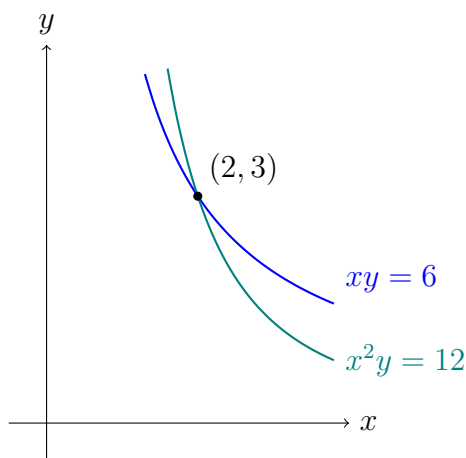
Final Answer

$$x + y - 3 = 0$$

Question 30 (i)

Find the angle of intersection between the curves $xy = 6$ and $x^2y = 12$.

TYPE 45a Angle of Intersection and Orthogonality of Curves – Finding Angle of Intersection Between Two Curves



Intersection of curves $xy = 6$ and $x^2y = 12$

Given: Curves $C_1 : xy = 6$ and $C_2 : x^2y = 12$.

Concept / Formula Used: θ m_1, m_2 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Solution:

First, we find the point(s) of intersection:

$$C_1 : xy = 6 \implies y = \frac{6}{x}$$

$$C_2 : x^2y = 12$$

Substitute $y = \frac{6}{x}$ into C_2 :

$$\begin{aligned}x^2 \left(\frac{6}{x} \right) &= 12 \\6x &= 12 \\x &= 2\end{aligned}$$

Substituting $x = 2$ back into $xy = 6$:

$$\begin{aligned}2y &= 6 \\y &= 3\end{aligned}$$

The point of intersection is $P(2, 3)$.

Now, find the slopes of the tangents to both curves at $P(2, 3)$:

For curve $C_1 : xy = 6$: Differentiating implicitly with respect to x :

$$\begin{aligned}y + x \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{y}{x}\end{aligned}$$

At $P(2, 3)$, slope m_1 :

$$m_1 = -\frac{3}{2}$$

For curve $C_2 : x^2y = 12$: Differentiating implicitly with respect to x :

$$\begin{aligned}2xy + x^2 \frac{dy}{dx} &= 0 \\ x^2 \frac{dy}{dx} &= -2xy \\ \frac{dy}{dx} &= -\frac{2y}{x}\end{aligned}$$

At $P(2, 3)$, slope m_2 :

$$m_2 = -\frac{2(3)}{2} = -3$$

Now compute the angle of intersection θ :

$$\begin{aligned}\tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{-\frac{3}{2} - (-3)}{1 + \left(-\frac{3}{2}\right)(-3)} \right| \\ &= \left| \frac{-\frac{3}{2} + 3}{1 + \frac{9}{2}} \right| \\ &= \left| \frac{\frac{3}{2}}{\frac{11}{2}} \right| \\ &= \frac{3}{11}\end{aligned}$$

Taking inverse tangent:

$$\theta = \tan^{-1} \left(\frac{3}{11} \right)$$

Final Answer

$$\theta = \tan^{-1} \left(\frac{3}{11} \right)$$

Question 30 (ii)

Find the angle of intersection between the curves $y = 4 - x^2$ and $y = x^2$.

TYPE 45a Angle of Intersection and Orthogonality of Curves – Finding Angle of Intersection Between Two Curves

★ Likely Exam Question

Given: Curves $C_1 : y = 4 - x^2$ and $C_2 : y = x^2$.

Concept / Formula Used: θ m_1, m_2 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Solution:

First, find the points of intersection by equating the two curves:

$$\begin{aligned} 4 - x^2 &= x^2 \\ 2x^2 &= 4 \\ x^2 &= 2 \\ x &= \pm\sqrt{2} \end{aligned}$$

When $x = \sqrt{2}$, $y = (\sqrt{2})^2 = 2$, giving point $P_1(\sqrt{2}, 2)$. When $x = -\sqrt{2}$, $y = (-\sqrt{2})^2 = 2$, giving point $P_2(-\sqrt{2}, 2)$.

Differentiating $C_1 : y = 4 - x^2$:

$$\frac{dy}{dx} = -2x$$

Differentiating $C_2 : y = x^2$:

$$\frac{dy}{dx} = 2x$$

At $P_1(\sqrt{2}, 2)$:

$$\begin{aligned} m_1 &= -2\sqrt{2} \\ m_2 &= 2\sqrt{2} \end{aligned}$$

Calculate $\tan \theta$:

$$\begin{aligned}\tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{-2\sqrt{2} - 2\sqrt{2}}{1 + (-2\sqrt{2})(2\sqrt{2})} \right| \\ &= \left| \frac{-4\sqrt{2}}{1 - 8} \right| \\ &= \left| \frac{-4\sqrt{2}}{-7} \right| \\ &= \frac{4\sqrt{2}}{7}\end{aligned}$$

Thus, $\theta = \tan^{-1} \left(\frac{4\sqrt{2}}{7} \right)$. By symmetry, the angle at $P_2(-\sqrt{2}, 2)$ is identical.

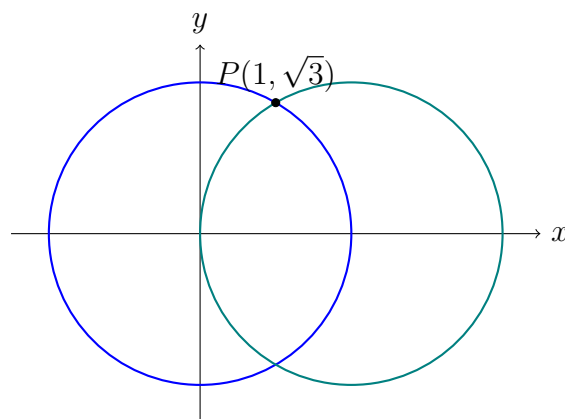
Final Answer

$$\theta = \tan^{-1} \left(\frac{4\sqrt{2}}{7} \right)$$

Question 31

Find the angle of intersection of the curves $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$ at the point in the first quadrant.

TYPE 45a Angle of Intersection and Orthogonality of Curves – Finding Angle of Intersection Between Two Curves



Intersection of circles $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$

Given: Circles $C_1 : x^2 + y^2 = 4$ and $C_2 : (x - 2)^2 + y^2 = 4$.

Concept / Formula Used: θ m_1, m_2 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Solution:

First, solve for the point of intersection in the first quadrant:

$$\begin{aligned}x^2 + y^2 &= 4 & \text{--- (1)} \\(x - 2)^2 + y^2 &= 4 & \text{--- (2)}\end{aligned}$$

Subtracting equation (2) from equation (1):

$$\begin{aligned}x^2 - (x - 2)^2 &= 0 \\x^2 - (x^2 - 4x + 4) &= 0 \\4x - 4 &= 0 \\4x &= 4 \\x &= 1\end{aligned}$$

Substitute $x = 1$ into equation (1):

$$\begin{aligned}1^2 + y^2 &= 4 \\y^2 &= 3 \\y &= \sqrt{3} \quad (\text{since } y > 0 \text{ in the first quadrant})\end{aligned}$$

The point of intersection in the first quadrant is $P(1, \sqrt{3})$.

Differentiating $C_1 : x^2 + y^2 = 4$ implicitly with respect to x :

$$\begin{aligned}2x + 2y \frac{dy}{dx} &= 0 \\\frac{dy}{dx} &= -\frac{x}{y}\end{aligned}$$

At $P(1, \sqrt{3})$, slope m_1 :

$$m_1 = -\frac{1}{\sqrt{3}}$$

Differentiating $C_2 : (x - 2)^2 + y^2 = 4$ implicitly with respect to x :

$$\begin{aligned}2(x - 2) + 2y \frac{dy}{dx} &= 0 \\\frac{dy}{dx} &= -\frac{x - 2}{y} = \frac{2 - x}{y}\end{aligned}$$

At $P(1, \sqrt{3})$, slope m_2 :

$$m_2 = \frac{2 - 1}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

Now compute $\tan \theta$:

$$\begin{aligned}
 \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\
 &= \left| \frac{-\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}}}{1 + \left(-\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right)} \right| \\
 &= \left| \frac{-\frac{2}{\sqrt{3}}}{1 - \frac{1}{3}} \right| \\
 &= \left| \frac{-\frac{2}{\sqrt{3}}}{\frac{2}{3}} \right| \\
 &= \frac{2}{\sqrt{3}} \cdot \frac{3}{2} \\
 &= \sqrt{3}
 \end{aligned}$$

Since $\tan \theta = \sqrt{3}$:

$$\theta = 60^\circ$$

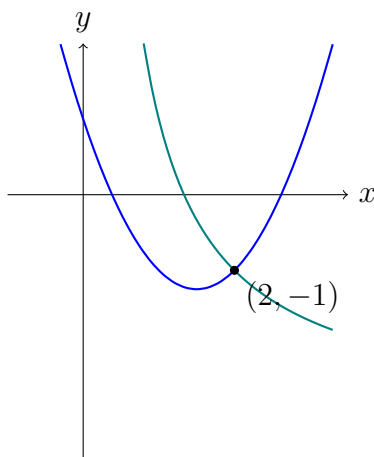
Final Answer

$\theta = 60^\circ$ (or $\frac{\pi}{3}$ radians)

Question 32

Prove that the curves $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ intersect at right angles at the point $(2, -1)$.

TYPE 45b Angle of Intersection and Orthogonality of Curves – Proving or Finding Condition for Orthogonal Intersection



Orthogonal intersection of $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ at $(2, -1)$

To Prove: The curves $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ intersect orthogonally at $(2, -1)$, i.e., $m_1 \cdot m_2 = -1$.

Concept / Formula Used: $m_1 \cdot m_2 = -1$

Solution:

First, check that $(2, -1)$ lies on both curves: For $C_1 : y = x^2 - 3x + 1$:

$$\text{RHS} = 2^2 - 3(2) + 1 = 4 - 6 + 1 = -1 = \text{LHS}$$

For $C_2 : x(y + 3) = 4$:

$$\text{LHS} = 2(-1 + 3) = 2(2) = 4 = \text{RHS}$$

Thus, $(2, -1)$ is a point of intersection of both curves.

Next, calculate slope m_1 for curve $C_1 : y = x^2 - 3x + 1$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^2 - 3x + 1) \\ &= 2x - 3 \end{aligned}$$

At point $(2, -1)$:

$$\begin{aligned} m_1 &= 2(2) - 3 \\ &= 4 - 3 \\ &= 1 \end{aligned}$$

Now, calculate slope m_2 for curve $C_2 : x(y + 3) = 4$: Rewriting C_2 :

$$\begin{aligned} y + 3 &= \frac{4}{x} \\ y &= \frac{4}{x} - 3 \end{aligned}$$

Differentiating with respect to x :

$$\frac{dy}{dx} = -\frac{4}{x^2}$$

At point $(2, -1)$:

$$\begin{aligned} m_2 &= -\frac{4}{2^2} \\ &= -\frac{4}{4} \\ &= -1 \end{aligned}$$

Now, find the product of slopes m_1 and m_2 :

$$\begin{aligned} m_1 \cdot m_2 &= (1) \cdot (-1) \\ &= -1 \end{aligned}$$

Since the product of the slopes of the tangents at $(2, -1)$ is -1 , the curves intersect at right angles.

Hence Proved

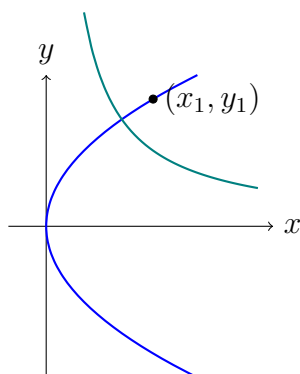
LHS = RHS. The given curves intersect orthogonally at $(2, -1)$. Hence proved.

Question 33

Find the condition that the curves $2x = y^2$ and $2xy = k$ intersect orthogonally.

TYPE 45b Angle of Intersection and Orthogonality of Curves – Proving or Finding Condition for Orthogonal Intersection

★ Likely Exam Question



Orthogonal intersection of $y^2 = 2x$ and $2xy = k$

Given: Curves $C_1 : y^2 = 2x$ and $C_2 : 2xy = k$.

Concept / Formula Used: $m_1 \cdot m_2 = -1$ at (x_1, y_1)

Solution:

Let (x_1, y_1) be a point of intersection of curves C_1 and C_2 . Then (x_1, y_1) satisfies both equations:

$$y_1^2 = 2x_1 \quad \text{--- (1)}$$

$$2x_1y_1 = k \quad \text{--- (2)}$$

Substitute $2x_1 = y_1^2$ from equation (1) into equation (2):

$$(y_1^2)y_1 = k$$

$$y_1^3 = k$$

$$y_1 = k^{1/3}$$

From equation (1):

$$2x_1 = (k^{1/3})^2$$

$$2x_1 = k^{2/3}$$

$$x_1 = \frac{k^{2/3}}{2}$$

Differentiating $C_1 : y^2 = 2x$ with respect to x :

$$2y \frac{dy}{dx} = 2$$

$$\frac{dy}{dx} = \frac{1}{y}$$

Slope of tangent m_1 at (x_1, y_1) :

$$m_1 = \frac{1}{y_1}$$

Differentiating $C_2 : 2xy = k \implies xy = \frac{k}{2}$ with respect to x :

$$y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

Slope of tangent m_2 at (x_1, y_1) :

$$m_2 = -\frac{y_1}{x_1}$$

For the curves to intersect orthogonally:

$$m_1 \cdot m_2 = -1$$

$$\left(\frac{1}{y_1}\right) \cdot \left(-\frac{y_1}{x_1}\right) = -1$$

$$-\frac{1}{x_1} = -1$$

$$x_1 = 1$$

Substitute $x_1 = 1$ into $x_1 = \frac{k^{2/3}}{2}$:

$$1 = \frac{k^{2/3}}{2}$$

$$k^{2/3} = 2$$

Cubing both sides:

$$(k^{2/3})^3 = 2^3$$

$$k^2 = 8$$

Final Answer

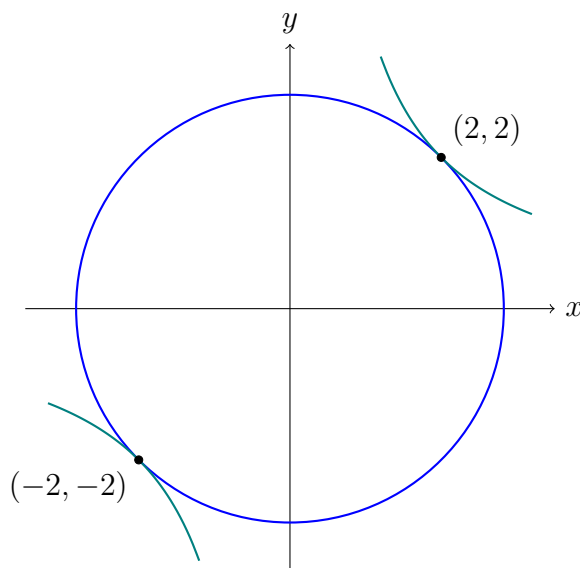
$$k^2 = 8$$

Question 34

Prove that the curves $xy = 4$ and $x^2 + y^2 = 8$ touch each other.

TYPE 45c Angle of Intersection and Orthogonality of Curves – Proving Two Curves Touch Each Other

★ Likely Exam Question



Curves $xy = 4$ and $x^2 + y^2 = 8$ touching at $(2, 2)$ and $(-2, -2)$

To Prove: The curves $xy = 4$ and $x^2 + y^2 = 8$ touch each other, i.e., they have equal tangent slopes ($m_1 = m_2$) at their common points of intersection.

Concept / Formula Used: $m_1 = m_2$

Solution:

First, find the points of intersection of $C_1 : xy = 4$ and $C_2 : x^2 + y^2 = 8$: From C_1 , $y = \frac{4}{x}$. Substitute into C_2 :

$$\begin{aligned} x^2 + \left(\frac{4}{x}\right)^2 &= 8 \\ x^2 + \frac{16}{x^2} &= 8 \\ x^4 + 16 &= 8x^2 \\ x^4 - 8x^2 + 16 &= 0 \\ (x^2 - 4)^2 &= 0 \\ x^2 - 4 &= 0 \\ x^2 &= 4 \\ x &= \pm 2 \end{aligned}$$

When $x = 2$, $y = \frac{4}{2} = 2$, giving point $P_1(2, 2)$. When $x = -2$, $y = \frac{4}{-2} = -2$, giving point $P_2(-2, -2)$.

Differentiating $C_1 : xy = 4$ with respect to x :

$$\begin{aligned} y + x \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{y}{x} \end{aligned}$$

Differentiating $C_2 : x^2 + y^2 = 8$ with respect to x :

$$2x + 2y \frac{dy}{dx} = 0$$
$$\frac{dy}{dx} = -\frac{x}{y}$$

At point $P_1(2, 2)$:

$$\text{Slope for } C_1 : m_1 = -\frac{2}{2} = -1$$

$$\text{Slope for } C_2 : m_2 = -\frac{2}{2} = -1$$

Since $m_1 = m_2 = -1$, both curves share a common tangent at $P_1(2, 2)$.

At point $P_2(-2, -2)$:

$$\text{Slope for } C_1 : m_1 = -\frac{-2}{-2} = -1$$

$$\text{Slope for } C_2 : m_2 = -\frac{-2}{-2} = -1$$

Since $m_1 = m_2 = -1$, both curves share a common tangent at $P_2(-2, -2)$.

Since the slopes of the tangents to both curves are equal at each point of intersection, the two curves touch each other.

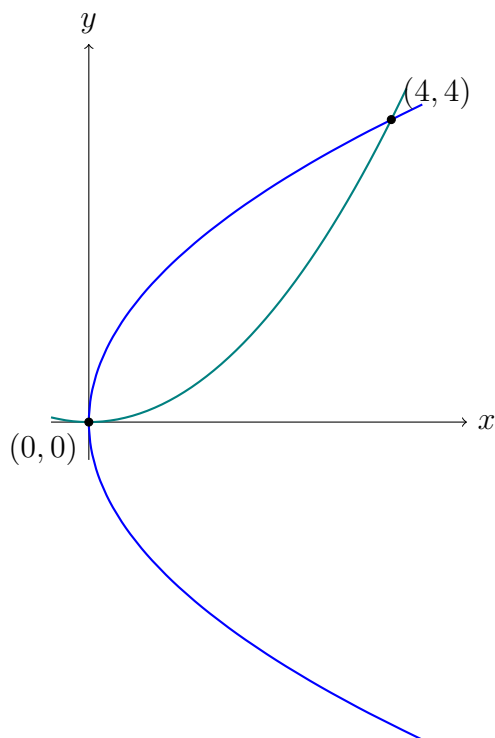
Hence Proved

Since $m_1 = m_2 = -1$ at $(2, 2)$ and $(-2, -2)$, the curves touch each other. Hence proved.

Question 35

Find the angle of intersection between the curves $y^2 = 4x$ and $x^2 = 4y$.

TYPE 45a Angle of Intersection and Orthogonality of Curves – Finding Angle of Intersection Between Two Curves



Intersection of parabolas $y^2 = 4x$ and $x^2 = 4y$

Given: Curves $C_1 : y^2 = 4x$ and $C_2 : x^2 = 4y$.

Concept / Formula Used: $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Solution:

First, find the points of intersection: From $C_2 : y = \frac{x^2}{4}$. Substitute into $C_1 : y^2 = 4x$:

$$\begin{aligned} \left(\frac{x^2}{4} \right)^2 &= 4x \\ \frac{x^4}{16} &= 4x \\ x^4 &= 64x \\ x^4 - 64x &= 0 \\ x(x^3 - 64) &= 0 \end{aligned}$$

So $x = 0$ or $x^3 = 64 \implies x = 4$. When $x = 0$, $y = 0$, giving $O(0,0)$. When $x = 4$, $y = \frac{4^2}{4} = 4$, giving $P(4,4)$.

Differentiating $C_1 : y^2 = 4x$:

$$\begin{aligned} 2y \frac{dy}{dx} &= 4 \\ \frac{dy}{dx} &= \frac{2}{y} \end{aligned}$$

Differentiating $C_2 : x^2 = 4y$:

$$2x = 4 \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{2}$$

Case 1: At $O(0,0)$: For C_1 , $\frac{dy}{dx} \rightarrow \infty$, so the tangent is the y-axis ($x = 0$). For C_2 , $\frac{dy}{dx} = \frac{0}{2} = 0$, so the tangent is the x-axis ($y = 0$). Since the x-axis and y-axis are perpendicular, the angle of intersection at $(0,0)$ is 90° (they cut orthogonally).

Case 2: At $P(4,4)$:

$$m_1 = \frac{2}{4} = \frac{1}{2}$$

$$m_2 = \frac{4}{2} = 2$$

Calculating $\tan \theta$:

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{\frac{1}{2} - 2}{1 + \left(\frac{1}{2}\right)(2)} \right| \\ &= \left| \frac{-\frac{3}{2}}{1 + 1} \right| \\ &= \left| \frac{-\frac{3}{2}}{2} \right| \\ &= \frac{3}{4} \end{aligned}$$

Thus, $\theta = \tan^{-1} \left(\frac{3}{4} \right)$.

Final Answer

At $(0,0)$, the curves cut orthogonally ($\theta = 90^\circ$); at $(4,4)$, the angle of intersection is $\theta = \tan^{-1} \left(\frac{3}{4} \right)$.

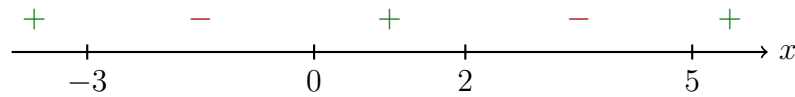
6.3 – Solving Inequalities via Sign-Scheme Method

Solve the following inequalities for $x \in \mathbb{R}$:

(i) $x(x-2)(x-5)(x+3) > 0$

(ii) $x^4 - 5x^2 + 4 \geq 0$

TYPE 48a Solving Inequalities via Sign-Scheme Method – Solving Polynomial and Reducible Quartic Inequalities via Sign



Sign schemes on the real number line for parts (i) and (ii)

To Find: The set of real values of x satisfying the given polynomial inequalities.

Concept / Formula Used: $P(x) = (x - a_1)(x - a_2) \dots (x - a_n) \quad a_1 < a_2 < \dots < a_n$
 $P(x)$

Solution:

Part (i): Solve $x(x-2)(x-5)(x+3) > 0$.

Step 1: Find the critical points by setting each factor to zero.

$$x = 0$$

$$x - 2 = 0 \implies x = 2$$

$$x - 5 = 0 \implies x = 5$$

$$x + 3 = 0 \implies x = -3$$

Step 2: Arrange the critical points in ascending order:

$$-3 < 0 < 2 < 5$$

Step 3: Test the sign of the product $f(x) = x(x-2)(x-5)(x+3)$ in each interval:

- For $x \in (5, \infty)$: Choose $x = 6 \implies (6)(4)(1)(9) = 216 > 0$ (+).
- For $x \in (2, 5)$: Choose $x = 3 \implies (3)(1)(-2)(6) = -36 < 0$ (-).
- For $x \in (0, 2)$: Choose $x = 1 \implies (1)(-1)(-4)(4) = 16 > 0$ (+).
- For $x \in (-3, 0)$: Choose $x = -1 \implies (-1)(-3)(-6)(2) = -36 < 0$ (-).
- For $x \in (-\infty, -3)$: Choose $x = -4 \implies (-4)(-6)(-9)(-1) = 216 > 0$ (+).

Step 4: Since the inequality is strictly positive (> 0), we select the positive intervals and exclude the critical points.

$$x \in (-\infty, -3) \cup (0, 2) \cup (5, \infty)$$

Part (ii): Solve $x^4 - 5x^2 + 4 \geq 0$.

Step 1: Let $y = x^2$. Substitute y into the inequality:

$$y^2 - 5y + 4 \geq 0$$

Step 2: Factorize the quadratic expression:

$$\begin{aligned}y^2 - 4y - y + 4 &\geq 0 \\y(y - 4) - 1(y - 4) &\geq 0 \\(y - 1)(y - 4) &\geq 0\end{aligned}$$

Step 3: Back-substitute $y = x^2$:

$$(x^2 - 1)(x^2 - 4) \geq 0$$

Step 4: Express as linear factors using $a^2 - b^2 = (a - b)(a + b)$:

$$(x - 1)(x + 1)(x - 2)(x + 2) \geq 0$$

Step 5: Identify critical points in ascending order:

$$-2 < -1 < 1 < 2$$

Step 6: Test the sign in each interval:

- For $x \in (2, \infty)$: $(+)(+)(+)(+) > 0$ (+).
- For $x \in (1, 2)$: $(+)(+)(-)(+) < 0$ (-).
- For $x \in (-1, 1)$: $(-)(+)(-)(+) > 0$ (+).
- For $x \in (-2, -1)$: $(-)(+)(-)(-) < 0$ (-).
- For $x \in (-\infty, -2)$: $(-)(-)(-)(-) > 0$ (+).

Step 7: Since the inequality is non-negative (≥ 0), include the critical points where the expression equals zero:

$$x \in (-\infty, -2] \cup [-1, 1] \cup [2, \infty)$$

Final Answer

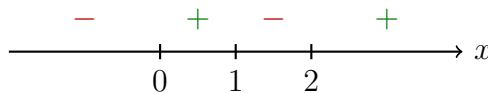
- (i) $x \in (-\infty, -3) \cup (0, 2) \cup (5, \infty)$
- (ii) $x \in (-\infty, -2] \cup [-1, 1] \cup [2, \infty)$

Common Mistake: In part (ii), forgetting to include the endpoints for non-strict inequalities ($\geq$) or failing to back-substitute $y = x^2$ into linear factors of x leads to incorrect solution sets.

Question 2

Find all real values of x which satisfy:

- (i) $x^3(x - 1)(x - 2) > 0$
- (ii) $x^2(x - 1)(x - 2) \leq 0$



Sign behavior around critical points with odd and even multiplicities

To Find: The set of all real values of x satisfying each given inequality.

Concept / Formula Used: $(x - a)^m$ m $x = a$ m $x = a$

Solution:

Part (i): Solve $x^3(x - 1)(x - 2) > 0$.

Step 1: Identify critical points:

$$\begin{aligned} x^3 &= 0 \implies x = 0 \quad (\text{multiplicity } 3, \text{ odd}) \\ x - 1 &= 0 \implies x = 1 \quad (\text{multiplicity } 1, \text{ odd}) \\ x - 2 &= 0 \implies x = 2 \quad (\text{multiplicity } 1, \text{ odd}) \end{aligned}$$

Step 2: Arrange critical points in ascending order:

$$0 < 1 < 2$$

Step 3: Test the sign of $f(x) = x^3(x - 1)(x - 2)$ across the intervals:

- For $x \in (2, \infty)$: Choose $x = 3 \implies (27)(2)(1) = 54 > 0$ (+).
- For $x \in (1, 2)$: Choose $x = 1.5 \implies (1.5)^3(0.5)(-0.5) = -0.84375 < 0$ (-).
- For $x \in (0, 1)$: Choose $x = 0.5 \implies (0.5)^3(-0.5)(-1.5) = 0.09375 > 0$ (+).
- For $x \in (-\infty, 0)$: Choose $x = -1 \implies (-1)^3(-2)(-3) = -6 < 0$ (-).

Step 4: Collect intervals where $f(x) > 0$:

$$x \in (0, 1) \cup (2, \infty)$$

Part (ii): Solve $x^2(x - 1)(x - 2) \leq 0$.

Step 1: Analyze the factor x^2 : Since $x^2 \geq 0$ for all real x , x^2 does not change the sign of the expression for $x \neq 0$.

Step 2: Consider two cases based on the value of x^2 :

Case 1: $x = 0$. Substituting $x = 0$ into the inequality:

$$0^2(0 - 1)(0 - 2) = 0 \leq 0$$

This is a true statement, so $x = 0$ is a valid solution.

Case 2: $x \neq 0$. Since $x^2 > 0$, we can divide both sides of the inequality by x^2 without changing the inequality sign:

$$(x - 1)(x - 2) \leq 0$$

Step 3: Solve $(x - 1)(x - 2) \leq 0$: The critical points are $x = 1$ and $x = 2$. The quadratic $(x - 1)(x - 2)$ is non-positive between its roots:

$$1 \leq x \leq 2 \implies x \in [1, 2]$$

Step 4: Combine the solutions from Case 1 and Case 2:

$$x \in \{0\} \cup [1, 2]$$

Final Answer

- (i) $x \in (0, 1) \cup (2, \infty)$
- (ii) $x \in \{0\} \cup [1, 2]$

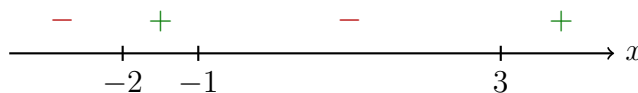
Common Mistake: In part (ii), students often divide by x^2 immediately and forget to check $x = 0$, thereby missing the isolated solution point $\{0\}$.

Question 3

Solve the following rational inequalities for $x \in \mathbb{R}$:

- (i) $\frac{1}{x - 2} \leq 1$
- (ii) $\frac{(x + 1)(x - 3)}{x + 2} \geq 0$

TYPE 48c Solving Inequalities via Sign-Scheme Method – Solving Rational Inequalities via Sign Scheme



Sign scheme on the real number line for part (ii)

To Find: The set of real values of x satisfying each rational inequality.

Concept / Formula Used: $\frac{N(x)}{D(x)} \geq 0 \quad N(x) = 0 \quad D(x) = 0$

Solution:

Part (i): Solve $\frac{1}{x - 2} \leq 1$.

Step 1: Identify domain restrictions. Denominator cannot be zero:

$$x - 2 \neq 0 \implies x \neq 2$$

Step 2: Transpose 1 to the left-hand side:

$$\frac{1}{x - 2} - 1 \leq 0$$

Step 3: Combine into a single fraction over the common denominator:

$$\begin{aligned}\frac{1 - 1(x - 2)}{x - 2} &\leq 0 \\ \frac{1 - x + 2}{x - 2} &\leq 0 \\ \frac{3 - x}{x - 2} &\leq 0\end{aligned}$$

Step 4: Multiply the numerator by -1 to make the coefficient of x positive (reversing the inequality sign):

$$\frac{x - 3}{x - 2} \geq 0$$

Step 5: Determine critical points:

$$\begin{aligned}\text{Numerator zero: } x - 3 = 0 &\implies x = 3 \\ \text{Denominator zero: } x - 2 = 0 &\implies x = 2 \quad (\text{excluded})\end{aligned}$$

Step 6: Test signs across the intervals $(-\infty, 2)$, $(2, 3)$, and $(3, \infty)$:

- For $x \in (3, \infty)$: Choose $x = 4 \implies \frac{4 - 3}{4 - 2} = \frac{1}{2} > 0 \quad (+)$.
- For $x \in (2, 3)$: Choose $x = 2.5 \implies \frac{2.5 - 3}{2.5 - 2} = \frac{-0.5}{0.5} = -1 < 0 \quad (-)$.
- For $x \in (-\infty, 2)$: Choose $x = 0 \implies \frac{0 - 3}{0 - 2} = \frac{3}{2} > 0 \quad (+)$.

Step 7: Select intervals where the expression is ≥ 0 , including $x = 3$ and excluding $x = 2$:

$$x \in (-\infty, 2) \cup [3, \infty)$$

Part (ii): Solve $\frac{(x + 1)(x - 3)}{x + 2} \geq 0$.

Step 1: Identify domain restrictions:

$$x + 2 \neq 0 \implies x \neq -2$$

Step 2: Determine all critical points from numerator and denominator:

$$\begin{aligned}x + 1 = 0 &\implies x = -1 \\ x - 3 = 0 &\implies x = 3 \\ x + 2 = 0 &\implies x = -2 \quad (\text{excluded})\end{aligned}$$

Step 3: Arrange critical points in ascending order:

$$-2 < -1 < 3$$

Step 4: Test the sign of $f(x) = \frac{(x + 1)(x - 3)}{x + 2}$ in each interval:

- For $x \in (3, \infty)$: Choose $x = 4 \implies \frac{(5)(1)}{6} = \frac{5}{6} > 0$ (+).
- For $x \in (-1, 3)$: Choose $x = 0 \implies \frac{(1)(-3)}{2} = -\frac{3}{2} < 0$ (-).
- For $x \in (-2, -1)$: Choose $x = -1.5 \implies \frac{(-0.5)(-4.5)}{0.5} = \frac{2.25}{0.5} = 4.5 > 0$ (+).
- For $x \in (-\infty, -2)$: Choose $x = -3 \implies \frac{(-2)(-6)}{-1} = -12 < 0$ (-).

Step 5: Select non-negative regions (≥ 0). Include numerator roots $x = -1$ and $x = 3$; exclude denominator root $x = -2$:

$$x \in (-2, -1] \cup [3, \infty)$$

Final Answer

- (i) $x \in (-\infty, 2) \cup [3, \infty)$
- (ii) $x \in (-2, -1] \cup [3, \infty)$

Common Mistake: Cross-multiplying by $(x - 2)$ in part (i) without considering its sign is a standard conceptual error; rational inequalities must always be rearranged to have 0 on one side.

6.4 – Increasing and Decreasing Functions

Question 1

Show that the following functions are strictly increasing on $\mathbb{R}$:

- (i) $f(x) = 3x + 17$
- (ii) $f(x) = 7x - 3$

TYPE 51a Proving Monotonicity on an Interval – Proving Monotonicity of Linear Functions

To Prove: Show that $f'(x) > 0$ for all $x \in \mathbb{R}$ for both given functions.

Concept / Formula Used: $f(x)$ I $f'(x) > 0$ $x \in I$

Solution:

Part (i): Let the given function be:

$$f(x) = 3x + 17$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(3x + 17) \\ &= 3 \cdot 1 + 0 \\ &= 3 \end{aligned}$$

Since $3 > 0$ for all $x \in \mathbb{R}$, we have:

$$f'(x) > 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x) = 3x + 17$ is strictly increasing on $\mathbb{R}$.

Part (ii): Let the given function be:

$$f(x) = 7x - 3$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(7x - 3) \\ &= 7 \cdot 1 - 0 \\ &= 7 \end{aligned}$$

Since $7 > 0$ for all $x \in \mathbb{R}$, we have:

$$f'(x) > 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x) = 7x - 3$ is strictly increasing on $\mathbb{R}$.

Hence Proved

Both functions $f(x) = 3x + 17$ and $f(x) = 7x - 3$ satisfy $f'(x) > 0$ for all $x \in \mathbb{R}$, so they are strictly increasing on $\mathbb{R}$.

Question 2

- (i) Show that the function f given by $f(x) = 3 - 7x$ is strictly decreasing.
- (ii) Show that the function $f(x) = 1 - \frac{1}{x}$ is strictly increasing.

TYPE 51b Proving Monotonicity on an Interval – Proving Monotonicity of Simple Linear and Rational Functions

To Prove:

- (i) Show that $f'(x) < 0$ for all $x \in \mathbb{R}$.
- (ii) Show that $f'(x) > 0$ for all $x \in \mathbb{R} \setminus \{0\}$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$

Solution:

Part (i): Given function:

$$f(x) = 3 - 7x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(3 - 7x) \\ &= 0 - 7 \\ &= -7 \end{aligned}$$

Since $-7 < 0$ for all real numbers x :

$$f'(x) < 0 \quad \forall x \in \mathbb{R}$$

Therefore, $f(x) = 3 - 7x$ is strictly decreasing on $\mathbb{R}$.

Part (ii): Given function:

$$f(x) = 1 - \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(1 - x^{-1}) \\ &= 0 - (-1)x^{-2} \\ &= \frac{1}{x^2} \end{aligned}$$

For any non-zero real number x :

$$x^2 > 0 \implies \frac{1}{x^2} > 0$$

Thus, $f'(x) > 0$ for all $x \in \mathbb{R} \setminus \{0\}$. Therefore, $f(x) = 1 - \frac{1}{x}$ is strictly increasing on its domain.

Hence Proved

Part (i) shows $f'(x) = -7 < 0$, so $f(x)$ is strictly decreasing. Part (ii) shows $f'(x) = \frac{1}{x^2} > 0$, so $f(x)$ is strictly increasing.

Question 3

- (i) Prove that the function $f(x) = 3 + \frac{1}{x}$ is strictly decreasing.
- (ii) Show that the function $f(x) = x - \frac{1}{x}$ is strictly increasing.

TYPE 51c Proving Monotonicity on an Interval – Proving Monotonicity of Simple Rational Functions

To Prove:

- (i) Show that $f'(x) < 0$ for all $x \neq 0$.
- (ii) Show that $f'(x) > 0$ for all $x \neq 0$.

Concept / Formula Used: $x \in \mathbb{R} \setminus \{0\}$ $x^2 > 0$

Solution:**Part (i):** Given function:

$$f(x) = 3 + \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (3 + x^{-1}) \\ &= 0 + (-1)x^{-2} \\ &= -\frac{1}{x^2} \end{aligned}$$

Since $x^2 > 0$ for all $x \neq 0$:

$$\frac{1}{x^2} > 0 \implies -\frac{1}{x^2} < 0$$

Thus, $f'(x) < 0$ for all $x \neq 0$. Hence, $f(x) = 3 + \frac{1}{x}$ is strictly decreasing on its domain.**Part (ii):** Given function:

$$f(x) = x - \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (x - x^{-1}) \\ &= 1 - (-1)x^{-2} \\ &= 1 + \frac{1}{x^2} \end{aligned}$$

Since $x^2 > 0$ for all $x \neq 0$, we have $\frac{1}{x^2} > 0$. Adding 1 to both sides:

$$1 + \frac{1}{x^2} > 1 > 0$$

Thus, $f'(x) > 0$ for all $x \neq 0$. Hence, $f(x) = x - \frac{1}{x}$ is strictly increasing on its domain.**Hence Proved**LHS derivations show $f'(x) < 0$ for part (i) and $f'(x) > 0$ for part (ii), establishing the required monotonicity.**Question 4**

- (i) Show that the function $f(x) = e^{-2x}$ is strictly decreasing on $\mathbb{R}$.
- (ii) Show that the function $f(x) = 5^x$ is strictly increasing.
- (iii) Show that the function $f(x) = \log x$ is strictly increasing.
- (iv) Show that the function $f(x) = -x + \cos x$ is decreasing.

To Prove: Verify the sign of $f'(x)$ on the respective domains for all four functions.

Concept / Formula Used: $\frac{d}{dx}(e^{kx}) = ke^{kx}$, $\frac{d}{dx}(a^x) = a^x \ln a$, $\frac{d}{dx}(\log x) = \frac{1}{x}$

Solution:

Part (i): Given $f(x) = e^{-2x}$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(e^{-2x}) \\ &= -2e^{-2x} \end{aligned}$$

Since exponential function $e^{-2x} > 0$ for all $x \in \mathbb{R}$:

$$-2e^{-2x} < 0 \implies f'(x) < 0 \quad \forall x \in \mathbb{R}$$

Thus, $f(x) = e^{-2x}$ is strictly decreasing on $\mathbb{R}$.

Part (ii): Given $f(x) = 5^x$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(5^x) \\ &= 5^x \log 5 \end{aligned}$$

Since $5^x > 0$ for all $x \in \mathbb{R}$ and $\log 5 > 0$:

$$f'(x) > 0 \quad \forall x \in \mathbb{R}$$

Thus, $f(x) = 5^x$ is strictly increasing on $\mathbb{R}$.

Part (iii): Given $f(x) = \log x$, defined for $x > 0$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(\log x) \\ &= \frac{1}{x} \end{aligned}$$

For all $x \in (0, \infty)$, $x > 0 \implies \frac{1}{x} > 0$. Thus, $f'(x) > 0$ for all $x > 0$. Hence, $f(x) = \log x$ is strictly increasing on its domain.

Part (iv): Given $f(x) = -x + \cos x$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(-x + \cos x) \\ &= -1 - \sin x \\ &= -(1 + \sin x) \end{aligned}$$

Using the standard trigonometric property $-1 \leq \sin x \leq 1$:

$$1 + \sin x \geq 0 \implies -(1 + \sin x) \leq 0$$

Thus, $f'(x) \leq 0$ for all $x \in \mathbb{R}$. Hence, $f(x) = -x + \cos x$ is a decreasing function on $\mathbb{R}$.

Hence Proved

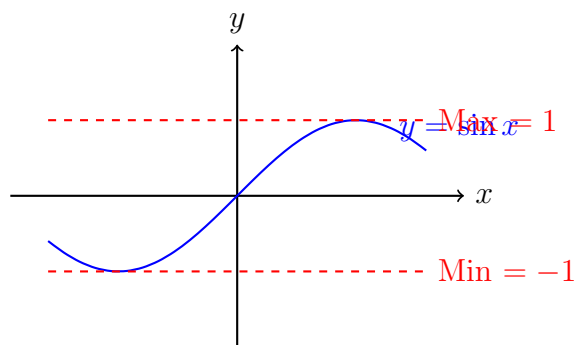
All four sub-parts are verified using derivative tests on their respective domains.

Question 5

Find the values of 'a' for which the function $f(x) = b - ax + \sin x$ is increasing on $\mathbb{R}$.

TYPE 55a Finding Parameter Range for Monotonicity – Finding Parameter Range for Monotonicity of Trigonometric-L

★ Likely Exam Question



Sign condition for parameter a

Given: Function $f(x) = b - ax + \sin x$ is increasing on $\mathbb{R}$, where b is a real constant.

To Find: The set of real values of a .

Concept / Formula Used: $f(x)$ $\mathbb{R}$ $f'(x) \geq 0$ $x \in \mathbb{R}$

Solution:

Given function:

$$f(x) = b - ax + \sin x$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(b - ax + \sin x) \\ &= 0 - a + \cos x \\ &= \cos x - a \end{aligned}$$

For $f(x)$ to be an increasing function on $\mathbb{R}$, we must have:

$$f'(x) \geq 0 \quad \forall x \in \mathbb{R}$$

Substituting $f'(x) = \cos x - a$:

$$\begin{aligned} \cos x - a &\geq 0 \quad \forall x \in \mathbb{R} \\ \cos x &\geq a \quad \forall x \in \mathbb{R} \end{aligned}$$

This inequality must hold true for all $x \in \mathbb{R}$. Therefore, a must be less than or equal to the minimum value of $\cos x$ on $\mathbb{R}$.

We know that:

$$-1 \leq \cos x \leq 1 \quad \forall x \in \mathbb{R}$$

Thus, the minimum value of $\cos x$ is -1 .

Setting $a \leq \min(\cos x)$:

$$a \leq -1$$

Final Answer

The function $f(x)$ is increasing on $\mathbb{R}$ for all $a \leq -1$, i.e., $a \in (-\infty, -1]$.

Question 6

- (i) Find the interval in which the function f given by $f(x) = 7 - 4x - x^2$ is strictly increasing.
- (ii) If $f(x) = x^3 - 48x$, find the interval where $f : \mathbb{R} \rightarrow \mathbb{R}$ is decreasing.
- (iii) Find the interval in which the function $f(x) = 2x^3 - 3x$ is strictly increasing.

TYPE 56a Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Quadratic and Cubic Polynomials

To Find: The intervals of monotonicity for each polynomial function.

Concept / Formula Used: $f'(x) > 0$ $f'(x) \leq 0$ $f'(x) < 0$

Solution:

Part (i): Given function:

$$f(x) = 7 - 4x - x^2$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(7 - 4x - x^2) \\ &= -4 - 2x \\ &= -2(x + 2) \end{aligned}$$

For $f(x)$ to be strictly increasing:

$$\begin{aligned} f'(x) &> 0 \\ -2(x + 2) &> 0 \end{aligned}$$

Dividing by -2 (and reversing the inequality sign):

$$\begin{aligned} x + 2 &< 0 \\ x &< -2 \end{aligned}$$

Thus, $x \in (-\infty, -2)$.

Part (ii): Given function:

$$f(x) = x^3 - 48x$$

Differentiating with respect to x :

$$\begin{aligned}f'(x) &= \frac{d}{dx}(x^3 - 48x) \\&= 3x^2 - 48 \\&= 3(x^2 - 16) \\&= 3(x - 4)(x + 4)\end{aligned}$$

For $f(x)$ to be decreasing:

$$\begin{aligned}f'(x) &\leq 0 \\3(x - 4)(x + 4) &\leq 0 \\(x - 4)(x + 4) &\leq 0\end{aligned}$$

Solving the quadratic inequality gives:

$$-4 \leq x \leq 4$$

Thus, $x \in [-4, 4]$.

Part (iii): Given function:

$$f(x) = 2x^3 - 3x$$

Differentiating with respect to x :

$$\begin{aligned}f'(x) &= \frac{d}{dx}(2x^3 - 3x) \\&= 6x^2 - 3 \\&= 3(2x^2 - 1)\end{aligned}$$

For $f(x)$ to be strictly increasing:

$$\begin{aligned}f'(x) &> 0 \\3(2x^2 - 1) &> 0 \\2x^2 - 1 &> 0 \\x^2 &> \frac{1}{2}\end{aligned}$$

Taking square root on both sides:

$$|x| > \frac{1}{\sqrt{2}} \implies x < -\frac{1}{\sqrt{2}} \quad \text{or} \quad x > \frac{1}{\sqrt{2}}$$

$$\text{Thus, } x \in \left(-\infty, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, \infty\right).$$

Final Answer

- (i) Strictly increasing in $(-\infty, -2)$
- (ii) Decreasing in $[-4, 4]$
- (iii) Strictly increasing in $\left(-\infty, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, \infty\right)$

Question 7

Show that the function f defined by $f(x) = (x - 1)e^x + 1$ is an increasing function for all $x > 0$.

TYPE 51e Proving Monotonicity on an Interval – Proving Monotonicity of Product of Polynomial and Exponential Function

To Prove: Show that $f'(x) > 0$ for all $x > 0$.

Concept / Formula Used: $\frac{d}{dx}(uv) = u'v + uv'$

Solution:

Given function:

$$f(x) = (x - 1)e^x + 1$$

Differentiating with respect to x using the product rule:

$$\begin{aligned} f'(x) &= \frac{d}{dx} [(x - 1)e^x + 1] \\ &= \frac{d}{dx}(x - 1) \cdot e^x + (x - 1) \frac{d}{dx}(e^x) + 0 \\ &= 1 \cdot e^x + (x - 1)e^x \\ &= e^x(1 + x - 1) \\ &= xe^x \end{aligned}$$

Now, examine the signs of the terms for $x > 0$:

$$\begin{aligned} x &> 0 \\ e^x &> 0 \quad \forall x \in \mathbb{R} \end{aligned}$$

Multiplying two positive quantities:

$$f'(x) = xe^x > 0 \quad \forall x > 0$$

Since $f'(x) > 0$ for all $x > 0$, the function $f(x)$ is strictly increasing for all $x > 0$.

Hence Proved

Since $f'(x) = xe^x > 0$ for all $x > 0$, $f(x) = (x - 1)e^x + 1$ is an increasing function for all $x > 0$.

Question 8

For what value of k , the function $f(x) = kx^3 + 5$ is decreasing?

TYPE 55b Finding Parameter Range for Monotonicity – Finding Parameter Range for Monotonicity of Cubic Polynomial

Given: Function $f(x) = kx^3 + 5$ is decreasing on $\mathbb{R}$.

To Find: The condition on parameter k .

Concept / Formula Used: $f(x) \quad f'(x) \leq 0 \quad x \in \mathbb{R}$

Solution:

Given function:

$$f(x) = kx^3 + 5$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(kx^3 + 5) \\ &= 3kx^2 \end{aligned}$$

For $f(x)$ to be decreasing on $\mathbb{R}$:

$$f'(x) \leq 0 \quad \forall x \in \mathbb{R}$$

Substituting $f'(x)$:

$$3kx^2 \leq 0 \quad \forall x \in \mathbb{R}$$

Since $x^2 \geq 0$ for all real numbers x :

- If $k < 0$, then $3k < 0$, so $3kx^2 \leq 0$ for all $x \in \mathbb{R}$.
- If $k = 0$, then $f(x) = 5$, a constant function, which is non-increasing.
- If $k > 0$, then $3kx^2 \geq 0$, which makes $f(x)$ increasing.

For non-trivial strict decreasing behavior (or general decreasing behavior):

$$k < 0 \quad (\text{or } k \leq 0)$$

Final Answer

$k < 0$ (or $k \leq 0$ for non-strictly decreasing).

Question 9

- Prove that the function $f(x) = ax + b$ is strictly decreasing iff $a < 0$.
- Prove that the function $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.

TYPE 51f Proving Monotonicity on an Interval – Proving Monotonicity of Parametric Linear and Greatest Integer Functions

To Prove:

- $f(x) = ax + b$ is strictly decreasing if and only if $a < 0$.
- $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.

Concept / Formula Used: $x \in [0, 1) \quad [x] = 0$

Solution:

Part (i): Let $f(x) = ax + b$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(ax + b) \\ &= a \end{aligned}$$

By definition of monotonicity:

- $f(x)$ is strictly decreasing $\iff f'(x) < 0 \quad \forall x \in \mathbb{R}$.
- Since $f'(x) = a$, this condition directly gives $a < 0$.

Thus, $f(x) = ax + b$ is strictly decreasing if and only if $a < 0$.

Part (ii): For $x \in [0, 1)$, by definition of the greatest integer function:

$$[x] = 0$$

Substitute $[x] = 0$ into $f(x)$:

$$\begin{aligned} f(x) &= 0 - x \\ &= -x \end{aligned}$$

Differentiating $f(x)$ with respect to x for $x \in (0, 1)$:

$$\begin{aligned} f'(x) &= \frac{d}{dx}(-x) \\ &= -1 \end{aligned}$$

Since $-1 < 0$ for all $x \in (0, 1)$:

$$f'(x) < 0 \quad \forall x \in (0, 1)$$

Alternatively, pick any $x_1, x_2 \in [0, 1)$ such that $x_1 < x_2$:

$$\begin{aligned} f(x_1) &= -x_1 \\ f(x_2) &= -x_2 \\ x_1 < x_2 &\implies -x_1 > -x_2 \implies f(x_1) > f(x_2) \end{aligned}$$

Hence, $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.

Hence Proved

Both assertions are proven: (i) $f'(x) = a < 0 \iff a < 0$, and (ii) $f(x) = -x$ on $[0, 1)$ has derivative $-1 < 0$.

Question 10

Prove that the function $f(x) = x^3 - 6x^2 + 12x + 5$ is increasing on $\mathbb{R}$.

TYPE 51g Proving Monotonicity on an Interval – Proving Monotonicity of Cubic Polynomial with Perfect Square Derivative

★ Likely Exam Question

To Prove: Show that $f'(x) \geq 0$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $(x - k)^2 \geq 0 \quad x$

Solution:

Given function:

$$f(x) = x^3 - 6x^2 + 12x + 5$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (x^3 - 6x^2 + 12x + 5) \\ &= 3x^2 - 12x + 12 \end{aligned}$$

Factoring out 3:

$$\begin{aligned} f'(x) &= 3(x^2 - 4x + 4) \\ &= 3(x - 2)^2 \end{aligned}$$

Since the square of any real number is non-negative:

$$(x - 2)^2 \geq 0 \quad \forall x \in \mathbb{R}$$

Multiplying by $3 > 0$:

$$3(x - 2)^2 \geq 0 \quad \forall x \in \mathbb{R}$$

Therefore:

$$f'(x) \geq 0 \quad \forall x \in \mathbb{R}$$

Since $f'(x) \geq 0$ for all $x \in \mathbb{R}$ and vanishes only at the point $x = 2$ (which does not contain any sub-interval), $f(x)$ is increasing on $\mathbb{R}$.

Hence Proved

$f'(x) = 3(x - 2)^2 \geq 0$ for all $x \in \mathbb{R}$, so $f(x)$ is an increasing function on $\mathbb{R}$.

Question 11

Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ is strictly increasing on $\mathbb{R}$.

TYPE 51g Proving Monotonicity on an Interval – Proving Monotonicity of Cubic Polynomial with Perfect Square Derivative

To Prove: Show that $f'(x) > 0$ for almost all $x \in \mathbb{R}$.

Concept / Formula Used: $a(bx - c)^2 \geq 0 \quad \mathbb{R}$

Solution:

Given function:

$$f(x) = 4x^3 - 18x^2 + 27x - 7$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (4x^3 - 18x^2 + 27x - 7) \\ &= 12x^2 - 36x + 27 \end{aligned}$$

Factoring out 3:

$$f'(x) = 3(4x^2 - 12x + 9)$$

Recognizing the perfect square inside the bracket:

$$4x^2 - 12x + 9 = (2x - 3)^2$$

Thus:

$$f'(x) = 3(2x - 3)^2$$

For all $x \in \mathbb{R}$:

$$(2x - 3)^2 \geq 0 \implies f'(x) \geq 0$$

Furthermore, $f'(x) = 0$ holds only at the isolated point:

$$2x - 3 = 0 \implies x = \frac{3}{2}$$

Since $f'(x) > 0$ for all $x \in \mathbb{R} \setminus \{\frac{3}{2}\}$, $f(x)$ does not remain constant on any interval. Hence, $f(x)$ is strictly increasing on $\mathbb{R}$.

Hence Proved

Since $f'(x) = 3(2x - 3)^2 \geq 0$ with equality at a single point, $f(x)$ is strictly increasing on $\mathbb{R}$.

Question 12

Prove that the function $f(x) = 5 - 3x + 3x^2 - x^3$ is decreasing on $\mathbb{R}$.

TYPE 51g Proving Monotonicity on an Interval – Proving Monotonicity of Cubic Polynomial with Perfect Square Derivative

To Prove: Show that $f'(x) \leq 0$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $f'(x) \leq 0 \implies f(x)$

Solution:

Given function:

$$f(x) = 5 - 3x + 3x^2 - x^3$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (5 - 3x + 3x^2 - x^3) \\ &= 0 - 3 + 6x - 3x^2 \\ &= -3(x^2 - 2x + 1) \end{aligned}$$

Recognizing the perfect square:

$$x^2 - 2x + 1 = (x - 1)^2$$

Thus:

$$f'(x) = -3(x - 1)^2$$

For all $x \in \mathbb{R}$:

$$(x - 1)^2 \geq 0$$

Multiplying by $-3 < 0$:

$$-3(x - 1)^2 \leq 0$$

Therefore:

$$f'(x) \leq 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x) = 5 - 3x + 3x^2 - x^3$ is a decreasing function on $\mathbb{R}$.

Hence Proved

Since $f'(x) = -3(x - 1)^2 \leq 0$ for all $x \in \mathbb{R}$, $f(x)$ is decreasing on $\mathbb{R}$.

Question 13

- (i) Show that $f(x) = 2x - \sin 2x$ is an increasing function.
 (ii) Show that the function $f(x) = \frac{16 \sin x}{4 + \cos x} - x$ is strictly decreasing in $\left(\frac{\pi}{2}, \pi\right)$.

TYPE 51h Proving Monotonicity on an Interval – Proving Monotonicity of Trigonometric and Rational Trigonometric Functions

To Prove:

- (i) $f'(x) \geq 0$ for all $x \in \mathbb{R}$.
 (ii) $f'(x) < 0$ for all $x \in \left(\frac{\pi}{2}, \pi\right)$.

Concept / Formula Used: $\frac{d}{dx} \left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$ $\cos^2 x + \sin^2 x = 1$

Solution:

Part (i): Given function:

$$f(x) = 2x - \sin 2x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(2x - \sin 2x) \\ &= 2 - 2 \cos 2x \\ &= 2(1 - \cos 2x) \end{aligned}$$

Using the standard identity $1 - \cos 2x = 2 \sin^2 x$:

$$\begin{aligned} f'(x) &= 2(2 \sin^2 x) \\ &= 4 \sin^2 x \end{aligned}$$

Since $\sin^2 x \geq 0$ for all $x \in \mathbb{R}$:

$$f'(x) = 4 \sin^2 x \geq 0 \quad \forall x \in \mathbb{R}$$

Hence, $f(x) = 2x - \sin 2x$ is an increasing function.

Part (ii): Given function:

$$f(x) = \frac{16 \sin x}{4 + \cos x} - x$$

Differentiating using the quotient rule:

$$\begin{aligned} f'(x) &= \frac{(4 + \cos x) \frac{d}{dx}(16 \sin x) - 16 \sin x \frac{d}{dx}(4 + \cos x)}{(4 + \cos x)^2} - 1 \\ &= \frac{(4 + \cos x)(16 \cos x) - 16 \sin x(-\sin x)}{(4 + \cos x)^2} - 1 \\ &= \frac{64 \cos x + 16 \cos^2 x + 16 \sin^2 x}{(4 + \cos x)^2} - 1 \end{aligned}$$

Using $\cos^2 x + \sin^2 x = 1$:

$$\begin{aligned} f'(x) &= \frac{64 \cos x + 16}{(4 + \cos x)^2} - 1 \\ &= \frac{64 \cos x + 16 - (4 + \cos x)^2}{(4 + \cos x)^2} \\ &= \frac{64 \cos x + 16 - (16 + 8 \cos x + \cos^2 x)}{(4 + \cos x)^2} \\ &= \frac{56 \cos x - \cos^2 x}{(4 + \cos x)^2} \\ &= \frac{\cos x(56 - \cos x)}{(4 + \cos x)^2} \end{aligned}$$

Now analyze the signs of the factors in the interval $x \in (\frac{\pi}{2}, \pi)$:

- In the second quadrant $(\frac{\pi}{2}, \pi)$, $\cos x < 0$.
- Since $-1 < \cos x < 0$, we have $56 - \cos x > 56 - 0 > 0$.
- The denominator $(4 + \cos x)^2 > 0$.

Combining these signs:

$$f'(x) = \frac{(\text{negative}) \cdot (\text{positive})}{\text{positive}} < 0$$

Thus, $f'(x) < 0$ for all $x \in (\frac{\pi}{2}, \pi)$. Hence, $f(x)$ is strictly decreasing in $(\frac{\pi}{2}, \pi)$.

Hence Proved

Both parts are proved using explicit derivative evaluation and interval sign analysis.

Question 14

- (i) Prove that the function f given by $f(x) = \log \cos x$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$.
- (ii) If $f(x) = a(\tan x - \cot x)$, where $a > 0$, then find whether $f(x)$ is increasing or decreasing function in its domain.

TYPE 62a Determining/Classifying Monotonicity of a Function – Determining Monotonicity of Logarithmic-Trigonometric**To Prove:**

- (i) Show $f'(x) < 0$ for $x \in \left(0, \frac{\pi}{2}\right)$.
- (ii) Determine the sign of $f'(x)$ for $f(x) = a(\tan x - \cot x)$ with $a > 0$.

Concept / Formula Used: $\frac{d}{dx}(\log(\cos x)) = \frac{1}{\cos x}(-\sin x) = -\tan x$
 $\frac{d}{dx}(\tan x) = \sec^2 x, \frac{d}{dx}(\cot x) = -\csc^2 x$

Solution:**Part (i):** Given function:

$$f(x) = \log \cos x$$

Differentiating with respect to x using chain rule:

$$\begin{aligned} f'(x) &= \frac{1}{\cos x} \cdot \frac{d}{dx}(\cos x) \\ &= \frac{1}{\cos x} \cdot (-\sin x) \\ &= -\tan x \end{aligned}$$

For $x \in \left(0, \frac{\pi}{2}\right)$:

$$\tan x > 0 \implies -\tan x < 0$$

Thus:

$$f'(x) < 0 \quad \forall x \in \left(0, \frac{\pi}{2}\right)$$

Hence, $f(x) = \log \cos x$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$.**Part (ii):** Given function:

$$f(x) = a(\tan x - \cot x), \quad a > 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= a \cdot \frac{d}{dx}(\tan x - \cot x) \\ &= a(\sec^2 x - (-\csc^2 x)) \\ &= a(\sec^2 x + \csc^2 x) \end{aligned}$$

Since $\sec^2 x > 0$ and $\csc^2 x > 0$ for all x in the domain of definition, and given $a > 0$:

$$f'(x) = a(\sec^2 x + \csc^2 x) > 0$$

Therefore, $f'(x) > 0$ for all x in its domain. Hence, $f(x)$ is an increasing function in its domain.

Final Answer

- (i) Proved that $f(x)$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$.
 (ii) $f(x)$ is an **increasing** function in its domain.

Question 15

Determine whether the following functions are increasing or decreasing for the stated values of x :

- (i) $f(x) = \frac{1}{x}$, $x < 0$
 (ii) $f(x) = \frac{1}{1+x^2}$, $x \geq 0$
 (iii) $f(x) = \cos^2 x$ in $\left[0, \frac{\pi}{2}\right]$
 (iv) $f(x) = -\frac{x}{2} + \sin x$, $-\frac{\pi}{3} < x < \frac{\pi}{3}$

TYPE 62b Determining/Classifying Monotonicity of a Function – Determining Monotonicity of Rational and Trigonometric Functions

To Find: Monotonicity behavior (increasing/decreasing) for each given function on its specified interval.

Concept / Formula Used: $f'(x) > 0$ < 0

Solution:

Part (i): Given $f(x) = \frac{1}{x} = x^{-1}$ for $x < 0$. Differentiating with respect to x :

$$f'(x) = -\frac{1}{x^2}$$

For any $x < 0$, $x^2 > 0 \implies \frac{1}{x^2} > 0 \implies -\frac{1}{x^2} < 0$. Thus, $f'(x) < 0$ for $x < 0$. So, $f(x)$ is **decreasing** for $x < 0$.

Part (ii): Given $f(x) = \frac{1}{1+x^2} = (1+x^2)^{-1}$ for $x \geq 0$. Differentiating using chain rule:

$$\begin{aligned} f'(x) &= -(1+x^2)^{-2} \cdot (2x) \\ &= -\frac{2x}{(1+x^2)^2} \end{aligned}$$

For $x > 0$: $2x > 0$ and $(1+x^2)^2 > 0$, so $f'(x) < 0$. At $x = 0$: $f'(0) = 0$. Thus, $f'(x) \leq 0$ for all $x \geq 0$. So, $f(x)$ is **decreasing** for $x \geq 0$.

Part (iii): Given $f(x) = \cos^2 x$ for $x \in \left[0, \frac{\pi}{2}\right]$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 2 \cos x (-\sin x) \\ &= -\sin 2x \end{aligned}$$

For $x \in (0, \frac{\pi}{2}) \implies 2x \in (0, \pi) \implies \sin 2x > 0$. Therefore, $f'(x) = -\sin 2x < 0$ for $x \in (0, \frac{\pi}{2})$. So, $f(x)$ is **decreasing** in $[0, \frac{\pi}{2}]$.

Part (iv): Given $f(x) = -\frac{x}{2} + \sin x$ for $x \in (-\frac{\pi}{3}, \frac{\pi}{3})$. Differentiating with respect to x :

$$f'(x) = -\frac{1}{2} + \cos x$$

For $x \in (-\frac{\pi}{3}, \frac{\pi}{3})$:

$$\cos x > \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Thus:

$$f'(x) = \cos x - \frac{1}{2} > 0$$

So, $f(x)$ is **increasing** in $(-\frac{\pi}{3}, \frac{\pi}{3})$.

Final Answer

- (i) Decreasing
- (ii) Decreasing
- (iii) Decreasing
- (iv) Increasing

Question 16

Which of the following functions are strictly decreasing on $(0, \frac{\pi}{2})$?

- (i) $\cos x$
- (ii) $\cos 2x$
- (iii) $\sin^2 x$
- (iv) $\tan x$

TYPE 62c Determining/Classifying Monotonicity of a Function – Identifying Strictly Decreasing Trigonometric Function

To Find: Identify which of the four given trigonometric functions are strictly decreasing on $(0, \frac{\pi}{2})$.

Concept / Formula Used: $f(x)$ $f'(x) < 0$

Solution:

Let us evaluate $f'(x)$ for each function on the open interval $x \in (0, \frac{\pi}{2})$:

Case (i): Let $f(x) = \cos x$.

$$f'(x) = -\sin x$$

In $(0, \frac{\pi}{2})$, $\sin x > 0$, so $f'(x) = -\sin x < 0$. Hence, $\cos x$ is **strictly decreasing** on $(0, \frac{\pi}{2})$.

Case (ii): Let $f(x) = \cos 2x$.

$$f'(x) = -2 \sin 2x$$

For $x \in (0, \frac{\pi}{2})$, we have $2x \in (0, \pi)$. Since $\sin 2x > 0$ for $2x \in (0, \pi)$, we get $f'(x) = -2\sin 2x < 0$. Hence, $\cos 2x$ is **strictly decreasing** on $(0, \frac{\pi}{2})$.

Case (iii): Let $f(x) = \sin^2 x$.

$$f'(x) = 2 \sin x \cos x = \sin 2x$$

In $(0, \frac{\pi}{2})$, $\sin 2x > 0$, so $f'(x) > 0$. Hence, $\sin^2 x$ is strictly increasing (not decreasing).

Case (iv): Let $f(x) = \tan x$.

$$f'(x) = \sec^2 x$$

Since $\sec^2 x > 0$ everywhere in $(0, \frac{\pi}{2})$, $f'(x) > 0$. Hence, $\tan x$ is strictly increasing (not decreasing).

Final Answer

The functions that are strictly decreasing on $(0, \frac{\pi}{2})$ are **(i) $\cos x$** and **(ii) $\cos 2x$** .

Question 17

Consider the statement "There exists at least one value of $b \in \mathbb{R}$ for which $f(x) = \frac{b}{x}$, $b \neq 0$ is strictly increasing in $\mathbb{R} - \{0\}$." State True or False. Justify.

TYPE 55c Finding Parameter Range for Monotonicity – Analyzing Monotonicity of Rational Function with Parameter

To Find: State whether the statement is True or False and provide mathematical justification.

Concept / Formula Used: $f(x) = \frac{b}{x}$ $f'(x) = -\frac{b}{x^2}$

Solution:

Given function:

$$f(x) = \frac{b}{x}, \quad b \neq 0, \quad x \in \mathbb{R} - \{0\}$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (bx^{-1}) \\ &= -bx^{-2} \\ &= -\frac{b}{x^2} \end{aligned}$$

For $f(x)$ to be strictly increasing on $\mathbb{R} - \{0\}$, we require:

$$f'(x) > 0 \quad \forall x \in \mathbb{R} - \{0\}$$

Substituting $f'(x) = -\frac{b}{x^2}$:

$$-\frac{b}{x^2} > 0$$

Since $x^2 > 0$ for all $x \neq 0$, dividing/multiplying by x^2 preserves the inequality direction:

$$-b > 0 \implies b < 0$$

Thus, if we choose any negative real number for b (e.g., $b = -1$), then:

$$f(x) = -\frac{1}{x} \implies f'(x) = \frac{1}{x^2} > 0 \quad \forall x \neq 0$$

This means there indeed exists at least one value of $b \in \mathbb{R}$ (in fact, any $b < 0$) for which $f(x)$ is strictly increasing on $\mathbb{R} - \{0\}$.

Therefore, the given statement is ****True****.

Final Answer

True. Justification: For any $b < 0$, $f'(x) = -\frac{b}{x^2} > 0$ for all $x \in \mathbb{R} - \{0\}$, making $f(x)$ strictly increasing.

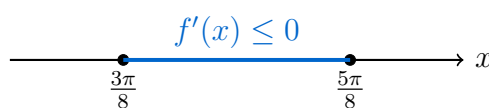
Question 18

Prove that the function $f(x) = \sin\left(2x + \frac{\pi}{4}\right)$ is decreasing for $\frac{3\pi}{8} \leq x \leq \frac{5\pi}{8}$.

TYPE 51i Proving Monotonicity on an Interval – Proving Monotonicity of Composite Sine Function on Closed Interval

To Prove: Function $f(x) = \sin\left(2x + \frac{\pi}{4}\right)$ is decreasing for $x \in \left[\frac{3\pi}{8}, \frac{5\pi}{8}\right]$.

Concept / Formula Used: $f(x)$ $[a, b]$ $f'(x) \leq 0$ $x \in (a, b)$



Interval evaluation for $f'(x)$

Solution:

Given function:

$$f(x) = \sin\left(2x + \frac{\pi}{4}\right)$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left[\sin\left(2x + \frac{\pi}{4}\right) \right] \\ &= \cos\left(2x + \frac{\pi}{4}\right) \cdot \frac{d}{dx} \left(2x + \frac{\pi}{4}\right) \\ &= 2 \cos\left(2x + \frac{\pi}{4}\right) \end{aligned}$$

Given interval for x :

$$\frac{3\pi}{8} \leq x \leq \frac{5\pi}{8}$$

Multiplying by 2:

$$\frac{3\pi}{4} \leq 2x \leq \frac{5\pi}{4}$$

Adding $\frac{\pi}{4}$ throughout:

$$\begin{aligned} \frac{3\pi}{4} + \frac{\pi}{4} &\leq 2x + \frac{\pi}{4} \leq \frac{5\pi}{4} + \frac{\pi}{4} \\ \pi &\leq 2x + \frac{\pi}{4} \leq \frac{3\pi}{2} \end{aligned}$$

For angle $\theta = 2x + \frac{\pi}{4}$ lying in the third quadrant $[\pi, \frac{3\pi}{2}]$:

$$\cos\left(2x + \frac{\pi}{4}\right) \leq 0$$

Multiplying by 2:

$$f'(x) = 2 \cos\left(2x + \frac{\pi}{4}\right) \leq 0$$

Since $f'(x) \leq 0$ for all $x \in [\frac{3\pi}{8}, \frac{5\pi}{8}]$, the function $f(x)$ is decreasing in this interval.

Hence Proved

Since $f'(x) \leq 0$ for all $x \in [\frac{3\pi}{8}, \frac{5\pi}{8}]$, $f(x) = \sin(2x + \frac{\pi}{4})$ is decreasing in the given interval.

Question 19

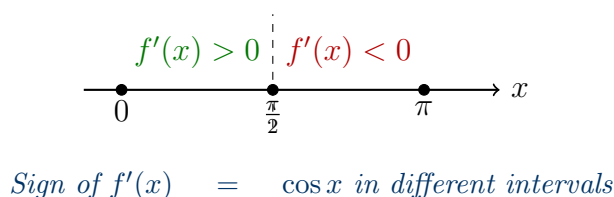
Prove that the function $f(x) = \sin x$ is:

- (i) strictly increasing in $(0, \frac{\pi}{2})$
- (ii) strictly decreasing in $(\frac{\pi}{2}, \pi)$
- (iii) neither increasing nor decreasing in $(0, \pi)$

TYPE 51j Proving Monotonicity on an Interval – Proving Monotonicity of Sine Function on Different Quadrants

To Prove: Monotonicity of $f(x) = \sin x$ on $(0, \frac{\pi}{2})$, $(\frac{\pi}{2}, \pi)$, and $(0, \pi)$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$



Solution:

The given function is:

$$f(x) = \sin x$$

Differentiating with respect to x :

$$f'(x) = \cos x$$

- (i) For $x \in (0, \frac{\pi}{2})$: This represents the first quadrant where cosine is positive.

$$\cos x > 0 \implies f'(x) > 0$$

Hence, $f(x) = \sin x$ is strictly increasing in $(0, \frac{\pi}{2})$.

- (ii) For $x \in (\frac{\pi}{2}, \pi)$: This represents the second quadrant where cosine is negative.

$$\cos x < 0 \implies f'(x) < 0$$

Hence, $f(x) = \sin x$ is strictly decreasing in $(\frac{\pi}{2}, \pi)$.

- (iii) For $x \in (0, \pi)$: In $(0, \frac{\pi}{2})$, $f'(x) > 0$, while in $(\frac{\pi}{2}, \pi)$, $f'(x) < 0$. Since $f'(x)$ changes sign over the interval $(0, \pi)$, $f(x)$ is neither strictly increasing nor strictly decreasing in $(0, \pi)$.

Hence Proved

The function $f(x) = \sin x$ is strictly increasing in $(0, \frac{\pi}{2})$, strictly decreasing in $(\frac{\pi}{2}, \pi)$, and neither increasing nor decreasing in $(0, \pi)$.

Question 20

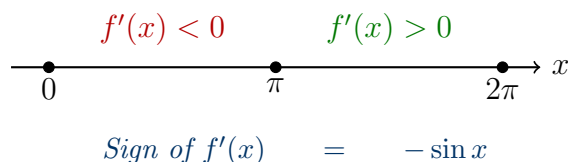
Prove that the function f defined by $f(x) = \cos x$ is:

- (i) strictly increasing in $(\pi, 2\pi)$
- (ii) strictly decreasing in $(0, \pi)$
- (iii) neither increasing nor decreasing in $(0, 2\pi)$

TYPE 51k Proving Monotonicity on an Interval – Proving Monotonicity of Cosine Function on Different Quadrants

To Prove: Monotonicity properties of $f(x) = \cos x$ on $(\pi, 2\pi)$, $(0, \pi)$, and $(0, 2\pi)$.

Concept / Formula Used: $f(x) = \cos x$ $f'(x) = -\sin x$ $f'(x) > 0$
 $f'(x) < 0$



Solution:

Given function:

$$f(x) = \cos x$$

Differentiating with respect to x :

$$f'(x) = -\sin x$$

(i) For $x \in (\pi, 2\pi)$: In the third and fourth quadrants, $\sin x < 0$.

$$f'(x) = -\sin x > 0$$

Hence, $f(x) = \cos x$ is strictly increasing in $(\pi, 2\pi)$.

(ii) For $x \in (0, \pi)$: In the first and second quadrants, $\sin x > 0$.

$$f'(x) = -\sin x < 0$$

Hence, $f(x) = \cos x$ is strictly decreasing in $(0, \pi)$.

(iii) For $x \in (0, 2\pi)$: $f'(x) < 0$ in $(0, \pi)$ and $f'(x) > 0$ in $(\pi, 2\pi)$. Since $f'(x)$ changes sign over $(0, 2\pi)$, $f(x)$ is neither increasing nor decreasing in $(0, 2\pi)$.

Hence Proved

The function $f(x) = \cos x$ is strictly increasing in $(\pi, 2\pi)$, strictly decreasing in $(0, \pi)$, and neither increasing nor decreasing in $(0, 2\pi)$.

Question 21

Prove that the following functions are neither increasing nor decreasing:

(i) $f(x) = x^2 - x + 1$ on $(-1, 1)$

(ii) $f(x) = \frac{1}{x^2+1}$ on $\mathbb{R}$

TYPE 69 Proving Non-Monotonicity of Quadratic and Rational Functions

To Prove: Show that $f'(x)$ changes sign within the given domains for both functions.

Concept / Formula Used: $f'(x)$ $f(x)$

$$\begin{array}{c} f' < 0 \qquad \qquad f' > 0 \\ \hline \qquad \qquad \qquad \bullet \qquad \qquad \qquad \rightarrow x \\ \qquad \qquad \qquad \frac{1}{2} \end{array}$$

Sign change of $f'(x)$ across critical point

Solution:

(i) Given $f(x) = x^2 - x + 1$ on $(-1, 1)$. Differentiating with respect to x :

$$f'(x) = 2x - 1$$

Setting $f'(x) = 0$:

$$2x - 1 = 0$$

$$x = \frac{1}{2}$$

Note that $x = \frac{1}{2} \in (-1, 1)$. For $x \in (-1, \frac{1}{2})$:

$$f'(x) = 2x - 1 < 0$$

For $x \in (\frac{1}{2}, 1)$:

$$f'(x) = 2x - 1 > 0$$

Since $f'(x)$ changes sign within $(-1, 1)$, $f(x)$ is neither increasing nor decreasing on $(-1, 1)$.

(ii) Given $f(x) = \frac{1}{x^2+1}$ on $\mathbb{R}$. Differentiating with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} [(x^2 + 1)^{-1}] \\ &= -(x^2 + 1)^{-2} \cdot 2x \\ &= -\frac{2x}{(x^2 + 1)^2} \end{aligned}$$

Since $(x^2 + 1)^2 > 0$ for all real x : For $x \in (-\infty, 0)$, $x < 0 \implies f'(x) > 0$. For $x \in (0, \infty)$, $x > 0 \implies f'(x) < 0$. Since $f'(x)$ changes sign at $x = 0$, $f(x)$ is neither increasing nor decreasing on $\mathbb{R}$.

Hence Proved

Both functions exhibit sign changes in $f'(x)$ within their domains, proving they are neither increasing nor decreasing.

Question 22

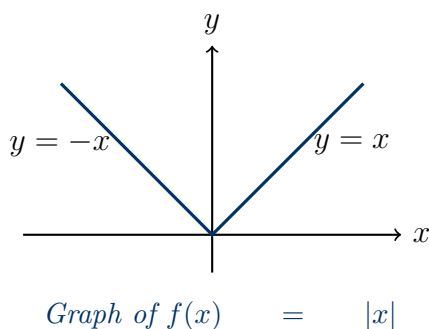
Prove that the function $f(x) = |x|$ is:

- (i) strictly increasing in $(0, \infty)$
- (ii) strictly decreasing in $(-\infty, 0)$

TYPE 511 Proving Monotonicity on an Interval – Proving Monotonicity of Absolute Value Function

To Prove: Monotonicity of $f(x) = |x|$ on $(0, \infty)$ and $(-\infty, 0)$.

Concept / Formula Used: $|x| = x$ $x > 0$ $|x| = -x$ $x < 0$



Solution:

The absolute value function is defined piecewise:

$$f(x) = |x| = \begin{cases} x, & \text{if } x > 0 \\ -x, & \text{if } x < 0 \end{cases}$$

(i) For $x \in (0, \infty)$: Here $f(x) = x$.

$$f'(x) = \frac{d}{dx}(x) = 1 > 0$$

Since $f'(x) > 0$ for all $x \in (0, \infty)$, $f(x) = |x|$ is strictly increasing in $(0, \infty)$.

(ii) For $x \in (-\infty, 0)$: Here $f(x) = -x$.

$$f'(x) = \frac{d}{dx}(-x) = -1 < 0$$

Since $f'(x) < 0$ for all $x \in (-\infty, 0)$, $f(x) = |x|$ is strictly decreasing in $(-\infty, 0)$.

Hence Proved

The function $f(x) = |x|$ is strictly increasing in $(0, \infty)$ and strictly decreasing in $(-\infty, 0)$.

Question 23

Prove that the function f given by $f(x) = \frac{4 \sin x}{2 + \cos x} - x$ is increasing in $[0, \frac{\pi}{2}]$.

TYPE 51m Proving Monotonicity on an Interval – Proving Monotonicity of Rational Trigonometric Function with Linear

To Prove: $f(x) = \frac{4 \sin x}{2 + \cos x} - x$ is increasing in $[0, \frac{\pi}{2}]$.

Concept / Formula Used: $\frac{d}{dx} \left[\frac{u}{v} \right] = \frac{vu' - uv'}{v^2}$ $f'(x) \geq 0$

$$\begin{array}{c} f'(x) \geq 0 \\ \bullet \text{-----} \bullet \rightarrow x \\ 0 \qquad \qquad \frac{\pi}{2} \end{array}$$

Domain interval for $f'(x)$

Solution:

Given function:

$$f(x) = \frac{4 \sin x}{2 + \cos x} - x$$

Differentiating with respect to x :

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} \left(\frac{4 \sin x}{2 + \cos x} \right) - \frac{d}{dx}(x) \\
 &= \frac{(2 + \cos x) \frac{d}{dx}(4 \sin x) - (4 \sin x) \frac{d}{dx}(2 + \cos x)}{(2 + \cos x)^2} - 1 \\
 &= \frac{(2 + \cos x)(4 \cos x) - (4 \sin x)(-\sin x)}{(2 + \cos x)^2} - 1 \\
 &= \frac{8 \cos x + 4 \cos^2 x + 4 \sin^2 x}{(2 + \cos x)^2} - 1
 \end{aligned}$$

Using $\cos^2 x + \sin^2 x = 1$:

$$\begin{aligned}
 f'(x) &= \frac{8 \cos x + 4}{(2 + \cos x)^2} - 1 \\
 &= \frac{8 \cos x + 4 - (2 + \cos x)^2}{(2 + \cos x)^2} \\
 &= \frac{8 \cos x + 4 - (4 + 4 \cos x + \cos^2 x)}{(2 + \cos x)^2} \\
 &= \frac{4 \cos x - \cos^2 x}{(2 + \cos x)^2} \\
 &= \frac{\cos x(4 - \cos x)}{(2 + \cos x)^2}
 \end{aligned}$$

Analyzing the sign of each factor on $[0, \frac{\pi}{2}]$:

$$\begin{aligned}
 \cos x &\geq 0 \quad \text{for all } x \in \left[0, \frac{\pi}{2}\right] \\
 4 - \cos x &\geq 3 > 0 \quad \text{since } -1 \leq \cos x \leq 1 \\
 (2 + \cos x)^2 &> 0 \quad \text{for all real } x
 \end{aligned}$$

Therefore:

$$f'(x) = \frac{\cos x(4 - \cos x)}{(2 + \cos x)^2} \geq 0 \quad \text{for all } x \in \left[0, \frac{\pi}{2}\right]$$

Hence Proved

Since $f'(x) \geq 0$ for all $x \in [0, \frac{\pi}{2}]$, the function $f(x)$ is increasing in $[0, \frac{\pi}{2}]$.

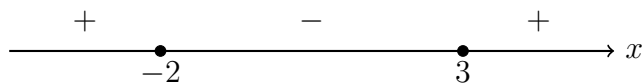
Question 24 (i)

Find the intervals in which the function $f(x) = 2x^3 - 3x^2 - 36x + 7$ is strictly increasing or strictly decreasing.

TYPE 56b Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Cubic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = 2x^3 - 3x^2 - 36x + 7$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$



$$\text{Sign scheme of } f'(x) = 6(x - 3)(x + 2)$$

Solution:

Given function:

$$f(x) = 2x^3 - 3x^2 - 36x + 7$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 6x^2 - 6x - 36 \\ &= 6(x^2 - x - 6) \\ &= 6(x - 3)(x + 2) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points:

$$x = -2, \quad x = 3$$

These points divide the real line into three intervals: $(-\infty, -2)$, $(-2, 3)$, and $(3, \infty)$.

$$\text{For } x \in (-\infty, -2) : (x - 3) < 0, (x + 2) < 0 \implies f'(x) > 0$$

$$\text{For } x \in (-2, 3) : (x - 3) < 0, (x + 2) > 0 \implies f'(x) < 0$$

$$\text{For } x \in (3, \infty) : (x - 3) > 0, (x + 2) > 0 \implies f'(x) > 0$$

Final Answer

Strictly increasing in $(-\infty, -2] \cup [3, \infty)$

Strictly decreasing in $[-2, 3]$

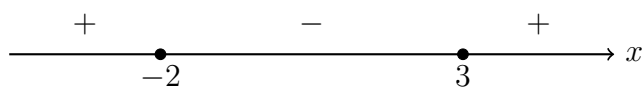
Question 24 (ii)

Find the intervals in which the function $f(x) = 4x^3 - 6x^2 - 72x + 30$ is strictly increasing or strictly decreasing.

TYPE 56b Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Cubic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = 4x^3 - 6x^2 - 72x + 30$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$



$$\text{Sign scheme for } f'(x) = 12(x - 3)(x + 2)$$

Solution:

Given function:

$$f(x) = 4x^3 - 6x^2 - 72x + 30$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 12x^2 - 12x - 72 \\ &= 12(x^2 - x - 6) \\ &= 12(x - 3)(x + 2) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points:

$$x = -2, \quad x = 3$$

Evaluating signs of $f'(x)$:

$$\begin{aligned} \text{On } (-\infty, -2) : \quad f'(x) &> 0 \\ \text{On } (-2, 3) : \quad f'(x) &< 0 \\ \text{On } (3, \infty) : \quad f'(x) &> 0 \end{aligned}$$

Final Answer

Strictly increasing in $(-\infty, -2] \cup [3, \infty)$
 Strictly decreasing in $[-2, 3]$

Question 24 (iii)

Find the intervals in which the function $f(x) = 2x^3 - 9x^2 + 12x + 15$ is strictly increasing or strictly decreasing.

TYPE 56b Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Cubic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = 2x^3 - 9x^2 + 12x + 15$.

Concept / Formula Used: $f'(x) = 0$ $f'(x)$

$$\begin{array}{ccccccc} & + & & - & & + & \\ & \xrightarrow{\hspace{1cm}} & & \xrightarrow{\hspace{1cm}} & & \xrightarrow{\hspace{1cm}} & \\ & 1 & & 2 & & & \\ \text{Sign scheme for } f'(x) & = & 6(x - 1)(x - 2) \end{array}$$

Solution:

Given function:

$$f(x) = 2x^3 - 9x^2 + 12x + 15$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 6x^2 - 18x + 12 \\ &= 6(x^2 - 3x + 2) \\ &= 6(x - 1)(x - 2) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points:

$$x = 1, \quad x = 2$$

Evaluating signs of $f'(x)$:

$$\text{On } (-\infty, 1) : f'(x) > 0$$

$$\text{On } (1, 2) : f'(x) < 0$$

$$\text{On } (2, \infty) : f'(x) > 0$$

Final Answer

Strictly increasing in $(-\infty, 1] \cup [2, \infty)$

Strictly decreasing in $[1, 2]$

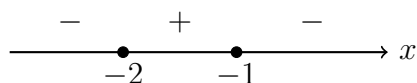
Question 24 (iv)

Find the intervals in which the function $f(x) = -2x^3 - 9x^2 - 12x + 1$ is strictly increasing or strictly decreasing.

TYPE 56b Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Cubic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = -2x^3 - 9x^2 - 12x + 1$.

Concept / Formula Used: $f'(x) = 0$



$$\text{Sign scheme for } f'(x) = -6(x+1)(x+2)$$

Solution:

Given function:

$$f(x) = -2x^3 - 9x^2 - 12x + 1$$

Differentiating with respect to x :

$$f'(x) = -6x^2 - 18x - 12$$

$$= -6(x^2 + 3x + 2)$$

$$= -6(x+1)(x+2)$$

Setting $f'(x) = 0$ gives critical points:

$$x = -2, \quad x = -1$$

Testing interval signs:

$$\text{For } x < -2 : (x+1) < 0, (x+2) < 0 \implies f'(x) < 0$$

$$\text{For } -2 < x < -1 : (x+1) < 0, (x+2) > 0 \implies f'(x) > 0$$

$$\text{For } x > -1 : (x+1) > 0, (x+2) > 0 \implies f'(x) < 0$$

Final AnswerStrictly increasing in $[-2, -1]$ Strictly decreasing in $(-\infty, -2] \cup [-1, \infty)$ **Question 24 (v)**Find the intervals in which the function $f(x) = x + \frac{1}{x}, x \neq 0$ is strictly increasing or strictly decreasing.**TYPE 56c** Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Rational Functions**To Find:** Intervals of strict monotonicity for $f(x) = x + \frac{1}{x}$.**Concept / Formula Used:** $f'(x) = \frac{d}{dx} \left(x + \frac{1}{x} \right) = 1 - \frac{1}{x^2}$

$$\begin{array}{ccccccc}
 & + & & - & & - & & + \\
 & \bullet & & \bullet & & \bullet & & \\
 & -1 & & 0 & & 1 & & x
 \end{array}$$

Sign scheme for $f'(x)$ = $\frac{x^2-1}{x^2}$

Solution:

Given function:

$$f(x) = x + \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned}
 f'(x) &= 1 - \frac{1}{x^2} \\
 &= \frac{x^2 - 1}{x^2} \\
 &= \frac{(x-1)(x+1)}{x^2}
 \end{aligned}$$

Since $x^2 > 0$ for all $x \neq 0$, the sign of $f'(x)$ is governed by $(x-1)(x+1)$. Setting $f'(x) = 0 \implies x = -1, 1$. Domain excludes $x = 0$.

$$\text{On } (-\infty, -1) : f'(x) > 0$$

$$\text{On } (-1, 0) : f'(x) < 0$$

$$\text{On } (0, 1) : f'(x) < 0$$

$$\text{On } (1, \infty) : f'(x) > 0$$

Final AnswerStrictly increasing in $(-\infty, -1] \cup [1, \infty)$ Strictly decreasing in $[-1, 0) \cup (0, 1]$

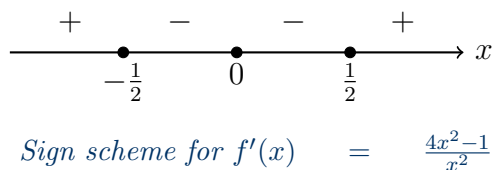
Question 24 (vi)

Find the intervals in which the function $f(x) = \frac{4x^2+1}{x}, x \neq 0$ is strictly increasing or strictly decreasing.

TYPE 56c Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Rational Functions

To Find: Intervals of strict monotonicity for $f(x) = \frac{4x^2+1}{x}$.

Concept / Formula Used: $f(x) = 4x + \frac{1}{x}$



Solution:

Rewrite the function:

$$f(x) = 4x + \frac{1}{x}, \quad x \neq 0$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 4 - \frac{1}{x^2} \\ &= \frac{4x^2 - 1}{x^2} \\ &= \frac{(2x - 1)(2x + 1)}{x^2} \end{aligned}$$

Critical points: $x = -\frac{1}{2}, \frac{1}{2}$. Point of discontinuity: $x = 0$. Since $x^2 > 0$ for $x \neq 0$:

$$\text{On } \left(-\infty, -\frac{1}{2}\right) : f'(x) > 0$$

$$\text{On } \left(-\frac{1}{2}, 0\right) : f'(x) < 0$$

$$\text{On } \left(0, \frac{1}{2}\right) : f'(x) < 0$$

$$\text{On } \left(\frac{1}{2}, \infty\right) : f'(x) > 0$$

Final Answer

Strictly increasing in $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$

Strictly decreasing in $\left[-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right]$

Question 24 (vii)

Find the intervals in which the function $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$ is strictly increasing or strictly decreasing.

TYPE 56d Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Quartic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$.

Concept / Formula Used: $f'(x) = 4x^3 - 24x^2 + 44x - 24$

$$\begin{array}{ccccccc} & - & & + & & - & & + \\ & \bullet & & \bullet & & \bullet & & \\ & 1 & & 2 & & 3 & & \\ \text{Sign scheme for } f'(x) & = & 4(x-1)(x-2)(x-3) \end{array}$$

Solution:

Given function:

$$f(x) = x^4 - 8x^3 + 22x^2 - 24x + 21$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 4x^3 - 24x^2 + 44x - 24 \\ &= 4(x^3 - 6x^2 + 11x - 6) \end{aligned}$$

By inspection, $x = 1$ is a root of $x^3 - 6x^2 + 11x - 6 = 0$. Factorizing:

$$\begin{aligned} f'(x) &= 4(x-1)(x^2 - 5x + 6) \\ &= 4(x-1)(x-2)(x-3) \end{aligned}$$

Critical points: $x = 1, 2, 3$. Testing interval signs:

$$\text{For } x \in (-\infty, 1) : f'(x) < 0$$

$$\text{For } x \in (1, 2) : f'(x) > 0$$

$$\text{For } x \in (2, 3) : f'(x) < 0$$

$$\text{For } x \in (3, \infty) : f'(x) > 0$$

Final Answer

Strictly increasing in $[1, 2] \cup [3, \infty)$

Strictly decreasing in $(-\infty, 1] \cup [2, 3]$

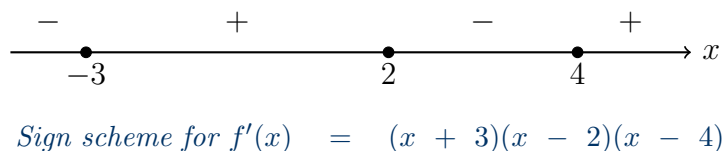
Question 24 (viii)

Find the intervals in which the function $f(x) = \frac{1}{4}x^4 - x^3 - 5x^2 + 24x + 12$ is strictly increasing or strictly decreasing.

TYPE 56d Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Quartic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = \frac{1}{4}x^4 - x^3 - 5x^2 + 24x + 12$.

Concept / Formula Used: $f'(x) = x^3 - 3x^2 - 10x + 24$



Solution:

Given function:

$$f(x) = \frac{1}{4}x^4 - x^3 - 5x^2 + 24x + 12$$

Differentiating with respect to x :

$$f'(x) = x^3 - 3x^2 - 10x + 24$$

Testing $x = 2$: $2^3 - 3(2)^2 - 10(2) + 24 = 8 - 12 - 20 + 24 = 0$. So $(x - 2)$ is a factor:

$$\begin{aligned} f'(x) &= (x - 2)(x^2 - x - 12) \\ &= (x - 2)(x - 4)(x + 3) \end{aligned}$$

Critical points: $x = -3, 2, 4$. Testing interval signs:

$$\text{For } x \in (-\infty, -3) : f'(x) < 0$$

$$\text{For } x \in (-3, 2) : f'(x) > 0$$

$$\text{For } x \in (2, 4) : f'(x) < 0$$

$$\text{For } x \in (4, \infty) : f'(x) > 0$$

Final Answer

Strictly increasing in $[-3, 2] \cup [4, \infty)$

Strictly decreasing in $(-\infty, -3] \cup [2, 4]$

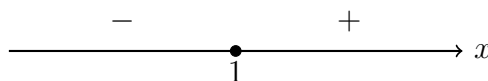
Question 24 (ix)

Find the intervals in which the function $f(x) = x^4 - 4x$ is strictly increasing or strictly decreasing.

TYPE 56d Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Quartic Polynomials

To Find: Intervals of strict monotonicity for $f(x) = x^4 - 4x$.

Concept / Formula Used: $f'(x) = 4(x - 1)(x^2 + x + 1)$



$$\text{Sign scheme for } f'(x) = 4(x-1)(x^2+x+1)$$

Solution:

Given function:

$$f(x) = x^4 - 4x$$

Differentiating with respect to x :

$$\begin{aligned} f'(x) &= 4x^3 - 4 \\ &= 4(x^3 - 1) \\ &= 4(x-1)(x^2+x+1) \end{aligned}$$

For the quadratic factor $x^2 + x + 1$:

$$\text{Discriminant } D = 1^2 - 4(1)(1) = -3 < 0$$

Thus $x^2 + x + 1 > 0$ for all real x . Hence, the sign of $f'(x)$ depends solely on $(x-1)$.

$$\text{For } x < 1 : f'(x) < 0$$

$$\text{For } x > 1 : f'(x) > 0$$

Final Answer

Strictly increasing in $[1, \infty)$

Strictly decreasing in $(-\infty, 1]$

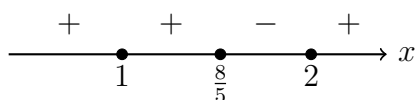
Question 24 (x)

Find the intervals in which the function $f(x) = (x-1)^3(x-2)^2$ is strictly increasing or strictly decreasing.

TYPE 56e Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Polynomials with Repeated Factors

To Find: Intervals of strict monotonicity for $f(x) = (x-1)^3(x-2)^2$.

Concept / Formula Used: $\frac{d}{dx}[uv] = u'v + uv'$



$$\text{Sign scheme for } f'(x) = (x-1)^2(x-2)(5x-8)$$

Solution:

Given function:

$$f(x) = (x-1)^3(x-2)^2$$

Differentiating with respect to x using product rule:

$$\begin{aligned} f'(x) &= 3(x-1)^2(x-2)^2 + (x-1)^3 \cdot 2(x-2) \\ &= (x-1)^2(x-2)[3(x-2) + 2(x-1)] \\ &= (x-1)^2(x-2)[3x-6+2x-2] \\ &= (x-1)^2(x-2)(5x-8) \end{aligned}$$

Critical points: $x = 1, \frac{8}{5}, 2$. Since $(x-1)^2 \geq 0$ for all real x , it does not change the sign of $f'(x)$. Evaluating the sign of $(x-2)(5x-8)$:

$$\text{For } x \in \left(-\infty, \frac{8}{5}\right) \setminus \{1\} : (x-2) < 0, (5x-8) < 0 \implies f'(x) > 0$$

$$\text{For } x \in \left(\frac{8}{5}, 2\right) : (x-2) < 0, (5x-8) > 0 \implies f'(x) < 0$$

$$\text{For } x \in (2, \infty) : (x-2) > 0, (5x-8) > 0 \implies f'(x) > 0$$

Final Answer

Strictly increasing in $\left(-\infty, \frac{8}{5}\right] \cup [2, \infty)$
 Strictly decreasing in $\left[\frac{8}{5}, 2\right]$

Question 25 (i)

Find the interval in which the function $f(x) = x^2e^{-x}$ is increasing.

TYPE 56f Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Product of Quadratic and Exponential Functions

To Find: Interval where $f(x) = x^2e^{-x}$ is increasing.

Concept / Formula Used: $\frac{d}{dx}[uv] = u'v + uv'$ $f(x)$ $f'(x) \geq 0$

$$\begin{array}{ccccccc} & - & & + & & - & \\ & & \bullet & & \bullet & & \\ & & 0 & & 2 & & \\ \text{Sign scheme for } f'(x) & = & x(2-x)e^{-x} & & & & \end{array}$$

Solution:

Given function:

$$f(x) = x^2e^{-x}$$

Differentiating with respect to x using product rule:

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^2) \cdot e^{-x} + x^2 \cdot \frac{d}{dx}(e^{-x}) \\ &= 2xe^{-x} + x^2(-e^{-x}) \\ &= e^{-x}(2x - x^2) \\ &= x(2-x)e^{-x} \end{aligned}$$

Since $e^{-x} > 0$ for all real x , the sign of $f'(x)$ depends on $x(2 - x)$. For $f(x)$ to be increasing:

$$\begin{aligned} f'(x) &\geq 0 \\ x(2 - x)e^{-x} &\geq 0 \\ x(2 - x) &\geq 0 \\ x(x - 2) &\leq 0 \\ 0 &\leq x \leq 2 \end{aligned}$$

Final Answer

The function $f(x) = x^2e^{-x}$ is increasing in $[0, 2]$.

Question 25 (ii)

Find the intervals in which the function given by $f(x) = x^2e^x$ is strictly increasing or strictly decreasing.

TYPE 56f Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Product of Quadratic and Exponential Functions

To Find: Intervals of strict monotonicity for $f(x) = x^2e^x$.

Concept / Formula Used: $f(x)$ $f'(x) > 0$ $f'(x) < 0$

$$\begin{array}{ccccccc} & + & & - & & + & \\ & \bullet & & \bullet & & & \\ & -2 & & 0 & & & \\ \text{Sign scheme for } f'(x) & = & x(x + 2)e^x \end{array}$$

Solution:

Given function:

$$f(x) = x^2e^x$$

Differentiating with respect to x using product rule:

$$\begin{aligned} f'(x) &= 2xe^x + x^2e^x \\ &= e^x(x^2 + 2x) \\ &= x(x + 2)e^x \end{aligned}$$

Since $e^x > 0$ for all real x , $f'(x) = 0 \implies x = 0, -2$. Testing interval signs:

$$\text{For } x \in (-\infty, -2) : x < 0, (x + 2) < 0 \implies f'(x) > 0$$

$$\text{For } x \in (-2, 0) : x < 0, (x + 2) > 0 \implies f'(x) < 0$$

$$\text{For } x \in (0, \infty) : x > 0, (x + 2) > 0 \implies f'(x) > 0$$

Final Answer

Strictly increasing in $(-\infty, -2] \cup [0, \infty)$

Strictly decreasing in $[-2, 0]$

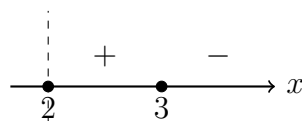
Question 26

Find the intervals in which the function $f(x) = 2 \log(x - 2) - x^2 + 4x - 5$ is strictly increasing or strictly decreasing.

TYPE 56g Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Logarithmic-Polynomial Com

To Find: Intervals of strict monotonicity for $f(x) = 2 \log(x - 2) - x^2 + 4x - 5$.

Concept / Formula Used: $x - 2 > 0 \implies x > 2$



Sign scheme of $f'(x)$ on domain $(2, \infty)$

Solution:

The domain of $f(x)$ requires:

$$x - 2 > 0 \implies x > 2$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} [2 \log(x - 2) - x^2 + 4x - 5] \\ &= \frac{2}{x - 2} - 2x + 4 \\ &= \frac{2}{x - 2} - 2(x - 2) \\ &= \frac{2 - 2(x - 2)^2}{x - 2} \\ &= \frac{2[1 - (x - 2)^2]}{x - 2} \\ &= \frac{2(1 - (x - 2))(1 + (x - 2))}{x - 2} \\ &= \frac{2(3 - x)(x - 1)}{x - 2} \end{aligned}$$

Since $x > 2$, both $(x - 2) > 0$ and $(x - 1) > 0$. Thus, the sign of $f'(x)$ depends solely on $(3 - x)$:

$$\text{For } x \in (2, 3) : 3 - x > 0 \implies f'(x) > 0$$

$$\text{For } x \in (3, \infty) : 3 - x < 0 \implies f'(x) < 0$$

Final Answer

Strictly increasing in $(2, 3]$

Strictly decreasing in $[3, \infty)$

Question 27

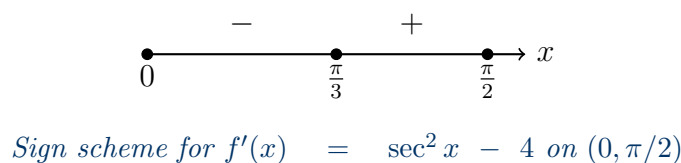
Find the intervals in which the function f given by $f(x) = \tan x - 4x, x \in (0, \frac{\pi}{2})$ is:

- (i) strictly increasing
- (ii) strictly decreasing

TYPE 56h Finding Intervals of Increase and Decrease – Finding Monotonicity Intervals for Trigonometric-Linear Functions

To Find: Intervals of strict monotonicity for $f(x) = \tan x - 4x$ on $(0, \frac{\pi}{2})$.

Concept / Formula Used: $\frac{d}{dx}(\tan x) = \sec^2 x$



Solution:

Given function:

$$f(x) = \tan x - 4x, \quad x \in \left(0, \frac{\pi}{2}\right)$$

Differentiating with respect to x :

$$f'(x) = \sec^2 x - 4$$

Setting $f'(x) = 0$:

$$\sec^2 x - 4 = 0$$

$$\sec^2 x = 4$$

$$\sec x = 2 \quad \left(\text{since } \sec x > 0 \text{ in } \left(0, \frac{\pi}{2}\right)\right)$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}$$

This critical point $x = \frac{\pi}{3}$ divides $(0, \frac{\pi}{2})$ into two sub-intervals:

(i) For $x \in (\frac{\pi}{3}, \frac{\pi}{2})$:

$$\cos x < \frac{1}{2} \implies \sec x > 2 \implies \sec^2 x > 4 \implies f'(x) > 0$$

Hence, $f(x)$ is strictly increasing in $[\frac{\pi}{3}, \frac{\pi}{2})$.

(ii) For $x \in (0, \frac{\pi}{3})$:

$$\cos x > \frac{1}{2} \implies \sec x < 2 \implies \sec^2 x < 4 \implies f'(x) < 0$$

Hence, $f(x)$ is strictly decreasing in $(0, \frac{\pi}{3}]$.

Final Answer

- (i) Strictly increasing in $\left[\frac{\pi}{3}, \frac{\pi}{2}\right)$
- (ii) Strictly decreasing in $\left(0, \frac{\pi}{3}\right]$

6.5 – Maxima and Minima Without Derivative Tests**Question 1**

Find the maximum and the minimum values (if any) of the following functions:

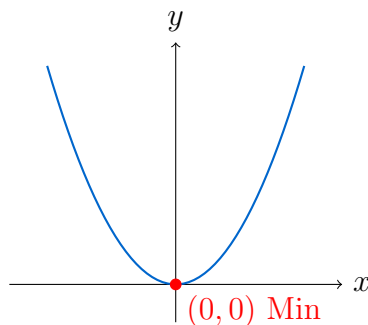
- (i) $f(x) = x^2$
- (ii) $f(x) = (2x - 1)^2 + 3$

TYPE 79a Finding Local Extrema by Inspection – Finding Extrema of Quadratic Functions via Inspection

Given: Functions $f(x) = x^2$ and $f(x) = (2x - 1)^2 + 3$ defined for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of $f(x)$.

Concept / Formula Used: $A \quad A^2 \geq 0 \quad 0 \quad A = 0 \quad A \rightarrow \pm\infty$



Graph of $y = x^2$ illustrating global minimum at origin $(0,0)$

Solution:

Part (i): $f(x) = x^2$

For all real numbers $x \in \mathbb{R}$, we have:

$$x^2 \geq 0$$

Thus, $f(x) \geq 0$ for all $x \in \mathbb{R}$.

The minimum value occurs when $x = 0$:

$$f(0) = 0^2 = 0$$

As $x \rightarrow \pm\infty$, $f(x) \rightarrow \infty$. Therefore, $f(x)$ has no maximum value.

Part (ii): $f(x) = (2x - 1)^2 + 3$

For all $x \in \mathbb{R}$, the squared term satisfies:

$$(2x - 1)^2 \geq 0$$

Adding 3 to both sides of the inequality:

$$\begin{aligned}(2x - 1)^2 + 3 &\geq 0 + 3 \\ f(x) &\geq 3\end{aligned}$$

The minimum value occurs when $2x - 1 = 0 \implies x = \frac{1}{2}$:

Evaluating $f(x)$ at $x = \frac{1}{2}$:

$$\begin{aligned}f\left(\frac{1}{2}\right) &= \left(2\left(\frac{1}{2}\right) - 1\right)^2 + 3 \\ &= (1 - 1)^2 + 3 \\ &= 0 + 3 = 3\end{aligned}$$

Since $(2x - 1)^2 \rightarrow \infty$ as $x \rightarrow \pm\infty$, there is no maximum value.

Final Answer

- (i) Minimum value = 0 at $x = 0$; No maximum value.
- (ii) Minimum value = 3 at $x = \frac{1}{2}$; No maximum value.

Question 2

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = -(x - 1)^2 + 10$
- (ii) $f(x) = 9x^2 + 12x + 2$

TYPE 79a Finding Local Extrema by Inspection – Finding Extrema of Quadratic Functions via Inspection

Given: Functions $f(x) = -(x - 1)^2 + 10$ and $f(x) = 9x^2 + 12x + 2$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $A \quad A^2 \geq 0 \implies -A^2 \leq 0 \quad ax^2 + bx + c$

Solution:

Part (i): $f(x) = -(x - 1)^2 + 10$

For all $x \in \mathbb{R}$, we have:

$$(x - 1)^2 \geq 0$$

Multiplying by -1 reverses the inequality:

$$-(x-1)^2 \leq 0$$

Adding 10 to both sides:

$$\begin{aligned} -(x-1)^2 + 10 &\leq 10 \\ f(x) &\leq 10 \end{aligned}$$

The maximum value is attained when $x-1=0 \implies x=1$:

$$f(1) = -(1-1)^2 + 10 = 10$$

Since $-(x-1)^2 \rightarrow -\infty$ as $x \rightarrow \pm\infty$, there is no minimum value.

Part (ii): $f(x) = 9x^2 + 12x + 2$

Completing the square for the quadratic expression:

$$\begin{aligned} f(x) &= (3x)^2 + 2(3x)(2) + 2^2 - 2^2 + 2 \\ &= (3x+2)^2 - 4 + 2 \\ &= (3x+2)^2 - 2 \end{aligned}$$

Since $(3x+2)^2 \geq 0$ for all $x \in \mathbb{R}$:

$$\begin{aligned} (3x+2)^2 - 2 &\geq -2 \\ f(x) &\geq -2 \end{aligned}$$

The minimum value occurs when $3x+2=0 \implies x=-\frac{2}{3}$:

$$f\left(-\frac{2}{3}\right) = \left(3\left(-\frac{2}{3}\right) + 2\right)^2 - 2 = 0 - 2 = -2$$

As $x \rightarrow \pm\infty$, $f(x) \rightarrow \infty$, so there is no maximum value.

Final Answer

- (i) Maximum value = 10 at $x=1$; No minimum value.
- (ii) Minimum value = -2 at $x=-\frac{2}{3}$; No maximum value.

Question 3

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = |x|$
- (ii) $f(x) = x^3 + 1$

TYPE 79b Finding Local Extrema by Inspection – Finding Extrema of Power and Absolute Value Functions

Given: Functions $f(x) = |x|$ and $f(x) = x^3 + 1$ defined for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $|x| \geq 0 \quad x \in \mathbb{R}$

Solution:

Part (i): $f(x) = |x|$

By definition of the absolute value function, for all $x \in \mathbb{R}$:

$$|x| \geq 0$$

Thus, $f(x) \geq 0$ for all $x \in \mathbb{R}$.

The minimum value occurs at $x = 0$:

$$f(0) = |0| = 0$$

As $x \rightarrow \pm\infty$, $|x| \rightarrow \infty$, so $f(x)$ has no maximum value.

Part (ii): $f(x) = x^3 + 1$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 + 1) \\ &= 3x^2 \end{aligned}$$

Since $3x^2 \geq 0$ for all $x \in \mathbb{R}$ and $f'(x) = 0$ only at $x = 0$, $f(x)$ is strictly increasing on $\mathbb{R}$.

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$, and as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

Therefore, $f(x)$ has neither a maximum value nor a minimum value.

Final Answer

- (i) Minimum value = 0 at $x = 0$; No maximum value.
- (ii) Neither maximum value nor minimum value exists.

Question 4

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = |x + 2| - 1$
- (ii) $f(x) = 3 + |x|$

TYPE 79b Finding Local Extrema by Inspection – Finding Extrema of Power and Absolute Value Functions

Given: Functions $f(x) = |x + 2| - 1$ and $f(x) = 3 + |x|$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $|A| \geq 0 \quad A \in \mathbb{R} \quad 0 \quad A = 0$

Solution:

Part (i): $f(x) = |x + 2| - 1$

For all $x \in \mathbb{R}$, we have:

$$|x + 2| \geq 0$$

Subtracting 1 from both sides:

$$\begin{aligned} |x + 2| - 1 &\geq 0 - 1 \\ f(x) &\geq -1 \end{aligned}$$

The minimum value occurs when $x + 2 = 0 \implies x = -2$:

$$f(-2) = |-2 + 2| - 1 = 0 - 1 = -1$$

Since $|x + 2| \rightarrow \infty$ as $x \rightarrow \pm\infty$, $f(x)$ has no maximum value.

Part (ii): $f(x) = 3 + |x|$

For all $x \in \mathbb{R}$, we have:

$$|x| \geq 0$$

Adding 3 to both sides:

$$\begin{aligned} 3 + |x| &\geq 3 + 0 \\ f(x) &\geq 3 \end{aligned}$$

The minimum value occurs when $x = 0$:

$$f(0) = 3 + |0| = 3$$

Since $|x| \rightarrow \infty$ as $x \rightarrow \pm\infty$, there is no maximum value.

Final Answer

- (i) Minimum value = -1 at $x = -2$; No maximum value.
- (ii) Minimum value = 3 at $x = 0$; No maximum value.

Question 5

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = -|x + 1| + 3$
- (ii) $f(x) = \sin 2x$

TYPE 79c Finding Local Extrema by Inspection – Finding Extrema of Trigonometric and Composite Absolute Value Functions

Given: Functions $f(x) = -|x + 1| + 3$ and $f(x) = \sin 2x$ defined on $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $|A| \geq 0 \implies -|A| \leq 0 \quad \theta \quad -1 \leq \sin \theta \leq 1$

Solution:**Part (i):** $f(x) = -|x + 1| + 3$ For all $x \in \mathbb{R}$, we have:

$$|x + 1| \geq 0$$

Multiplying by -1 reverses the inequality:

$$-|x + 1| \leq 0$$

Adding 3 to both sides:

$$\begin{aligned} -|x + 1| + 3 &\leq 3 \\ f(x) &\leq 3 \end{aligned}$$

The maximum value occurs when $x + 1 = 0 \implies x = -1$:

$$f(-1) = -|-1 + 1| + 3 = 0 + 3 = 3$$

As $x \rightarrow \pm\infty$, $-|x + 1| \rightarrow -\infty$, so there is no minimum value.**Part (ii):** $f(x) = \sin 2x$ For all $x \in \mathbb{R}$, the standard range of the sine function is $[-1, 1]$:

$$-1 \leq \sin 2x \leq 1$$

Therefore:

$$\text{Maximum value of } f(x) = 1$$

$$\text{Minimum value of } f(x) = -1$$

Maximum value 1 occurs at $2x = \frac{\pi}{2} + 2k\pi \implies x = \frac{\pi}{4} + k\pi, k \in \mathbb{Z}$.Minimum value -1 occurs at $2x = \frac{3\pi}{2} + 2k\pi \implies x = \frac{3\pi}{4} + k\pi, k \in \mathbb{Z}$.**Final Answer**

- (i) Maximum value = 3 at $x = -1$; No minimum value.
 (ii) Maximum value = 1, Minimum value = -1 .

Question 6

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = \sin 2x + 5$
 (ii) $f(x) = |\sin 4x + 3|$

TYPE 79c Finding Local Extrema by Inspection – Finding Extrema of Trigonometric and Composite Absolute Value Functions**Given:** Functions $f(x) = \sin 2x + 5$ and $f(x) = |\sin 4x + 3|$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $\theta \quad -1 \leq \sin \theta \leq 1$

Solution:

Part (i): $f(x) = \sin 2x + 5$

For all $x \in \mathbb{R}$, we know:

$$-1 \leq \sin 2x \leq 1$$

Adding 5 across the inequality:

$$\begin{aligned} -1 + 5 &\leq \sin 2x + 5 \leq 1 + 5 \\ 4 &\leq f(x) \leq 6 \end{aligned}$$

Therefore:

Maximum value = 6

Minimum value = 4

Part (ii): $f(x) = |\sin 4x + 3|$

For all $x \in \mathbb{R}$, we know:

$$-1 \leq \sin 4x \leq 1$$

Adding 3 across the inequality:

$$\begin{aligned} -1 + 3 &\leq \sin 4x + 3 \leq 1 + 3 \\ 2 &\leq \sin 4x + 3 \leq 4 \end{aligned}$$

Since the range $[2, 4]$ consists entirely of positive numbers, taking the absolute value preserves the bounds:

$$2 \leq |\sin 4x + 3| \leq 4$$

Therefore:

Maximum value = 4

Minimum value = 2

Final Answer

- (i) Maximum value = 6, Minimum value = 4.
- (ii) Maximum value = 4, Minimum value = 2.

Question 7

Find the maximum and the minimum values (if any) of the following functions:

- (i) $f(x) = |\sin 3x| - 3$

$$(ii) f(x) = 4 - |\cos 3x|$$

TYPE 79c Finding Local Extrema by Inspection – Finding Extrema of Trigonometric and Composite Absolute Value Functions

Given: Functions $f(x) = |\sin 3x| - 3$ and $f(x) = 4 - |\cos 3x|$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values (if any) of each function.

Concept / Formula Used: $-1 \leq \sin \theta \leq 1$ $-1 \leq \cos \theta \leq 1$ $0 \leq |\sin \theta| \leq 1$ $0 \leq |\cos \theta| \leq 1$

Solution:

Part (i): $f(x) = |\sin 3x| - 3$

For all $x \in \mathbb{R}$, the absolute value of sine is bounded as:

$$0 \leq |\sin 3x| \leq 1$$

Subtracting 3 across the inequality:

$$\begin{aligned} 0 - 3 &\leq |\sin 3x| - 3 \leq 1 - 3 \\ -3 &\leq f(x) \leq -2 \end{aligned}$$

Therefore:

$$\text{Maximum value} = -2$$

$$\text{Minimum value} = -3$$

Part (ii): $f(x) = 4 - |\cos 3x|$

For all $x \in \mathbb{R}$, the absolute value of cosine is bounded as:

$$0 \leq |\cos 3x| \leq 1$$

Multiplying by -1 reverses the inequality:

$$-1 \leq -|\cos 3x| \leq 0$$

Adding 4 across the inequality:

$$\begin{aligned} 4 - 1 &\leq 4 - |\cos 3x| \leq 4 + 0 \\ 3 &\leq f(x) \leq 4 \end{aligned}$$

Therefore:

$$\text{Maximum value} = 4$$

$$\text{Minimum value} = 3$$

Final Answer

(i) Maximum value = -2 , Minimum value = -3 .

(ii) Maximum value = 4 , Minimum value = 3 .

Question 8

- (i) Is the minimum value of $\cos x$ zero?
 (ii) What are the minimum and maximum values of $2 - 3 \cos x$?

TYPE 79c Finding Local Extrema by Inspection – Finding Extrema of Trigonometric and Composite Absolute Value Functions

Given: Expressions $\cos x$ and $g(x) = 2 - 3 \cos x$ for real $x \in \mathbb{R}$.

To Find: (i) Verification of whether the minimum value of $\cos x$ is zero. (ii) Extreme values of $2 - 3 \cos x$.

Concept / Formula Used: $x \quad -1 \leq \cos x \leq 1$

Solution:

Part (i):

For all real numbers $x \in \mathbb{R}$, the domain-wide bounds of $\cos x$ are:

$$-1 \leq \cos x \leq 1$$

The minimum value attained by $\cos x$ is -1 , which occurs at $x = \pi + 2k\pi, k \in \mathbb{Z}$.

Hence, the minimum value of $\cos x$ is NOT zero; it is -1 .

Part (ii):

Let $g(x) = 2 - 3 \cos x$. We start from the standard inequality:

$$-1 \leq \cos x \leq 1$$

Multiplying all parts by -3 (reversing inequality direction):

$$\begin{aligned} (-3)(1) &\leq -3 \cos x \leq (-3)(-1) \\ -3 &\leq -3 \cos x \leq 3 \end{aligned}$$

Adding 2 to all parts:

$$\begin{aligned} 2 - 3 &\leq 2 - 3 \cos x \leq 2 + 3 \\ -1 &\leq 2 - 3 \cos x \leq 5 \end{aligned}$$

Therefore, the minimum value is -1 and the maximum value is 5 .

Final Answer

- (i) No, the minimum value of $\cos x$ is -1 .
 (ii) Minimum value $= -1$, Maximum value $= 5$.

Question 9

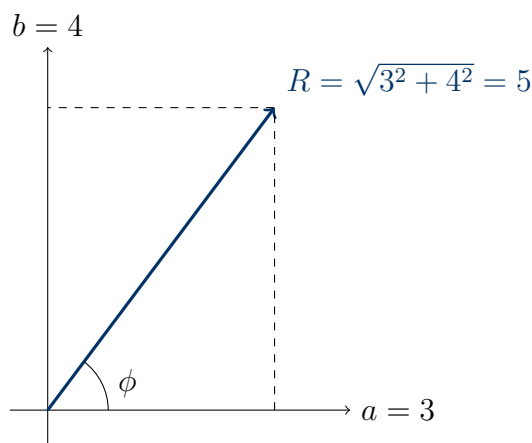
What are the maximum and minimum values of $3 \sin x + 4 \cos x$?

TYPE 79d Finding Local Extrema by Inspection – Finding Extrema of Linear Combination of Sine and Cosine

Given: Function $f(x) = 3 \sin x + 4 \cos x$ for $x \in \mathbb{R}$.

To Find: The maximum and minimum values of $f(x)$.

Concept / Formula Used: $a \sin x + b \cos x = R \sin(x + \phi)$ where $R = \sqrt{a^2 + b^2}$ and $\phi \in \mathbb{R}$.



Phasor representation of $3 \sin x + 4 \cos x$ showing amplitude $R = \sqrt{3^2 + 4^2} = 5$

Solution:

We express $3 \sin x + 4 \cos x$ in single harmonic form $R \sin(x + \phi)$.

Comparing $3 \sin x + 4 \cos x$ with $a \sin x + b \cos x$:

$$a = 3$$

$$b = 4$$

The amplitude R is calculated as:

$$\begin{aligned} R &= \sqrt{a^2 + b^2} \\ &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} = 5 \end{aligned}$$

Let $3 = 5 \cos \phi$ and $4 = 5 \sin \phi$, where $\tan \phi = \frac{4}{3}$.

Substituting these into $f(x)$:

$$\begin{aligned} f(x) &= 5 \cos \phi \sin x + 5 \sin \phi \cos x \\ &= 5(\sin x \cos \phi + \cos x \sin \phi) \\ &= 5 \sin(x + \phi) \end{aligned}$$

Since $-1 \leq \sin(x + \phi) \leq 1$ for all real x :

$$\begin{aligned} -5(1) &\leq 5 \sin(x + \phi) \leq 5(1) \\ -5 &\leq f(x) \leq 5 \end{aligned}$$

Thus:

Maximum value = 5

Minimum value = -5

Final Answer

Maximum value = 5, Minimum value = -5.

Question 10

Find the maximum and minimum values, if any, of the following functions:

(i) $f(x) = x$ in $(0, 1)$

(ii) $f(x) = x$ in $[0, 1]$

TYPE 79e Finding Local Extrema by Inspection – Analyzing Extrema of Identity Function on Intervals

Given: Function $f(x) = x$ defined on (i) open interval $(0, 1)$, and (ii) closed interval $[0, 1]$.

To Find: The maximum and minimum values of $f(x)$ in each domain.

Concept / Formula Used: $[a, b]$ (a, b)

Solution:

Part (i): $f(x) = x$ on $(0, 1)$

The domain is the open interval $x \in (0, 1)$.

For any $x \in (0, 1)$, we have $0 < f(x) < 1$.

Suppose $f(x)$ attains a minimum value at some point $c \in (0, 1)$. Then $f(c) = c > 0$.

However, choosing $x_1 = \frac{c}{2} \in (0, 1)$ gives:

$$f(x_1) = \frac{c}{2} < c = f(c)$$

Thus, no element in $(0, 1)$ can serve as a minimum value.

Similarly, suppose $f(x)$ attains a maximum value at $d \in (0, 1)$. Then $f(d) = d < 1$.

Choosing $x_2 = \frac{d+1}{2} \in (0, 1)$ gives:

$$f(x_2) = \frac{d+1}{2} > d = f(d)$$

Thus, no element in $(0, 1)$ can serve as a maximum value.

Therefore, $f(x)$ has neither a maximum value nor a minimum value on $(0, 1)$.

Part (ii): $f(x) = x$ on $[0, 1]$

The domain is the closed interval $x \in [0, 1]$.

Since $0 \leq x \leq 1$ for all $x \in [0, 1]$:

$$0 \leq f(x) \leq 1$$

The boundary points are included in the domain:

$$f(0) = 0 \quad (\text{Minimum value at } x = 0)$$

$$f(1) = 1 \quad (\text{Maximum value at } x = 1)$$

Final Answer

- (i) No maximum value, no minimum value in $(0, 1)$.
- (ii) Maximum value = 1 at $x = 1$, Minimum value = 0 at $x = 0$.

Question 11

Prove that the function $f(x) = x^3 + x^2 + x + 1$ does not have maxima or minima.

TYPE 84 Proving Non-Existence of Extrema via Strictly Monotonic Derivative

To Prove: The function $f(x) = x^3 + x^2 + x + 1$ has no local or global maxima or minima on $\mathbb{R}$.

Concept / Formula Used: $x = c \quad f'(c) = 0 \quad f'(x) > 0 \quad \mathbb{R} \quad f(x)$

Solution:

The given function is:

$$f(x) = x^3 + x^2 + x + 1$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 + x^2 + x + 1) \\ &= 3x^2 + 2x + 1 \end{aligned}$$

To check for critical points, set $f'(x) = 0$:

$$3x^2 + 2x + 1 = 0$$

Calculating the discriminant $D = b^2 - 4ac$ for this derivative quadratic equation:

$$\begin{aligned} D &= 2^2 - 4(3)(1) \\ &= 4 - 12 \\ &= -8 < 0 \end{aligned}$$

Since the coefficient of x^2 is $a = 3 > 0$ and the discriminant $D < 0$, the quadratic derivative $3x^2 + 2x + 1$ is strictly positive for all real numbers $x \in \mathbb{R}$.

Alternatively, completing the square for $f'(x)$:

$$\begin{aligned} f'(x) &= 3 \left(x^2 + \frac{2}{3}x \right) + 1 \\ &= 3 \left(x^2 + \frac{2}{3}x + \frac{1}{9} \right) + 1 - 3 \left(\frac{1}{9} \right) \\ &= 3 \left(x + \frac{1}{3} \right)^2 + 1 - \frac{1}{3} \\ &= 3 \left(x + \frac{1}{3} \right)^2 + \frac{2}{3} \end{aligned}$$

Since $\left(x + \frac{1}{3}\right)^2 \geq 0$, we get:

$$f'(x) \geq \frac{2}{3} > 0 \quad \forall x \in \mathbb{R}$$

Because $f'(x) \neq 0$ for all $x \in \mathbb{R}$, there exist no critical points.

Hence Proved

Since $f'(x) > 0$ for all real x , $f(x)$ has no critical points and is strictly increasing on $\mathbb{R}$. Therefore, $f(x)$ does not have maxima or minima. Hence proved.

Question 12

Find the (absolute) maximum and the (absolute) minimum values of the following functions in the indicated intervals. Also find the points of (absolute) maxima and minima:

- (i) $f(x) = (x - 1)^2 + 3$ in $[-3, 1]$
- (ii) $f(x) = 2x^3 - 15x^2 + 36x + 1$ on the interval $[1, 5]$
- (iii) $f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$ in $[0, 3]$
- (iv) $f(x) = x + \sin x$ in $[0, 2\pi]$
- (v) $f(x) = \sin x + \sqrt{3} \cos x$ in $[0, \pi]$
- (vi) $f(x) = 3 + |x + 1|$ in $[-2, 3]$

TYPE 85a Finding Absolute Extrema on a Closed Interval – Finding Absolute Extrema of Diverse Functions on Closed I

Given: Functions and continuous domains for subparts (i) through (vi).

To Find: Absolute maximum value, absolute minimum value, and corresponding points of absolute maxima and minima for each part.

Concept / Formula Used: $f(x)$ $[a, b]$ $c \in (a, b)$ $f'(c) = 0$ $f'(c)$ $f(x)$ a, b

Solution:

Part (i): $f(x) = (x - 1)^2 + 3$ on $[-3, 1]$

Differentiating $f(x)$:

$$f'(x) = 2(x - 1)$$

Setting $f'(x) = 0 \implies 2(x - 1) = 0 \implies x = 1$.

The evaluation points in $[-3, 1]$ are $x = -3$ and $x = 1$.

Evaluating $f(x)$:

$$\begin{aligned} f(-3) &= (-3 - 1)^2 + 3 = (-4)^2 + 3 = 16 + 3 = 19 \\ f(1) &= (1 - 1)^2 + 3 = 0 + 3 = 3 \end{aligned}$$

Comparing values:

Absolute Maximum Value = 19 at point of maximum $x = -3$

Absolute Minimum Value = 3 at point of minimum $x = 1$

Part (ii): $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$

Differentiating $f(x)$:

$$\begin{aligned} f'(x) &= 6x^2 - 30x + 36 \\ &= 6(x^2 - 5x + 6) \\ &= 6(x - 2)(x - 3) \end{aligned}$$

Setting $f'(x) = 0 \implies x = 2$ or $x = 3$. Both critical points lie in $[1, 5]$.

Evaluating $f(x)$ at critical points $x = 2, 3$ and boundary points $x = 1, 5$:

$$\begin{aligned} f(1) &= 2(1)^3 - 15(1)^2 + 36(1) + 1 = 2 - 15 + 36 + 1 = 24 \\ f(2) &= 2(2)^3 - 15(2)^2 + 36(2) + 1 = 16 - 60 + 72 + 1 = 29 \\ f(3) &= 2(3)^3 - 15(3)^2 + 36(3) + 1 = 54 - 135 + 108 + 1 = 28 \\ f(5) &= 2(5)^3 - 15(5)^2 + 36(5) + 1 = 250 - 375 + 180 + 1 = 56 \end{aligned}$$

Comparing values:

Absolute Maximum Value = 56 at point of maximum $x = 5$

Absolute Minimum Value = 24 at point of minimum $x = 1$

Part (iii): $f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$ on $[0, 3]$

Differentiating $f(x)$:

$$\begin{aligned} f'(x) &= 12x^3 - 24x^2 + 24x - 48 \\ &= 12(x^3 - 2x^2 + 2x - 4) \\ &= 12[x^2(x - 2) + 2(x - 2)] \\ &= 12(x - 2)(x^2 + 2) \end{aligned}$$

Setting $f'(x) = 0 \implies (x - 2)(x^2 + 2) = 0$.

Since $x^2 + 2 > 0$ for all real x , the only real critical point is $x = 2 \in [0, 3]$.

Evaluating $f(x)$ at $x = 0, 2, 3$:

$$f(0) = 3(0)^4 - 8(0)^3 + 12(0)^2 - 48(0) + 25 = 25$$

$$f(2) = 3(2)^4 - 8(2)^3 + 12(2)^2 - 48(2) + 25$$

$$= 3(16) - 8(8) + 12(4) - 96 + 25$$

$$= 48 - 64 + 48 - 96 + 25 = -39$$

$$f(3) = 3(3)^4 - 8(3)^3 + 12(3)^2 - 48(3) + 25$$

$$= 3(81) - 8(27) + 12(9) - 144 + 25$$

$$= 243 - 216 + 108 - 144 + 25 = 16$$

Comparing values:

Absolute Maximum Value = 25 at point of maximum $x = 0$

Absolute Minimum Value = -39 at point of minimum $x = 2$

Part (iv): $f(x) = x + \sin x$ on $[0, 2\pi]$

Differentiating $f(x)$:

$$f'(x) = 1 + \cos x$$

Setting $f'(x) = 0 \implies 1 + \cos x = 0 \implies \cos x = -1$.

In $[0, 2\pi]$, $\cos x = -1$ gives critical point $x = \pi$.

Evaluating $f(x)$ at $x = 0, \pi, 2\pi$:

$$f(0) = 0 + \sin 0 = 0$$

$$f(\pi) = \pi + \sin \pi = \pi + 0 = \pi$$

$$f(2\pi) = 2\pi + \sin 2\pi = 2\pi + 0 = 2\pi$$

Comparing values:

Absolute Maximum Value = 2π at point of maximum $x = 2\pi$

Absolute Minimum Value = 0 at point of minimum $x = 0$

Part (v): $f(x) = \sin x + \sqrt{3} \cos x$ on $[0, \pi]$

Differentiating $f(x)$:

$$f'(x) = \cos x - \sqrt{3} \sin x$$

Setting $f'(x) = 0$:

$$\cos x - \sqrt{3} \sin x = 0$$

$$\sqrt{3} \sin x = \cos x$$

$$\tan x = \frac{1}{\sqrt{3}}$$

In $[0, \pi]$, $\tan x = \frac{1}{\sqrt{3}}$ gives critical point $x = \frac{\pi}{6}$.

Evaluating $f(x)$ at $x = 0, \frac{\pi}{6}, \pi$:

$$\begin{aligned} f(0) &= \sin 0 + \sqrt{3} \cos 0 = 0 + \sqrt{3}(1) = \sqrt{3} \\ f\left(\frac{\pi}{6}\right) &= \sin \frac{\pi}{6} + \sqrt{3} \cos \frac{\pi}{6} = \frac{1}{2} + \sqrt{3} \left(\frac{\sqrt{3}}{2}\right) = \frac{1}{2} + \frac{3}{2} = 2 \\ f(\pi) &= \sin \pi + \sqrt{3} \cos \pi = 0 + \sqrt{3}(-1) = -\sqrt{3} \end{aligned}$$

Comparing values:

$$\text{Absolute Maximum Value} = 2 \quad \text{at point of maximum } x = \frac{\pi}{6}$$

$$\text{Absolute Minimum Value} = -\sqrt{3} \quad \text{at point of minimum } x = \pi$$

Part (vi): $f(x) = 3 + |x + 1|$ on $[-2, 3]$

The function is continuous on $[-2, 3]$ but non-differentiable at $x = -1$, making $x = -1$ a critical point.

Evaluating $f(x)$ at boundary points $x = -2, 3$ and critical point $x = -1$:

$$\begin{aligned} f(-2) &= 3 + |-2 + 1| = 3 + |-1| = 3 + 1 = 4 \\ f(-1) &= 3 + |-1 + 1| = 3 + |0| = 3 \\ f(3) &= 3 + |3 + 1| = 3 + |4| = 3 + 4 = 7 \end{aligned}$$

Comparing values:

$$\text{Absolute Maximum Value} = 7 \quad \text{at point of maximum } x = 3$$

$$\text{Absolute Minimum Value} = 3 \quad \text{at point of minimum } x = -1$$

Final Answer

- (i) Max = 19 at $x = -3$; Min = 3 at $x = 1$
- (ii) Max = 56 at $x = 5$; Min = 24 at $x = 1$
- (iii) Max = 25 at $x = 0$; Min = -39 at $x = 2$
- (iv) Max = 2π at $x = 2\pi$; Min = 0 at $x = 0$
- (v) Max = 2 at $x = \frac{\pi}{6}$; Min = $-\sqrt{3}$ at $x = \pi$
- (vi) Max = 7 at $x = 3$; Min = 3 at $x = -1$

Question 13

Find the maximum and minimum values of $x + \sin 2x$ on $[0, 2\pi]$. Also find points of maxima and minima.

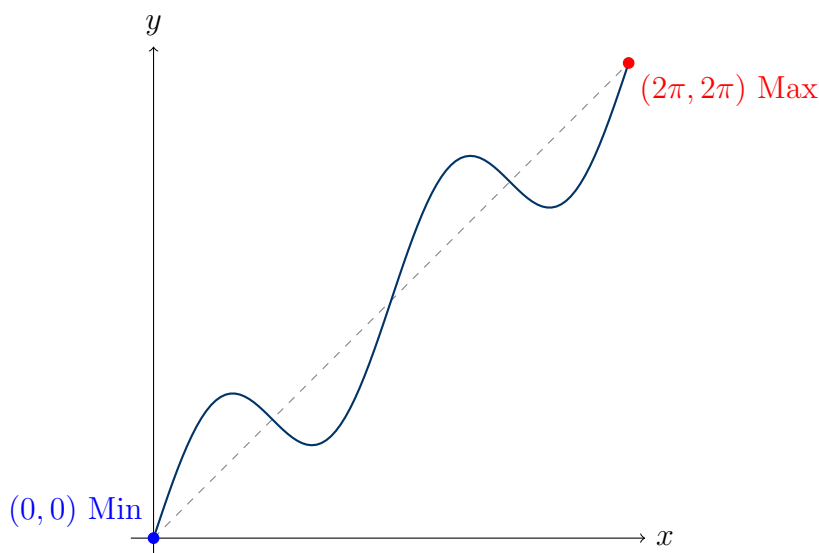
TYPE 85b Finding Absolute Extrema on a Closed Interval – Finding Absolute Extrema of Trigonometric-Linear Functions

★ Likely Exam Question

Given: Function $f(x) = x + \sin 2x$ defined on the closed interval $x \in [0, 2\pi]$.

To Find: Absolute maximum value, absolute minimum value, and corresponding points of maxima and minima.

Concept / Formula Used: $f(x)$ $[a, b]$ $f'(x) = 0$ $x = a, b$



Sketch of $f(x) = x + \sin 2x$ on $[0, 2\pi]$ highlighting boundary extrema

Solution:

The given function is:

$$f(x) = x + \sin 2x, \quad x \in [0, 2\pi]$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x + \sin 2x) \\ &= 1 + 2 \cos 2x \end{aligned}$$

To find the critical points, set $f'(x) = 0$:

$$\begin{aligned} 1 + 2 \cos 2x &= 0 \\ 2 \cos 2x &= -1 \\ \cos 2x &= -\frac{1}{2} \end{aligned}$$

Since $x \in [0, 2\pi]$, the transformed variable $2x \in [0, 4\pi]$.

The solutions for $\cos 2x = -\frac{1}{2}$ in $[0, 4\pi]$ are:

$$\begin{aligned} 2x &= \pi - \frac{\pi}{3}, \quad \pi + \frac{\pi}{3}, \quad 3\pi - \frac{\pi}{3}, \quad 3\pi + \frac{\pi}{3} \\ 2x &= \frac{2\pi}{3}, \quad \frac{4\pi}{3}, \quad \frac{8\pi}{3}, \quad \frac{10\pi}{3} \end{aligned}$$

Dividing by 2, the critical points $x \in [0, 2\pi]$ are:

$$x = \frac{\pi}{3}, \quad \frac{2\pi}{3}, \quad \frac{4\pi}{3}, \quad \frac{5\pi}{3}$$

Evaluating $f(x)$ at all critical points and boundary points $x = 0, 2\pi$:

1. At $x = 0$:

$$f(0) = 0 + \sin 0 = 0$$

2. At $x = \frac{\pi}{3}$:

$$f\left(\frac{\pi}{3}\right) = \frac{\pi}{3} + \sin\left(\frac{2\pi}{3}\right) = \frac{\pi}{3} + \frac{\sqrt{3}}{2} \approx 1.913$$

3. At $x = \frac{2\pi}{3}$:

$$f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + \sin\left(\frac{4\pi}{3}\right) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \approx 1.228$$

4. At $x = \frac{4\pi}{3}$:

$$f\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} + \sin\left(\frac{8\pi}{3}\right) = \frac{4\pi}{3} + \frac{\sqrt{3}}{2} \approx 5.055$$

5. At $x = \frac{5\pi}{3}$:

$$f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{3} + \sin\left(\frac{10\pi}{3}\right) = \frac{5\pi}{3} - \frac{\sqrt{3}}{2} \approx 4.370$$

6. At $x = 2\pi$:

$$f(2\pi) = 2\pi + \sin(4\pi) = 2\pi + 0 = 2\pi \approx 6.283$$

Comparing all candidate values:

Absolute Maximum Value = 2π at point of maximum $x = 2\pi$

Absolute Minimum Value = 0 at point of minimum $x = 0$

Final Answer

Maximum value = 2π at point of maxima $x = 2\pi$.

Minimum value = 0 at point of minima $x = 0$.

Question 14

Find the absolute maximum and absolute minimum values of the function $f(x) = \cos^2 x + \sin x$, $x \in [0, \pi]$.

Given: Function $f(x) = \cos^2 x + \sin x$ for $x \in [0, \pi]$.

To Find: The absolute maximum and absolute minimum values of $f(x)$.

Concept / Formula Used: $\cos^2 x = 1 - \sin^2 x$ $f(x)$ $[a, b]$ $f'(x) = 0$ a, b

Solution:

Using the identity $\cos^2 x = 1 - \sin^2 x$, we re-write $f(x)$:

$$f(x) = 1 - \sin^2 x + \sin x, \quad x \in [0, \pi]$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= -2 \sin x \cos x + \cos x \\ &= \cos x(1 - 2 \sin x) \end{aligned}$$

Setting $f'(x) = 0$ to determine critical points:

$$\cos x(1 - 2 \sin x) = 0$$

This yields two factor equations:

$$\text{Case 1: } \cos x = 0 \implies x = \frac{\pi}{2} \in [0, \pi]$$

$$\text{Case 2: } 1 - 2 \sin x = 0 \implies \sin x = \frac{1}{2} \implies x = \frac{\pi}{6}, \frac{5\pi}{6} \in [0, \pi]$$

Thus, the critical points in $[0, \pi]$ are $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$.

Evaluating $f(x)$ at critical points and domain boundary points $x = 0, \pi$:

1. At $x = 0$:

$$f(0) = \cos^2 0 + \sin 0 = 1^2 + 0 = 1$$

2. At $x = \frac{\pi}{6}$:

$$\begin{aligned} f\left(\frac{\pi}{6}\right) &= \cos^2\left(\frac{\pi}{6}\right) + \sin\left(\frac{\pi}{6}\right) \\ &= \left(\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2} \\ &= \frac{3}{4} + \frac{2}{4} = \frac{5}{4} \end{aligned}$$

3. At $x = \frac{\pi}{2}$:

$$\begin{aligned} f\left(\frac{\pi}{2}\right) &= \cos^2\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right) \\ &= 0^2 + 1 = 1 \end{aligned}$$

4. At $x = \frac{5\pi}{6}$:

$$\begin{aligned} f\left(\frac{5\pi}{6}\right) &= \cos^2\left(\frac{5\pi}{6}\right) + \sin\left(\frac{5\pi}{6}\right) \\ &= \left(-\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2} \\ &= \frac{3}{4} + \frac{2}{4} = \frac{5}{4} \end{aligned}$$

5. At $x = \pi$:

$$f(\pi) = \cos^2 \pi + \sin \pi = (-1)^2 + 0 = 1$$

Comparing all calculated values:

$$\text{Absolute Maximum Value} = \frac{5}{4} \quad \text{at } x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\text{Absolute Minimum Value} = 1 \quad \text{at } x = 0, \frac{\pi}{2}, \pi$$

Final Answer

Absolute maximum value = $\frac{5}{4}$.
Absolute minimum value = 1.

6.6 – Maxima and Minima Using Derivative Tests

Question 1

Find the turning points of the following functions and distinguish between them. Also find the local maximum and minimum values of the functions:

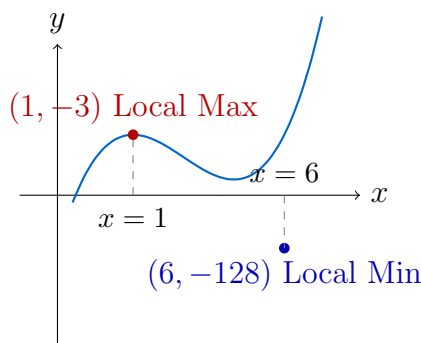
(i) $f(x) = 2x^3 - 21x^2 + 36x - 20$

(ii) $f(x) = x^3 - 3x^2 + 3x$

TYPE 88a Finding and Classifying Local Extrema via Derivative Tests – Finding and Classifying Local Extrema of Cubic Functions

Given: Functions $f(x) = 2x^3 - 21x^2 + 36x - 20$ and $f(x) = x^3 - 3x^2 + 3x$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x_0) < 0$ x_0 $f''(x_0) > 0$
0 x_0



Turning points and behavior of cubic polynomial $f(x) = 2x^3 - 21x^2 + 36x - 20$

Solution:

Part (i): Given function:

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(2x^3 - 21x^2 + 36x - 20) \\ &= 6x^2 - 42x + 36 \\ &= 6(x^2 - 7x + 6) \\ &= 6(x - 1)(x - 6) \end{aligned}$$

To find the turning points, set $f'(x) = 0$:

$$\begin{aligned} 6(x - 1)(x - 6) &= 0 \\ x &= 1 \quad \text{or} \quad x = 6 \end{aligned}$$

Differentiating $f'(x)$ to find the second derivative:

$$\begin{aligned} f''(x) &= \frac{d}{dx}(6x^2 - 42x + 36) \\ &= 12x - 42 \end{aligned}$$

Testing at critical point $x = 1$:

$$\begin{aligned} f''(1) &= 12(1) - 42 \\ &= -30 < 0 \end{aligned}$$

Since $f''(1) < 0$, $x = 1$ is a point of local maximum. Evaluating the function at $x = 1$:

$$\begin{aligned} f(1) &= 2(1)^3 - 21(1)^2 + 36(1) - 20 \\ &= 2 - 21 + 36 - 20 \\ &= -3 \end{aligned}$$

Turning point 1: $(1, -3)$ is a point of local maximum.

Testing at critical point $x = 6$:

$$\begin{aligned}f''(6) &= 12(6) - 42 \\&= 72 - 42 \\&= 30 > 0\end{aligned}$$

Since $f''(6) > 0$, $x = 6$ is a point of local minimum. Evaluating the function at $x = 6$:

$$\begin{aligned}f(6) &= 2(6)^3 - 21(6)^2 + 36(6) - 20 \\&= 2(216) - 21(36) + 216 - 20 \\&= 432 - 756 + 216 - 20 \\&= -128\end{aligned}$$

Turning point 2: $(6, -128)$ is a point of local minimum.

Part (ii): Given function:

$$f(x) = x^3 - 3x^2 + 3x$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned}f'(x) &= \frac{d}{dx}(x^3 - 3x^2 + 3x) \\&= 3x^2 - 6x + 3 \\&= 3(x^2 - 2x + 1) \\&= 3(x - 1)^2\end{aligned}$$

To find the turning points, set $f'(x) = 0$:

$$\begin{aligned}3(x - 1)^2 &= 0 \\x &= 1\end{aligned}$$

Differentiating $f'(x)$ to find the second derivative:

$$\begin{aligned}f''(x) &= \frac{d}{dx}(3x^2 - 6x + 3) \\&= 6x - 6\end{aligned}$$

Testing at critical point $x = 1$:

$$\begin{aligned}f''(1) &= 6(1) - 6 \\&= 0\end{aligned}$$

Since $f''(1) = 0$, we test the sign of $f'(x)$ in the neighborhood of $x = 1$: For $x < 1$, $f'(x) = 3(x - 1)^2 > 0$. For $x > 1$, $f'(x) = 3(x - 1)^2 > 0$. Since $f'(x)$ does not change sign as x increases through 1, $x = 1$ is neither a local maximum nor a local minimum; it is a point of inflexion.

Evaluating the function at $x = 1$:

$$\begin{aligned} f(1) &= (1)^3 - 3(1)^2 + 3(1) \\ &= 1 - 3 + 3 \\ &= 1 \end{aligned}$$

Final Answer

(i) Turning points are $(1, -3)$ (Local Maxima) and $(6, -128)$ (Local Minima).

Local maximum value = -3 , Local minimum value = -128 .

(ii) Turning point is $(1, 1)$, which is a point of inflexion (neither local maximum nor local minimum).

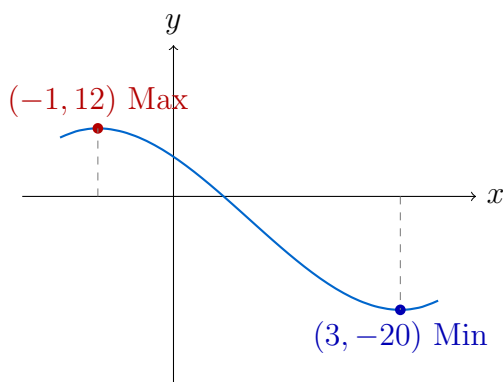
Question 2

Find the coordinates of stationary points on the curve $y = x^3 - 3x^2 - 9x + 7$, and distinguish between points of local maxima and minima.

TYPE 88a Finding and Classifying Local Extrema via Derivative Tests – Finding and Classifying Local Extrema of Cubic

Given: Curve equation $y = x^3 - 3x^2 - 9x + 7$.

Concept / Formula Used: $\frac{dy}{dx} = 0$ $\frac{d^2y}{dx^2} < 0$ $\frac{d^2y}{dx^2} > 0$



Stationary points on curve $y = x^3 - 3x^2 - 9x + 7$

Solution:

Given equation of the curve:

$$y = x^3 - 3x^2 - 9x + 7$$

Differentiating y with respect to x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^3 - 3x^2 - 9x + 7) \\ &= 3x^2 - 6x - 9 \\ &= 3(x^2 - 2x - 3) \\ &= 3(x - 3)(x + 1)\end{aligned}$$

For stationary points, set $\frac{dy}{dx} = 0$:

$$\begin{aligned}3(x - 3)(x + 1) &= 0 \\ x = 3 \quad \text{or} \quad x &= -1\end{aligned}$$

Differentiating $\frac{dy}{dx}$ to get the second derivative:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx}(3x^2 - 6x - 9) \\ &= 6x - 6\end{aligned}$$

At $x = -1$:

$$\begin{aligned}\frac{d^2y}{dx^2} &= 6(-1) - 6 \\ &= -12 < 0\end{aligned}$$

Since $\frac{d^2y}{dx^2} < 0$, $x = -1$ is a point of local maximum. Calculating the y -coordinate at $x = -1$:

$$\begin{aligned}y &= (-1)^3 - 3(-1)^2 - 9(-1) + 7 \\ &= -1 - 3 + 9 + 7 \\ &= 12\end{aligned}$$

Stationary point $(-1, 12)$ is a point of local maximum.

At $x = 3$:

$$\begin{aligned}\frac{d^2y}{dx^2} &= 6(3) - 6 \\ &= 18 - 6 \\ &= 12 > 0\end{aligned}$$

Since $\frac{d^2y}{dx^2} > 0$, $x = 3$ is a point of local minimum. Calculating the y -coordinate at $x = 3$:

$$\begin{aligned}y &= (3)^3 - 3(3)^2 - 9(3) + 7 \\ &= 27 - 27 - 27 + 7 \\ &= -20\end{aligned}$$

Stationary point $(3, -20)$ is a point of local minimum.

Final Answer

Point of local maximum is $(-1, 12)$.

Point of local minimum is $(3, -20)$.

Question 3

Find the points of local maxima and minima (if any) of each of the following functions. Find also the local maximum and minimum values:

- (i) $f(x) = x^2$
- (ii) $f(x) = x^3 - 3x$
- (iii) $f(x) = x^3 - 12x$
- (iv) $f(x) = x^3 - 6x^2 + 9x + 15$
- (v) $f(x) = x^3 - 3x + 3$
- (vi) $f(x) = 2x^3 - 6x^2 + 6x + 5$
- (vii) $f(x) = \frac{x}{2} + \frac{2}{x}, \quad x > 0$
- (viii) $f(x) = \frac{1}{x^2 + 2}$
- (ix) $f(x) = 2 \sin x - x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$
- (x) $f(x) = \sin x + \cos x, \quad 0 < x < \frac{\pi}{2}$
- (xi) $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$

TYPE 88b Finding and Classifying Local Extrema via Derivative Tests – Finding Local Extrema of Polynomial, Rational

Given: Various functions $f(x)$ listed in subparts (i) to (xi).

Concept / Formula Used: $f(x) \quad f'(x) = 0 \quad f''(x) > 0 \implies f''(x) < 0 \implies$

Solution:

(i) $f(x) = x^2$

$$f'(x) = 2x = 0 \implies x = 0$$

$$f''(x) = 2 > 0 \quad \text{for all } x$$

Point of local minimum at $x = 0$. Local minimum value $f(0) = 0$. No point of local maximum.

(ii) $f(x) = x^3 - 3x$

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \implies x = 1, -1$$

$$f''(x) = 6x$$

At $x = -1$: $f''(-1) = -6 < 0 \implies$ point of local maximum at $x = -1$. Local maximum value $f(-1) = (-1)^3 - 3(-1) = 2$. At $x = 1$: $f''(1) = 6 > 0 \implies$ point of local minimum

at $x = 1$. Local minimum value $f(1) = 1^3 - 3(1) = -2$.

(iii) $f(x) = x^3 - 12x$

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 0 \implies x = 2, -2$$

$$f''(x) = 6x$$

At $x = -2$: $f''(-2) = -12 < 0 \implies$ point of local maximum at $x = -2$. Local maximum value $f(-2) = (-2)^3 - 12(-2) = 16$. At $x = 2$: $f''(2) = 12 > 0 \implies$ point of local minimum at $x = 2$. Local minimum value $f(2) = 2^3 - 12(2) = -16$.

(iv) $f(x) = x^3 - 6x^2 + 9x + 15$

$$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) = 0 \implies x = 1, 3$$

$$f''(x) = 6x - 12$$

At $x = 1$: $f''(1) = -6 < 0 \implies$ point of local maximum at $x = 1$. Local maximum value $f(1) = 1 - 6 + 9 + 15 = 19$. At $x = 3$: $f''(3) = 6 > 0 \implies$ point of local minimum at $x = 3$. Local minimum value $f(3) = 27 - 54 + 27 + 15 = 15$.

(v) $f(x) = x^3 - 3x + 3$

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \implies x = 1, -1$$

$$f''(x) = 6x$$

At $x = -1$: $f''(-1) = -6 < 0 \implies$ point of local maximum at $x = -1$. Local maximum value $f(-1) = -1 + 3 + 3 = 5$. At $x = 1$: $f''(1) = 6 > 0 \implies$ point of local minimum at $x = 1$. Local minimum value $f(1) = 1 - 3 + 3 = 1$.

(vi) $f(x) = 2x^3 - 6x^2 + 6x + 5$

$$f'(x) = 6x^2 - 12x + 6 = 6(x - 1)^2 = 0 \implies x = 1$$

$$f''(x) = 12(x - 1)$$

At $x = 1$, $f''(1) = 0$. Also, $f'(x) \geq 0$ for all x , so $f'(x)$ does not change sign across $x = 1$. Hence, $x = 1$ is a point of inflexion. There are no points of local maximum or local minimum.

(vii) $f(x) = \frac{x}{2} + \frac{2}{x}, \quad x > 0$

$$f'(x) = \frac{1}{2} - \frac{2}{x^2} = 0 \implies x^2 = 4 \implies x = 2 \quad (\text{since } x > 0)$$

$$f''(x) = \frac{4}{x^3}$$

At $x = 2$: $f''(2) = \frac{4}{8} = \frac{1}{2} > 0 \implies$ point of local minimum at $x = 2$. Local minimum value $f(2) = \frac{2}{2} + \frac{2}{2} = 2$. No point of local maximum.

$$\text{(viii)} \quad f(x) = \frac{1}{x^2 + 2}$$

$$f'(x) = -\frac{2x}{(x^2 + 2)^2} = 0 \implies x = 0$$

$$f''(x) = \frac{-2(x^2 + 2)^2 - (-2x) \cdot 2(x^2 + 2)(2x)}{(x^2 + 2)^4} = \frac{6x^2 - 4}{(x^2 + 2)^3}$$

At $x = 0$: $f''(0) = \frac{-4}{8} = -\frac{1}{2} < 0 \implies$ point of local maximum at $x = 0$. Local maximum value $f(0) = \frac{1}{0+2} = \frac{1}{2}$. No point of local minimum.

$$\text{(ix)} \quad f(x) = 2 \sin x - x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$f'(x) = 2 \cos x - 1 = 0 \implies \cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3} \quad \text{or} \quad x = -\frac{\pi}{3} \quad \text{in} \quad \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$f''(x) = -2 \sin x$$

At $x = \frac{\pi}{3}$: $f''\left(\frac{\pi}{3}\right) = -2 \sin \frac{\pi}{3} = -\sqrt{3} < 0 \implies$ point of local maximum at $x = \frac{\pi}{3}$. Local maximum value $f\left(\frac{\pi}{3}\right) = 2\left(\frac{\sqrt{3}}{2}\right) - \frac{\pi}{3} = \sqrt{3} - \frac{\pi}{3}$.

At $x = -\frac{\pi}{3}$: $f''\left(-\frac{\pi}{3}\right) = -2 \sin\left(-\frac{\pi}{3}\right) = \sqrt{3} > 0 \implies$ point of local minimum at $x = -\frac{\pi}{3}$. Local minimum value $f\left(-\frac{\pi}{3}\right) = 2\left(-\frac{\sqrt{3}}{2}\right) - \left(-\frac{\pi}{3}\right) = -\sqrt{3} + \frac{\pi}{3}$.

$$\text{(x)} \quad f(x) = \sin x + \cos x, \quad 0 < x < \frac{\pi}{2}$$

$$f'(x) = \cos x - \sin x = 0 \implies \tan x = 1 \implies x = \frac{\pi}{4}$$

$$f''(x) = -\sin x - \cos x$$

At $x = \frac{\pi}{4}$: $f''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\sqrt{2} < 0 \implies$ point of local maximum at $x = \frac{\pi}{4}$. Local maximum value $f\left(\frac{\pi}{4}\right) = \sin \frac{\pi}{4} + \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$. No point of local minimum.

$$\text{(xi)} \quad f(x) = 3x^4 + 4x^3 - 12x^2 + 12$$

$$f'(x) = 12x^3 + 12x^2 - 24x = 12x(x^2 + x - 2) = 12x(x + 2)(x - 1) = 0$$

$$x = 0, 1, -2$$

$$f''(x) = 36x^2 + 24x - 24$$

At $x = 0$: $f''(0) = -24 < 0 \implies$ point of local maximum at $x = 0$. Local maximum value $f(0) = 12$.

At $x = 1$: $f''(1) = 36 + 24 - 24 = 36 > 0 \implies$ point of local minimum at $x = 1$. Local minimum value $f(1) = 3(1)^4 + 4(1)^3 - 12(1)^2 + 12 = 7$.

At $x = -2$: $f''(-2) = 36(4) + 24(-2) - 24 = 72 > 0 \implies$ point of local minimum at $x = -2$. Local minimum value $f(-2) = 3(16) + 4(-8) - 12(4) + 12 = -20$.

Final Answer

Subpart	Local Max at	Local Min at	Max Value	Min Value
(i)	None	$x = 0$	–	0
(ii)	$x = -1$	$x = 1$	2	–2
(iii)	$x = -2$	$x = 2$	16	–16
(iv)	$x = 1$	$x = 3$	19	15
(v)	$x = -1$	$x = 1$	5	1
(vi)	None	None	–	–
(vii)	None	$x = 2$	–	2
(viii)	$x = 0$	None	$\frac{1}{2}$	–
(ix)	$x = \frac{\pi}{3}$	$x = -\frac{\pi}{3}$	$\sqrt{3} - \frac{\pi}{3}$	$-\sqrt{3} + \frac{\pi}{3}$
(x)	$x = \frac{\pi}{4}$	None	$\sqrt{2}$	–
(xi)	$x = 0$	$x = 1, -2$	12	7, –20

Question 4

For what values of a and b , the function $f(x) = x^3 + ax^2 + bx - 3$ has local maximum value at $x = 0$ and local minimum value at $x = 1$?

TYPE 90a Finding Function Parameters from Given Extrema – Finding Polynomial Parameters from Given Local Extrema

★ Likely Exam Question

Given: Function $f(x) = x^3 + ax^2 + bx - 3$ with critical points at $x = 0$ (local max) and $x = 1$ (local min).

Concept / Formula Used: $f(x)$ $x = x_1$ $x = x_2$ $f'(x_1) = 0$ $f'(x_2) = 0$

Solution:

Given function:

$$f(x) = x^3 + ax^2 + bx - 3$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 + ax^2 + bx - 3) \\ &= 3x^2 + 2ax + b \end{aligned}$$

Since $f(x)$ has a local maximum at $x = 0$, $x = 0$ is a stationary point, so $f'(0) = 0$:

$$\begin{aligned} 3(0)^2 + 2a(0) + b &= 0 \\ b &= 0 \end{aligned}$$

Since $f(x)$ has a local minimum at $x = 1$, $x = 1$ is a stationary point, so $f'(1) = 0$:

$$3(1)^2 + 2a(1) + b = 0$$

Substituting $b = 0$:

$$\begin{aligned} 3 + 2a + 0 &= 0 \\ 2a &= -3 \\ a &= -\frac{3}{2} \end{aligned}$$

Verification using second derivative test:

$$\begin{aligned} f''(x) &= \frac{d}{dx}(3x^2 + 2ax + b) \\ &= 6x + 2a \\ &= 6x + 2\left(-\frac{3}{2}\right) \\ &= 6x - 3 \end{aligned}$$

At $x = 0$: $f''(0) = -3 < 0$, which confirms local maximum at $x = 0$.

At $x = 1$: $f''(1) = 6(1) - 3 = 3 > 0$, which confirms local minimum at $x = 1$.

Final Answer

$$a = -\frac{3}{2}, \quad b = 0$$

Question 5

Given $2f(x) = \log|x|^a - bx^2 + x$, if $f(x)$ has extreme values at $x = -1$ and $x = 2$, find the value of a and b .

TYPE 90b Finding Function Parameters from Given Extrema – Finding Function Parameters from Given Extreme Values

Given: Function relation $2f(x) = \log|x|^a - bx^2 + x$ with extreme values at $x = -1$ and $x = 2$.

Concept / Formula Used: $x = -1 \quad x = 2 \quad f'(-1) = 0 \quad f'(2) = 0$
 $\log|x|^a = a \log|x|$

Solution:

Given equation:

$$2f(x) = a \log|x| - bx^2 + x$$

Dividing by 2:

$$f(x) = \frac{a}{2} \log|x| - \frac{b}{2}x^2 + \frac{x}{2}$$

Differentiating $f(x)$ with respect to x :

$$\begin{aligned}f'(x) &= \frac{a}{2x} - \frac{b}{2}(2x) + \frac{1}{2} \\&= \frac{a}{2x} - bx + \frac{1}{2}\end{aligned}$$

Since $f(x)$ has extreme values at $x = -1$ and $x = 2$, we have $f'(-1) = 0$ and $f'(2) = 0$.

Setting $f'(-1) = 0$:

$$\begin{aligned}\frac{a}{2(-1)} - b(-1) + \frac{1}{2} &= 0 \\-\frac{a}{2} + b + \frac{1}{2} &= 0\end{aligned}$$

Multiplying the entire equation by 2:

$$\begin{aligned}-a + 2b + 1 &= 0 \\a - 2b &= 1 \quad \text{--- (Equation 1)}\end{aligned}$$

Setting $f'(2) = 0$:

$$\begin{aligned}\frac{a}{2(2)} - b(2) + \frac{1}{2} &= 0 \\\frac{a}{4} - 2b + \frac{1}{2} &= 0\end{aligned}$$

Multiplying the entire equation by 4:

$$\begin{aligned}a - 8b + 2 &= 0 \\a - 8b &= -2 \quad \text{--- (Equation 2)}\end{aligned}$$

Subtracting Equation 2 from Equation 1 to solve the simultaneous system:

$$\begin{aligned}(a - 2b) - (a - 8b) &= 1 - (-2) \\6b &= 3 \\b &= \frac{3}{6} \\b &= \frac{1}{2}\end{aligned}$$

Substituting $b = \frac{1}{2}$ back into Equation 1:

$$\begin{aligned}a - 2\left(\frac{1}{2}\right) &= 1 \\a - 1 &= 1 \\a &= 2\end{aligned}$$

Final Answer

$$a = 2, \quad b = \frac{1}{2}$$

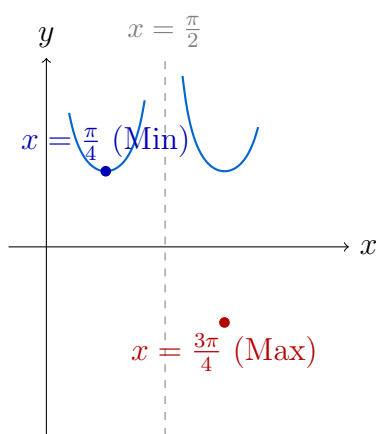
Question 6

Aditya hires a speed boat in an island to do water sports, which is moving along a curve $y = \frac{1}{\cos x \sin x}$, $x \in (0, \pi)$ in water. Find the number of stationary points.

TYPE 92 Finding the Number of Stationary Points of a Trigonometric Function

Given: Curve equation $y = \frac{1}{\cos x \sin x}$ for domain $x \in (0, \pi)$.

Concept / Formula Used: $2 \sin x \cos x = \sin 2x$ $\frac{dy}{dx} = 0$



Sketch of curve $y = 2 \csc 2x$ for $x \in (0, \pi)$ showing stationary points

Solution:

Given curve:

$$y = \frac{1}{\cos x \sin x}, \quad x \in (0, \pi)$$

Multiplying numerator and denominator by 2:

$$\begin{aligned} y &= \frac{2}{2 \sin x \cos x} \\ &= \frac{2}{\sin 2x} \\ &= 2 \csc 2x \end{aligned}$$

Differentiating y with respect to x :

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(2 \csc 2x) \\ &= 2(-\csc 2x \cot 2x) \cdot \frac{d}{dx}(2x) \\ &= -4 \csc 2x \cot 2x \end{aligned}$$

For stationary points, set $\frac{dy}{dx} = 0$:

$$-4 \csc 2x \cot 2x = 0$$

Since $\csc 2x \neq 0$ for any $x \in (0, \pi)$, we must have:

$$\cot 2x = 0$$

Given $x \in (0, \pi)$, the argument $2x \in (0, 2\pi)$. The values of $2x \in (0, 2\pi)$ for which $\cot 2x = 0$ are:

$$\begin{aligned} 2x = \frac{\pi}{2} &\implies x = \frac{\pi}{4} \\ 2x = \frac{3\pi}{2} &\implies x = \frac{3\pi}{4} \end{aligned}$$

Both $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ lie within the open interval $(0, \pi)$. Therefore, there are exactly 2 stationary points.

Final Answer

Number of stationary points = 2 (at $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$).

Question 7

Anu enjoys a roller coaster ride by first going downwards and then upwards to the maximum height. The relation between the distance travelled (cm) with respect to the time taken to complete the ride by Anu is given by the following equation:

$$y = 4x - \frac{1}{2}x^2$$

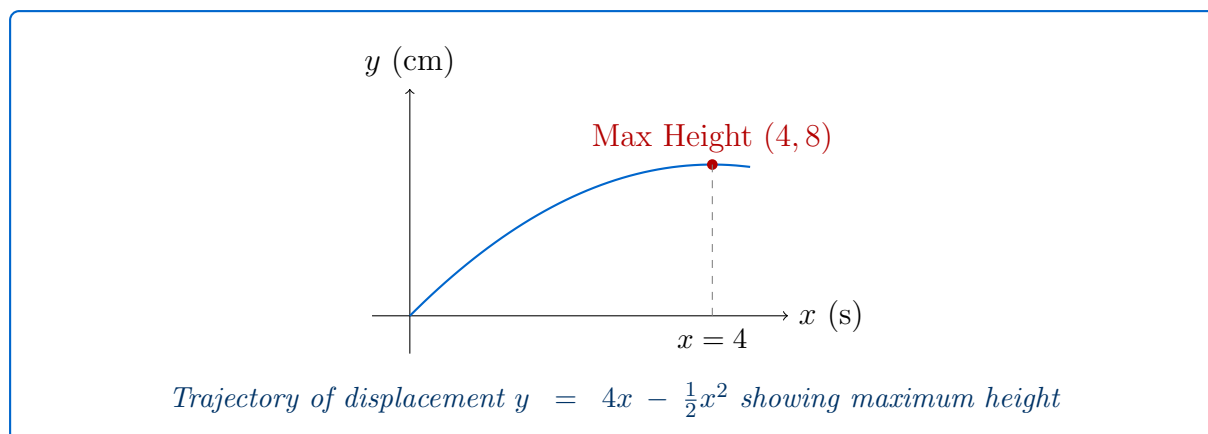
where x = time in seconds.

- What is the rate of change of displacement with respect to the time?
- How many seconds will it take her to go to its maximum height?

TYPE 93 Solving Kinematics Maxima Problems using First Derivative

Given: Displacement relation $y = 4x - \frac{1}{2}x^2$, where x is time in seconds.

Concept / Formula Used: $\frac{dy}{dx} = 0$ $\frac{d^2y}{dx^2} < 0$



Solution:

Part (a): Rate of change of displacement with respect to time Displacement equation:

$$y = 4x - \frac{1}{2}x^2$$

Differentiating y with respect to time x :

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(4x - \frac{1}{2}x^2 \right) \\ &= 4 - \frac{1}{2}(2x) \\ &= 4 - x\end{aligned}$$

Hence, the rate of change of displacement with respect to time is $4 - x$.

Part (b): Time taken to reach maximum height At maximum height, the rate of change of displacement $\frac{dy}{dx}$ equals zero:

$$\begin{aligned}\frac{dy}{dx} &= 0 \\ 4 - x &= 0 \\ x &= 4 \text{ seconds}\end{aligned}$$

Verification of maximum using the second derivative test:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx}(4 - x) \\ &= -1 < 0\end{aligned}$$

Since $\frac{d^2y}{dx^2} < 0$, y is indeed maximum at $x = 4$ seconds.

Final Answer

(a) Rate of change of displacement $= 4 - x$

(b) Time taken to reach maximum height $= 4$ seconds

6.7 – Applications of Maxima and Minima

Question 1

Find two numbers whose sum is 24 and whose product is as large as possible.

TYPE 94 Maximizing Product of Two Numbers with Constant Sum

Given: Two numbers x and y such that $x + y = 24$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) < 0$

Solution:

Let the two numbers be x and y . We are given:

$$\begin{aligned}x + y &= 24 \\y &= 24 - x\end{aligned}$$

Let P be the product of the two numbers:

$$P = xy$$

Substituting $y = 24 - x$ into the product equation:

$$\begin{aligned}P(x) &= x(24 - x) \\P(x) &= 24x - x^2\end{aligned}$$

Differentiating $P(x)$ with respect to x :

$$\begin{aligned}\frac{dP}{dx} &= \frac{d}{dx}(24x - x^2) \\\frac{dP}{dx} &= 24 - 2x\end{aligned}$$

For critical points, we set $\frac{dP}{dx} = 0$:

$$\begin{aligned}24 - 2x &= 0 \\2x &= 24 \\x &= 12\end{aligned}$$

Now, we find the second derivative to test for maximum:

$$\begin{aligned}\frac{d^2P}{dx^2} &= \frac{d}{dx}(24 - 2x) \\\frac{d^2P}{dx^2} &= -2\end{aligned}$$

Since $\frac{d^2P}{dx^2} = -2 < 0$, the product P is maximum at $x = 12$.

Now, back-substitute $x = 12$ to find y :

$$y = 24 - 12$$

$$y = 12$$

Thus, the two numbers are 12 and 12.

Final Answer

The two numbers are 12 and 12.

Question 2

The sum of two positive numbers is 20. Find the numbers

- (i) if the sum of their squares is minimum.
- (ii) if the product of the square of one and the cube of the other is maximum.

TYPE 95 Optimizing Sum of Squares or Products with Constant Sum

Given: Two positive numbers x and y such that $x + y = 20$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0$ $f'(x) = 0$
 $f''(x) < 0$

Solution:

Let the two positive numbers be x and y . We are given:

$$x + y = 20$$

$$y = 20 - x$$

(i) If the sum of their squares is minimum:

Let S be the sum of the squares of the two numbers:

$$S = x^2 + y^2$$

Substituting $y = 20 - x$:

$$S(x) = x^2 + (20 - x)^2$$

$$S(x) = x^2 + 400 - 40x + x^2$$

$$S(x) = 2x^2 - 40x + 400$$

Differentiating $S(x)$ with respect to x :

$$\frac{dS}{dx} = \frac{d}{dx}(2x^2 - 40x + 400)$$

$$\frac{dS}{dx} = 4x - 40$$

For critical points, set $\frac{dS}{dx} = 0$:

$$\begin{aligned}4x - 40 &= 0 \\4x &= 40 \\x &= 10\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2S}{dx^2} &= \frac{d}{dx}(4x - 40) \\ \frac{d^2S}{dx^2} &= 4\end{aligned}$$

Since $\frac{d^2S}{dx^2} = 4 > 0$, the sum of squares S is minimum at $x = 10$.

Back-substituting $x = 10$ to find y :

$$\begin{aligned}y &= 20 - 10 \\ y &= 10\end{aligned}$$

Thus, the numbers are 10 and 10.

(ii) If the product of the square of one and the cube of the other is maximum:

Let P be the product of the square of x and the cube of y :

$$P = x^2y^3$$

Substituting $y = 20 - x$:

$$P(x) = x^2(20 - x)^3$$

Differentiating $P(x)$ with respect to x using the product rule:

$$\begin{aligned}\frac{dP}{dx} &= \frac{d}{dx} [x^2(20 - x)^3] \\ &= 2x(20 - x)^3 + x^2 \cdot 3(20 - x)^2 \cdot (-1) \\ &= 2x(20 - x)^3 - 3x^2(20 - x)^2 \\ &= x(20 - x)^2 [2(20 - x) - 3x] \\ &= x(20 - x)^2 (40 - 2x - 3x) \\ &= x(20 - x)^2 (40 - 5x)\end{aligned}$$

For critical points, set $\frac{dP}{dx} = 0$:

$$x(20 - x)^2(40 - 5x) = 0$$

Since x and y are positive numbers, $x \neq 0$ and $x \neq 20$. Thus:

$$\begin{aligned}40 - 5x &= 0 \\ 5x &= 40 \\ x &= 8\end{aligned}$$

Now, let us find the second derivative to confirm maximum:

$$\frac{d^2P}{dx^2} = \frac{d}{dx} [(40x - 5x^2)(20 - x)^2]$$

Using the product rule at $x = 8$:

$$\frac{d^2P}{dx^2} = (40 - 10x)(20 - x)^2 + (40x - 5x^2) \cdot 2(20 - x)(-1)$$

Substitute $x = 8$:

$$\begin{aligned} \left. \frac{d^2P}{dx^2} \right|_{x=8} &= (40 - 80)(12)^2 + (320 - 320) \cdot 2(12)(-1) \\ &= -40 \cdot 144 + 0 \\ &= -5760 < 0 \end{aligned}$$

Since the second derivative is negative, P is maximum at $x = 8$.

Back-substituting $x = 8$ to find y :

$$\begin{aligned} y &= 20 - 8 \\ y &= 12 \end{aligned}$$

Thus, the numbers are 8 and 12.

Final Answer

- (i) The numbers are 10 and 10.
- (ii) The numbers are 8 and 12.

Question 3

The product of two positive numbers is 16. Find the numbers

- (i) if their sum is least.
- (ii) if the sum of one and the square of the other is least.

TYPE 96 Minimizing Sum of Two Numbers with Constant Product

Given: Two positive numbers x and y such that $xy = 16$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0$

Solution:

Let the two positive numbers be x and y . We are given:

$$\begin{aligned} xy &= 16 \\ y &= \frac{16}{x} \end{aligned}$$

(i) If their sum is least:

Let S be the sum of the two numbers:

$$S = x + y$$

Substituting $y = \frac{16}{x}$:

$$S(x) = x + \frac{16}{x}$$

Differentiating $S(x)$ with respect to x :

$$\begin{aligned}\frac{dS}{dx} &= \frac{d}{dx} \left(x + \frac{16}{x} \right) \\ \frac{dS}{dx} &= 1 - \frac{16}{x^2}\end{aligned}$$

For critical points, set $\frac{dS}{dx} = 0$:

$$\begin{aligned}1 - \frac{16}{x^2} &= 0 \\ x^2 &= 16 \\ x &= 4 \quad (\text{since } x > 0)\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2S}{dx^2} &= \frac{d}{dx} \left(1 - \frac{16}{x^2} \right) \\ \frac{d^2S}{dx^2} &= \frac{32}{x^3}\end{aligned}$$

At $x = 4$:

$$\left. \frac{d^2S}{dx^2} \right|_{x=4} = \frac{32}{4^3} = \frac{32}{64} = \frac{1}{2} > 0$$

Since the second derivative is positive, the sum S is minimum at $x = 4$.

Back-substituting $x = 4$ to find y :

$$\begin{aligned}y &= \frac{16}{4} \\ y &= 4\end{aligned}$$

Thus, the numbers are 4 and 4.

(ii) If the sum of one and the square of the other is least:

Let S be the sum of x and the square of y :

$$S = x + y^2$$

Substituting $y = \frac{16}{x}$:

$$\begin{aligned}S(x) &= x + \left(\frac{16}{x} \right)^2 \\ S(x) &= x + \frac{256}{x^2}\end{aligned}$$

Differentiating $S(x)$ with respect to x :

$$\begin{aligned}\frac{dS}{dx} &= \frac{d}{dx} \left(x + \frac{256}{x^2} \right) \\ \frac{dS}{dx} &= 1 - \frac{512}{x^3}\end{aligned}$$

For critical points, set $\frac{dS}{dx} = 0$:

$$\begin{aligned}1 - \frac{512}{x^3} &= 0 \\ x^3 &= 512 \\ x &= 8\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2S}{dx^2} &= \frac{d}{dx} \left(1 - \frac{512}{x^3} \right) \\ \frac{d^2S}{dx^2} &= \frac{1536}{x^4}\end{aligned}$$

At $x = 8$:

$$\left. \frac{d^2S}{dx^2} \right|_{x=8} = \frac{1536}{8^4} = \frac{1536}{4096} = \frac{3}{8} > 0$$

Since the second derivative is positive, the sum S is minimum at $x = 8$.

Back-substituting $x = 8$ to find y :

$$\begin{aligned}y &= \frac{16}{8} \\ y &= 2\end{aligned}$$

Thus, the numbers are 8 and 2.

Final Answer

- (i) The numbers are 4 and 4.
- (ii) The numbers are 8 and 2.

Question 4

Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.

TYPE 97 Maximizing Product of Powers of Two Numbers with Constant Sum

Given: Two positive numbers x and y such that $x + y = 60$.

Concept / Formula Used: $f(y)$ $f'(y) = 0$ $f''(y) < 0$

Solution:

Let the two positive numbers be x and y . We are given:

$$\begin{aligned}x + y &= 60 \\x &= 60 - y\end{aligned}$$

Let P be the product xy^3 :

$$P = xy^3$$

Substituting $x = 60 - y$:

$$\begin{aligned}P(y) &= (60 - y)y^3 \\P(y) &= 60y^3 - y^4\end{aligned}$$

Differentiating $P(y)$ with respect to y :

$$\begin{aligned}\frac{dP}{dy} &= \frac{d}{dy}(60y^3 - y^4) \\ \frac{dP}{dy} &= 180y^2 - 4y^3 \\ \frac{dP}{dy} &= 4y^2(45 - y)\end{aligned}$$

For critical points, set $\frac{dP}{dy} = 0$:

$$4y^2(45 - y) = 0$$

Since y is a positive number, $y \neq 0$. Thus:

$$\begin{aligned}45 - y &= 0 \\y &= 45\end{aligned}$$

Now, find the second derivative to confirm maximum:

$$\begin{aligned}\frac{d^2P}{dy^2} &= \frac{d}{dy}(180y^2 - 4y^3) \\ \frac{d^2P}{dy^2} &= 360y - 12y^2\end{aligned}$$

At $y = 45$:

$$\begin{aligned}\left. \frac{d^2P}{dy^2} \right|_{y=45} &= 360(45) - 12(45)^2 \\ &= 45(360 - 540) \\ &= 45(-180) \\ &= -8100 < 0\end{aligned}$$

Since the second derivative is negative, the product P is maximum at $y = 45$.

Back-substituting $y = 45$ to find x :

$$\begin{aligned}x &= 60 - 45 \\x &= 15\end{aligned}$$

Thus, the two positive numbers are 15 and 45.

Final Answer

The numbers are $x = 15$ and $y = 45$.

Question 5

Find two positive numbers x and y such that their sum is 35 and the product x^2y^5 is maximum.

TYPE 97 Maximizing Product of Powers of Two Numbers with Constant Sum

Given: Two positive numbers x and y such that $x + y = 35$.

Concept / Formula Used: $f(y)$ $f'(y) = 0$ $f''(y) < 0$

Solution:

Let the two positive numbers be x and y . We are given:

$$\begin{aligned}x + y &= 35 \\x &= 35 - y\end{aligned}$$

Let P be the product x^2y^5 :

$$P = x^2y^5$$

Substituting $x = 35 - y$:

$$P(y) = (35 - y)^2y^5$$

Differentiating $P(y)$ with respect to y using the product rule:

$$\begin{aligned}\frac{dP}{dy} &= \frac{d}{dy} [(35 - y)^2y^5] \\&= 2(35 - y)(-1)y^5 + (35 - y)^2 \cdot 5y^4 \\&= -2y^5(35 - y) + 5y^4(35 - y)^2 \\&= y^4(35 - y)[-2y + 5(35 - y)] \\&= y^4(35 - y)(175 - 7y)\end{aligned}$$

For critical points, set $\frac{dP}{dy} = 0$:

$$y^4(35 - y)(175 - 7y) = 0$$

Since x and y are positive numbers, $y \neq 0$ and $y \neq 35$. Thus:

$$\begin{aligned}175 - 7y &= 0 \\7y &= 175 \\y &= 25\end{aligned}$$

Let us check the sign of $\frac{dP}{dy}$ around $y = 25$:

- For $y < 25$ (slightly), $175 - 7y > 0 \implies \frac{dP}{dy} > 0$.
- For $y > 25$ (slightly), $175 - 7y < 0 \implies \frac{dP}{dy} < 0$.

Since $\frac{dP}{dy}$ changes sign from positive to negative as y increases through 25, the product P is maximum at $y = 25$.

Back-substituting $y = 25$ to find x :

$$x = 35 - 25$$

$$x = 10$$

Thus, the two positive numbers are 10 and 25.

Final Answer

The numbers are $x = 10$ and $y = 25$.

Question 6

Sum of two numbers is 5. If the sum of the cubes of these numbers is least, then find the sum of the squares of these numbers.

TYPE 98 Minimizing Sum of Cubes with Constant Sum

Given: Two numbers x and y such that $x + y = 5$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0$

Solution:

Let the two numbers be x and y . We are given:

$$x + y = 5$$

$$y = 5 - x$$

Let S be the sum of the cubes of the two numbers:

$$S = x^3 + y^3$$

Substituting $y = 5 - x$:

$$S(x) = x^3 + (5 - x)^3$$

Differentiating $S(x)$ with respect to x :

$$\begin{aligned}\frac{dS}{dx} &= \frac{d}{dx} [x^3 + (5 - x)^3] \\ &= 3x^2 + 3(5 - x)^2 \cdot (-1) \\ &= 3x^2 - 3(25 - 10x + x^2) \\ &= 3x^2 - 75 + 30x - 3x^2 \\ &= 30x - 75\end{aligned}$$

For critical points, set $\frac{dS}{dx} = 0$:

$$\begin{aligned} 30x - 75 &= 0 \\ 30x &= 75 \\ x &= \frac{75}{30} \\ x &= \frac{5}{2} \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2S}{dx^2} &= \frac{d}{dx}(30x - 75) \\ \frac{d^2S}{dx^2} &= 30 \end{aligned}$$

Since $\frac{d^2S}{dx^2} = 30 > 0$, the sum of cubes S is minimum at $x = \frac{5}{2}$.

Back-substituting $x = \frac{5}{2}$ to find y :

$$\begin{aligned} y &= 5 - \frac{5}{2} \\ y &= \frac{5}{2} \end{aligned}$$

Now, we need to find the sum of the squares of these numbers, which is $x^2 + y^2$:

$$\begin{aligned} x^2 + y^2 &= \left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2 \\ &= \frac{25}{4} + \frac{25}{4} \\ &= \frac{50}{4} \\ &= \frac{25}{2} \end{aligned}$$

Final Answer

The sum of the squares of the numbers is $\frac{25}{2}$ (or 12.5).

Question 7

Three numbers are given whose sum is 180 and the ratio of first two of them is 1 : 2. If the product of the numbers is greatest, find the numbers.

TYPE 99 Maximizing Product of Three Numbers with Constant Sum

Given: Three numbers x, y, z such that $x + y + z = 180$ and $x : y = 1 : 2$.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) < 0$

Solution:

Let the three numbers be x, y , and z . We are given:

$$x + y + z = 180 \quad \text{--- (1)}$$

$$\frac{x}{y} = \frac{1}{2} \implies y = 2x \quad \text{--- (2)}$$

Substituting $y = 2x$ into equation (1):

$$x + 2x + z = 180$$

$$3x + z = 180$$

$$z = 180 - 3x \quad \text{--- (3)}$$

Let P be the product of the three numbers:

$$P = xyz$$

Substituting $y = 2x$ and $z = 180 - 3x$:

$$P(x) = x(2x)(180 - 3x)$$

$$P(x) = 2x^2(180 - 3x)$$

$$P(x) = 360x^2 - 6x^3$$

Differentiating $P(x)$ with respect to x :

$$\frac{dP}{dx} = \frac{d}{dx}(360x^2 - 6x^3)$$

$$\frac{dP}{dx} = 720x - 18x^2$$

$$\frac{dP}{dx} = 18x(40 - x)$$

For critical points, set $\frac{dP}{dx} = 0$:

$$18x(40 - x) = 0$$

Since the numbers must be positive for a non-zero maximum product, $x \neq 0$. Thus:

$$40 - x = 0$$

$$x = 40$$

Finding the second derivative:

$$\frac{d^2P}{dx^2} = \frac{d}{dx}(720x - 18x^2)$$

$$\frac{d^2P}{dx^2} = 720 - 36x$$

At $x = 40$:

$$\begin{aligned} \left. \frac{d^2P}{dx^2} \right|_{x=40} &= 720 - 36(40) \\ &= 720 - 1440 \\ &= -720 < 0 \end{aligned}$$

Since the second derivative is negative, the product P is maximum at $x = 40$.

Now, back-substitute $x = 40$ to find y and z :

$$y = 2(40) = 80$$

$$z = 180 - 3(40) = 180 - 120 = 60$$

Thus, the three numbers are 40, 80, and 60.

Final Answer

The numbers are 40, 80, and 60.

Question 8

Find the maximum profit that a company can make, if the profit function is given by $p(x) = 41 + 72x - 18x^2$.

TYPE 100 Finding Maximum of Quadratic Profit Function

Given: Profit function $p(x) = 41 + 72x - 18x^2$.

Concept / Formula Used: $p(x)$ $p'(x) = 0$ $p''(x) < 0$

Solution:

The profit function is given by:

$$p(x) = 41 + 72x - 18x^2$$

Differentiating $p(x)$ with respect to x :

$$p'(x) = \frac{d}{dx}(41 + 72x - 18x^2)$$

$$p'(x) = 72 - 36x$$

For critical points, set $p'(x) = 0$:

$$72 - 36x = 0$$

$$36x = 72$$

$$x = 2$$

Finding the second derivative:

$$p''(x) = \frac{d}{dx}(72 - 36x)$$

$$p''(x) = -36$$

Since $p''(x) = -36 < 0$, the profit is maximum at $x = 2$.

Now, substitute $x = 2$ into the profit function to find the maximum profit:

$$\begin{aligned} p(2) &= 41 + 72(2) - 18(2)^2 \\ &= 41 + 144 - 18(4) \\ &= 41 + 144 - 72 \\ &= 41 + 72 \\ &= 113 \end{aligned}$$

Final Answer

The maximum profit is 113.

Question 9

The cost (in Rs.) per article C , of manufacturing a certain article is given by the formula $C = 5 + \frac{48}{x} + 3x^2$, where x is the number of articles manufactured per hour. Find the minimum value of C .

TYPE 101 Finding Minimum of Rational-Quadratic Cost Function

Given: Cost function $C(x) = 5 + \frac{48}{x} + 3x^2$ for $x > 0$.

Concept / Formula Used: $C(x)$ $C'(x) = 0$ $C''(x) > 0$

Solution:

The cost function is:

$$C(x) = 5 + \frac{48}{x} + 3x^2$$

Differentiating $C(x)$ with respect to x :

$$\begin{aligned} \frac{dC}{dx} &= \frac{d}{dx} \left(5 + \frac{48}{x} + 3x^2 \right) \\ \frac{dC}{dx} &= -\frac{48}{x^2} + 6x \end{aligned}$$

For critical points, set $\frac{dC}{dx} = 0$:

$$\begin{aligned} -\frac{48}{x^2} + 6x &= 0 \\ 6x &= \frac{48}{x^2} \\ x^3 &= 8 \\ x &= 2 \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2C}{dx^2} &= \frac{d}{dx} \left(-\frac{48}{x^2} + 6x \right) \\ \frac{d^2C}{dx^2} &= \frac{96}{x^3} + 6 \end{aligned}$$

At $x = 2$:

$$\begin{aligned}\left.\frac{d^2C}{dx^2}\right|_{x=2} &= \frac{96}{2^3} + 6 \\ &= \frac{96}{8} + 6 \\ &= 12 + 6 \\ &= 18 > 0\end{aligned}$$

Since the second derivative is positive, the cost C is minimum at $x = 2$.

Now, substitute $x = 2$ into the cost function to find the minimum cost:

$$\begin{aligned}C(2) &= 5 + \frac{48}{2} + 3(2)^2 \\ &= 5 + 24 + 3(4) \\ &= 5 + 24 + 12 \\ &= 41\end{aligned}$$

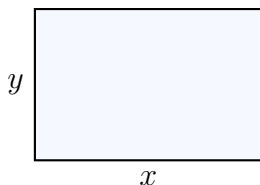
Final Answer

The minimum value of C is 41.

Question 10

Prove that of all rectangles with given area, the square has the smallest perimeter.

TYPE 102 Minimizing Perimeter of Rectangle with Constant Area



Rectangle with dimensions x and y

To Prove: The perimeter of a rectangle with given area A is minimum when it is a square ($x = y$).

Concept / Formula Used: $P = 2(x + y)$ $A = xy$

Solution:

Let the sides of the rectangle be x and y . The area A is given and is constant:

$$\begin{aligned}A &= xy \\ y &= \frac{A}{x}\end{aligned}$$

The perimeter P of the rectangle is:

$$P = 2(x + y)$$

Substituting $y = \frac{A}{x}$ into the perimeter equation:

$$P(x) = 2 \left(x + \frac{A}{x} \right)$$

Differentiating $P(x)$ with respect to x :

$$\frac{dP}{dx} = 2 \left(1 - \frac{A}{x^2} \right)$$

For critical points, set $\frac{dP}{dx} = 0$:

$$\begin{aligned} 2 \left(1 - \frac{A}{x^2} \right) &= 0 \\ 1 - \frac{A}{x^2} &= 0 \\ x^2 &= A \\ x &= \sqrt{A} \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2P}{dx^2} &= \frac{d}{dx} \left[2 \left(1 - \frac{A}{x^2} \right) \right] \\ \frac{d^2P}{dx^2} &= \frac{4A}{x^3} \end{aligned}$$

At $x = \sqrt{A}$:

$$\left. \frac{d^2P}{dx^2} \right|_{x=\sqrt{A}} = \frac{4A}{(\sqrt{A})^3} = \frac{4}{\sqrt{A}} > 0$$

Since the second derivative is positive, the perimeter P is minimum when $x = \sqrt{A}$.

Now, back-substitute $x = \sqrt{A}$ to find y :

$$\begin{aligned} y &= \frac{A}{\sqrt{A}} \\ y &= \sqrt{A} \end{aligned}$$

Since $x = y = \sqrt{A}$, the rectangle is a square.

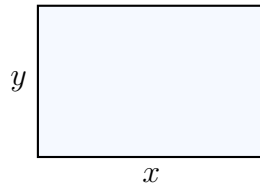
Hence Proved

LHS = RHS, hence proved that the rectangle with given area has the smallest perimeter when it is a square.

Question 11

Of all rectangles each of which has perimeter 24 cm, find the one having maximum area. Also find that area.

TYPE 103 Maximizing Area of Rectangle with Constant Perimeter



Rectangle with dimensions x and y

Given: Perimeter of the rectangle $P = 24$ cm.

Concept / Formula Used: $P = 2(x + y)$ $A = xy$

Solution:

Let the sides of the rectangle be x and y . We are given:

$$\begin{aligned}2(x + y) &= 24 \\x + y &= 12 \\y &= 12 - x\end{aligned}$$

The area A of the rectangle is:

$$A = xy$$

Substituting $y = 12 - x$:

$$\begin{aligned}A(x) &= x(12 - x) \\A(x) &= 12x - x^2\end{aligned}$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned}\frac{dA}{dx} &= \frac{d}{dx}(12x - x^2) \\\frac{dA}{dx} &= 12 - 2x\end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned}12 - 2x &= 0 \\2x &= 12 \\x &= 6\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dx^2} &= \frac{d}{dx}(12 - 2x) \\\frac{d^2A}{dx^2} &= -2\end{aligned}$$

Since $\frac{d^2A}{dx^2} = -2 < 0$, the area is maximum when $x = 6$ cm.

Back-substituting $x = 6$ to find y :

$$y = 12 - 6$$

$$y = 6 \text{ cm}$$

Thus, the rectangle is a square of side 6 cm.

The maximum area is:

$$\begin{aligned} A &= 6 \times 6 \\ &= 36 \text{ cm}^2 \end{aligned}$$

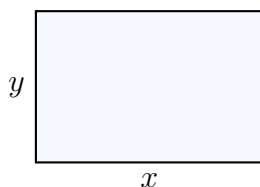
Final Answer

The rectangle has dimensions 6 cm $\times$ 6 cm and its maximum area is 36 cm².

Question 12

Prove that of all rectangles with given perimeter, the square has the largest area.

TYPE 103 Maximizing Area of Rectangle with Constant Perimeter



Rectangle with dimensions x and y

To Prove: The area of a rectangle with given perimeter P is maximum when it is a square ($x = y$).

Concept / Formula Used: $P = 2(x + y)$ $A = xy$

Solution:

Let the sides of the rectangle be x and y . The perimeter P is given and is constant:

$$\begin{aligned} 2(x + y) &= P \\ x + y &= \frac{P}{2} \\ y &= \frac{P}{2} - x \end{aligned}$$

Let A be the area of the rectangle:

$$A = xy$$

Substituting $y = \frac{P}{2} - x$:

$$\begin{aligned} A(x) &= x \left(\frac{P}{2} - x \right) \\ A(x) &= \frac{P}{2}x - x^2 \end{aligned}$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned}\frac{dA}{dx} &= \frac{d}{dx} \left(\frac{P}{2}x - x^2 \right) \\ \frac{dA}{dx} &= \frac{P}{2} - 2x\end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned}\frac{P}{2} - 2x &= 0 \\ 2x &= \frac{P}{2} \\ x &= \frac{P}{4}\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dx^2} &= \frac{d}{dx} \left(\frac{P}{2} - 2x \right) \\ \frac{d^2A}{dx^2} &= -2\end{aligned}$$

Since $\frac{d^2A}{dx^2} = -2 < 0$, the area A is maximum when $x = \frac{P}{4}$.

Back-substituting $x = \frac{P}{4}$ to find y :

$$\begin{aligned}y &= \frac{P}{2} - \frac{P}{4} \\ y &= \frac{P}{4}\end{aligned}$$

Since $x = y = \frac{P}{4}$, the rectangle is a square.

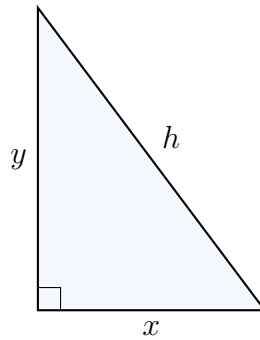
Hence Proved

LHS = RHS, hence proved that the rectangle with given perimeter has the largest area when it is a square.

Question 13

Prove that the perimeter of a right-angled triangle of given hypotenuse is maximum when the triangle is isosceles.

TYPE 104 Maximizing Perimeter of Right-Angled Triangle with Constant Hypotenuse



Right-angled triangle with hypotenuse h

To Prove: The perimeter of a right-angled triangle with given hypotenuse h is maximum when the other two sides are equal ($x = y$).

Concept / Formula Used: $x^2 + y^2 = h^2$ $P = x + y + h$

Solution:

Let the two legs of the right-angled triangle be x and y , and the hypotenuse be h (constant). By Pythagoras' theorem:

$$\begin{aligned} x^2 + y^2 &= h^2 \\ y &= \sqrt{h^2 - x^2} \end{aligned}$$

The perimeter P of the triangle is:

$$P = x + y + h$$

Substituting $y = \sqrt{h^2 - x^2}$:

$$P(x) = x + \sqrt{h^2 - x^2} + h$$

Differentiating $P(x)$ with respect to x :

$$\begin{aligned} \frac{dP}{dx} &= \frac{d}{dx} \left(x + \sqrt{h^2 - x^2} + h \right) \\ &= 1 + \frac{1}{2\sqrt{h^2 - x^2}} \cdot (-2x) \\ &= 1 - \frac{x}{\sqrt{h^2 - x^2}} \end{aligned}$$

For critical points, set $\frac{dP}{dx} = 0$:

$$\begin{aligned} 1 - \frac{x}{\sqrt{h^2 - x^2}} &= 0 \\ \sqrt{h^2 - x^2} &= x \end{aligned}$$

Squaring both sides:

$$\begin{aligned}h^2 - x^2 &= x^2 \\2x^2 &= h^2 \\x^2 &= \frac{h^2}{2} \\x &= \frac{h}{\sqrt{2}} \quad (\text{since } x > 0)\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2P}{dx^2} &= \frac{d}{dx} \left(1 - \frac{x}{\sqrt{h^2 - x^2}} \right) \\&= - \frac{\sqrt{h^2 - x^2} \cdot 1 - x \cdot \frac{-2x}{2\sqrt{h^2 - x^2}}}{h^2 - x^2} \\&= - \frac{\sqrt{h^2 - x^2} + \frac{x^2}{\sqrt{h^2 - x^2}}}{h^2 - x^2} \\&= - \frac{h^2 - x^2 + x^2}{(h^2 - x^2)^{3/2}} \\&= - \frac{h^2}{(h^2 - x^2)^{3/2}}\end{aligned}$$

At $x = \frac{h}{\sqrt{2}}$:

$$\begin{aligned}\left. \frac{d^2P}{dx^2} \right|_{x=\frac{h}{\sqrt{2}}} &= - \frac{h^2}{\left(h^2 - \frac{h^2}{2}\right)^{3/2}} \\&= - \frac{h^2}{\left(\frac{h^2}{2}\right)^{3/2}} \\&= - \frac{2\sqrt{2}}{h} < 0\end{aligned}$$

Since the second derivative is negative, the perimeter is maximum when $x = \frac{h}{\sqrt{2}}$.

Back-substituting $x = \frac{h}{\sqrt{2}}$ to find y :

$$\begin{aligned}y &= \sqrt{h^2 - \left(\frac{h}{\sqrt{2}}\right)^2} \\&= \sqrt{h^2 - \frac{h^2}{2}} \\&= \frac{h}{\sqrt{2}}\end{aligned}$$

Since $x = y = \frac{h}{\sqrt{2}}$, the triangle is isosceles.

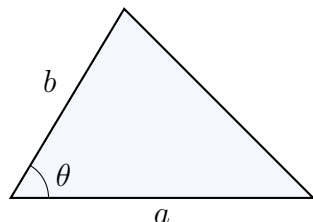
Hence Proved

LHS = RHS, hence proved that the perimeter is maximum when the triangle is isosceles.

Question 14

Two sides of a triangle have lengths a and b , and the angle between them is θ . What value of θ will maximize the area of the triangle? Also find the maximum area of the triangle.

TYPE 105 Maximizing Area of Triangle with Variable Angle



Triangle with sides a, b and angle θ

Given: Two sides of a triangle of lengths a and b with included angle θ .

Concept / Formula Used: $A = \frac{1}{2}ab \sin \theta$

Solution:

The area A of the triangle is given by:

$$A(\theta) = \frac{1}{2}ab \sin \theta$$

Differentiating $A(\theta)$ with respect to θ :

$$\begin{aligned}\frac{dA}{d\theta} &= \frac{d}{d\theta} \left(\frac{1}{2}ab \sin \theta \right) \\ \frac{dA}{d\theta} &= \frac{1}{2}ab \cos \theta\end{aligned}$$

For critical points, set $\frac{dA}{d\theta} = 0$:

$$\frac{1}{2}ab \cos \theta = 0$$

Since $a, b > 0$, we must have:

$$\begin{aligned}\cos \theta &= 0 \\ \theta &= \frac{\pi}{2} \quad (\text{since } 0 < \theta < \pi)\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{d\theta^2} &= \frac{d}{d\theta} \left(\frac{1}{2}ab \cos \theta \right) \\ \frac{d^2A}{d\theta^2} &= -\frac{1}{2}ab \sin \theta\end{aligned}$$

At $\theta = \frac{\pi}{2}$:

$$\begin{aligned}\left. \frac{d^2 A}{d\theta^2} \right|_{\theta=\frac{\pi}{2}} &= -\frac{1}{2}ab \sin\left(\frac{\pi}{2}\right) \\ &= -\frac{1}{2}ab < 0\end{aligned}$$

Since the second derivative is negative, the area is maximum when $\theta = \frac{\pi}{2}$ (or 90°).

The maximum area of the triangle is:

$$\begin{aligned}A_{\max} &= \frac{1}{2}ab \sin\left(\frac{\pi}{2}\right) \\ &= \frac{1}{2}ab\end{aligned}$$

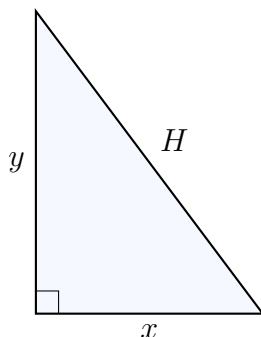
Final Answer

The area is maximum when $\theta = \frac{\pi}{2}$ and the maximum area is $\frac{1}{2}ab$.

Question 15

A right-angled triangle with constant area is given. Prove that the hypotenuse of the triangle is least when the triangle is isosceles.

TYPE 106 Minimizing Hypotenuse of Right-Angled Triangle with Constant Area



Right-angled triangle with legs x, y

To Prove: The hypotenuse of a right-angled triangle with constant area A is minimum when the legs are equal ($x = y$).

Concept / Formula Used: $A = \frac{1}{2}xy$ $H^2 = x^2 + y^2$

Solution:

Let the legs of the right-angled triangle be x and y . The area A is constant:

$$\begin{aligned}A &= \frac{1}{2}xy \\ y &= \frac{2A}{x}\end{aligned}$$

Let Z be the square of the hypotenuse:

$$Z = x^2 + y^2$$

Minimizing the hypotenuse is equivalent to minimizing its square Z . Substituting $y = \frac{2A}{x}$:

$$Z(x) = x^2 + \left(\frac{2A}{x}\right)^2$$

$$Z(x) = x^2 + \frac{4A^2}{x^2}$$

Differentiating $Z(x)$ with respect to x :

$$\frac{dZ}{dx} = \frac{d}{dx} \left(x^2 + \frac{4A^2}{x^2} \right)$$

$$\frac{dZ}{dx} = 2x - \frac{8A^2}{x^3}$$

For critical points, set $\frac{dZ}{dx} = 0$:

$$2x - \frac{8A^2}{x^3} = 0$$

$$2x = \frac{8A^2}{x^3}$$

$$x^4 = 4A^2$$

$$x^2 = 2A$$

$$x = \sqrt{2A} \quad (\text{since } x > 0)$$

Finding the second derivative:

$$\frac{d^2Z}{dx^2} = \frac{d}{dx} \left(2x - \frac{8A^2}{x^3} \right)$$

$$\frac{d^2Z}{dx^2} = 2 + \frac{24A^2}{x^4}$$

At $x = \sqrt{2A}$:

$$\left. \frac{d^2Z}{dx^2} \right|_{x=\sqrt{2A}} = 2 + \frac{24A^2}{(\sqrt{2A})^4}$$

$$= 2 + \frac{24A^2}{4A^2}$$

$$= 2 + 6$$

$$= 8 > 0$$

Since the second derivative is positive, the square of the hypotenuse Z (and thus the hypotenuse itself) is minimum when $x = \sqrt{2A}$.

Back-substituting $x = \sqrt{2A}$ to find y :

$$y = \frac{2A}{\sqrt{2A}}$$

$$y = \sqrt{2A}$$

Since $x = y = \sqrt{2A}$, the triangle is isosceles.

Hence Proved

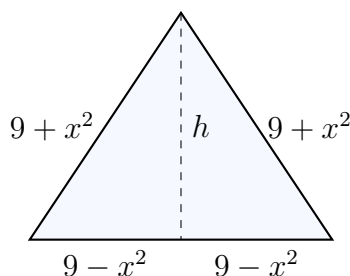
LHS = RHS, hence proved that the hypotenuse is least when the triangle is isosceles.

Question 16

The lengths of the sides of a triangle are $9 + x^2$, $9 + x^2$ and $18 - 2x^2$. Calculate:

- the area of the triangle in terms of x .
- the value of x for which this area is maximum.

TYPE 107 Maximizing Area of Isosceles Triangle with Variable Side Lengths



Isosceles triangle with sides $9 + x^2, 9 + x^2, 18 - 2x^2$

Given: An isosceles triangle with sides $a = 9 + x^2$, $b = 9 + x^2$, and base $c = 18 - 2x^2$.

Concept / Formula Used: $A = \frac{1}{2} \times \text{base} \times \text{height}$

Solution:

Let the base of the isosceles triangle be $c = 18 - 2x^2$. The altitude h drawn from the vertex to the base bisects the base. Therefore, half of the base is:

$$\begin{aligned} \frac{c}{2} &= \frac{18 - 2x^2}{2} \\ &= 9 - x^2 \end{aligned}$$

Using Pythagoras' theorem on one of the right-angled halves of the triangle:

$$\begin{aligned} h^2 + \left(\frac{c}{2}\right)^2 &= a^2 \\ h^2 + (9 - x^2)^2 &= (9 + x^2)^2 \\ h^2 &= (9 + x^2)^2 - (9 - x^2)^2 \end{aligned}$$

Using the algebraic identity $(A + B)^2 - (A - B)^2 = 4AB$:

$$\begin{aligned} h^2 &= 4(9)(x^2) \\ h^2 &= 36x^2 \\ h &= 6x \quad (\text{since } x > 0) \end{aligned}$$

(i) Area of the triangle in terms of x :

$$\begin{aligned} A &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2}(18 - 2x^2)(6x) \\ &= (9 - x^2)(6x) \\ &= 54x - 6x^3 \end{aligned}$$

(ii) Value of x for which this area is maximum:

Differentiating $A(x)$ with respect to x :

$$\begin{aligned} \frac{dA}{dx} &= \frac{d}{dx}(54x - 6x^3) \\ \frac{dA}{dx} &= 54 - 18x^2 \end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned} 54 - 18x^2 &= 0 \\ 18x^2 &= 54 \\ x^2 &= 3 \\ x &= \sqrt{3} \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2A}{dx^2} &= \frac{d}{dx}(54 - 18x^2) \\ \frac{d^2A}{dx^2} &= -36x \end{aligned}$$

At $x = \sqrt{3}$:

$$\left. \frac{d^2A}{dx^2} \right|_{x=\sqrt{3}} = -36\sqrt{3} < 0$$

Since the second derivative is negative, the area is maximum when $x = \sqrt{3}$.

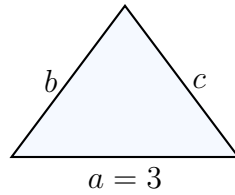
Final Answer

- (i) Area $A = 54x - 6x^3$.
(ii) $x = \sqrt{3}$.

Question 17

The perimeter of a triangle is 8 cm. If one of the sides is 3 cm, what are the lengths of the other sides for maximum area of the triangle?

TYPE 108 Maximizing Area of Triangle with Constant Perimeter



Triangle with sides a, b, c

Given: Perimeter of the triangle $P = 8$ cm and one side $a = 3$ cm.

Concept / Formula Used: $A = \sqrt{s(s-a)(s-b)(s-c)}$ $s = \frac{P}{2}$

Solution:

Let the sides of the triangle be a, b , and c . We are given:

$$\begin{aligned} a &= 3 \text{ cm} \\ a + b + c &= 8 \\ 3 + b + c &= 8 \\ b + c &= 5 \\ c &= 5 - b \end{aligned}$$

The semi-perimeter s is:

$$s = \frac{a + b + c}{2} = \frac{8}{2} = 4 \text{ cm}$$

Using Heron's formula, the area A of the triangle is:

$$\begin{aligned} A &= \sqrt{s(s-a)(s-b)(s-c)} \\ A &= \sqrt{4(4-3)(4-b)(4-c)} \\ A &= \sqrt{4(1)(4-b)(4-(5-b))} \\ A &= \sqrt{4(4-b)(b-1)} \end{aligned}$$

To maximize A , we can maximize its square, $Z = A^2$:

$$\begin{aligned} Z(b) &= 4(4-b)(b-1) \\ Z(b) &= 4(4b - 4 - b^2 + b) \\ Z(b) &= 4(-b^2 + 5b - 4) \end{aligned}$$

Differentiating $Z(b)$ with respect to b :

$$\begin{aligned} \frac{dZ}{db} &= \frac{d}{db} [4(-b^2 + 5b - 4)] \\ \frac{dZ}{db} &= 4(-2b + 5) \end{aligned}$$

For critical points, set $\frac{dZ}{db} = 0$:

$$\begin{aligned} 4(-2b + 5) &= 0 \\ -2b + 5 &= 0 \\ 2b &= 5 \\ b &= 2.5 \text{ cm} \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2Z}{db^2} &= \frac{d}{db} [4(-2b + 5)] \\ \frac{d^2Z}{db^2} &= -8 \end{aligned}$$

Since $\frac{d^2Z}{db^2} = -8 < 0$, the area is maximum when $b = 2.5$ cm.

Back-substituting $b = 2.5$ cm to find c :

$$\begin{aligned} c &= 5 - 2.5 \\ c &= 2.5 \text{ cm} \end{aligned}$$

Thus, the lengths of the other two sides are 2.5 cm and 2.5 cm.

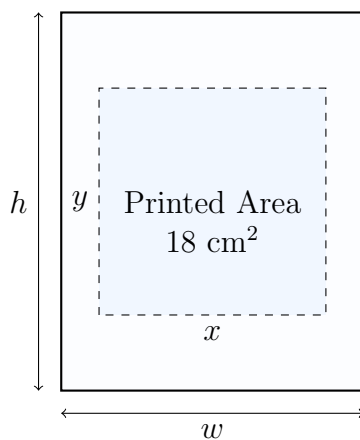
Final Answer

The other sides are 2.5 cm and 2.5 cm.

Question 18

A sheet of paper is to contain 18 cm^2 of printed matter. The margins at the top and bottom are 2 cm each, and at the sides 1 cm each. Find the dimensions of the sheet which require the least amount of paper.

TYPE 109 Minimizing Area of Rectangular Sheet with Margins



Sheet of paper with margins

Given: Printed area $A_p = 18 \text{ cm}^2$. Margins: top/bottom = 2 cm, sides = 1 cm.

Concept / Formula Used: $A = wh$ $w = x + 2$ $h = y + 4$

Solution:

Let the dimensions of the printed area be x (width) and y (height). We are given:

$$xy = 18$$

$$y = \frac{18}{x}$$

The total width of the sheet of paper is:

$$w = x + 2(1) = x + 2$$

The total height of the sheet of paper is:

$$h = y + 2(2) = y + 4$$

The total area A of the sheet of paper is:

$$A = wh$$

$$A = (x + 2)(y + 4)$$

Substituting $y = \frac{18}{x}$:

$$\begin{aligned} A(x) &= (x + 2) \left(\frac{18}{x} + 4 \right) \\ &= 18 + 4x + \frac{36}{x} + 8 \\ &= 26 + 4x + \frac{36}{x} \end{aligned}$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned} \frac{dA}{dx} &= \frac{d}{dx} \left(26 + 4x + \frac{36}{x} \right) \\ \frac{dA}{dx} &= 4 - \frac{36}{x^2} \end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned} 4 - \frac{36}{x^2} &= 0 \\ 4 &= \frac{36}{x^2} \\ x^2 &= 9 \\ x &= 3 \text{ cm} \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2A}{dx^2} &= \frac{d}{dx} \left(4 - \frac{36}{x^2} \right) \\ \frac{d^2A}{dx^2} &= \frac{72}{x^3} \end{aligned}$$

At $x = 3$:

$$\left. \frac{d^2 A}{dx^2} \right|_{x=3} = \frac{72}{3^3} = \frac{72}{27} = \frac{8}{3} > 0$$

Since the second derivative is positive, the total area is minimum when $x = 3$ cm.

Now, find the dimensions of the sheet:

$$\begin{aligned} w &= x + 2 = 3 + 2 = 5 \text{ cm} \\ y &= \frac{18}{3} = 6 \text{ cm} \\ h &= y + 4 = 6 + 4 = 10 \text{ cm} \end{aligned}$$

Thus, the dimensions of the sheet of paper are 5 cm by 10 cm.

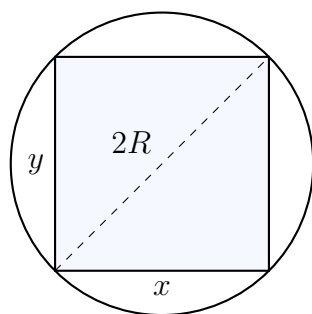
Final Answer

The dimensions of the sheet are 5 cm $\times$ 10 cm.

Question 19

Show that of all rectangles inscribed in a given fixed circle the square has the maximum area.

TYPE 110 Maximizing Area of Rectangle Inscribed in a Circle



Rectangle inscribed in a circle

To Prove: The area of an inscribed rectangle in a circle of fixed radius R is maximum when the rectangle is a square ($x = y$).

Concept / Formula Used: $x^2 + y^2 = (2R)^2 = 4R^2$ $A = xy$

Solution:

Let the rectangle inscribed in a circle of fixed radius R have length x and width y . The diagonal of the rectangle is the diameter of the circle, which is $2R$. By Pythagoras' theorem:

$$\begin{aligned} x^2 + y^2 &= (2R)^2 \\ x^2 + y^2 &= 4R^2 \\ y &= \sqrt{4R^2 - x^2} \end{aligned}$$

The area A of the rectangle is:

$$A = xy$$

To maximize A , we can maximize its square, $Z = A^2$:

$$Z(x) = x^2y^2$$

$$Z(x) = x^2(4R^2 - x^2)$$

$$Z(x) = 4R^2x^2 - x^4$$

Differentiating $Z(x)$ with respect to x :

$$\frac{dZ}{dx} = \frac{d}{dx}(4R^2x^2 - x^4)$$

$$\frac{dZ}{dx} = 8R^2x - 4x^3$$

$$\frac{dZ}{dx} = 4x(2R^2 - x^2)$$

For critical points, set $\frac{dZ}{dx} = 0$:

$$4x(2R^2 - x^2) = 0$$

Since $x > 0$, we must have:

$$2R^2 - x^2 = 0$$

$$x^2 = 2R^2$$

$$x = \sqrt{2}R$$

Finding the second derivative:

$$\frac{d^2Z}{dx^2} = \frac{d}{dx}(8R^2x - 4x^3)$$

$$\frac{d^2Z}{dx^2} = 8R^2 - 12x^2$$

At $x = \sqrt{2}R$:

$$\begin{aligned} \left. \frac{d^2Z}{dx^2} \right|_{x=\sqrt{2}R} &= 8R^2 - 12(2R^2) \\ &= 8R^2 - 24R^2 \\ &= -16R^2 < 0 \end{aligned}$$

Since the second derivative is negative, the area is maximum when $x = \sqrt{2}R$.

Back-substituting $x = \sqrt{2}R$ to find y :

$$\begin{aligned} y &= \sqrt{4R^2 - (\sqrt{2}R)^2} \\ &= \sqrt{4R^2 - 2R^2} \\ &= \sqrt{2R^2} \\ &= \sqrt{2}R \end{aligned}$$

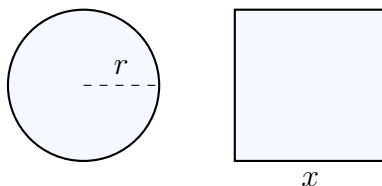
Since $x = y = \sqrt{2}R$, the rectangle is a square.

Hence Proved

LHS = RHS, hence proved that the inscribed rectangle of maximum area is a square.

Question 20

- (i) The sum of perimeters of a circle and a square is k , where k is some constant. Prove that the sum of their areas is least when the side of the square is double the radius of the circle.
- (ii) Given the sum of perimeters of a circle and a square, show that the sum of areas is least when the diameter of the circle is equal to the side of the square.

TYPE 111 Minimizing Combined Area of Circle and Square*Circle and Square***To Prove:**

- (i) Sum of areas is least when $x = 2r$.
- (ii) Sum of areas is least when $2r = x$.

Concept / Formula Used: $P_c = 2\pi r$ $P_s = 4x$ $A_c = \pi r^2$ $A_s = x^2$

Solution:

Let the radius of the circle be r and the side of the square be x . We are given that the sum of their perimeters is a constant k :

$$\begin{aligned} 2\pi r + 4x &= k \\ 4x &= k - 2\pi r \\ x &= \frac{k - 2\pi r}{4} \end{aligned}$$

The sum of their areas A is:

$$A = \pi r^2 + x^2$$

Substituting $x = \frac{k - 2\pi r}{4}$:

$$\begin{aligned} A(r) &= \pi r^2 + \left(\frac{k - 2\pi r}{4} \right)^2 \\ A(r) &= \pi r^2 + \frac{(k - 2\pi r)^2}{16} \end{aligned}$$

Differentiating $A(r)$ with respect to r :

$$\begin{aligned}\frac{dA}{dr} &= \frac{d}{dr} \left[\pi r^2 + \frac{(k - 2\pi r)^2}{16} \right] \\ &= 2\pi r + \frac{2(k - 2\pi r) \cdot (-2\pi)}{16} \\ &= 2\pi r - \frac{\pi(k - 2\pi r)}{4}\end{aligned}$$

For critical points, set $\frac{dA}{dr} = 0$:

$$2\pi r - \frac{\pi(k - 2\pi r)}{4} = 0$$

Dividing both sides by π :

$$\begin{aligned}2r - \frac{k - 2\pi r}{4} &= 0 \\ 8r - (k - 2\pi r) &= 0 \\ 8r + 2\pi r &= k \\ 2r(4 + \pi) &= k \\ r &= \frac{k}{2(4 + \pi)}\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dr^2} &= \frac{d}{dr} \left[2\pi r - \frac{\pi(k - 2\pi r)}{4} \right] \\ &= 2\pi - \frac{\pi(-2\pi)}{4} \\ &= 2\pi + \frac{\pi^2}{2}\end{aligned}$$

Since $\frac{d^2A}{dr^2} = 2\pi + \frac{\pi^2}{2} > 0$, the sum of areas A is minimum (least) at this value of r .

Now, let us find the relation between x and r at this minimum. We know:

$$4x = k - 2\pi r$$

Substituting $k = 2r(4 + \pi)$:

$$\begin{aligned}4x &= 2r(4 + \pi) - 2\pi r \\ 4x &= 8r + 2\pi r - 2\pi r \\ 4x &= 8r \\ x &= 2r\end{aligned}$$

(i) This proves that the side of the square (x) is double the radius of the circle ($2r$).

(ii) Since the diameter of the circle is $d = 2r$, the relation $x = 2r$ implies:

$$x = d$$

This proves that the sum of areas is least when the diameter of the circle is equal to the side of the square.

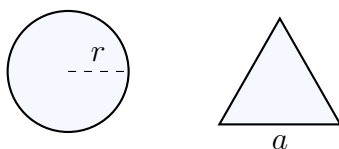
Hence Proved

$LHS = RHS$, hence proved for both parts (i) and (ii).

Question 21

A wire 10 metres long is cut into two parts. One part is bent into the shape of a circle and the other into the shape of an equilateral triangle. How should the wire be cut so that the combined area of the two figures is as small as possible?

TYPE 112 Minimizing Combined Area of Circle and Triangle



Circle and Equilateral Triangle

Given: Total length of wire $L = 10$ m.

Concept / Formula Used: $A_c = \pi r^2$ $A_t = \frac{\sqrt{3}}{4} a^2$

Solution:

Let the wire of length 10 m be cut into two pieces of length x and $10 - x$. Let the piece of length x be bent into a circle of radius r :

$$\begin{aligned} 2\pi r &= x \\ r &= \frac{x}{2\pi} \end{aligned}$$

Let the other piece of length $10 - x$ be bent into an equilateral triangle of side a :

$$\begin{aligned} 3a &= 10 - x \\ a &= \frac{10 - x}{3} \end{aligned}$$

The combined area A of the circle and the equilateral triangle is:

$$A = \pi r^2 + \frac{\sqrt{3}}{4} a^2$$

Substituting $r = \frac{x}{2\pi}$ and $a = \frac{10-x}{3}$:

$$\begin{aligned} A(x) &= \pi \left(\frac{x}{2\pi} \right)^2 + \frac{\sqrt{3}}{4} \left(\frac{10-x}{3} \right)^2 \\ A(x) &= \frac{x^2}{4\pi} + \frac{\sqrt{3}}{36} (10-x)^2 \end{aligned}$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned}\frac{dA}{dx} &= \frac{d}{dx} \left[\frac{x^2}{4\pi} + \frac{\sqrt{3}}{36}(10-x)^2 \right] \\ &= \frac{2x}{4\pi} + \frac{\sqrt{3}}{36} \cdot 2(10-x) \cdot (-1) \\ &= \frac{x}{2\pi} - \frac{\sqrt{3}}{18}(10-x)\end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned}\frac{x}{2\pi} - \frac{\sqrt{3}}{18}(10-x) &= 0 \\ \frac{x}{2\pi} &= \frac{\sqrt{3}}{18}(10-x) \\ 18x &= 2\pi\sqrt{3}(10-x) \\ 9x &= \pi\sqrt{3}(10-x) \\ 9x &= 10\pi\sqrt{3} - \pi\sqrt{3}x \\ x(9 + \pi\sqrt{3}) &= 10\pi\sqrt{3} \\ x &= \frac{10\pi\sqrt{3}}{9 + \pi\sqrt{3}} \text{ m}\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dx^2} &= \frac{d}{dx} \left[\frac{x}{2\pi} - \frac{\sqrt{3}}{18}(10-x) \right] \\ &= \frac{1}{2\pi} + \frac{\sqrt{3}}{18}\end{aligned}$$

Since $\frac{d^2A}{dx^2} = \frac{1}{2\pi} + \frac{\sqrt{3}}{18} > 0$, the combined area is minimum at this value of x .

The length of the other piece is:

$$\begin{aligned}10 - x &= 10 - \frac{10\pi\sqrt{3}}{9 + \pi\sqrt{3}} \\ &= \frac{10(9 + \pi\sqrt{3}) - 10\pi\sqrt{3}}{9 + \pi\sqrt{3}} \\ &= \frac{90 + 10\pi\sqrt{3} - 10\pi\sqrt{3}}{9 + \pi\sqrt{3}} \\ &= \frac{90}{9 + \pi\sqrt{3}} \text{ m}\end{aligned}$$

Thus, the wire should be cut into two pieces of lengths $\frac{10\pi\sqrt{3}}{9 + \pi\sqrt{3}}$ m and $\frac{90}{9 + \pi\sqrt{3}}$ m.

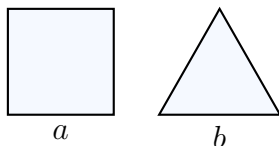
Final Answer

The lengths of the two pieces are $\frac{10\pi\sqrt{3}}{9 + \pi\sqrt{3}}$ m and $\frac{90}{9 + \pi\sqrt{3}}$ m.

Question 22

A wire of length 36 cm is cut into two pieces. One of the piece is turned into the form of a square and the other in the form of an equilateral triangle. Find the length of each piece so that the sum of the areas of the two figures be minimum.

TYPE 113 Minimizing Combined Area of Square and Triangle



Square and Equilateral Triangle

Given: Total length of wire $L = 36$ cm.

Concept / Formula Used: $A_s = a^2$ $A_t = \frac{\sqrt{3}}{4}b^2$

Solution:

Let the wire of length 36 cm be cut into two pieces of length x and $36 - x$. Let the piece of length x be bent into a square of side a :

$$\begin{aligned} 4a &= x \\ a &= \frac{x}{4} \end{aligned}$$

Let the other piece of length $36 - x$ be bent into an equilateral triangle of side b :

$$\begin{aligned} 3b &= 36 - x \\ b &= \frac{36 - x}{3} \end{aligned}$$

The combined area A of the square and the equilateral triangle is:

$$A = a^2 + \frac{\sqrt{3}}{4}b^2$$

Substituting $a = \frac{x}{4}$ and $b = \frac{36-x}{3}$:

$$\begin{aligned} A(x) &= \left(\frac{x}{4}\right)^2 + \frac{\sqrt{3}}{4} \left(\frac{36-x}{3}\right)^2 \\ A(x) &= \frac{x^2}{16} + \frac{\sqrt{3}}{36}(36-x)^2 \end{aligned}$$

Differentiating $A(x)$ with respect to x :

$$\begin{aligned}\frac{dA}{dx} &= \frac{d}{dx} \left[\frac{x^2}{16} + \frac{\sqrt{3}}{36}(36-x)^2 \right] \\ &= \frac{2x}{16} + \frac{\sqrt{3}}{36} \cdot 2(36-x) \cdot (-1) \\ &= \frac{x}{8} - \frac{\sqrt{3}}{18}(36-x)\end{aligned}$$

For critical points, set $\frac{dA}{dx} = 0$:

$$\begin{aligned}\frac{x}{8} - \frac{\sqrt{3}}{18}(36-x) &= 0 \\ \frac{x}{8} &= \frac{\sqrt{3}}{18}(36-x) \\ 18x &= 8\sqrt{3}(36-x) \\ 9x &= 4\sqrt{3}(36-x) \\ 9x &= 144\sqrt{3} - 4\sqrt{3}x \\ x(9+4\sqrt{3}) &= 144\sqrt{3} \\ x &= \frac{144\sqrt{3}}{9+4\sqrt{3}} \text{ cm}\end{aligned}$$

Finding the second derivative:

$$\begin{aligned}\frac{d^2A}{dx^2} &= \frac{d}{dx} \left[\frac{x}{8} - \frac{\sqrt{3}}{18}(36-x) \right] \\ &= \frac{1}{8} + \frac{\sqrt{3}}{18}\end{aligned}$$

Since $\frac{d^2A}{dx^2} = \frac{1}{8} + \frac{\sqrt{3}}{18} > 0$, the combined area is minimum at this value of x .

The length of the other piece is:

$$\begin{aligned}36-x &= 36 - \frac{144\sqrt{3}}{9+4\sqrt{3}} \\ &= \frac{36(9+4\sqrt{3}) - 144\sqrt{3}}{9+4\sqrt{3}} \\ &= \frac{324 + 144\sqrt{3} - 144\sqrt{3}}{9+4\sqrt{3}} \\ &= \frac{324}{9+4\sqrt{3}} \text{ cm}\end{aligned}$$

Thus, the lengths of the two pieces are $\frac{144\sqrt{3}}{9+4\sqrt{3}}$ cm and $\frac{324}{9+4\sqrt{3}}$ cm.

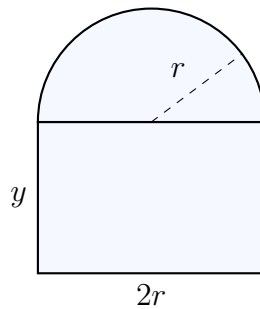
Final Answer

The lengths of the two pieces are $\frac{144\sqrt{3}}{9+4\sqrt{3}}$ cm and $\frac{324}{9+4\sqrt{3}}$ cm.

Question 23

A window is in the form of a rectangle above which there is a semicircle. If the perimeter of the window is p cm, show that the window will admit maximum possible light only when the radius of semicircle is $\frac{p}{\pi + 4}$ cm.

TYPE 114 Maximizing Area of Norman Window with Constant Perimeter



Window with rectangular base and semicircular top

To Prove: The area of the window is maximum when the radius of the semicircle is $r = \frac{p}{\pi + 4}$ cm.

Concept / Formula Used: $P = 2r + 2y + \pi r = p$ $A = 2ry + \frac{1}{2}\pi r^2$

Solution:

Let the radius of the semicircular top be r . Then the width of the rectangular base is $2r$. Let the height of the rectangular part be y . The perimeter P of the window consists of the bottom side ($2r$), two vertical sides ($2y$), and the semicircular arc (πr):

$$P = 2r + 2y + \pi r$$

$$p = (\pi + 2)r + 2y$$

$$2y = p - (\pi + 2)r$$

$$y = \frac{p - (\pi + 2)r}{2}$$

The area A of the window (which determines the amount of light admitted) is:

$$A = \text{Area of rectangle} + \text{Area of semicircle}$$

$$A = (2r)y + \frac{1}{2}\pi r^2$$

Substituting $y = \frac{p-(\pi+2)r}{2}$:

$$\begin{aligned} A(r) &= 2r \left[\frac{p-(\pi+2)r}{2} \right] + \frac{1}{2}\pi r^2 \\ &= r[p-(\pi+2)r] + \frac{1}{2}\pi r^2 \\ &= pr - (\pi+2)r^2 + \frac{1}{2}\pi r^2 \\ &= pr - \pi r^2 - 2r^2 + \frac{1}{2}\pi r^2 \\ &= pr - \left(\frac{\pi}{2} + 2 \right) r^2 \end{aligned}$$

Differentiating $A(r)$ with respect to r :

$$\begin{aligned} \frac{dA}{dr} &= \frac{d}{dr} \left[pr - \left(\frac{\pi}{2} + 2 \right) r^2 \right] \\ &= p - 2 \left(\frac{\pi}{2} + 2 \right) r \\ &= p - (\pi + 4)r \end{aligned}$$

For critical points, set $\frac{dA}{dr} = 0$:

$$\begin{aligned} p - (\pi + 4)r &= 0 \\ (\pi + 4)r &= p \\ r &= \frac{p}{\pi + 4} \text{ cm} \end{aligned}$$

Finding the second derivative:

$$\begin{aligned} \frac{d^2A}{dr^2} &= \frac{d}{dr} [p - (\pi + 4)r] \\ &= -(\pi + 4) \end{aligned}$$

Since $\frac{d^2A}{dr^2} = -(\pi + 4) < 0$, the area is maximum when $r = \frac{p}{\pi+4}$ cm.

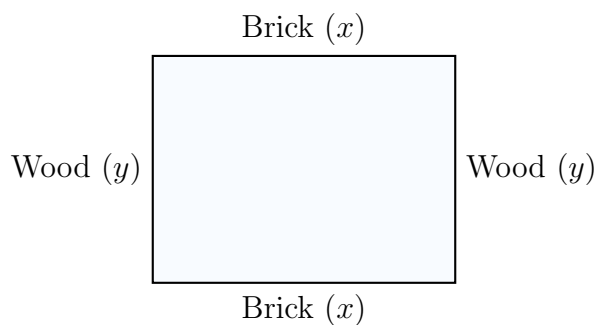
Hence Proved

LHS = RHS, hence proved that the window admits maximum light when the radius of the semicircle is $\frac{p}{\pi + 4}$ cm.

Question 24

A rectangular area of 9000 m² is to be surrounded by a fence with two opposite sides being made of bricks and the other two of wood. One metre of wooden fencing costs Rs. 250 while one metre of brick walling costs Rs. 100. What is the least amount of money that must be allotted for the construction of such a fence?

TYPE 116 Minimizing Cost of Fencing Rectangular Area



Fenced rectangular area with brick and wooden sides

Given: Area $A = 9000 \text{ m}^2$, cost of brick walling is Rs. 100/m, cost of wooden fencing is Rs. 250/m.

To Find: The least amount of money required for the construction of the fence.

Concept / Formula Used: $f(x)$ $f'(x) = 0$ $f''(x) > 0$

Solution:

Let the length of the brick sides be x metres and the length of the wooden sides be y metres.

The area of the rectangular region is given as:

$$xy = 9000$$

Solving for y in terms of x :

$$y = \frac{9000}{x}$$

The cost of the two opposite brick sides of length x is:

$$\text{Cost}_{\text{brick}} = 2 \times 100 \times x = 200x$$

The cost of the two opposite wooden sides of length y is:

$$\text{Cost}_{\text{wood}} = 2 \times 250 \times y = 500y$$

The total cost C is the sum of these two costs:

$$C = 200x + 500y$$

Substitute $y = \frac{9000}{x}$ into the cost equation:

$$\begin{aligned} C(x) &= 200x + 500 \left(\frac{9000}{x} \right) \\ &= 200x + \frac{4500000}{x} \end{aligned}$$

Differentiating $C(x)$ with respect to x :

$$\frac{dC}{dx} = 200 - \frac{4500000}{x^2}$$

For the minimum cost, set $\frac{dC}{dx} = 0$:

$$\begin{aligned}200 - \frac{4500000}{x^2} &= 0 \\200 &= \frac{4500000}{x^2} \\x^2 &= \frac{4500000}{200} \\x^2 &= 22500 \\x &= 150\end{aligned}$$

To verify if this is a minimum, find the second derivative:

$$\frac{d^2C}{dx^2} = \frac{9000000}{x^3}$$

At $x = 150$:

$$\begin{aligned}\frac{d^2C}{dx^2} &= \frac{9000000}{150^3} \\&= \frac{9000000}{3375000} \\&= \frac{8}{3} > 0\end{aligned}$$

Since the second derivative is positive, the cost is minimum at $x = 150$.

Now, calculate the minimum cost:

$$\begin{aligned}C(150) &= 200(150) + \frac{4500000}{150} \\&= 30000 + 30000 \\&= 60000\end{aligned}$$

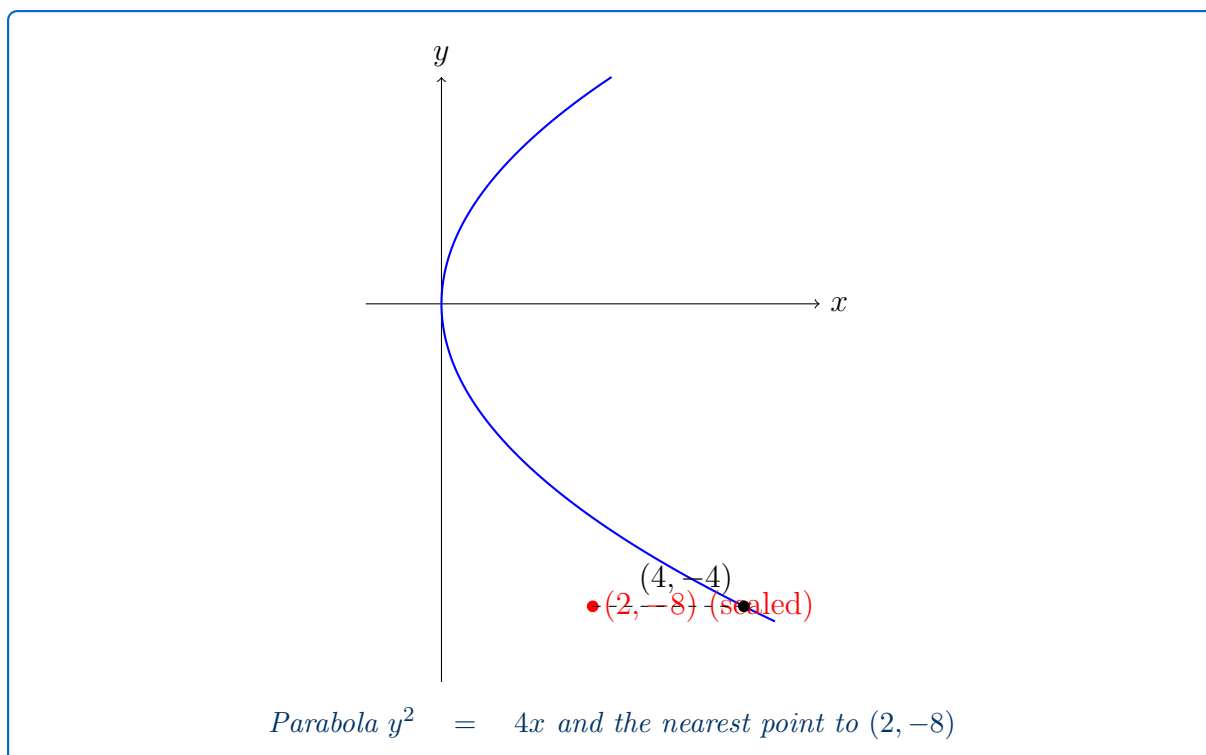
Final Answer

Rs. 60000

Question 25

- (i) Find the point on the parabola $y^2 = 4x$ which is nearest to the point $(2, -8)$.
- (ii) Find the point on the curve $x^2 = 8y$ which is nearest to the point $(2, 4)$.

TYPE 117 Finding Point on Parabola Nearest to a Given Point



Given: The curves (i) $y^2 = 4x$ and (ii) $x^2 = 8y$, and the respective points (i) $(2, -8)$ and (ii) $(2, 4)$.

To Find: The coordinates of the nearest point on each curve.

Concept	/	Formula	Used:	D	(x_1, y_1)	(x_2, y_2)	D	=
		$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	D	$Z = D^2$				

Solution:

(i) Let $Q(x, y)$ be any point on the parabola $y^2 = 4x$.

Expressing x in terms of y :

$$x = \frac{y^2}{4}$$

The distance D between $Q(x, y)$ and the point $P(2, -8)$ is:

$$D = \sqrt{(x - 2)^2 + (y + 8)^2}$$

Let $Z = D^2$:

$$Z = (x - 2)^2 + (y + 8)^2$$

Substitute $x = \frac{y^2}{4}$ into the equation for Z :

$$\begin{aligned} Z(y) &= \left(\frac{y^2}{4} - 2\right)^2 + (y + 8)^2 \\ &= \frac{y^4}{16} - y^2 + 4 + y^2 + 16y + 64 \\ &= \frac{y^4}{16} + 16y + 68 \end{aligned}$$

Differentiating $Z(y)$ with respect to y :

$$\frac{dZ}{dy} = \frac{4y^3}{16} + 16 = \frac{y^3}{4} + 16$$

For the minimum distance, set $\frac{dZ}{dy} = 0$:

$$\begin{aligned} \frac{y^3}{4} + 16 &= 0 \\ \frac{y^3}{4} &= -16 \\ y^3 &= -64 \\ y &= -4 \end{aligned}$$

Finding the second derivative:

$$\frac{d^2Z}{dy^2} = \frac{3y^2}{4}$$

At $y = -4$:

$$\frac{d^2Z}{dy^2} = \frac{3(-4)^2}{4} = 12 > 0$$

Since the second derivative is positive, the distance is minimum at $y = -4$.

Now, find the corresponding x -coordinate:

$$x = \frac{(-4)^2}{4} = 4$$

Thus, the nearest point is $(4, -4)$.

(ii) Let $Q(x, y)$ be any point on the curve $x^2 = 8y$.

Expressing y in terms of x :

$$y = \frac{x^2}{8}$$

The distance D between $Q(x, y)$ and the point $P(2, 4)$ is:

$$D = \sqrt{(x - 2)^2 + (y - 4)^2}$$

Let $Z = D^2$:

$$Z = (x - 2)^2 + (y - 4)^2$$

Substitute $y = \frac{x^2}{8}$ into the equation for Z :

$$\begin{aligned} Z(x) &= (x - 2)^2 + \left(\frac{x^2}{8} - 4\right)^2 \\ &= x^2 - 4x + 4 + \frac{x^4}{64} - x^2 + 16 \\ &= \frac{x^4}{64} - 4x + 20 \end{aligned}$$

Differentiating $Z(x)$ with respect to x :

$$\frac{dZ}{dx} = \frac{4x^3}{64} - 4 = \frac{x^3}{16} - 4$$

For the minimum distance, set $\frac{dZ}{dx} = 0$:

$$\begin{aligned} \frac{x^3}{16} - 4 &= 0 \\ \frac{x^3}{16} &= 4 \\ x^3 &= 64 \\ x &= 4 \end{aligned}$$

Finding the second derivative:

$$\frac{d^2Z}{dx^2} = \frac{3x^2}{16}$$

At $x = 4$:

$$\frac{d^2Z}{dx^2} = \frac{3(4)^2}{16} = 3 > 0$$

Since the second derivative is positive, the distance is minimum at $x = 4$.

Now, find the corresponding y -coordinate:

$$y = \frac{4^2}{8} = 2$$

Thus, the nearest point is $(4, 2)$.

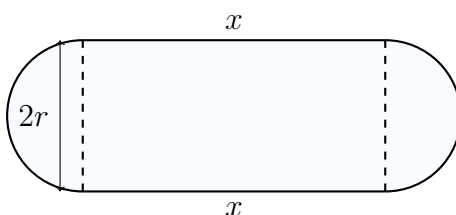
Final Answer

- (i) $(4, -4)$
- (ii) $(4, 2)$

Question 26

A running track of 440 m is to be laid out enclosing a football field. The football field is in the shape of a rectangle with semicircle at each end. If the area of the rectangular portion is to be maximum, then find the lengths of its sides. Also calculate the area of the football field.

TYPE 118 Maximizing Rectangular Area of Stadium with Semicircular Ends



Football field with rectangular portion and semicircular ends

Given: Total perimeter of the track $P = 440$ m.

To Find: The lengths of the sides of the rectangular portion for maximum area, and the total area of the football field.

Concept / Formula Used: $P = 2x + 2\pi r$ $A = 2xr$ $A_{\text{total}} = 2xr + \pi r^2$

Solution:

Let the length of the rectangular portion be x and the width be $2r$, where r is the radius of the semicircular ends.

The total perimeter of the track is given by:

$$P = 2x + 2\pi r = 440$$

Dividing by 2:

$$x + \pi r = 220$$

Solving for x :

$$x = 220 - \pi r$$

The area A of the rectangular portion is:

$$A = x(2r) = 2r(220 - \pi r) = 440r - 2\pi r^2$$

Differentiating A with respect to r :

$$\frac{dA}{dr} = 440 - 4\pi r$$

For maximum area, set $\frac{dA}{dr} = 0$:

$$\begin{aligned} 440 - 4\pi r &= 0 \\ 4\pi r &= 440 \\ r &= \frac{110}{\pi} \end{aligned}$$

Finding the second derivative:

$$\frac{d^2A}{dr^2} = -4\pi$$

Since $\frac{d^2A}{dr^2} < 0$, the area of the rectangular portion is maximum when $r = \frac{110}{\pi}$.

Now, find the dimensions of the rectangular portion: - Width ($2r$):

$$2r = 2 \left(\frac{110}{\pi} \right) = \frac{220}{\pi} \text{ m}$$

Using $\pi = \frac{22}{7}$:

$$2r = \frac{220}{\frac{22}{7}} = 70 \text{ m}$$

- Length (x):

$$x = 220 - \pi \left(\frac{110}{\pi} \right) = 110 \text{ m}$$

Now, calculate the total area of the football field:

$$\begin{aligned} A_{\text{total}} &= \text{Area of rectangle} + \text{Area of two semicircles} \\ &= 2xr + \pi r^2 \\ &= 2(110) \left(\frac{110}{\pi} \right) + \pi \left(\frac{110}{\pi} \right)^2 \\ &= \frac{24200}{\pi} + \frac{12100}{\pi} \\ &= \frac{36300}{\pi} \end{aligned}$$

Using $\pi = \frac{22}{7}$:

$$\begin{aligned} A_{\text{total}} &= 36300 \times \frac{7}{22} \\ &= 1650 \times 7 \\ &= 11550 \text{ m}^2 \end{aligned}$$

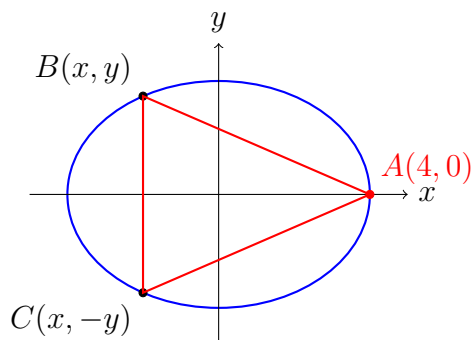
Final Answer

Lengths of sides: 110 m and 70 m; Total area: 11550 m²

Question 27

Find the maximum area of an isosceles triangle inscribed in the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with its vertex at one end of the major axis.

TYPE 119 Maximizing Area of Isosceles Triangle Inscribed in an Ellipse



Isosceles triangle inscribed in an ellipse

Given: Equation of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, vertex of the triangle at $(4, 0)$.

To Find: The maximum area of the inscribed isosceles triangle.

Concept / Formula Used: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a \cos \theta, b \sin \theta)$ $A = \frac{1}{2} \times \text{base} \times \text{height}$

Solution:

Let the equation of the ellipse be $\frac{x^2}{16} + \frac{y^2}{9} = 1$. Here, $a = 4$ and $b = 3$.

Let one vertex of the isosceles triangle be at $A(4, 0)$, which is one end of the major axis. Let the other two vertices be $B(4 \cos \theta, 3 \sin \theta)$ and $C(4 \cos \theta, -3 \sin \theta)$, where $\theta \in (0, \pi)$.

The base of the triangle is the length of BC :

$$BC = 3 \sin \theta - (-3 \sin \theta) = 6 \sin \theta$$

The height of the triangle is the distance from $A(4, 0)$ to the line $x = 4 \cos \theta$:

$$h = 4 - 4 \cos \theta = 4(1 - \cos \theta)$$

The area A of the triangle is:

$$\begin{aligned} A &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} (6 \sin \theta) [4(1 - \cos \theta)] \\ &= 12 \sin \theta (1 - \cos \theta) \end{aligned}$$

Differentiating A with respect to θ :

$$\begin{aligned}\frac{dA}{d\theta} &= 12 [\cos \theta (1 - \cos \theta) + \sin \theta (\sin \theta)] \\ &= 12 [\cos \theta - \cos^2 \theta + \sin^2 \theta] \\ &= 12 [\cos \theta - \cos^2 \theta + (1 - \cos^2 \theta)] \\ &= 12 [1 + \cos \theta - 2 \cos^2 \theta]\end{aligned}$$

Set $\frac{dA}{d\theta} = 0$ for critical points:

$$\begin{aligned}1 + \cos \theta - 2 \cos^2 \theta &= 0 \\ 2 \cos^2 \theta - \cos \theta - 1 &= 0 \\ (2 \cos \theta + 1)(\cos \theta - 1) &= 0\end{aligned}$$

This gives:

$$\cos \theta = 1 \quad \text{or} \quad \cos \theta = -\frac{1}{2}$$

Since $\theta \in (0, \pi)$, $\cos \theta = 1$ is not possible. Thus:

$$\cos \theta = -\frac{1}{2} \implies \theta = \frac{2\pi}{3}$$

To check for maximum, find the second derivative:

$$\frac{d^2 A}{d\theta^2} = 12(-\sin \theta + 4 \cos \theta \sin \theta) = 12 \sin \theta (4 \cos \theta - 1)$$

At $\theta = \frac{2\pi}{3}$:

$$\sin \left(\frac{2\pi}{3} \right) = \frac{\sqrt{3}}{2}, \quad \cos \left(\frac{2\pi}{3} \right) = -\frac{1}{2}$$

$$\begin{aligned}\frac{d^2 A}{d\theta^2} &= 12 \left(\frac{\sqrt{3}}{2} \right) \left(4 \left(-\frac{1}{2} \right) - 1 \right) \\ &= 6\sqrt{3}(-3) \\ &= -18\sqrt{3} < 0\end{aligned}$$

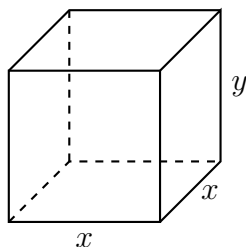
Since the second derivative is negative, the area is maximum at $\theta = \frac{2\pi}{3}$.

Calculate the maximum area:

$$\begin{aligned}A_{\max} &= 12 \sin \left(\frac{2\pi}{3} \right) \left(1 - \cos \left(\frac{2\pi}{3} \right) \right) \\ &= 12 \left(\frac{\sqrt{3}}{2} \right) \left(1 - \left(-\frac{1}{2} \right) \right) \\ &= 6\sqrt{3} \left(\frac{3}{2} \right) \\ &= 9\sqrt{3} \text{ sq. units}\end{aligned}$$

Final Answer $9\sqrt{3}$ sq. units**Question 28**

Show that the surface area of a closed cuboid with square base and given volume is minimum when it is a cube.

TYPE 120 Minimizing Surface Area of Closed Cuboid*Closed cuboid with square base*

To Prove: The surface area of the closed cuboid is minimum when it is a cube (i.e., $x = y$).

Concept / Formula Used: $V = x^2y$ $S = 2x^2 + 4xy$

Solution:

Let the side of the square base of the cuboid be x and its height be y .

The volume V of the cuboid is given to be constant:

$$V = x^2y$$

Solving for y in terms of x :

$$y = \frac{V}{x^2}$$

The total surface area S of the closed cuboid is:

$$S = 2x^2 + 4xy$$

Substitute $y = \frac{V}{x^2}$ into the surface area equation:

$$\begin{aligned} S(x) &= 2x^2 + 4x \left(\frac{V}{x^2} \right) \\ &= 2x^2 + \frac{4V}{x} \end{aligned}$$

Differentiating $S(x)$ with respect to x :

$$\frac{dS}{dx} = 4x - \frac{4V}{x^2}$$

For the minimum surface area, set $\frac{dS}{dx} = 0$:

$$\begin{aligned} 4x - \frac{4V}{x^2} &= 0 \\ 4x^3 &= 4V \\ x^3 &= V \end{aligned}$$

Substitute $V = x^2y$ back into the equation:

$$\begin{aligned} x^3 &= x^2y \\ x &= y \end{aligned}$$

To confirm that this is a minimum, find the second derivative:

$$\frac{d^2S}{dx^2} = 4 + \frac{8V}{x^3}$$

Since $x > 0$ and $V > 0$, the second derivative is positive:

$$\frac{d^2S}{dx^2} > 0$$

Thus, the surface area is minimum when $x = y$, which means the cuboid is a cube.

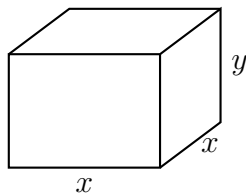
Hence Proved

The surface area is minimum when the cuboid is a cube.

Question 29

An open box with a square base is to be made of given iron sheet of area 27 sq. m. Show that the maximum volume of the box is 13.5 cu. m.

TYPE 121 Maximizing Volume of Open Box



Open box with square base

To Prove: The maximum volume of the box is 13.5 m^3 .

Concept / Formula Used: $S = x^2 + 4xy = 27$ $V = x^2y$

Solution:

Let the side of the square base of the open box be x and its height be y .

The surface area of the open box (which has no top) is given as:

$$S = x^2 + 4xy = 27$$

Solving for y in terms of x :

$$\begin{aligned} 4xy &= 27 - x^2 \\ y &= \frac{27 - x^2}{4x} \end{aligned}$$

The volume V of the box is:

$$\begin{aligned} V &= x^2y \\ &= x^2 \left(\frac{27 - x^2}{4x} \right) \\ &= \frac{x(27 - x^2)}{4} \\ &= \frac{27x - x^3}{4} \end{aligned}$$

Differentiating V with respect to x :

$$\frac{dV}{dx} = \frac{27 - 3x^2}{4}$$

For the maximum volume, set $\frac{dV}{dx} = 0$:

$$\begin{aligned} \frac{27 - 3x^2}{4} &= 0 \\ 27 - 3x^2 &= 0 \\ 3x^2 &= 27 \\ x^2 &= 9 \\ x &= 3 \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\frac{d^2V}{dx^2} = \frac{-6x}{4} = -1.5x$$

At $x = 3$:

$$\frac{d^2V}{dx^2} = -1.5(3) = -4.5 < 0$$

Since the second derivative is negative, the volume is maximum at $x = 3$.

Calculate the maximum volume:

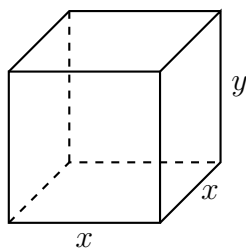
$$\begin{aligned} V_{\max} &= \frac{27(3) - 3^3}{4} \\ &= \frac{81 - 27}{4} \\ &= \frac{54}{4} \\ &= 13.5 \text{ m}^3 \end{aligned}$$

Hence Proved

The maximum volume of the box is 13.5 cu. m.

Question 30

The volume of a closed metal box with a square base is 4096 cm^3 . The cost of polishing the outer surface of the box is Rs. 4 per cm^2 . Find the dimensions of the box for the minimum cost of polishing it.

TYPE 122 Minimizing Polishing Cost of Closed Box

Closed box with square base

Given: Volume $V = 4096 \text{ cm}^3$, cost rate is Rs. 4/ cm^2 .

To Find: The dimensions of the box for minimum cost.

Concept / Formula Used: $V = x^2y$ $S = 2x^2 + 4xy$ $C = 4S$

Solution:

Let the side of the square base of the closed box be x cm and its height be y cm.

The volume of the box is:

$$V = x^2y = 4096 \implies y = \frac{4096}{x^2}$$

The total surface area S of the closed box is:

$$S = 2x^2 + 4xy$$

The cost of polishing C is:

$$C = 4S = 4(2x^2 + 4xy) = 8x^2 + 16xy$$

Substitute $y = \frac{4096}{x^2}$ into the cost equation:

$$\begin{aligned} C(x) &= 8x^2 + 16x \left(\frac{4096}{x^2} \right) \\ &= 8x^2 + \frac{65536}{x} \end{aligned}$$

Differentiating $C(x)$ with respect to x :

$$\frac{dC}{dx} = 16x - \frac{65536}{x^2}$$

For the minimum cost, set $\frac{dC}{dx} = 0$:

$$\begin{aligned}16x - \frac{65536}{x^2} &= 0 \\16x^3 &= 65536 \\x^3 &= 4096 \\x &= 16 \text{ cm}\end{aligned}$$

Finding the second derivative:

$$\frac{d^2C}{dx^2} = 16 + \frac{131072}{x^3}$$

At $x = 16$:

$$\begin{aligned}\frac{d^2C}{dx^2} &= 16 + \frac{131072}{16^3} \\&= 16 + \frac{131072}{4096} \\&= 16 + 32 \\&= 48 > 0\end{aligned}$$

Since the second derivative is positive, the cost is minimum at $x = 16$.

Now, find the height y :

$$y = \frac{4096}{16^2} = \frac{4096}{256} = 16 \text{ cm}$$

Thus, the dimensions of the box for minimum cost are $16 \text{ cm} \times 16 \text{ cm} \times 16 \text{ cm}$.

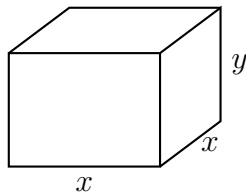
Final Answer

$16 \text{ cm} \times 16 \text{ cm} \times 16 \text{ cm}$

Question 31

An open tank with a square base and vertical sides is to be constructed from a metal sheet so as to hold a given quantity of water. Show that the cost of material will be least when the depth of the tank is half of its width.

TYPE 123 Minimizing Surface Area of Open Tank



Open tank with square base

To Prove: The cost of material (surface area) is least when the depth y is half of its width x (i.e., $y = \frac{x}{2}$).

Concept / Formula Used: $V = x^2y = \text{constant}$ $S = x^2 + 4xy$

Solution:

Let the side of the square base (width) of the open tank be x and its depth be y .

The volume V of water to be held is constant:

$$V = x^2y \implies y = \frac{V}{x^2}$$

The cost of material is proportional to the total surface area S of the open tank:

$$S = x^2 + 4xy$$

Substitute $y = \frac{V}{x^2}$ into the surface area equation:

$$\begin{aligned} S(x) &= x^2 + 4x \left(\frac{V}{x^2} \right) \\ &= x^2 + \frac{4V}{x} \end{aligned}$$

Differentiating $S(x)$ with respect to x :

$$\frac{dS}{dx} = 2x - \frac{4V}{x^2}$$

For the minimum surface area, set $\frac{dS}{dx} = 0$:

$$\begin{aligned} 2x - \frac{4V}{x^2} &= 0 \\ 2x^3 &= 4V \\ x^3 &= 2V \end{aligned}$$

Substitute $V = x^2y$ back into the equation:

$$\begin{aligned} x^3 &= 2(x^2y) \\ x &= 2y \\ y &= \frac{x}{2} \end{aligned}$$

To confirm that this is a minimum, find the second derivative:

$$\frac{d^2S}{dx^2} = 2 + \frac{8V}{x^3}$$

Since $x > 0$ and $V > 0$, the second derivative is positive:

$$\frac{d^2S}{dx^2} > 0$$

Thus, the cost of material is least when the depth of the tank is half of its width.

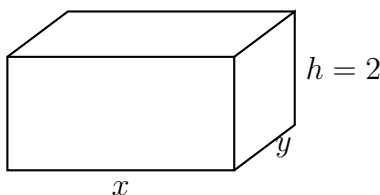
Hence Proved

The cost of material is least when the depth is half of its width.

Question 32

A tank with rectangular base and rectangular sides, open at the top, is to be constructed so that its depth is 2 m and volume is 8 m^3 . If building of tank costs Rs. 70 per square metre for the base and Rs. 45 per square metre for the sides, what is the cost of least expensive tank?

TYPE 124 Minimizing Construction Cost of Open Rectangular Tank



Open rectangular tank

Given: Depth $h = 2 \text{ m}$, volume $V = 8 \text{ m}^3$, cost of base is Rs. $70/\text{m}^2$, cost of sides is Rs. $45/\text{m}^2$.

To Find: The cost of the least expensive tank.

Concept / Formula Used: $V = xyh$ $C = 70(xy) + 45(2xh + 2yh)$

Solution:

Let the length and width of the rectangular base be x and y respectively. The depth is given as $h = 2 \text{ m}$.

The volume of the tank is:

$$V = xyh = 2xy = 8 \implies xy = 4 \implies y = \frac{4}{x}$$

The cost of building the base is:

$$\text{Cost}_{\text{base}} = 70 \times xy = 70(4) = 280$$

The cost of building the four sides is:

$$\text{Cost}_{\text{sides}} = 45 \times (2xh + 2yh) = 45 \times (4x + 4y) = 180(x + y)$$

The total cost C is:

$$C = 280 + 180(x + y)$$

Substitute $y = \frac{4}{x}$ into the cost equation:

$$C(x) = 280 + 180 \left(x + \frac{4}{x} \right)$$

Differentiating $C(x)$ with respect to x :

$$\frac{dC}{dx} = 180 \left(1 - \frac{4}{x^2} \right)$$

For the minimum cost, set $\frac{dC}{dx} = 0$:

$$\begin{aligned} 180 \left(1 - \frac{4}{x^2} \right) &= 0 \\ 1 - \frac{4}{x^2} &= 0 \\ x^2 &= 4 \\ x &= 2 \quad (\text{since } x > 0) \end{aligned}$$

Finding the second derivative:

$$\frac{d^2C}{dx^2} = 180 \left(\frac{8}{x^3} \right)$$

At $x = 2$:

$$\frac{d^2C}{dx^2} = 180 \left(\frac{8}{8} \right) = 180 > 0$$

Since the second derivative is positive, the cost is minimum at $x = 2$.

Calculate the minimum cost:

$$\begin{aligned} C(2) &= 280 + 180 \left(2 + \frac{4}{2} \right) \\ &= 280 + 180(4) \\ &= 280 + 720 \\ &= 1000 \end{aligned}$$

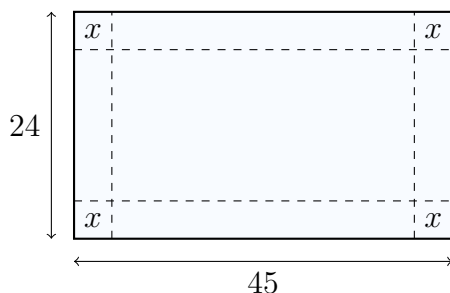
Final Answer

Rs. 1000

Question 33

A rectangular sheet of tin 45 cm by 24 cm is to be made into a box without top by cutting off squares from the corners and folding up the flaps. What should be the side of the square in order that the volume of the box is maximum?

TYPE 125 Maximizing Volume of Box from Rectangular Sheet



Tin sheet with cut-out corners

Given: Dimensions of the sheet are 45 cm $\times$ 24 cm.

To Find: The side of the square x to be cut off for maximum volume.

Concept / Formula Used: $V = \text{length} \times \text{width} \times \text{height} = (45 - 2x)(24 - 2x)x$

Solution:

Let the side of the square cut off from each corner be x cm.

The dimensions of the open box will be: - Length: $l = 45 - 2x$ - Width: $w = 24 - 2x$ - Height: $h = x$

For the dimensions to be positive, we must have:

$$24 - 2x > 0 \implies x < 12$$

The volume V of the box is:

$$\begin{aligned} V(x) &= x(45 - 2x)(24 - 2x) \\ &= x(1080 - 90x - 48x + 4x^2) \\ &= x(1080 - 138x + 4x^2) \\ &= 4x^3 - 138x^2 + 1080x \end{aligned}$$

Differentiating $V(x)$ with respect to x :

$$\frac{dV}{dx} = 12x^2 - 276x + 1080$$

For maximum volume, set $\frac{dV}{dx} = 0$:

$$12x^2 - 276x + 1080 = 0$$

$$12(x^2 - 23x + 90) = 0$$

$$x^2 - 23x + 90 = 0$$

$$(x - 18)(x - 5) = 0$$

This gives $x = 18$ or $x = 5$. Since $x < 12$, we reject $x = 18$. Thus, $x = 5$.

Finding the second derivative:

$$\frac{d^2V}{dx^2} = 24x - 276$$

At $x = 5$:

$$\frac{d^2V}{dx^2} = 24(5) - 276 = 120 - 276 = -156 < 0$$

Since the second derivative is negative, the volume is maximum when $x = 5$ cm.

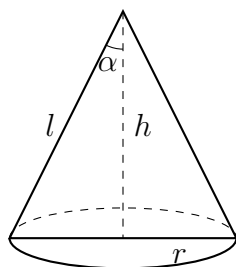
Final Answer

5 cm

Question 34

Show that the semi-vertical angle of a cone of the maximum volume and of given slant height is $\cos^{-1} \frac{1}{\sqrt{3}}$.

TYPE 126 Maximizing Volume of Cone with Constant Slant Height



Cone with semi-vertical angle α

To Prove: The semi-vertical angle α of the cone of maximum volume is $\cos^{-1} \frac{1}{\sqrt{3}}$.

Concept / Formula Used: $V = \frac{1}{3}\pi r^2 h$ $r = l \sin \alpha$ $h = l \cos \alpha$

Solution:

Let l be the given slant height of the cone, and α be its semi-vertical angle.

The radius r and height h of the cone are:

$$r = l \sin \alpha, \quad h = l \cos \alpha$$

The volume V of the cone is:

$$\begin{aligned} V &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3}\pi (l \sin \alpha)^2 (l \cos \alpha) \\ &= \frac{1}{3}\pi l^3 \sin^2 \alpha \cos \alpha \end{aligned}$$

Using the identity $\sin^2 \alpha = 1 - \cos^2 \alpha$:

$$V = \frac{1}{3}\pi l^3 (1 - \cos^2 \alpha) \cos \alpha = \frac{1}{3}\pi l^3 (\cos \alpha - \cos^3 \alpha)$$

Differentiating V with respect to α :

$$\begin{aligned} \frac{dV}{d\alpha} &= \frac{1}{3}\pi l^3 (-\sin \alpha - 3 \cos^2 \alpha (-\sin \alpha)) \\ &= \frac{1}{3}\pi l^3 \sin \alpha (3 \cos^2 \alpha - 1) \end{aligned}$$

For maximum volume, set $\frac{dV}{d\alpha} = 0$: Since $\alpha \in (0, \pi/2)$, $\sin \alpha \neq 0$. Thus:

$$\begin{aligned} 3 \cos^2 \alpha - 1 &= 0 \\ \cos^2 \alpha &= \frac{1}{3} \\ \cos \alpha &= \frac{1}{\sqrt{3}} \\ \alpha &= \cos^{-1} \left(\frac{1}{\sqrt{3}} \right) \end{aligned}$$

To confirm that this is a maximum, find the second derivative:

$$\begin{aligned} \frac{d^2V}{d\alpha^2} &= \frac{1}{3}\pi l^3 \frac{d}{d\alpha} [\sin \alpha (3 \cos^2 \alpha - 1)] \\ &= \frac{1}{3}\pi l^3 [\cos \alpha (3 \cos^2 \alpha - 1) + \sin \alpha (-6 \cos \alpha \sin \alpha)] \\ &= \frac{1}{3}\pi l^3 [3 \cos^3 \alpha - \cos \alpha - 6 \cos \alpha \sin^2 \alpha] \end{aligned}$$

At $\cos \alpha = \frac{1}{\sqrt{3}}$ and $\sin^2 \alpha = 1 - \cos^2 \alpha = \frac{2}{3}$:

$$\begin{aligned} \frac{d^2V}{d\alpha^2} &= \frac{1}{3}\pi l^3 \left[3 \left(\frac{1}{3\sqrt{3}} \right) - \frac{1}{\sqrt{3}} - 6 \left(\frac{1}{\sqrt{3}} \right) \left(\frac{2}{3} \right) \right] \\ &= \frac{1}{3}\pi l^3 \left[\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}} - \frac{4}{\sqrt{3}} \right] \\ &= -\frac{4\pi l^3}{3\sqrt{3}} < 0 \end{aligned}$$

Since the second derivative is negative, the volume is maximum when $\alpha = \cos^{-1} \frac{1}{\sqrt{3}}$.

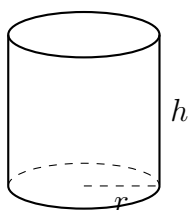
Hence Proved

The semi-vertical angle is $\cos^{-1} \frac{1}{\sqrt{3}}$.

Question 35

Show that a right circular cylinder of given volume, which is open at the top, has minimum total surface area if its height is equal to radius of its base.

TYPE 127 Minimizing Surface Area of Open Cylinder with Constant Volume



Open cylinder

To Prove: The total surface area of the open cylinder is minimum when $h = r$.

Concept / Formula Used: $V = \pi r^2 h = \text{constant}$ $S = \pi r^2 + 2\pi r h$

Solution:

Let r be the radius of the base and h be the height of the open cylinder.

The volume V is given to be constant:

$$V = \pi r^2 h \implies h = \frac{V}{\pi r^2}$$

The total surface area S of the cylinder (open at the top) is:

$$S = \pi r^2 + 2\pi r h$$

Substitute $h = \frac{V}{\pi r^2}$ into the surface area equation:

$$\begin{aligned} S(r) &= \pi r^2 + 2\pi r \left(\frac{V}{\pi r^2} \right) \\ &= \pi r^2 + \frac{2V}{r} \end{aligned}$$

Differentiating $S(r)$ with respect to r :

$$\frac{dS}{dr} = 2\pi r - \frac{2V}{r^2}$$

For the minimum surface area, set $\frac{dS}{dr} = 0$:

$$\begin{aligned} 2\pi r - \frac{2V}{r^2} &= 0 \\ 2\pi r^3 &= 2V \\ \pi r^3 &= V \end{aligned}$$

Substitute $V = \pi r^2 h$ back into the equation:

$$\begin{aligned} \pi r^3 &= \pi r^2 h \\ r &= h \end{aligned}$$

To confirm that this is a minimum, find the second derivative:

$$\frac{d^2 S}{dr^2} = 2\pi + \frac{4V}{r^3}$$

Since $r > 0$ and $V > 0$, the second derivative is positive:

$$\frac{d^2 S}{dr^2} > 0$$

Thus, the total surface area is minimum when the height is equal to the radius of its base ($h = r$).

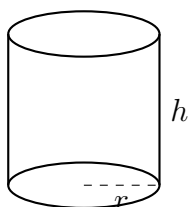
Hence Proved

The total surface area is minimum when $h = r$.

Question 36

- (i) Show that a closed cylindrical vessel of given volume has the least (total) surface area when its height is twice its radius.
- (ii) A closed right circular cylinder has a volume of 2156 cu. cm. What will be the radius of its base so that its total surface area is minimum? Find the height of the cylinder when its total surface area is minimum. Take $\pi = \frac{22}{7}$.

TYPE 128 Minimizing Surface Area of Closed Cylinder with Constant Volume



Closed cylinder

Given: For (ii), Volume $V = 2156 \text{ cm}^3$ and $\pi = \frac{22}{7}$.

To Find: For (ii), the radius r and height h for minimum surface area.

Concept / Formula Used: $V = \pi r^2 h$ $S = 2\pi r^2 + 2\pi r h$

Solution:

(i) Let r be the radius and h be the height of the closed cylinder.

The volume V is constant:

$$V = \pi r^2 h \implies h = \frac{V}{\pi r^2}$$

The total surface area S is:

$$S = 2\pi r^2 + 2\pi r h$$

Substitute $h = \frac{V}{\pi r^2}$ into the surface area equation:

$$\begin{aligned} S(r) &= 2\pi r^2 + 2\pi r \left(\frac{V}{\pi r^2} \right) \\ &= 2\pi r^2 + \frac{2V}{r} \end{aligned}$$

Differentiating $S(r)$ with respect to r :

$$\frac{dS}{dr} = 4\pi r - \frac{2V}{r^2}$$

For the minimum surface area, set $\frac{dS}{dr} = 0$:

$$\begin{aligned} 4\pi r - \frac{2V}{r^2} &= 0 \\ 4\pi r^3 &= 2V \\ 2\pi r^3 &= V \end{aligned}$$

Substitute $V = \pi r^2 h$ back into the equation:

$$\begin{aligned} 2\pi r^3 &= \pi r^2 h \\ 2r &= h \end{aligned}$$

Finding the second derivative:

$$\frac{d^2 S}{dr^2} = 4\pi + \frac{4V}{r^3}$$

Since $r > 0$ and $V > 0$, the second derivative is positive:

$$\frac{d^2 S}{dr^2} > 0$$

Thus, the surface area is minimum when $h = 2r$.

(ii) From part (i), the surface area is minimum when:

$$2\pi r^3 = V$$

Substitute $V = 2156$ and $\pi = \frac{22}{7}$:

$$\begin{aligned} 2 \times \frac{22}{7} \times r^3 &= 2156 \\ \frac{44}{7} r^3 &= 2156 \\ r^3 &= 2156 \times \frac{7}{44} \\ r^3 &= 49 \times 7 \\ r^3 &= 343 \\ r &= 7 \text{ cm} \end{aligned}$$

Now, find the height h :

$$h = 2r = 2(7) = 14 \text{ cm}$$

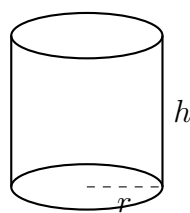
Final Answer

- (i) Proved that $h = 2r$
- (ii) Radius $r = 7$ cm, Height $h = 14$ cm

Question 37

Show that the radius of a closed right circular cylinder of given surface area and maximum volume is equal to half of its height.

TYPE 129 Maximizing Volume of Closed Cylinder with Constant Surface Area



Closed cylinder

To Prove: The radius r of the cylinder of maximum volume is equal to half of its height (i.e., $r = \frac{h}{2}$).

Concept / Formula Used: $S = 2\pi r^2 + 2\pi rh = \text{constant}$ $V = \pi r^2 h$

Solution:

Let r be the radius and h be the height of the closed cylinder.

The total surface area S is constant:

$$S = 2\pi r^2 + 2\pi rh \implies h = \frac{S - 2\pi r^2}{2\pi r}$$

The volume V of the cylinder is:

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi r^2 \left(\frac{S - 2\pi r^2}{2\pi r} \right) \\ &= \frac{r(S - 2\pi r^2)}{2} \\ &= \frac{Sr - 2\pi r^3}{2} \end{aligned}$$

Differentiating V with respect to r :

$$\frac{dV}{dr} = \frac{S - 6\pi r^2}{2}$$

For maximum volume, set $\frac{dV}{dr} = 0$:

$$\begin{aligned} \frac{S - 6\pi r^2}{2} &= 0 \\ S &= 6\pi r^2 \end{aligned}$$

Substitute $S = 2\pi r^2 + 2\pi rh$ back into the equation:

$$\begin{aligned} 2\pi r^2 + 2\pi rh &= 6\pi r^2 \\ 2\pi rh &= 4\pi r^2 \\ h &= 2r \\ r &= \frac{h}{2} \end{aligned}$$

To confirm that this is a maximum, find the second derivative:

$$\frac{d^2V}{dr^2} = \frac{-12\pi r}{2} = -6\pi r$$

Since $r > 0$, the second derivative is negative:

$$\frac{d^2V}{dr^2} < 0$$

Thus, the volume is maximum when the radius is equal to half of its height.

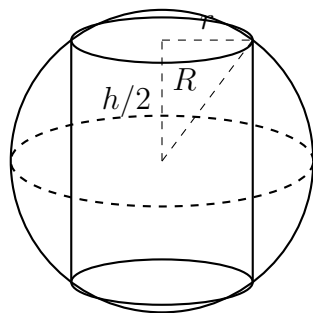
Hence Proved

The radius is equal to half of its height.

Question 38

Find the maximum volume of the cylinder which can be inscribed in a sphere of radius $3\sqrt{3}$ cm. (Leave the answer in terms of π).

TYPE 130 Maximizing Volume of Cylinder Inscribed in a Sphere



Cylinder inscribed in a sphere

Given: Radius of the sphere $R = 3\sqrt{3}$ cm.

To Find: The maximum volume of the inscribed cylinder.

Concept / Formula Used: $r^2 + \left(\frac{h}{2}\right)^2 = R^2$ $V = \pi r^2 h$

Solution:

Let $R = 3\sqrt{3}$ cm be the radius of the sphere. Let r and h be the radius and height of the inscribed cylinder.

From the right-angled triangle formed by the center of the sphere, the center of the cylinder's top face, and a point on the rim of the cylinder's top face:

$$r^2 + \left(\frac{h}{2}\right)^2 = R^2 \implies r^2 = R^2 - \frac{h^2}{4}$$

Substitute $R = 3\sqrt{3}$:

$$r^2 = (3\sqrt{3})^2 - \frac{h^2}{4} = 27 - \frac{h^2}{4}$$

The volume V of the cylinder is:

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi \left(27 - \frac{h^2}{4}\right) h \\ &= \pi \left(27h - \frac{h^3}{4}\right) \end{aligned}$$

Differentiating V with respect to h :

$$\frac{dV}{dh} = \pi \left(27 - \frac{3h^2}{4} \right)$$

For maximum volume, set $\frac{dV}{dh} = 0$:

$$\begin{aligned} 27 - \frac{3h^2}{4} &= 0 \\ \frac{3h^2}{4} &= 27 \\ h^2 &= 36 \\ h &= 6 \text{ cm} \quad (\text{since } h > 0) \end{aligned}$$

Finding the second derivative:

$$\frac{d^2V}{dh^2} = \pi \left(-\frac{6h}{4} \right) = -1.5\pi h$$

At $h = 6$:

$$\frac{d^2V}{dh^2} = -9\pi < 0$$

Since the second derivative is negative, the volume is maximum when $h = 6$ cm.

Calculate the maximum volume:

$$\begin{aligned} V_{\max} &= \pi \left(27(6) - \frac{6^3}{4} \right) \\ &= \pi \left(162 - \frac{216}{4} \right) \\ &= \pi(162 - 54) \\ &= 108\pi \text{ cm}^3 \end{aligned}$$

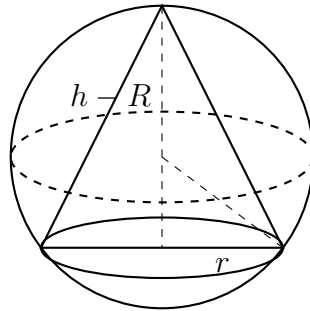
Final Answer

$108\pi \text{ cm}^3$

Question 39

A cone is inscribed in a sphere of radius 12 cm. If the volume of the cone is maximum, find its height.

TYPE 131 Maximizing Volume of Cone Inscribed in a Sphere



Cone inscribed in a sphere

Given: Radius of the sphere $R = 12$ cm.

To Find: The height h of the cone for maximum volume.

Concept / Formula Used: $r^2 + (h - R)^2 = R^2 \implies r^2 = 2Rh - h^2$ $V = \frac{1}{3}\pi r^2 h$

Solution:

Let $R = 12$ cm be the radius of the sphere. Let r and h be the radius and height of the inscribed cone.

From the geometry of the inscribed cone in the sphere:

$$r^2 + (h - R)^2 = R^2 \implies r^2 = 2Rh - h^2$$

Substitute $R = 12$:

$$r^2 = 24h - h^2$$

The volume V of the cone is:

$$\begin{aligned} V &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3}\pi(24h - h^2)h \\ &= \frac{1}{3}\pi(24h^2 - h^3) \end{aligned}$$

Differentiating V with respect to h :

$$\begin{aligned} \frac{dV}{dh} &= \frac{1}{3}\pi(48h - 3h^2) \\ &= \pi(16h - h^2) \end{aligned}$$

For maximum volume, set $\frac{dV}{dh} = 0$:

$$\begin{aligned} \pi(16h - h^2) &= 0 \\ h(16 - h) &= 0 \end{aligned}$$

Since $h > 0$, we have:

$$h = 16 \text{ cm}$$

Finding the second derivative:

$$\frac{d^2V}{dh^2} = \pi(16 - 2h)$$

At $h = 16$:

$$\frac{d^2V}{dh^2} = \pi(16 - 32) = -16\pi < 0$$

Since the second derivative is negative, the volume is maximum when $h = 16$ cm.

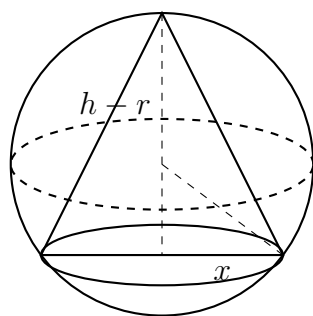
Final Answer

16 cm

Question 40

Show that the altitude of a right circular cone of maximum curved surface area which can be inscribed in a sphere of radius r is $\frac{4r}{3}$.

TYPE 132 Maximizing Curved Surface Area of Cone Inscribed in a Sphere



Cone inscribed in a sphere

To Prove: The altitude h of the cone of maximum curved surface area is $\frac{4r}{3}$.

Concept / Formula Used: $x^2 + (h - r)^2 = r^2 \implies x^2 = 2rh - h^2$ $l^2 = x^2 + h^2 = 2rh$ $C = \pi xl$

Solution:

Let r be the radius of the sphere. Let x , h , and l be the base radius, height (altitude), and slant height of the inscribed cone respectively.

From the right-angled triangle in the sphere:

$$x^2 + (h - r)^2 = r^2 \implies x^2 = 2rh - h^2$$

The slant height l is given by:

$$l^2 = x^2 + h^2 = (2rh - h^2) + h^2 = 2rh$$

The curved surface area C of the cone is:

$$C = \pi x l$$

To maximize C , we can maximize $Z = C^2$:

$$\begin{aligned} Z &= \pi^2 x^2 l^2 \\ &= \pi^2 (2rh - h^2)(2rh) \\ &= 2\pi^2 r(2rh^2 - h^3) \end{aligned}$$

Differentiating Z with respect to h :

$$\frac{dZ}{dh} = 2\pi^2 r(4rh - 3h^2)$$

For maximum curved surface area, set $\frac{dZ}{dh} = 0$:

$$\begin{aligned} 4rh - 3h^2 &= 0 \\ h(4r - 3h) &= 0 \end{aligned}$$

Since $h > 0$, we have:

$$h = \frac{4r}{3}$$

Finding the second derivative:

$$\frac{d^2Z}{dh^2} = 2\pi^2 r(4r - 6h)$$

At $h = \frac{4r}{3}$:

$$\begin{aligned} \frac{d^2Z}{dh^2} &= 2\pi^2 r \left(4r - 6 \left(\frac{4r}{3} \right) \right) \\ &= 2\pi^2 r(4r - 8r) \\ &= -8\pi^2 r^2 < 0 \end{aligned}$$

Since the second derivative is negative, the curved surface area is maximum when the altitude is $h = \frac{4r}{3}$.

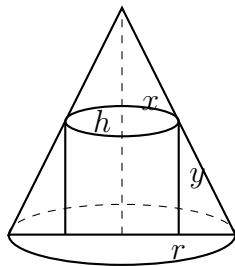
Hence Proved

The altitude of the cone is $\frac{4r}{3}$.

Question 41

Show that the height of a right circular cylinder of greatest volume which can be inscribed in a right circular cone of height h and radius r is one-third of the height of the cone and the greatest volume of the cylinder is $\frac{4}{9}$ times the volume of the cone.

TYPE 133 Maximizing Volume of Cylinder Inscribed in a Cone



Cylinder inscribed in a cone

To Prove: The height of the cylinder of greatest volume is $\frac{h}{3}$ and its volume is $\frac{4}{9}$ times the volume of the cone.

Concept / Formula Used: $\frac{h-y}{h} = \frac{x}{r} \implies y = h \left(1 - \frac{x}{r}\right)$ $V = \pi x^2 y$ $V_{\text{cone}} = \frac{1}{3} \pi r^2 h$

Solution:

Let r and h be the radius and height of the given cone. Let x and y be the radius and height of the inscribed cylinder.

Using similar triangles:

$$\frac{h-y}{h} = \frac{x}{r} \implies y = h \left(1 - \frac{x}{r}\right)$$

The volume V of the cylinder is

$$V = \pi x^2 y$$

Substituting $y = h \left(1 - \frac{x}{r}\right)$:

$$V = \pi x^2 \cdot h \left(1 - \frac{x}{r}\right) = \pi h \left(x^2 - \frac{x^3}{r}\right)$$

Differentiating with respect to x :

$$\frac{dV}{dx} = \pi h \left(2x - \frac{3x^2}{r}\right) = \pi h x \left(2 - \frac{3x}{r}\right)$$

Setting $\frac{dV}{dx} = 0$:

$$\pi h x \left(2 - \frac{3x}{r}\right) = 0$$

Since $x \neq 0$ (the cylinder has positive radius), we require

$$2 - \frac{3x}{r} = 0 \implies x = \frac{2r}{3}$$

Differentiating again to test for a maximum:

$$\frac{d^2V}{dx^2} = \pi h \left(2 - \frac{6x}{r}\right)$$

At $x = \frac{2r}{3}$:

$$\frac{d^2V}{dx^2} = \pi h \left(2 - \frac{6}{r} \cdot \frac{2r}{3} \right) = \pi h(2 - 4) = -2\pi h < 0$$

Since $\frac{d^2V}{dx^2} < 0$, the volume is maximum at $x = \frac{2r}{3}$.

Substituting back into $y = h \left(1 - \frac{x}{r} \right)$:

$$y = h \left(1 - \frac{2}{3} \right) = \frac{h}{3}$$

This proves the height of the cylinder of greatest volume is one-third the height of the cone.

Now computing the maximum volume:

$$\begin{aligned} V_{\max} &= \pi x^2 y = \pi \left(\frac{2r}{3} \right)^2 \cdot \frac{h}{3} \\ &= \pi \cdot \frac{4r^2}{9} \cdot \frac{h}{3} \\ &= \frac{4\pi r^2 h}{27} \end{aligned}$$

Comparing with the volume of the cone, $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$:

$$\begin{aligned} \frac{V_{\max}}{V_{\text{cone}}} &= \frac{\frac{4\pi r^2 h}{27}}{\frac{1}{3}\pi r^2 h} = \frac{4}{27} \times 3 = \frac{4}{9} \\ \therefore V_{\max} &= \frac{4}{9} V_{\text{cone}} \end{aligned}$$

Hence Proved

The height of the cylinder of greatest volume inscribed in the cone is $\frac{h}{3}$, and its greatest volume is $\frac{4}{9}$ times the volume of the cone.

Question 42

The sum of the surface areas of a rectangular parallelepiped with sides x , $2x$ and $\frac{x}{3}$ and a sphere is given to be constant. Prove that the sum of their volumes is minimum if x is equal to three times the radius of the sphere. Also find the minimum value of the sum of their volumes.

TYPE 115 Minimizing Combined Volume of Cuboid and Sphere

Given: Sides of the rectangular parallelepiped are x , $2x$, and $\frac{x}{3}$. Let the radius of the sphere be r . The sum of their surface areas is a constant, say S .

To Prove: The sum of their volumes V is minimum when $x = 3r$.

To Find: The minimum value of the sum of their volumes V .

Concept / Formula Used:

- Surface area of a rectangular parallelopiped: $S_{\text{cuboid}} = 2(ab + bc + ca)$
- Volume of a rectangular parallelopiped: $V_{\text{cuboid}} = abc$
- Surface area of a sphere: $S_{\text{sphere}} = 4\pi r^2$
- Volume of a sphere: $V_{\text{sphere}} = \frac{4}{3}\pi r^3$

Solution:

Let the sides of the rectangular parallelopiped be $a = x$, $b = 2x$, and $c = \frac{x}{3}$. Let the radius of the sphere be r .

The surface area of the rectangular parallelopiped is:

$$\begin{aligned} S_{\text{cuboid}} &= 2(ab + bc + ca) \\ &= 2\left(x \cdot 2x + 2x \cdot \frac{x}{3} + \frac{x}{3} \cdot x\right) \\ &= 2\left(2x^2 + \frac{2}{3}x^2 + \frac{1}{3}x^2\right) \\ &= 2(3x^2) \\ &= 6x^2 \end{aligned}$$

The surface area of the sphere is:

$$S_{\text{sphere}} = 4\pi r^2$$

The sum of their surface areas is given to be constant, say S :

$$S = 6x^2 + 4\pi r^2 = \text{constant} \quad (1)$$

Differentiating equation (1) with respect to x :

$$\begin{aligned} \frac{d}{dx}(6x^2 + 4\pi r^2) &= 0 \\ 12x + 8\pi r \frac{dr}{dx} &= 0 \\ 8\pi r \frac{dr}{dx} &= -12x \\ \frac{dr}{dx} &= -\frac{3x}{2\pi r} \end{aligned} \quad (2)$$

Now, let V be the sum of their volumes:

$$\begin{aligned} V &= V_{\text{cuboid}} + V_{\text{sphere}} \\ &= x \cdot 2x \cdot \frac{x}{3} + \frac{4}{3}\pi r^3 \\ &= \frac{2}{3}x^3 + \frac{4}{3}\pi r^3 \end{aligned} \quad (3)$$

Differentiating V with respect to x :

$$\begin{aligned}\frac{dV}{dx} &= \frac{d}{dx} \left(\frac{2}{3}x^3 + \frac{4}{3}\pi r^3 \right) \\ &= 2x^2 + 4\pi r^2 \frac{dr}{dx}\end{aligned}$$

Substituting the value of $\frac{dr}{dx}$ from equation (2):

$$\begin{aligned}\frac{dV}{dx} &= 2x^2 + 4\pi r^2 \left(-\frac{3x}{2\pi r} \right) \\ &= 2x^2 - 6xr \\ &= 2x(x - 3r)\end{aligned}$$

For the volume to be minimum, we must have $\frac{dV}{dx} = 0$:

$$2x(x - 3r) = 0$$

Since $x > 0$ (as it represents a physical dimension), we have:

$$\begin{aligned}x - 3r &= 0 \\ x &= 3r\end{aligned}$$

To confirm that this point corresponds to a minimum, we find the second derivative $\frac{d^2V}{dx^2}$:

$$\begin{aligned}\frac{d^2V}{dx^2} &= \frac{d}{dx} (2x^2 - 6xr) \\ &= 4x - 6 \left(r + x \frac{dr}{dx} \right)\end{aligned}$$

At $x = 3r$, substituting $\frac{dr}{dx} = -\frac{3(3r)}{2\pi r} = -\frac{9}{2\pi}$:

$$\begin{aligned}\frac{d^2V}{dx^2} &= 4(3r) - 6 \left(r + 3r \left(-\frac{9}{2\pi} \right) \right) \\ &= 12r - 6r + \frac{81r}{\pi} \\ &= 6r + \frac{81r}{\pi} \\ &= r \left(6 + \frac{81}{\pi} \right)\end{aligned}$$

Since $r > 0$, we have $\frac{d^2V}{dx^2} > 0$. Thus, the sum of their volumes is minimum when $x = 3r$.

Now, we find the minimum value of the sum of their volumes by substituting $x = 3r$ into equation (3):

$$\begin{aligned} V_{\min} &= \frac{2}{3}(3r)^3 + \frac{4}{3}\pi r^3 \\ &= \frac{2}{3}(27r^3) + \frac{4}{3}\pi r^3 \\ &= 18r^3 + \frac{4}{3}\pi r^3 \\ &= \frac{2}{3}(27r^3 + 2\pi r^3) \\ &= \frac{2}{3}(27 + 2\pi)r^3 \end{aligned}$$

Final Answer

The sum of the volumes is minimum when $x = 3r$ (i.e., x is equal to three times the radius of the sphere).

The minimum value of the sum of their volumes is:

$$V_{\min} = \frac{2}{3}(27 + 2\pi)r^3 \text{ cubic units}$$

where r is the radius of the sphere.

Question 43

A manufacturer plans to construct a cylindrical can to hold one cubic metre of oil. If the cost of constructing top and bottom of the can is twice the cost of constructing the side, what are the dimensions of the most economical can?

TYPE 134 Minimizing Cost of Cylindrical Can

Given: Volume of the cylindrical can $V = 1 \text{ m}^3$. Cost per unit area of the top and bottom is twice the cost per unit area of the curved side surface.

To Find: The radius r and height h of the cylinder that minimize the total cost of construction.

Concept / Formula Used: Volume of a cylinder $V = \pi r^2 h$, Curved surface area $S_{\text{side}} = 2\pi r h$, Area of top and bottom $S_{\text{ends}} = 2\pi r^2$.

Solution:

Let the radius of the cylindrical can be r metres and its height be h metres.

The volume V of the cylinder is given by:

$$\begin{aligned} V &= \pi r^2 h \\ 1 &= \pi r^2 h \end{aligned}$$

From this, we can express the height h in terms of r :

$$h = \frac{1}{\pi r^2} \quad (\text{i})$$

Let the cost of constructing the side surface per unit area be k (where $k > 0$). Then, the cost of constructing the top and bottom per unit area is $2k$.

The total cost of construction C is given by:

$$C = (\text{Cost of top and bottom}) + (\text{Cost of side})$$

$$C = (2k) \cdot (2\pi r^2) + k \cdot (2\pi r h)$$

$$C = 4k\pi r^2 + 2k\pi r h$$

$$C = 2k(2\pi r^2 + \pi r h)$$

To minimize the total cost C , we need to minimize the function:

$$f(r) = 2\pi r^2 + \pi r h$$

Substituting the value of h from equation (i) into $f(r)$:

$$f(r) = 2\pi r^2 + \pi r \left(\frac{1}{\pi r^2} \right)$$

$$f(r) = 2\pi r^2 + \frac{1}{r}$$

Now, we differentiate $f(r)$ with respect to r :

$$\frac{df}{dr} = \frac{d}{dr} (2\pi r^2 + r^{-1})$$

$$\frac{df}{dr} = 4\pi r - \frac{1}{r^2}$$

For a local extremum, we set $\frac{df}{dr} = 0$:

$$4\pi r - \frac{1}{r^2} = 0$$

$$4\pi r = \frac{1}{r^2}$$

$$r^3 = \frac{1}{4\pi}$$

$$r = \left(\frac{1}{4\pi} \right)^{1/3}$$

To verify whether this value of r gives a minimum, we find the second derivative $\frac{d^2f}{dr^2}$:

$$\frac{d^2f}{dr^2} = \frac{d}{dr} (4\pi r - r^{-2})$$

$$\frac{d^2f}{dr^2} = 4\pi + \frac{2}{r^3}$$

Since $r > 0$, we have:

$$\begin{aligned}\frac{d^2f}{dr^2} &= 4\pi + \frac{2}{\frac{1}{4\pi}} \\ &= 4\pi + 8\pi \\ &= 12\pi > 0\end{aligned}$$

Since the second derivative is positive, the cost is minimum when the radius is:

$$r = \left(\frac{1}{4\pi}\right)^{1/3} \text{ m}$$

Now, we substitute this value of r back into equation (i) to find the height h :

$$\begin{aligned}h &= \frac{1}{\pi r^2} \\ h &= \frac{1}{\pi \left[\left(\frac{1}{4\pi}\right)^{1/3}\right]^2} \\ h &= \frac{1}{\pi \left(\frac{1}{4\pi}\right)^{2/3}} \\ h &= \frac{1}{\pi \cdot \frac{1}{(4\pi)^{2/3}}} \\ h &= \frac{(4\pi)^{2/3}}{\pi} \\ h &= \frac{4^{2/3} \cdot \pi^{2/3}}{\pi} \\ h &= \frac{16^{1/3}}{\pi^{1/3}} \\ h &= \left(\frac{16}{\pi}\right)^{1/3} \text{ m}\end{aligned}$$

Thus, the dimensions of the most economical can are:

$$\text{Radius } r = \left(\frac{1}{4\pi}\right)^{1/3} \text{ metre}$$

$$\text{Height } h = \left(\frac{16}{\pi}\right)^{1/3} \text{ metre}$$

Final Answer

$$\text{Radius } r = \left(\frac{1}{4\pi}\right)^{1/3} \text{ m}$$

$$\text{Height } h = \left(\frac{16}{\pi}\right)^{1/3} \text{ m}$$

Common Mistake: A common error is setting the cost function directly as the total surface area $2\pi r^2 + 2\pi rh$ without accounting for the different cost factors of the top/bottom and the side. Always write the cost function with the given cost ratios incorporated.